MA'ADEN معادن
Saudi Arabian Mining Company شركة التعدين العربية السعودية

Project **Mansourah Massarah Gold Project**
Customer: **MA'ADEN GOLD AND BASE METALS (MA'ADEN)**

Title: **CALCULATION**
3650 POX ACID WASH,ELTN. & CARBON REGN. DESIGN OF MONORAIL STRUCTURE FDN

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DEFINITIONS AND ABBREVIATIONS

Document	:	Information and the medium on which it is contained (ISO 9000:2015)
EMPLOYER	:	'EMPLOYER' means MA'ADEN GOLD AND BASE METALS (MGBM) and the PMC American Bechtel Inc. or their representatives
Contractor	:	Contractor' means consortium composed of LARSEN & TOUBRO LIMITED, Saudi Arabia AND OUTOTEC SAUDI ARABIA LLC, Saudi Arabia (for IK Contract) and LARSEN & TOUBRO LIMITED – ECC Construction Group – (SHJ BR) and OUTOTEC ENGINEERING DMCC, Dubai (for OOK Contract).
'PMT'	:	Ma'aden Project Management Team
'MMGP Project'	:	Mansourah Massarah Gold Processing Project
Attribute	:	Information element that describes a document or tag.
Supplier	:	An entity (supplier, consultant, sub-contractor) under the direct project management of Contractor
Sub-Supplier	:	A sub entity (supplier, consultant, sub-contractor) under the direct project management of Contractor.
QA	:	Quality Assurance
QC	:	Quality Control
Works	:	Full scope of the EPC Contract including the engineering, procurement, construction, commissioning and associated activities
ELTN.	:	Elution
REGN.	:	Regeneration
FDN	:	Foundation

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1. INTENT

This document is to prepare design of foundation for **Pox Acid Wash/Elution/Carbon Regeneration monorail support structures.**The analysis & design is carried out in accordance with project specifications and codes & standards. The design calculation ensures the safety and serviceability of the foundation system.

2.EXECUTIVE SUMMARY

The Saudi Arabian Mining Company (Ma'aden) has announced that its wholly-owned subsidiary, Ma'aden Gold and Base Metals Company, have issued a Notice of Award to Outotec and Larsen & Toubro Consortium to complete the engineering, procurement, construction, pre-commissioning, commissioning and start-up assistance of the Mansourah & Massarah (MMGP) Gold Project.

MMGP project will become Ma'aden's largest ever gold project, producing an average of 250,000 ounces of gold per year over the life of mine. Mansourah-Massarah is situated within the Central Arabian Gold Region (CAGR), approximately 320km northeast of Taif towards Riyadh and about 35 km southeast of Hufairah, which is a small village situated on Jeddah - Riyadh highway. Mansourah - Massarah gold deposits are located 22°38'6.40"N; 42°44'22.90"E, and 22°42'1.40"N; 42°41'58.40"E respectively, approximately 537 km east of Jeddah and 77 km southeast of the town of Zalim.

The plant throughput capacity is 4000000t/a and is required to operate on a continuous basis at nominally 24 hours per day and 7 days a week.

3. SCOPE

The analysis has been carried out in STAAD PRO. For design of raft force values are directly taken from STAAD & designed in excel sheet. Column is designed in STAAD.

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4. DESIGN REQUIREMENTS

4.1 Design Load

All Design Loads are indicated in Attachment - 1. Also referred Civil, Structural and Architectural Design Criteria (Doc. No. MD-216-1000-EG-ST-CRT-1001) Clause 4.2.

5. CONCRETE FOUNDATIONS

5.1 Design

Design of concrete structures and foundations shall be in accordance with SBC-304, ACI 318M.

5.2 Materials

Concrete Strength of 28 days

- Foundation and Structural Members: 35 MPa
- Grade slab and pavement: 30 MPa
- Blinding Concrete below foundation bases: 15 Mpa

Reinforcement Grade

- Deformed Bar : ASTM A 615 / A 615M Grade 60 (fy = 420MPa)
- Welded Wire Fabric: ASTM A 185(Plain) / A 497(Deformed)

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5.3 Minimum Concrete Cover

Concrete Exposure	Member	Minimum Cover(mm)
Exposed to weather or contact with ground	Foundation	50
Exposed to weather or contact with ground	Column	50

5.4 Allowable Soil Bearing Capacity

The net safe bearing capacity of soil for foundation design has been considered 375 Kn/m^2 at 1.2 m below FGL. (Refer Table 5 - BH-11 of Geotechnical Report : MD-216-3000-EG-CV-RPT-1008).

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6. DESIGN CALCULATION FOR FOUNDATION

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DESIGN CALCULATION OF FOOTING F1

Size of Columns at node	Along-Z	Along-X
522	0.60m	0.60m

Size of fdn. raft:	2.00m	2.00m
Thickness of raft:	0.50	m
Original Gr. Level:	0.00	m
Finished Gr. Level:	0.00	m
U/S of B/Pl Above GL:	0.30	m
U/S level of fdn:	-1.20	m
Unit weight of conc:	24	kN/m^3
Unit weight of soil:	20	kN/m^3

Calculation of Weight of Foundation :

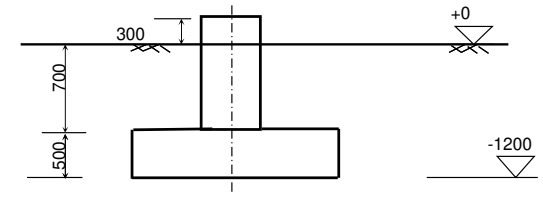
Weight of raft = (2*2*0.5*24) = 48.00KN

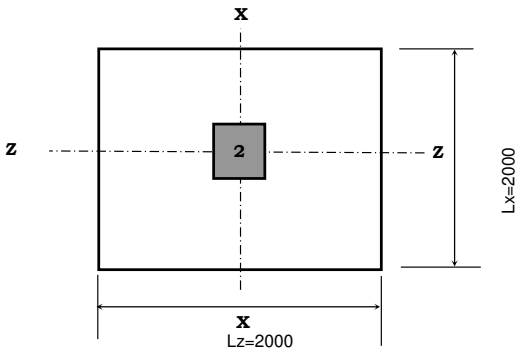
Weight of soil & Surcharge = 100 + ((2*2)-(0.6*0.6))*0.7*20 = 150.96KN

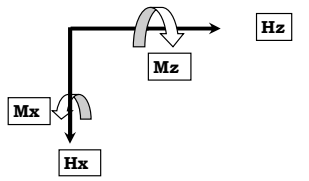
Total = 198.96KN

Design Input:

Characteristic strength of concrete, f'_c=	35 MPa
Yield stress of steel reinforcement, f_y=	420 MPa
Clear cover for main reinforcement bars =	50mm







H_x=Force along X-X direction

Hz=Force along Z-Z direction

M_x=Moment about X-X direction

Mz=Moment about Z-Z direction

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Check for overturning and sliding of footing F1

Load at bottom of fooring	Node 522				
L/C No.	V (kN)	Hx (kN)	Mz (kN-m)	Hz (kN)	Mx (kN-m)
201 1.0DL	41.20	0.03	-0.05	-5.00	-7.49
202 1.0DL + 1.0LL	174.32	-0.10	0.14	-30.00	-44.94
203 1.0DL + 1.0WLX1	56.40	-8.67	13.00	-5.02	-7.51
204 1.0DL + 1.0WLX2	26.00	8.73	-13.10	-4.98	-7.46
205 1.0DL + 1.0WLZ1	35.20	0.47	-0.71	-10.80	-16.19
206 1.0DL + 1.0WLZ2	47.20	-0.41	0.61	0.80	1.22
207 1.0DL + 0.75WLX1 + 0.75LL	152.44	-6.59	9.88	-23.77	-35.59
208 1.0DL + 0.75WLX2 + 0.75LL	129.64	6.46	-9.69	-23.74	-35.56
209 1.0DL + 0.75WLZ1 + 0.75LL	136.54	0.27	-0.40	-28.10	-42.10
210 1.0DL + 0.75WLZ2 + 0.75LL	145.54	-0.39	0.59	-19.40	-29.05
211 0.6DL + 1.0WLX1	39.92	-8.68	13.02	-3.02	-4.52
212 0.6DL + 1.0WLX2	9.52	8.72	-13.08	-2.98	-4.47
213 0.6DL + 1.0WLZ1	18.72	0.46	-0.69	-8.80	-13.19
214 0.6DL + 1.0WLZ2	30.72	-0.42	0.63	2.80	4.21
215 1.0DL + 1.0LL + 1.0TL1	172.32	1.00	-1.51	37.40	56.16
216 1.0DL + 1.0LL + 1.0TL2	176.32	-1.20	1.79	-97.40	-146.04
217 0.6DL + 0.7ELX	27.07	-0.68	1.02	-3.01	-4.50
218 0.6DL + 0.7ELZ	23.72	0.11	-0.16	-3.53	-5.28

Check for overturning and sliding:

L/C No.	Resultant Load at bottom of footing								
	V (kN)	Hz (kN)	Mx (kN-m)	Hx (kN)	Mz (kN-m)	Overturning		Sliding	
						F.O.S. Mz	F.O.S. Mx	F.O.S. Hx	F.O.S. Hz
201 1.0DL	240.2	-5.00	-7.5	0.0	-0.1	4709.0	32.1	2825.4	19.2
202 1.0DL + 1.0LL	373.3	-30.00	-44.9	-0.1	0.1	2592.2	8.3	1555.3	5.0
203 1.0DL + 1.0WLX1	255.4	-5.02	-7.5	-8.7	13.0	19.6	34.0	11.8	20.3
204 1.0DL + 1.0WLX2	225.0	-4.98	-7.5	8.7	-13.1	17.2	30.1	10.3	18.1
205 1.0DL + 1.0WLZ1	234.2	-10.80	-16.2	0.5	-0.7	329.3	14.5	197.6	8.7
206 1.0DL + 1.0WLZ2	246.2	0.80	1.2	-0.4	0.6	404.2	202.6	242.5	123.1
207 1.0DL + 0.75WLX1 + 0.75LL	351.4	-23.77	-35.6	-6.6	9.9	35.6	9.9	21.3	5.9
208 1.0DL + 0.75WLX2 + 0.75LL	328.6	-23.74	-35.6	6.5	-9.7	33.9	9.2	20.3	5.5
209 1.0DL + 0.75WLZ1 + 0.75LL	335.5	-28.10	-42.1	0.3	-0.4	838.8	8.0	502.6	4.8
210 1.0DL + 0.75WLZ2 + 0.75LL	344.5	-19.40	-29.0	-0.4	0.6	583.9	11.9	349.7	7.1
211 0.6DL + 1.0WLX1	238.9	-3.02	-4.5	-8.7	13.0	18.3	52.9	11.0	31.6
212 0.6DL + 1.0WLX2	208.5	-2.98	-4.5	8.7	-13.1	15.9	46.7	9.6	28.0
213 0.6DL + 1.0WLZ1	217.7	-8.80	-13.2	0.5	-0.7	315.0	16.5	189.3	9.9
214 0.6DL + 1.0WLZ2	229.7	2.80	4.2	-0.4	0.6	365.2	54.6	218.7	32.8
215 1.0DL + 1.0LL + 1.0TL1	371.3	37.40	56.2	1.0	-1.5	246.5	6.6	147.9	4.0
216 1.0DL + 1.0LL + 1.0TL2	375.3	-97.40	-146.0	-1.2	1.8	209.2	2.6	125.5	1.5
217 0.6DL + 0.7ELX	226.0	-3.01	-4.5	-0.7	1.0	221.8	50.2	133.0	30.1
218 0.6DL + 0.7ELZ	222.7	-3.53	-5.3	0.1	-0.2	1357.8	42.2	817.2	25.3

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Base pressure check of footing F1

	Node	L/C	Horizontal Fx kN/m2	Vertical Fy kN/m2	Horizontal Fz kN/m2
Max Px	456	201	0	60.71	0
Min Px	456	201	0	60.71	0
Max Py	487	216	0	199.92	0
Min Py	797	216	0	0.00	0
Max Pz	456	201	0	60.71	0
Min Pz	456	201	0	60.71	0

Gross Maximum Base pressure, p_{max} = 199.9 KN/m^2

Net Allowable Safe Bearing Capacity = 350 KN/m^2
Depth of soil above footing = 0.7 m
Unit weight of soil = 20 KN/m^3

Gross Safe Bearing Capacity = 364 KN/m^2
> 199.9 KN/m^2
Hence OK

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Design of foundation raft F1

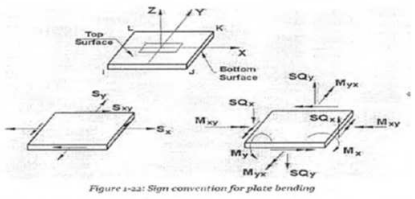
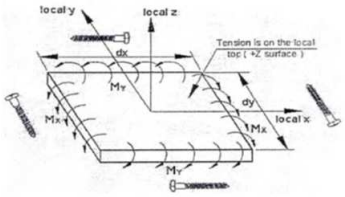


Figure 1-22: Sign convention for plate bending



Sign convention :
Local Z axis of plate is perpendicular to surface & positive towards upward direction.

Moment :
Mx : BM along X direction, +ve means tension on +Z surface of plate
My : BM along Y direction, +ve means tension on +Z surface of plate
For -ve Mx & My it is vice-virsa

Shear force :
SQx : Shear at face perpendicular to X axis
SQy : Shear at face perpendicular to Y axis

Axial force :
+ve Sx means tension along local X axis of plate & -ve Sx means compression along local X axis. for Sy this is similar.

Reinforcement in Foundation:
PLATE CENTER STRESS : FDN_F1_MOMENT At the face of the column

		Shear		Membrane			Bending Moment			M + I M_{xy} I		
	Plate	Qx	Qy	Sx	Sy	Sxy	Mx	My	Mxy	Mx	My	L/C
		(N/mm²)	(N/mm²)	(N/mm²)	(N/mm²)	(N/mm²)	(kN-m/m)	(kN-m/m)	(kN-m/m)	(kN-m/m)	(kN-m/m)	
Max Qx	469	0.281	-0.242	0	0	0	-27.66	-35.616	3.028	-30.691	-38.644	116:1.4DL + 1.7LL + 1.0 TL2
Min Qx	466	-0.279	-0.238	0	0	0	-27.24	-35.384	-2.779	-30.017	-38.163	116:1.4DL + 1.7LL + 1.0 TL2
Max Qy	507	-0.051	0.227	0	0	0	-31.61	-22.448	-2.594	-34.208	-25.042	115:1.4DL + 1.7LL + 1.0 TL1
Min Qy	468	0.065	-0.402	0	0	0	-39.71	-49.631	0.126	-39.839	-49.757	116:1.4DL + 1.7LL + 1.0 TL2
Max Sx	806	-0.001	0.001	0	0	0	-0.282	-0.248	-0.197	-0.479	-0.445	101:1.4DL
Min Sx	806	-0.001	0.001	0	0	0	-0.282	-0.248	-0.197	-0.479	-0.445	101:1.4DL
Max Sy	806	-0.001	0.001	0	0	0	-0.282	-0.248	-0.197	-0.479	-0.445	101:1.4DL
Min Sy	806	-0.001	0.001	0	0	0	-0.282	-0.248	-0.197	-0.479	-0.445	101:1.4DL
Max Sxy	806	-0.001	0.001	0	0	0	-0.282	-0.248	-0.197	-0.479	-0.445	101:1.4DL
Min Sxy	806	-0.001	0.001	0	0	0	-0.282	-0.248	-0.197	-0.479	-0.445	101:1.4DL
Max Mx	517	-0.031	-0.008	0	0	0	5.293	0.025	1.604	6.897	1.629	108:0.9DL + 1.6WLX2
Min Mx	467	-0.069	-0.4	0	0	0	-39.96	-49.55	-0.012	-39.968	-49.562	116:1.4DL + 1.7LL + 1.0 TL2
Max My	507	0.073	-0.112	0	0	0	-17.83	32.5	-12.131	-29.961	44.631	116:1.4DL + 1.7LL + 1.0 TL2
Min My	468	0.065	-0.402	0	0	0	-39.71	-49.631	0.126	-39.839	-49.757	116:1.4DL + 1.7LL + 1.0 TL2
Max Mxy	509	-0.09	-0.009	0	0	0	-3.187	8.423	19.674	-22.861	28.097	116:1.4DL + 1.7LL + 1.0 TL2
Min Mxy	506	0.092	-0.013	0	0	0	-2.678	8.693	-19.939	-22.617	28.632	116:1.4DL + 1.7LL + 1.0 TL2

PLATE CENTER STRESS : FDN_F1_SHEAR At a distance "d" from the face of the column

		Shear		Membrane			Bending Moment			M + I M_{xy} I		
	Plate	Qx	Qy	Sx	Sy	Sxy	Mx	My	Mxy	Mx	My	L/C
		(N/mm²)	(N/mm²)	(N/mm²)	(N/mm²)	(N/mm²)	(kN-m/m)	(kN-m/m)	(kN-m/m)	(kN-m/m)	(kN-m/m)	
Max Qx	526	0.124	0.005	0	0	0	-5.416	-26.842	6.254	-11.67	-33.096	116:1.4DL + 1.7LL + 1.0 TL2
Min Qx	465	-0.122	-0.05	0	0	0	-6.922	-19.814	0.752	-7.674	-20.566	116:1.4DL + 1.7LL + 1.0 TL2
Max Qy	499	-0.028	0.109	0	0	0	-22.53	-6.683	-1.875	-24.401	-8.558	115:1.4DL + 1.7LL + 1.0 TL1
Min Qy	476	0.049	-0.196	0	0	0	-32.39	-14.655	-1.409	-33.798	-16.064	116:1.4DL + 1.7LL + 1.0 TL2
Max Sx	806	-0.001	0.001	0	0	0	-0.282	-0.248	-0.197	-0.479	-0.445	101:1.4DL
Min Sx	806	-0.001	0.001	0	0	0	-0.282	-0.248	-0.197	-0.479	-0.445	101:1.4DL
Max Sy	806	-0.001	0.001	0	0	0	-0.282	-0.248	-0.197	-0.479	-0.445	101:1.4DL
Min Sy	806	-0.001	0.001	0	0	0	-0.282	-0.248	-0.197	-0.479	-0.445	101:1.4DL
Max Sxy	806	-0.001	0.001	0	0	0	-0.282	-0.248	-0.197	-0.479	-0.445	101:1.4DL
Min Sxy	806	-0.001	0.001	0	0	0	-0.282	-0.248	-0.197	-0.479	-0.445	101:1.4DL
Max Mx	518	-0.015	-0.002	0	0	0	1.693	0.915	0.525	2.218	1.44	108:0.9DL + 1.6WLX2
Min Mx	476	0.049	-0.196	0	0	0	-32.39	-14.655	-1.409	-33.798	-16.064	116:1.4DL + 1.7LL + 1.0 TL2
Max My	499	0.012	-0.059	0	0	0	-7.316	11.056	-2.825	-10.141	13.881	116:1.4DL + 1.7LL + 1.0 TL2
Min My	526	0.124	0.005	0	0	0	-5.416	-26.842	6.254	-11.67	-33.096	116:1.4DL + 1.7LL + 1.0 TL2
Max Mxy	518	0.054	0.058	0	0	0	-3.695	-14.704	10.998	-14.693	-25.702	116:1.4DL + 1.7LL + 1.0 TL2
Min Mxy	513	-0.051	0.057	0	0	0	-3.356	-14.412	-11.015	-14.371	-25.427	116:1.4DL + 1.7LL + 1.0 TL2

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Material Property :

Grade of concrete =		fc'35	
Grade of reinf. =		fy420	
Clear cover =	bottom face	50	mm
	top face	50	mm
Thicknes of slab =		500	mm

Design Constants :

φ for Moment =	0.9
φ for shear =	0.75
β_1	0.8
ε_t limit (for tension controlled section) =	0.005
Concrete compressive strain, ε_{cu} =	0.003

Reinforcement Along X:

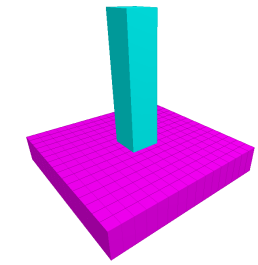
Force Values			Depth		Reinforcement Calculation													
Load condition	Factored Mt, Mx	Tension face due to bending	clear cover	d	M_u	Kn = Mu1/bd2 ϕ	ρ	$A_{s\ required}$	$A_{s,min}$	Dia of bar	$S_{required}$	$S_{provided}$	$A_{s,\ provided}$	a	c = a/β_1	ε_t = ε_{cu}/c *(d-c)	Remarks
	KN-m/m		mm	mm	KN-m/m	(N/mm2	mm	mm^2	mm^2	mm	mm c/c	mm c/c	mm^2	mm	mm		
108:0.9D	6.90	top face	50	444	6.897	0.05	0.0001	49.4	532.8	12	212.36	150	754.3	10.64	13.29	0.097	tension control
116:1.4D	-39.97	bottom face	50	442	39.968	0.27	0.0007	288.4		16	377.52	150	1341	18.91	23.64	0.053	tension control
116:1.4D	-22.86	bottom face	50	442	22.861	0.16	0.0004	164.6		16	377.52	150	1341	18.91	23.64	0.053	tension control
116:1.4D	-22.62	bottom face	50	442	22.617	0.15	0.0004	162.9		16	377.52	150	1341	18.91	23.64	0.053	tension control

Shear Check		Qx	FOS	Vu	Mu	$\rho_{w,pro}$	bd	V_{ca}	V_{cb}	V_{cc}	$V_{c,design}$	$\varphi(V_c+0.66\sqrt{f_c'}bd)\geq V_u$			Rema rks	φV_c	Remar ks	Remarks :Min shear reinf
		Mpa		KN	KN-m/m		m^2	KN	KN	KN	KN	KN				KN		
Load condition	Max Qx	0.124	1	62	11.67	0.002	0.444	450.526	433.101	761.754	433.1	1625	> 62	ok	324.8	φVc>Vu	not required
	Min Qx	0.122	1	61	7.674			465.535			433.1	1625	> 61	ok	324.8	φVc>Vu	not required

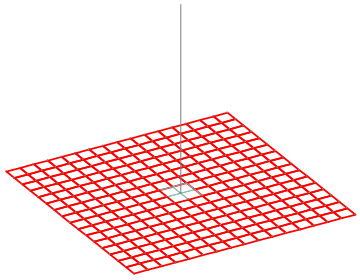
Reinforcement Along Z:

Force Values			Depth		Reinforcement Calculation												
Load condition	Factored Mt, My	Tension face due to bending	clear cover	d	M_u	Kn = Mu1/bd2 ϕ	ρ	$A_{s\ required}$	$A_{s,min}$	Dia of bar	$S_{required}$	$S_{provided}$	$A_{s,\ provided}$	a	c = a/β_1	ε_t = ε_{cu}/c *(d-c)	Remarks
	KN-m/m		mm	mm	KN-m/m	(N/mm2	mm	mm^2	mm^2	mm	mm c/c	mm c/c	mm^2	mm	mm		
116:1.4D	44.63	top face	50	444	44.631	0.30	0.0007	320.7	532.8	12	212.36	150	754.3	10.64	13.29	0.097	tension control
116:1.4D	-49.76	bottom face	50	442	49.757	0.34	0.0008	359.4		16	377.52	150	1341	18.91	23.64	0.053	tension control
116:1.4D	28.10	top face	50	444	28.097	0.19	0.0005	201.5		12	212.36	150	754.3	10.64	13.29	0.097	tension control
116:1.4D	28.63	top face	50	444	28.632	0.19	0.0005	205.4		12	212.36	150	754.3	10.64	13.29	0.097	tension control

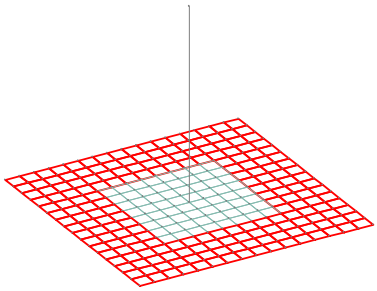
Shear Check		Qx	FOS	Vu	Mu	$\rho_{w,pro}$	bd	V_{ca}	V_{cb}	V_{cc}	$V_{c,design}$	$\varphi(V_c+0.66\sqrt{f_c'}bd)\geq V_u$			Rema rks	φV_c	Remar ks	Remarks :Min shear reinf
		Mpa		KN	T-m/m		m^2	KN	KN	KN	KN	KN				KN		
Load condition	Max Qy	0.109	1	54.5	8.558	0.002	0.444	456.536	433.101	fc'761.8	433.1	455	> 54.5	ok	324.8	φVc>Vu	not required
	Min Qy	0.196	1	98	16.064			455.012			433.1	455	> 98	ok	324.8	φVc>Vu	not required



3D rendered view



Highlighted plate for Moment at face of column



Highlighted plate for Shear at 'd' distance from face of column

SUMMARY OF REINFORCEMENT IN RAFT:

Along X Direction :
At Bottom **16ϕT** **150** c/c
At Top **12ϕT** **150** c/c
Along Z Direction :
At Bottom **16ϕT** **150** c/c
At Top **12ϕT** **150** c/c

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DESIGN CALCULATION OF FOOTING F2

Size of Columns at node	Along-Z	Along-X
522	0.60m	0.60m

Size of fdn. raft:	2.00m	2.00m
Thickness of raft:	0.50	m
Original Gr. Level:	0.00	m
Finished Gr. Level:	0.00	m
U/S of B/Pl Above GL:	0.30	m
U/S level of fdn:	-1.20	m
Unit weight of conc:	24	kN/m^3
Unit weight of soil:	20	kN/m^3

Calculation of Weight of Foundation :

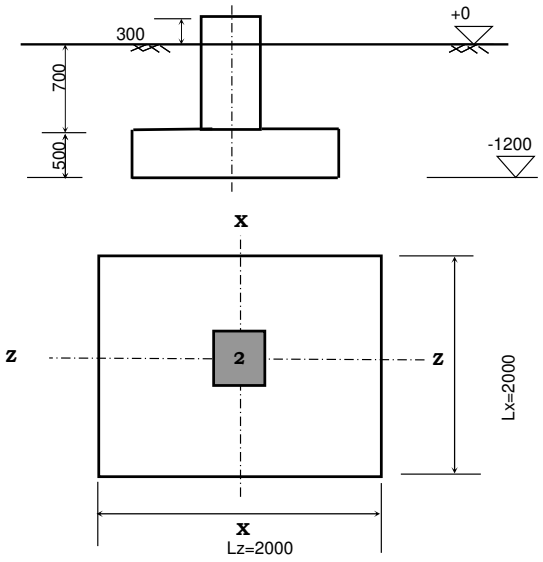
Weight of raft = (2*2*0.5*24) = 48.00KN

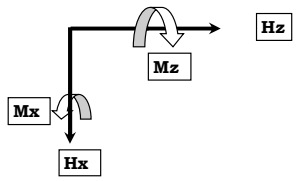
Weight of soil & Surcharge = 100 + ((2*2)-(0.6*0.6))*0.7*20 = 150.96KN

Total = 198.96KN

Design Input:

Characteristic strength of concrete, f'_c=	35 MPa
Yield stress of steel reinforcement, f_y=	420 MPa
Clear cover for main reinforcement bars =	50mm





H_x=Force along X-X direction

Hz=Force along Z-Z direction

M_x=Moment about X-X direction

Mz=Moment about Z-Z direction

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Check for overturning and sliding of footing F2

Load at bottom of fooring	Node 522				
L/C No.	V (kN)	Hx (kN)	Mz (kN-m)	Hz (kN)	Mx (kN-m)
201 1.0DL	41.20	0.03	-0.05	-5.00	-7.49
202 1.0DL + 1.0LL	174.32	-0.10	0.14	-30.00	-44.94
203 1.0DL + 1.0WLX1	56.40	-8.67	13.00	-5.02	-7.51
204 1.0DL + 1.0WLX2	26.00	8.73	-13.10	-4.98	-7.46
205 1.0DL + 1.0WLZ1	35.20	0.47	-0.71	-10.80	-16.19
206 1.0DL + 1.0WLZ2	47.20	-0.41	0.61	0.80	1.22
207 1.0DL + 0.75WLX1 + 0.75LL	152.44	-6.59	9.88	-23.77	-35.59
208 1.0DL + 0.75WLX2 + 0.75LL	129.64	6.46	-9.69	-23.74	-35.56
209 1.0DL + 0.75WLZ1 + 0.75LL	136.54	0.27	-0.40	-28.10	-42.10
210 1.0DL + 0.75WLZ2 + 0.75LL	145.54	-0.39	0.59	-19.40	-29.05
211 0.6DL + 1.0WLX1	39.92	-8.68	13.02	-3.02	-4.52
212 0.6DL + 1.0WLX2	9.52	8.72	-13.08	-2.98	-4.47
213 0.6DL + 1.0WLZ1	18.72	0.46	-0.69	-8.80	-13.19
214 0.6DL + 1.0WLZ2	30.72	-0.42	0.63	2.80	4.21
215 1.0DL + 1.0LL + 1.0TL1	172.32	1.00	-1.51	37.40	56.16
216 1.0DL + 1.0LL + 1.0TL2	176.32	-1.20	1.79	-97.40	-146.04
217 0.6DL + 0.7ELX	27.07	-0.68	1.02	-3.01	-4.50
218 0.6DL + 0.7ELZ	23.72	0.11	-0.16	-3.53	-5.28

Check for overturning and sliding:

L/C No.	Resultant Load at bottom of footing								
	V (kN)	Hz (kN)	Mx (kN-m)	Hx (kN)	Mz (kN-m)	Overturning		Sliding	
						F.O.S. Mz	F.O.S. Mx	F.O.S. Hx	F.O.S. Hz
201 1.0DL	240.2	-5.00	-7.5	0.0	-0.1	4709.0	32.1	2825.4	19.2
202 1.0DL + 1.0LL	373.3	-30.00	-44.9	-0.1	0.1	2592.2	8.3	1555.3	5.0
203 1.0DL + 1.0WLX1	255.4	-5.02	-7.5	-8.7	13.0	19.6	34.0	11.8	20.3
204 1.0DL + 1.0WLX2	225.0	-4.98	-7.5	8.7	-13.1	17.2	30.1	10.3	18.1
205 1.0DL + 1.0WLZ1	234.2	-10.80	-16.2	0.5	-0.7	329.3	14.5	197.6	8.7
206 1.0DL + 1.0WLZ2	246.2	0.80	1.2	-0.4	0.6	404.2	202.6	242.5	123.1
207 1.0DL + 0.75WLX1 + 0.75LL	351.4	-23.77	-35.6	-6.6	9.9	35.6	9.9	21.3	5.9
208 1.0DL + 0.75WLX2 + 0.75LL	328.6	-23.74	-35.6	6.5	-9.7	33.9	9.2	20.3	5.5
209 1.0DL + 0.75WLZ1 + 0.75LL	335.5	-28.10	-42.1	0.3	-0.4	838.8	8.0	502.6	4.8
210 1.0DL + 0.75WLZ2 + 0.75LL	344.5	-19.40	-29.0	-0.4	0.6	583.9	11.9	349.7	7.1
211 0.6DL + 1.0WLX1	238.9	-3.02	-4.5	-8.7	13.0	18.3	52.9	11.0	31.6
212 0.6DL + 1.0WLX2	208.5	-2.98	-4.5	8.7	-13.1	15.9	46.7	9.6	28.0
213 0.6DL + 1.0WLZ1	217.7	-8.80	-13.2	0.5	-0.7	315.0	16.5	189.3	9.9
214 0.6DL + 1.0WLZ2	229.7	2.80	4.2	-0.4	0.6	365.2	54.6	218.7	32.8
215 1.0DL + 1.0LL + 1.0TL1	371.3	37.40	56.2	1.0	-1.5	246.5	6.6	147.9	4.0
216 1.0DL + 1.0LL + 1.0TL2	375.3	-97.40	-146.0	-1.2	1.8	209.2	2.6	125.5	1.5
217 0.6DL + 0.7ELX	226.0	-3.01	-4.5	-0.7	1.0	221.8	50.2	133.0	30.1
218 0.6DL + 0.7ELZ	222.7	-3.53	-5.3	0.1	-0.2	1357.8	42.2	817.2	25.3

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Base pressure check of footing F2

	Node	L/C	Horizontal Fx kN/m2	Vertical Fy kN/m2	Horizontal Fz kN/m2
Max Px	456	201	0	57.02	0
Min Px	456	201	0	57.02	0
Max Py	489	216	0	238.26	0
Min Py	494	216	0	0.00	0
Max Pz	456	201	0	57.02	0
Min Pz	456	201	0	57.02	0

Gross Maximum Base pressure, p_{max} = 238.3 KN/m^2

Net Allowable Safe Bearing Capacity = 350 KN/m^2

Depth of soil above footing = 0.7 m

Unit weight of soil = 20 KN/m^3

Gross Safe Bearing Capacity = 364 KN/m^2

> 238.3 KN/m^2

Hence OK

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Design of foundation raft F2

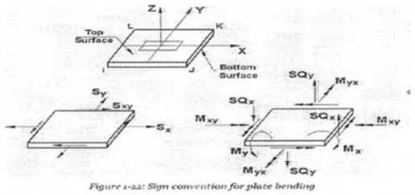
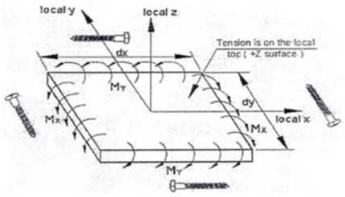


Figure 1-22: Sign convention for plate bending



Sign convention :
Local Z axis of plate is perpendicular to surface & positive towards upward direction.

Moment :
Mx : BM along X direction, +ve means tension on +Z surface of plate
My : BM along Y direction, +ve means tension on +Z surface of plate
For -ve Mx & My it is vice-virsa

Shear force :
SQx : Shear at face perpendicular to X axis
SQy : Shear at face perpendicular to Y axis

Axial force :
+ve Sx means tension along local X axis of plate & -ve Sx means compression along local X axis. for Sy this is similar.

Reinforcement in Foundation:
PLATE CENTER STRESS : FDN_F2_MOMENT At the face of the column

		Shear		Membrane			Bending Moment			M + I M_{xy} I		
	Plate	Qx	Qy	Sx	Sy	Sxy	Mx	My	Mxy	Mx	My	L/C
		(N/mm²)	(N/mm²)	(N/mm²)	(N/mm²)	(N/mm²)	(kN-m/m)	(kN-m/m)	(kN-m/m)	(kN-m/m)	(kN-m/m)	
Max Qx	469	0.261	-0.246	0	0	0	-21.5	-39.339	-1.447	-22.947	-40.786	116:1.4DL + 1.7LL + 1.0 TL2
Min Qx	466	-0.273	-0.24	0	0	0	-25.11	-35.119	-6.871	-31.978	-41.99	116:1.4DL + 1.7LL + 1.0 TL2
Max Qy	508	0.056	0.228	0	0	0	-26.96	-23.335	2.434	-29.398	-25.769	115:1.4DL + 1.7LL + 1.0 TL1
Min Qy	468	0.064	-0.406	0	0	0	-34	-51.448	-4.349	-38.347	-55.797	116:1.4DL + 1.7LL + 1.0 TL2
Max Sx	814	0.001	0.001	0	0	0	-0.173	-0.238	-0.239	-0.412	-0.477	101:1.4DL
Min Sx	814	0.001	0.001	0	0	0	-0.173	-0.238	-0.239	-0.412	-0.477	101:1.4DL
Max Sy	814	0.001	0.001	0	0	0	-0.173	-0.238	-0.239	-0.412	-0.477	101:1.4DL
Min Sy	814	0.001	0.001	0	0	0	-0.173	-0.238	-0.239	-0.412	-0.477	101:1.4DL
Max Sxy	814	0.001	0.001	0	0	0	-0.173	-0.238	-0.239	-0.412	-0.477	101:1.4DL
Min Sxy	814	0.001	0.001	0	0	0	-0.173	-0.238	-0.239	-0.412	-0.477	101:1.4DL
Max Mx	514	0.003	-0.002	0	0	0	4.811	-5.153	-2.644	7.455	-7.797	107:0.9DL + 1.6WLX1
Min Mx	467	-0.062	-0.404	0	0	0	-35.91	-50.17	-4.336	-40.242	-54.506	116:1.4DL + 1.7LL + 1.0 TL2
Max My	508	-0.066	-0.108	0	0	0	-13.69	32.247	4.653	-18.345	36.9	116:1.4DL + 1.7LL + 1.0 TL2
Min My	468	0.064	-0.406	0	0	0	-34	-51.448	-4.349	-38.347	-55.797	116:1.4DL + 1.7LL + 1.0 TL2
Max Mxy	509	-0.089	0	0	0	0	1.095	8.617	12.047	13.142	20.664	116:1.4DL + 1.7LL + 1.0 TL2
Min Mxy	506	0.108	-0.018	0	0	0	-2.236	8.31	-26.331	-28.567	34.641	116:1.4DL + 1.7LL + 1.0 TL2

PLATE CENTER STRESS : FDN_F2_SHEAR At a distance "d" from the face of the column

		Shear		Membrane			Bending Moment			M + I M_{xy} I		
	Plate	Qx	Qy	Sx	Sy	Sxy	Mx	My	Mxy	Mx	My	L/C
		(N/mm²)	(N/mm²)	(N/mm²)	(N/mm²)	(N/mm²)	(kN-m/m)	(kN-m/m)	(kN-m/m)	(kN-m/m)	(kN-m/m)	
Max Qx	786	0.091	-0.016	0	0	0	-5.884	1.319	-6.138	-12.022	7.457	116:1.4DL + 1.7LL + 1.0 TL2
Min Qx	465	-0.126	-0.048	0	0	0	-6.03	-18.515	-3.179	-9.209	-21.694	116:1.4DL + 1.7LL + 1.0 TL2
Max Qy	500	0.035	0.111	0	0	0	-17.68	-7.218	1.101	-18.785	-8.319	115:1.4DL + 1.7LL + 1.0 TL1
Min Qy	476	0.048	-0.202	0	0	0	-26.53	-15.708	-4.468	-30.995	-20.176	116:1.4DL + 1.7LL + 1.0 TL2
Max Sx	814	0.001	0.001	0	0	0	-0.173	-0.238	-0.239	-0.412	-0.477	101:1.4DL
Min Sx	814	0.001	0.001	0	0	0	-0.173	-0.238	-0.239	-0.412	-0.477	101:1.4DL
Max Sy	814	0.001	0.001	0	0	0	-0.173	-0.238	-0.239	-0.412	-0.477	101:1.4DL
Min Sy	814	0.001	0.001	0	0	0	-0.173	-0.238	-0.239	-0.412	-0.477	101:1.4DL
Max Sxy	814	0.001	0.001	0	0	0	-0.173	-0.238	-0.239	-0.412	-0.477	101:1.4DL
Min Sxy	814	0.001	0.001	0	0	0	-0.173	-0.238	-0.239	-0.412	-0.477	101:1.4DL
Max Mx	513	0.004	0.002	0	0	0	2.432	-2.638	-1.165	3.597	-3.803	107:0.9DL + 1.6WLX1
Min Mx	475	-0.045	-0.198	0	0	0	-28	-14.978	-1.557	-29.554	-16.535	116:1.4DL + 1.7LL + 1.0 TL2
Max My	500	0.006	-0.059	0	0	0	-3.663	11.221	-4.136	-7.799	15.357	116:1.4DL + 1.7LL + 1.0 TL2
Min My	526	0.074	0.034	0	0	0	-2.019	-33.806	-0.251	-2.27	-34.057	116:1.4DL + 1.7LL + 1.0 TL2
Max Mxy	465	-0.044	-0.043	0	0	0	-5.69	-11.016	7.594	-13.284	-18.61	115:1.4DL + 1.7LL + 1.0 TL1
Min Mxy	513	-0.047	0.05	0	0	0	-3.71	-14.23	-16.832	-20.542	-31.062	116:1.4DL + 1.7LL + 1.0 TL2

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Material Property :

Grade of concrete =		fc'35	
Grade of reinf. =		fy420	
Clear cover =	bottom face	50	mm
	top face	50	mm
Thicknes of slab =		500	mm

Design Constants :

φ for Moment =	0.9
φ for shear =	0.75
β_1	0.8
ε_t limit (for tension controlled section) =	0.005
Concrete compressive strain, ε_{cu} =	0.003

Reinforcement Along X:

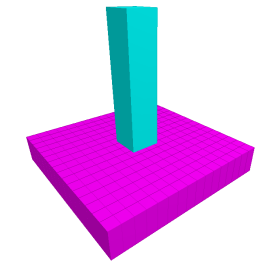
Force Values			Depth		Reinforcement Calculation													
Load condition	Factored Mt, Mx	Tension face due to bending	clear cover	d	M_u	Kn = Mu1/bd2 ϕ	ρ	$A_{s\ required}$	$A_{s,min}$	Dia of bar	$S_{required}$	$S_{provided}$	$A_{s,\ provided}$	a	c = a/β_1	ε_t = ε_{cu}/c *(d-c)	Remarks
	KN-m/m		mm	mm	KN-m/m	(N/mm2	mm	mm^2	mm^2	mm	mm c/c	mm c/c	mm^2	mm	mm		
107:0.9D	7.46	top face	50	444	7.455	0.05	0.0001	53.3	532.8	12	212.36	150	754.3	10.64	13.29	0.097	tension control
116:1.4D	-40.24	bottom face	50	442	40.242	0.27	0.0007	290.4		16	377.52	150	1341	18.91	23.64	0.053	tension control
116:1.4D	13.14	top face	50	444	13.142	0.09	0.0002	94.1		12	212.36	150	754.3	10.64	13.29	0.097	tension control
116:1.4D	-28.57	bottom face	50	442	28.567	0.19	0.0005	205.9		16	377.52	150	1341	18.91	23.64	0.053	tension control

Shear Check		Qx	FOS	Vu	Mu	$\rho_{w,pro}$	bd	V_{ca}	V_{cb}	V_{cc}	$V_{c,design}$	$\varphi(V_c+0.66\sqrt{f_c'}bd)\geq V_u$			Remarks	φV_c	Remarks	Remarks :Min shear reinf
		Mpa		KN	KN-m/m		m^2	KN	KN	KN	KN	KN				KN		
Load condition	Max Qx	0.091	1	45.5	12.022	0.002	0.444	441.826	433.101	761.754	433.1	1625	> 45.5	ok	324.8	φVc>Vu	not required
	Min Qx	0.126	1	63	9.209			459.228			433.1	1625	> 63	ok	324.8	φVc>Vu	not required

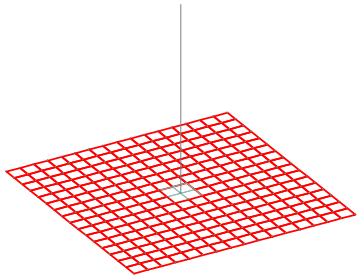
Reinforcement Along Z:

Force Values			Depth		Reinforcement Calculation												
Load condition	Factored Mt, My	Tension face due to bending	clear cover	d	M_u	Kn = Mu1/bd2 ϕ	ρ	$A_{s\ required}$	$A_{s,min}$	Dia of bar	$S_{required}$	$S_{provided}$	$A_{s,\ provided}$	a	c = a/β_1	ε_t = ε_{cu}/c *(d-c)	Remarks
	KN-m/m		mm	mm	KN-m/m	(N/mm2	mm	mm^2	mm^2	mm	mm c/c	mm c/c	mm^2	mm	mm		
116:1.4D	36.90	top face	50	444	36.9	0.25	0.0006	265.0	532.8	12	212.36	150	754.3	10.64	13.29	0.097	tension control
116:1.4D	-55.80	bottom face	50	442	55.797	0.38	0.0009	403.4		16	377.52	150	1341	18.91	23.64	0.053	tension control
116:1.4D	20.66	top face	50	444	20.664	0.14	0.0003	148.1		12	212.36	150	754.3	10.64	13.29	0.097	tension control
116:1.4D	34.64	top face	50	444	34.641	0.23	0.0006	248.7		12	212.36	150	754.3	10.64	13.29	0.097	tension control

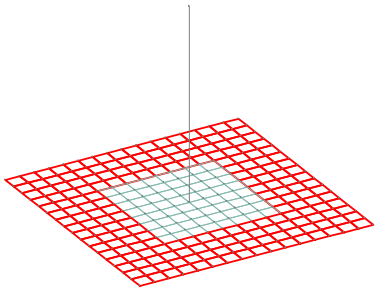
Shear Check		Qx	FOS	Vu	Mu	$\rho_{w,pro}$	bd	V_{ca}	V_{cb}	V_{cc}	$V_{c,design}$	$\varphi(V_c+0.66\sqrt{f_c'}bd)\geq V_u$			Remarks	φV_c	Remarks	Remarks :Min shear reinf
		Mpa		KN	T-m/m		m^2	KN	KN	KN	KN	KN				KN		
Load condition	Max Qy	0.111	1	55.5	8.319	0.002	0.444	458.262	433.101	fc'761.8	433.1	455	> 55.5	ok	324.8	φVc>Vu	not required
	Min Qy	0.202	1	101	20.176			448.779			433.1	455	> 101	ok	324.8	φVc>Vu	not required



3D rendered view



Highlighted plate for Moment at face of column



Highlighted plate for Shear at 'd' distance from face of column

SUMMARY OF REINFORCEMENT IN RAFT:

Along X Direction :
At Bottom **16ϕT** **150** c/c
At Top **12ϕT** **150** c/c
Along Z Direction :
At Bottom **16ϕT** **150** c/c
At Top **12ϕT** **150** c/c

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DESIGN CALCULATION OF FOOTING F3

Size of Columns at node	Along-Z	Along-X
522	0.60m	0.60m

Size of fdn. raft:	2.00m	2.00m
Thickness of raft:	0.50	m
Original Gr. Level:	0.00	m
Finished Gr. Level:	0.00	m
U/S of B/Pl Above GL:	0.30	m
U/S level of fdn:	-1.20	m
Unit weight of conc:	24	kN/m^3
Unit weight of soil:	20	kN/m^3

Calculation of Weight of Foundation :

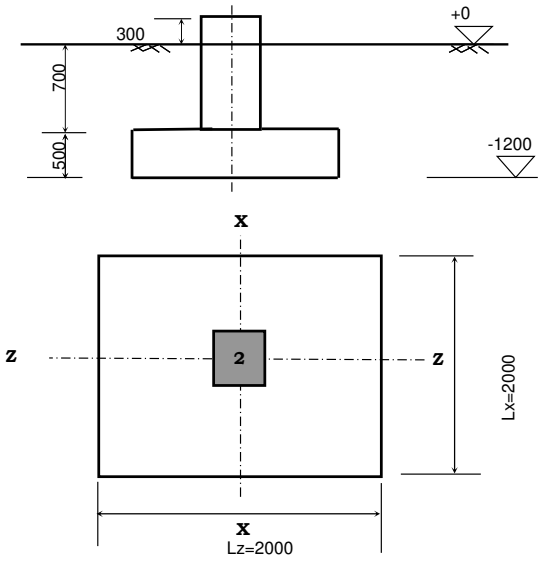
Weight of raft = (2*2*0.5*24) = 48.00KN

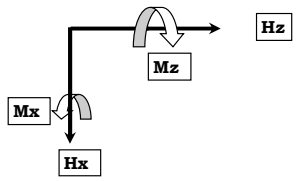
Weight of soil & Surcharge = 100 + ((2*2)-(0.6*0.6))*0.7*20 = 150.96KN

Total = 198.96KN

Design Input:

Characteristic strength of concrete, f'_c=	35 MPa
Yield stress of steel reinforcement, f_y=	420 MPa
Clear cover for main reinforcement bars =	50mm





H_x=Force along X-X direction

Hz=Force along Z-Z direction

M_x=Moment about X-X direction

Mz=Moment about Z-Z direction

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Check for overturning and sliding of footing F3

Load at bottom of fooring	Node 522				
L/C No.	V (kN)	Hx (kN)	Mz (kN-m)	Hz (kN)	Mx (kN-m)
201 1.0DL	41.20	0.03	-0.05	-5.00	-7.49
202 1.0DL + 1.0LL	174.32	-0.10	0.14	-30.00	-44.94
203 1.0DL + 1.0WLX1	56.40	-8.67	13.00	-5.02	-7.51
204 1.0DL + 1.0WLX2	26.00	8.73	-13.10	-4.98	-7.46
205 1.0DL + 1.0WLZ1	35.20	0.47	-0.71	-10.80	-16.19
206 1.0DL + 1.0WLZ2	47.20	-0.41	0.61	0.80	1.22
207 1.0DL + 0.75WLX1 + 0.75LL	152.44	-6.59	9.88	-23.77	-35.59
208 1.0DL + 0.75WLX2 + 0.75LL	129.64	6.46	-9.69	-23.74	-35.56
209 1.0DL + 0.75WLZ1 + 0.75LL	136.54	0.27	-0.40	-28.10	-42.10
210 1.0DL + 0.75WLZ2 + 0.75LL	145.54	-0.39	0.59	-19.40	-29.05
211 0.6DL + 1.0WLX1	39.92	-8.68	13.02	-3.02	-4.52
212 0.6DL + 1.0WLX2	9.52	8.72	-13.08	-2.98	-4.47
213 0.6DL + 1.0WLZ1	18.72	0.46	-0.69	-8.80	-13.19
214 0.6DL + 1.0WLZ2	30.72	-0.42	0.63	2.80	4.21
215 1.0DL + 1.0LL + 1.0TL1	172.32	1.00	-1.51	37.40	56.16
216 1.0DL + 1.0LL + 1.0TL2	176.32	-1.20	1.79	-97.40	-146.04
217 0.6DL + 0.7ELX	27.07	-0.68	1.02	-3.01	-4.50
218 0.6DL + 0.7ELZ	23.72	0.11	-0.16	-3.53	-5.28

Check for overturning and sliding:

L/C No.	Resultant Load at bottom of footing								
	V (kN)	Hz (kN)	Mx (kN-m)	Hx (kN)	Mz (kN-m)	Overturning		Sliding	
						F.O.S. Mz	F.O.S. Mx	F.O.S. Hx	F.O.S. Hz
201 1.0DL	240.2	-5.00	-7.5	0.0	-0.1	4709.0	32.1	2825.4	19.2
202 1.0DL + 1.0LL	373.3	-30.00	-44.9	-0.1	0.1	2592.2	8.3	1555.3	5.0
203 1.0DL + 1.0WLX1	255.4	-5.02	-7.5	-8.7	13.0	19.6	34.0	11.8	20.3
204 1.0DL + 1.0WLX2	225.0	-4.98	-7.5	8.7	-13.1	17.2	30.1	10.3	18.1
205 1.0DL + 1.0WLZ1	234.2	-10.80	-16.2	0.5	-0.7	329.3	14.5	197.6	8.7
206 1.0DL + 1.0WLZ2	246.2	0.80	1.2	-0.4	0.6	404.2	202.6	242.5	123.1
207 1.0DL + 0.75WLX1 + 0.75LL	351.4	-23.77	-35.6	-6.6	9.9	35.6	9.9	21.3	5.9
208 1.0DL + 0.75WLX2 + 0.75LL	328.6	-23.74	-35.6	6.5	-9.7	33.9	9.2	20.3	5.5
209 1.0DL + 0.75WLZ1 + 0.75LL	335.5	-28.10	-42.1	0.3	-0.4	838.8	8.0	502.6	4.8
210 1.0DL + 0.75WLZ2 + 0.75LL	344.5	-19.40	-29.0	-0.4	0.6	583.9	11.9	349.7	7.1
211 0.6DL + 1.0WLX1	238.9	-3.02	-4.5	-8.7	13.0	18.3	52.9	11.0	31.6
212 0.6DL + 1.0WLX2	208.5	-2.98	-4.5	8.7	-13.1	15.9	46.7	9.6	28.0
213 0.6DL + 1.0WLZ1	217.7	-8.80	-13.2	0.5	-0.7	315.0	16.5	189.3	9.9
214 0.6DL + 1.0WLZ2	229.7	2.80	4.2	-0.4	0.6	365.2	54.6	218.7	32.8
215 1.0DL + 1.0LL + 1.0TL1	371.3	37.40	56.2	1.0	-1.5	246.5	6.6	147.9	4.0
216 1.0DL + 1.0LL + 1.0TL2	375.3	-97.40	-146.0	-1.2	1.8	209.2	2.6	125.5	1.5
217 0.6DL + 0.7ELX	226.0	-3.01	-4.5	-0.7	1.0	221.8	50.2	133.0	30.1
218 0.6DL + 0.7ELZ	222.7	-3.53	-5.3	0.1	-0.2	1357.8	42.2	817.2	25.3

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Base pressure check of footing F3

	Node	L/C	Horizontal Fx kN/m2	Vertical Fy kN/m2	Horizontal Fz kN/m2
Max Px	457	201	0	75.95	0
Min Px	457	201	0	75.95	0
Max Py	483	216	0	256.04	0
Min Py	841	215	0	0.00	0
Max Pz	457	201	0	75.95	0
Min Pz	457	201	0	75.95	0

Gross Maximum Base pressure, p_{max} = 256.0 KN/m^2

Net Allowable Safe Bearing Capacity = 350 KN/m^2
Depth of soil above footing = 0.7 m
Unit weight of soil = 20 KN/m^3

Gross Safe Bearing Capacity = 364 KN/m^2
> 256.0 KN/m^2
Hence OK

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Design of foundation raft F3

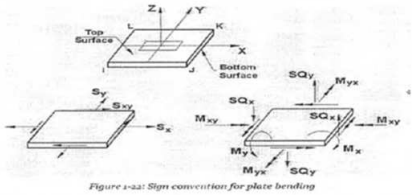
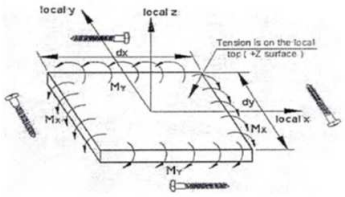


Figure 1-22: Sign convention for plate bending



Sign convention :
Local Z axis of plate is perpendicular to surface & positive towards upward direction.

Moment :
Mx : BM along X direction, +ve means tension on +Z surface of plate
My : BM along Y direction, +ve means tension on +Z surface of plate
For -ve Mx & My it is vice-virsa

Shear force :
SQx : Shear at face perpendicular to X axis
SQy : Shear at face perpendicular to Y axis

Axial force :
+ve Sx means tension along local X axis of plate & -ve Sx means compression along local X axis. for Sy this is similar.

Reinforcement in Foundation:
PLATE CENTER STRESS : FDN_F3_MOMENT At the face of the column

		Shear		Membrane			Bending Moment			M + I M_{xy} I		
	Plate	Qx	Qy	Sx	Sy	Sxy	Mx	My	Mxy	Mx	My	L/C
		(N/mm²)	(N/mm²)	(N/mm²)	(N/mm²)	(N/mm²)	(kN-m/m)	(kN-m/m)	(kN-m/m)	(kN-m/m)	(kN-m/m)	
Max Qx	469	0.277	-0.248	0	0	0	-11.99	-38.86	8.25	-20.238	-47.11	116:1.4DL + 1.7LL + 1.0 TL2
Min Qx	869	-0.235	-0.236	0	0	0	-12.14	-49.842	-8.417	-20.552	-58.259	116:1.4DL + 1.7LL + 1.0 TL2
Max Qy	873	-0.036	0.534	0	0	0	-3.106	-62.855	19.123	-22.229	-81.978	116:1.4DL + 1.7LL + 1.0 TL2
Min Qy	467	-0.04	-0.422	0	0	0	-18.23	-59.61	0.371	-18.603	-59.981	116:1.4DL + 1.7LL + 1.0 TL2
Max Sx	878	0	0.004	0	0	0	0.056	0.376	0.251	0.307	0.627	101:1.4DL
Min Sx	878	0	0.004	0	0	0	0.056	0.376	0.251	0.307	0.627	101:1.4DL
Max Sy	878	0	0.004	0	0	0	0.056	0.376	0.251	0.307	0.627	101:1.4DL
Min Sy	878	0	0.004	0	0	0	0.056	0.376	0.251	0.307	0.627	101:1.4DL
Max Sxy	878	0	0.004	0	0	0	0.056	0.376	0.251	0.307	0.627	101:1.4DL
Min Sxy	878	0	0.004	0	0	0	0.056	0.376	0.251	0.307	0.627	101:1.4DL
Max Mx	517	0.042	0.04	0	0	0	13.75	-17.852	7.476	21.226	-25.328	104:1.2DL + 1.6WLX2 + 1.0 LL
Min Mx	468	0.07	-0.413	0	0	0	-20.22	-56.475	4.032	-24.248	-60.507	116:1.4DL + 1.7LL + 1.0 TL2
Max My	875	0.078	-0.059	0	0	0	-10.51	36.806	7.802	-18.308	44.608	116:1.4DL + 1.7LL + 1.0 TL2
Min My	873	-0.036	0.534	0	0	0	-3.106	-62.855	19.123	-22.229	-81.978	116:1.4DL + 1.7LL + 1.0 TL2
Max Mxy	509	-0.11	-0.006	0	0	0	7.777	8.546	33.69	41.467	42.236	116:1.4DL + 1.7LL + 1.0 TL2
Min Mxy	469	0.04	-0.07	0	0	0	2.292	-8.789	-22.81	25.102	-31.599	115:1.4DL + 1.7LL + 1.0 TL1

PLATE CENTER STRESS : FDN_F3_SHEAR At a distance "d" from the face of the column

		Shear		Membrane			Bending Moment			M + I M_{xy} I		
	Plate	Qx	Qy	Sx	Sy	Sxy	Mx	My	Mxy	Mx	My	L/C
		(N/mm²)	(N/mm²)	(N/mm²)	(N/mm²)	(N/mm²)	(kN-m/m)	(kN-m/m)	(kN-m/m)	(kN-m/m)	(kN-m/m)	
Max Qx	470	0.124	-0.05	0	0	0	5.442	-20.367	4.803	10.245	-25.17	116:1.4DL + 1.7LL + 1.0 TL2
Min Qx	790	-0.154	-0.007	0	0	0	0.671	0.36	10.25	10.921	10.61	116:1.4DL + 1.7LL + 1.0 TL2
Max Qy	877	0.005	0.182	0	0	0	0.277	-12.433	2.067	2.344	-14.5	115:1.4DL + 1.7LL + 1.0 TL1
Min Qy	475	-0.041	-0.209	0	0	0	-13.57	-19.398	0.272	-13.839	-19.67	116:1.4DL + 1.7LL + 1.0 TL2
Max Sx	878	0	0.004	0	0	0	0.056	0.376	0.251	0.307	0.627	101:1.4DL
Min Sx	878	0	0.004	0	0	0	0.056	0.376	0.251	0.307	0.627	101:1.4DL
Max Sy	878	0	0.004	0	0	0	0.056	0.376	0.251	0.307	0.627	101:1.4DL
Min Sy	878	0	0.004	0	0	0	0.056	0.376	0.251	0.307	0.627	101:1.4DL
Max Sxy	878	0	0.004	0	0	0	0.056	0.376	0.251	0.307	0.627	101:1.4DL
Min Sxy	878	0	0.004	0	0	0	0.056	0.376	0.251	0.307	0.627	101:1.4DL
Max Mx	827	0.014	0.034	0	0	0	13.631	-10.382	-2.306	15.937	-12.688	115:1.4DL + 1.7LL + 1.0 TL1
Min Mx	476	0.044	-0.206	0	0	0	-14.48	-17.71	-0.362	-14.84	-18.072	116:1.4DL + 1.7LL + 1.0 TL2
Max My	877	-0.029	-0.092	0	0	0	-1.171	25.728	4.662	-5.833	30.39	116:1.4DL + 1.7LL + 1.0 TL2
Min My	526	0.124	0.003	0	0	0	6.242	-27.279	14.732	20.974	-42.011	116:1.4DL + 1.7LL + 1.0 TL2
Max Mxy	510	-0.024	0.046	0	0	0	3.757	-7.13	22.332	26.089	-29.462	116:1.4DL + 1.7LL + 1.0 TL2
Min Mxy	470	0.023	-0.03	0	0	0	6.842	-11.196	-17.451	24.293	-28.647	115:1.4DL + 1.7LL + 1.0 TL1

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Material Property :

Grade of concrete =		fc'35	
Grade of reinf. =		fy420	
Clear cover =	bottom face	50	mm
	top face	50	mm
Thicknes of slab =		500	mm

Design Constants :

φ for Moment =	0.9
φ for shear =	0.75
β_1	0.8
ε_t limit (for tension controlled section) =	0.005
Concrete compressive strain, ε_{cu} =	0.003

Reinforcement Along X:

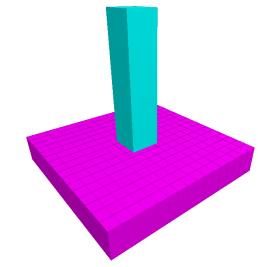
Force Values			Depth		Reinforcement Calculation													
Load condition	Factored Mt, Mx	Tension face due to bending	clear cover	d	M_u	Kn = Mu1/bd2 ϕ	ρ	$A_{s\ required}$	$A_{s,min}$	Dia of bar	$S_{required}$	$S_{provided}$	$A_{s,\ provided}$	a	c = a/β_1	ε_t = ε_{cu}/c *(d-c)	Remarks
	KN-m/m		mm	mm	KN-m/m	(N/mm2	mm	mm^2	mm^2	mm	mm c/c	mm c/c	mm^2	mm	mm		
104:1.2D	21.23	top face	50	444	21.226	0.14	0.0003	152.1	532.8	12	212.36	150	754.3	10.64	13.29	0.097	tension control
116:1.4D	-24.25	bottom face	50	442	24.248	0.17	0.0004	174.6		16	377.52	150	1341	18.91	23.64	0.053	tension control
116:1.4D	41.47	top face	50	444	41.467	0.28	0.0007	297.9		12	212.36	150	754.3	10.64	13.29	0.097	tension control
115:1.4D	25.10	top face	50	444	25.102	0.17	0.0004	180.0		12	212.36	150	754.3	10.64	13.29	0.097	tension control

Shear Check		Qx	FOS	Vu	Mu	$\rho_{w,pro}$	bd	V_{ca}	V_{cb}	V_{cc}	$V_{c,design}$	$\varphi(V_c+0.66\sqrt{f_c'}bd)\geq V_u$			Remarks	φVc	Remarks	Remarks :Min shear reinf
		Mpa		KN	KN-m/m		m^2	KN	KN	KN	KN	KN				KN		
Load condition	Max Qx	0.124	1	62	10.245	0.002	0.444	454.734	433.101	761.754	433.1	1625	> 62	ok	324.8	φVc>Vu	not required
	Min Qx	0.154	1	77	10.921			460.421			433.1	1625	> 77	ok	324.8	φVc>Vu	not required

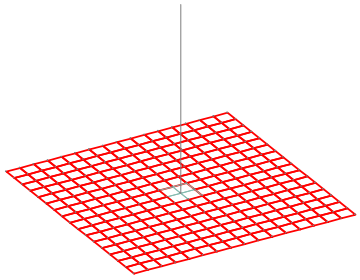
Reinforcement Along Z:

Force Values			Depth		Reinforcement Calculation												
Load condition	Factored Mt, My	Tension face due to bending	clear cover	d	M_u	Kn = Mu1/bd2 ϕ	ρ	$A_{s\ required}$	$A_{s,min}$	Dia of bar	$S_{required}$	$S_{provided}$	$A_{s,\ provided}$	a	c = a/β_1	ε_t = ε_{cu}/c *(d-c)	Remarks
	KN-m/m		mm	mm	KN-m/m	(N/mm2	mm	mm^2	mm^2	mm	mm c/c	mm c/c	mm^2	mm	mm		
116:1.4D	44.61	top face	50	444	44.608	0.30	0.0007	320.6	532.8	12	212.36	150	754.3	10.64	13.29	0.097	tension control
116:1.4D	-81.98	bottom face	50	442	81.978	0.56	0.0013	594.4		16	338.37	150	1341	18.91	23.64	0.053	tension control
116:1.4D	42.24	top face	50	444	42.236	0.29	0.0007	303.5		12	212.36	150	754.3	10.64	13.29	0.097	tension control
115:1.4D	-31.60	bottom face	50	442	31.599	0.22	0.0005	227.8		16	377.52	150	1341	18.91	23.64	0.053	tension control

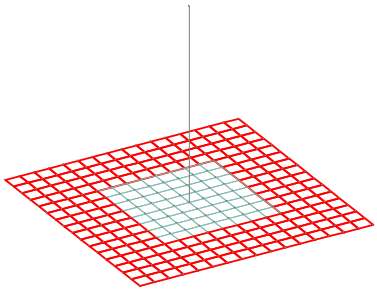
Shear Check		Qx	FOS	Vu	Mu	$\rho_{w,pro}$	bd	V_{ca}	V_{cb}	V_{cc}	$V_{c,design}$	$\varphi(V_c+0.66\sqrt{f_c'}bd)\geq V_u$			Remarks	φVc	Remarks	Remarks :Min shear reinf
		Mpa		KN	T-m/m		m^2	KN	KN	KN	KN	KN				KN		
Load condition	Max Qy	0.182	1	91	14.5	0.002	0.444	456.01	433.101	fc'761.8	433.1	455	> 91	ok	324.8	φVc>Vu	not required
	Min Qy	0.209	1	104.5	19.67			450.526			433.1	455	> 104.5	ok	324.8	φVc>Vu	not required



3D rendered view



Highlighted plate for Moment at face of column



Highlighted plate for Shear at 'd' distance from face of column

SUMMARY OF REINFORCEMENT IN RAFT:

Along X Direction :
At Bottom **16ϕT** **150** c/c
At Top **12ϕT** **150** c/c
Along Z Direction :
At Bottom **16ϕT** **150** c/c
At Top **12ϕT** **150** c/c

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REFERENCES

Reference is made to codes, standards and specifications as indicated in Civil, Structural and Architectural Design Criteria (Doc. No. MD-216-1000-EG-ST-CRT-1001) Clause 4.1.

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ATTACHMENT 1 INPUT DRAWINGS & DOCUMENTS

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MA'ADEN معادن
Saudi Arabian Mining Company شركة التعدين العربية السعودية

Outotec
LARSEN & TOUBRO

Project: Mansourah Massarah Gold Project

Customer: MA'ADEN GOLD AND BASE METALS (MA'ADEN)

Title: CALCULATION

3650 POX ACID WASH, ELUTN. AND CARBON REGN.

DESIGN CALCULATION

ISSUED FOR REFERENCE

MD-216-3650-EG-ST-CAL-2302

Rev	Date	Revision Text	Prepared By	Reviewed By	Approved By	Employer
A01	06-06-2020	FIRST ISSUE	APL	SYC	PRI	

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L&T Document Number:

O19045 450 S K 26060 002 0

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Global Project ID	900707
Delivery Type	ENGINEERING PROCUREMENT CONSTRUCTION
Project Name	MANSOURAH MASSARAH GOLD PROJECT
Customer	MA'ADEN GOLD AND BASE METALS (MA'ADEN)
Project Country	KINGDOM OF SAUDI ARABIA

Ma'aden Revision No:	A01		Outotec / L&T Revision No.:	0	
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Prepared By	APL		Prepared By	APL	
Checked	SYC		Checked	SYC	
Approved	PRI		Approved	PRI	

REFERENCE DOCUMENTS		
DOCUMENT NUMBER	DOCUMENT DESCRIPTION	
MD-216-1000-EG-ST-CRT-1001	Civil Structural Design Criteria	
SBC 301 :2007	Saudi Structural - Loading & Forces Code	
SBC 306 :2007	Saudi Steel Structures Code	
AISC LRFD	Building Code Requirement of Steel Structural Building	
AISC Steel Design Guide 1 - 2nd Edition	Base Plate and Anchor Rod Design	
EN 1991- 4 : 2006	Actions on silos	

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DEFINITIONS AND ABBREVIATIONS

Document	:	Information and the medium on which it is contained (ISO 9000:2015)
EMPLOYER	:	EMPLOYER' means MA'ADEN GOLD AND BASE METALS (MGBM) and the PMC American Bechtel Inc. or their representatives
Contractor	:	Contractor' means consortium composed of LARSEN & TOUBRO LIMITED, Saudi Arabia AND OUTOTEC SAUDI ARABIA LLC, Saudi Arabia (for IK Contract) and LARSEN & TOUBRO LIMITED – ECC Construction Group – (SHJ BR) and OUTOTEC ENGINEERING DMCC, Dubai (for OOK Contract).
'PMT'	:	Ma'aden Project Management Team
'MMGP Project'	:	Mansourah Massarah Gold Processing Project
Attribute	:	Information element that describes a document or tag.
Supplier	:	An entity (supplier, consultant, sub-contractor) under the direct project management of Contractor
Sub-Supplier	:	A sub entity (supplier, consultant, sub-contractor) under the direct project management of Contractor.
QA	:	Quality Assurance
QC	:	Quality Control
Works	:	Full scope of the EPC Contract including the engineering, procurement, construction, commissioning and associated activities

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Approved	PRI	Approved	PRI

1. INTENT

The document is prepared for Structural design Calculation of POX acid wash, Elution and carbon regeneration monorail supporting structure. This structure supports 3MT capacity monorail.

2. EXECUTIVE SUMMARY

The Saudi Arabian Mining Company (Ma'aden) has announced that its wholly-owned subsidiary, Ma'aden Gold and Base Metals Company, have issued a Notice of Award to Outotec and Larsen & Toubro Consortium to complete the engineering, procurement, construction, pre-commissioning, commissioning and start-up assistance of the Mansourah & Massarah (MMGP) Gold Project.
MMGP project will become Ma'aden's largest ever gold project, producing an average of 250,000 ounces of gold per year over the life of mine. Mansourah-Massarah is situated within the Central Arabian Gold Region (CAGR), approximately 320km northeast of Taif towards Riyadh and about 35 km southeast of Hufairah, which is a small village situated on Jeddah - Riyadh highway. Mansourah - Massarah gold deposits are located 22°38'6.40"N; 42°44'22.90"E, and 22°42'1.40"N; 42°41'58.40"E respectively, approximately 537 km east of Jeddah and 77 km southeast of the town of Zalim.
The plant throughput capacity is 4000000t/a and is required to operate on a continuous basis at nominally 24 hours per day and 7 days a week.

3. SCOPE

This document is prepared for design of Steel Structure of 3650 Pox acid wash monorail supporting structure.
All the loads applied to the structure for analysis & design followed Civil -Structural Design Criteria and SBC 301:2007
3D- Model and analysis have been performed in Staad-Pro Software. All Major components of structure have been design in Staad-Pro by AISC-LRFD Standards.
Monorail & Base plate design have been performed using MS-Excel as Per SBC-306:2007 & AISC-LRFD.

4 STEEL STRUCTURE

4.1 Design — Design of Steel structures and its components shall be in accordance with SBC-306 and AISC-LRFD.

4.2 Materials

- For Rolled Section like, HE, IPE, UPN & Angle Section	:	EN 10025 S275 JR
- For Flat & Plates	:	ASTM A36 / EN 10025-2 S275 JR
- For Chequred Plate	:	ASTM A786/ASTM A786M Pattern 4/5

4.3 Reference — Reference is made to codes, standards and specifications as indicated in Civil, Structural and Architectural Design Criteria (Doc. No. MD-216-1000-EG-ST-CRT-1001) Clause 4.1.

5 LOAD CALCULATONS

5.1 DEAD LOAD

Selfweight of structure is applied through Staad Pro.

Dead load on each floor :

Selfweight of FRP Grating (38mm thick) = 0.2 kN/m^2

All other dead loads are applied as applicable.

Dead load of stairs (approximately 2m length has been taken)

Selfweight of FRP Grating (38mm thick)	=	0.2	kN/m^2
Selfweight of stringer	=	0.253	kN/m
Selfweight of handrail	=	0.15	kN/m
Stair width	=	1.2	m
Stringer Span	=	2	m
Concentrated load on beam supporting stringer	=	1.2 x 2 x 0.2 / 4 + 0.253 x 2 / 2 + 0.15 x 2 / 2 =	0.5 kN

5.2 LIVE LOAD

Live load on each floor :

Floor live load on FRP Grating = 5 kN/m^2

All other live loads are applied as applicable.

Monorail Load (Cap. of 3 MT) = Capacity + Wt.of Hoist = 30+5 = 35 kN
Impact Factor = 20 %
So, Monorail Load = = 1.2 x 35 = 42 kN

Live load on stair (approximately 2m length has been taken)	=	5	kN/m^2
Stair width	=	1.2	m
Stringer Span	=	2	m
Stringer Length	=	2	m
Concentrated load on beam supporting stringer	=	1.2 x 2 x 5 / 4 =	3 kN

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Checked	SYC							Checked	SYC				
Approved	PRI							Approved	PRI				

5.3 WIND LOAD

WIND LOAD CALCULATION OF 3650-POX ACID WASH MONORAIL SUPPORTING STRUCTURE :

Wind Load Parameters (As per design Criteria) :

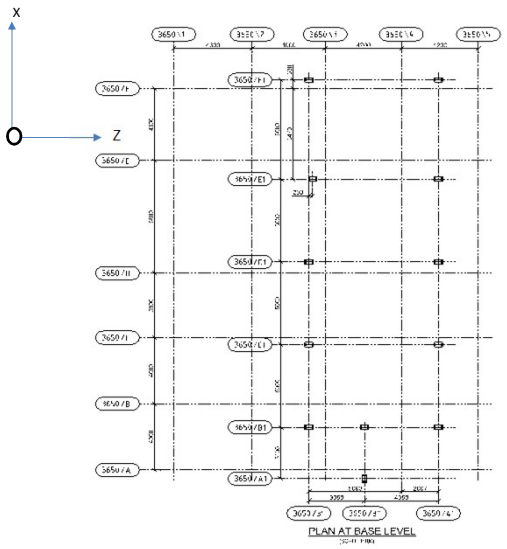
Basic wind speed, V_b	=	43	m/s	=	155	Km/h
Exposure category	=	C	(For Process Plant Area)			
Building Category	=	II	(Ref. Table 1.6-1 : SBC 301-2007)			
Wind directionablity factor, K_d	=	0.85	(Ref. Table 6.4-1 : SBC 301-2007)			
Topography Factor, K_{zt}	=	1	(As The project is not located in isolated hills or escarpments)			
Importance Factor, I	=	1	(Ref. Table 6.5-1 : SBC 301-2007)			
Velocity Pressure Co-efficient	=	Kz	(Ref. Table 7.2-2 : SBC 301-2007)			

Wind Velocity Pressure, qz = $0.0473 \times 10^{-3} K_z K_{zt} K_d V^2 I$ (KN/m^2) (Ref. Eq. 7.2-12 : SBC 301-2007)

Level	Height	kz	Pressure	Unit
EL +923.0	10m	1.0	0.966	kN/m^2
EL +933.0	20m	1.16	1.120	kN/m^2

Average Pressure upto 5.8m, qz = **0.966** kN/m^2

Gust Effect Factor, G = 0.85 (For Rigid Structure)
(Ref. Section 7.2.7.1 : SBC 301-2007)



WIND LOAD IN +X DIRECTION (on Grids 3650/F1, 3650/E1, 3650/D1, 3650/C1, 3650/B1 & 3650/A1)

1) Wind Load due to Structural Building Structure

Area for Structural Frame work at Axis 3650/7'

Member	W (m)	L (m)	A (m^2)	Nos	Total Area (m2)
Column	0.36	5.80	2.09	2	4.18
Top beam	0.36	7.40	2.66	1	2.66
Bottom beam	0.45	7.40	3.33	1	3.33
			Total Effective Projected Area =	A_f =	**10.17**

Height of Building	=	5.8	m
Frontal Area	=	45.0	m^2
Solidity ratio, Φ	=	0.23	
Force Co-efficient, C_f	=	1.80	(Ref. Table 7.2-17 : SBC 301-2007)
Design Wind Load	F =	$q_z G C_f A_f$	(Ref. Eq. 7.2-22 : SBC 301-2007)
Total Wind load	=	15.0	KN

	Force	Unit
Wind load on column	1.30	kN/m

WIND LOAD IN +Z DIRECTION (on Grids 3650/3', 3650/3'', 3650/4' & 3650/4'')

1) Wind Load due to Structural Building Structure

Area for Structural Frame work between Axis 3650/E1 & 3650/F1

Member	W (m)	L (m)	A (m^2)	Nos	Total Area (m^2)
Column	0.17	5.85	0.99	2	1.99
Top beam	0.15	6.00	0.90	1	0.90
Bottom beam	0.3	8.50	2.55	1	2.55
Bracing	0.1	4.00	0.40	4	1.60
Monorail beam	0.36	10.00	3.60	1	3.60
			Total Effective Projected Area =	A_f =	**10.64**

Height of Building	=	5.85	m
Frontal Area	=	36.1	m^2
Solidity ratio, Φ	=	0.29	
Force Co-efficient, C_f	=	1.80	(Ref. Table 7.2-17 : SBC 301-2007)
Design Wind Load	F =	$q_z G C_f A_f$	(Ref. Eq. 7.2-22 : SBC 301-2007)
Total Wind load	=	15.7	KN

Wind load on column	1.34	kN/m

Area for Structural Frame work between Axis 3650/D1 to 3650/A1

Member	W (m)	L (m)	A (m^2)	Nos	Total Area (m^2)
Column	0.17	5.85	0.99	4	3.98
Top beam	0.15	13.10	1.97	1	1.97
Bottom beam	0.24	7.50	1.80	1	1.80
Bracing	0.1	4.70	0.47	4	1.88
Monorail beam	0.36	15.00	5.40	1	5.40
			Total Effective Projected Area =	A_f =	**15.02**

Height of Building	=	5.85	m
Frontal Area	=	77.6	m^2
Solidity ratio, Φ	=	0.19	
Force Co-efficient, C_f	=	1.80	(Ref. Table 7.2-17 : SBC 301-2007)
Design Wind Load	F =	$q_z G C_f A_f$	(Ref. Eq. 7.2-22 : SBC 301-2007)
Total Wind load	=	22.2	KN

Wind load on column	0.95	kN/m

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Ma'aden Revision No:	A01		Outotec / L&T Revision No.:	0	
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Revision Text	FIRST ISSUE		Revision Text	FIRST ISSUE	
Prepared By	APL		Prepared By	APL	
Checked	SYC		Checked	SYC	
Approved	PRI		Approved	PRI	

5.4 TEMPERATURE LOAD

There are two temperature Load case is considered that is TL1 & TL2

TL1 - For Temperature Difference + 25^0 C

TL2 - For Temperature Difference -25^0 C

5.5 EARTHQUAKE LOAD

Seismic Design Category		=	A				
Site class		=	B	It is corresponds to Site Class 2 in Staad Pro			
Spectral response acceleration (1-sec)	S1	=	0.02				
Spectral response acceleration (.02-sec)	Ss	=	0.1				
Importance factor	I	=	1	(From Table 9.5, SBC 301:2007)			
Response Reduction factor along X	Rx	=	4	(From Table 10.2, SBC 301:2007)			
Response Reduction factor along Z	Rz	=	4	(From Table 10.2, SBC 301:2007)			
Height of building	h	=	5.85	m			
Time period along X	Tx	=	0.207	(eqn. 10.9.3.2-1, SBC 301:2007)			
Time period along Z	Tz	=	0.207	(eqn. 10.9.3.2-1, SBC 301:2007)			
Factor Fa	Fa	=	1				
Factor Fv	Fv	=	1	Seismic Response co-eff, Cs =SDS x I /R=	0.017	(eqn. 10.9.2.1-1, SBC 301:2007)	
SM1	Fa x S1	=	0.02	Max. value of $C_S = S_{D1}$*I /(T * R) =	0.016	(eqn. 10.9.2.1-2, SBC 301:2007)	
SMS	Fa x Ss	=	0.1	Min. value of CS = 0.044 * SDS * I =	0.003	(eqn. 10.9.2.1-3, SBC 301:2007)	
Design spectral acceleration (1-sec)	SD1 = 2/3 x SM1	=	0.0133	Total Deal load + Applicable live load =	557	kN	(From Staad output file)
Design spectral acceleration (0.2-sec)	SDS = 2/3 x SMS	=	0.0667	Seismic Base shear, V = Cs x W =	8.97	kN	(eqn. 10.9.2-1, SBC 301:2007)

Total horizontal wind force = 73 kN (From Staad output file) > Seismic Base shear mentioned above

So, in this case wind forcce is governing. Though load combinations due to earthquake have been considered here in this calculation.

The above parameters are used in the staad input to find the final earthquake load of the structure as per IBC-2003

5.6 LOAD COMBINATIONS

5.6.1 Factored Load Combination for Structural design

(As Per Section 2.3 of SBC 301-2007)

LOAD CASES	Reference
LOAD 201 1.4 DL	(Eq. 2.3.2-1 of SBC 301)
LOAD 202 1.2 (DL + TL1) + 1.6 LL LOAD 203 1.2 (DL + TL2) + 1.6 LL	(Eq. 2.3.2-2 of SBC 301)
LOAD 204 1.2 DL+ 1.6 WL(+Z) + 0.5 LL LOAD 205 1.2 DL+ 1.6 WL(-Z) + 0.5 LL LOAD 206 1.2 DL+ 1.6 WL(+X) + 0.5 LL LOAD 207 1.2 DL+ 1.6 WL(-X) + 0.5 LL	(Eq. 2.3.2-4 of SBC 301)
LOAD 208 1.2 DL+ 1.0 EL(+X) + 0.5 LL LOAD 209 1.2 DL+ 1.0 EL(-X) + 0.5 LL LOAD 210 1.2 DL+ 1.0 EL(+Z) + 0.5 LL LOAD 211 1.2 DL+ 1.0 EL(-Z) + 0.5 LL	(Eq. 2.3.2-5 of SBC 301)
LOAD 212 0.9 DL + 1.6 WL(+Z) LOAD 213 0.9 DL + 1.6 WL(-Z) LOAD 214 0.9 DL + 1.6 WL(+X) LOAD 215 0.9 DL + 1.6 WL(-X)	(Eq. 2.3.2-6 of SBC 301)
LOAD 216 0.9 DL+ 1.0 EL(+X) LOAD 217 0.9 DL+ 1.0 EL(-X) LOAD 218 0.9 DL+ 1.0 EL(+Z) LOAD 219 0.9 DL+ 1.0 EL(-Z)	(Eq. 2.3.2-7 of SBC 301)

5.7.2 Load Combination for Servicibilty check

LOAD CASES	Reference
LOAD 101 DL	(Eq. 2.4.1-1 of SBC301)
LOAD 102 (DL + TL1) + LL LOAD 103 (DL + TL2) + LL	(Eq. 2.4.1-2 of SBC 301)
LOAD 104 DL+ 0.75 WL(+Z) + 0.75 LL LOAD 105 DL+ 0.75 WL(-Z) + 0.75 LL LOAD 106 DL+ 0.75 WL(+X) + 0.75 LL LOAD 107 DL+ 0.75 WL(-X) + 0.75 LL	(Eq. 2.4.1-6 of SBC 301)
LOAD 108 DL + 0.525 EL(+X) + 0.75 LL LOAD 109 DL + 0.525 EL(-X) + 0.75 LL LOAD 110 DL + 0.525 EL(+Z) + 0.75 LL LOAD 111 DL + 0.525 EL(-Z) + 0.75 LL	(Eq. 2.4.1-6 of SBC 301)
LOAD 112 0.6 DL + 1.0 WL(+Z) LOAD 113 0.6 DL + 1.0 WL(-Z) LOAD 114 0.6 DL + 1.0 WL(+X) LOAD 115 0.6 DL + 1.0 WL(-X)	(Eq. 2.4.1-7 of SBC 301)
LOAD 116 0.6 DL + 0.7 EL(+X) LOAD 117 0.6 DL + 0.7 EL(-X) LOAD 118 0.6 DL + 0.7 EL(+Z) LOAD 119 0.6 DL + 0.7 EL(-Z)	(Eq. 2.4.1-8 of SBC 301)

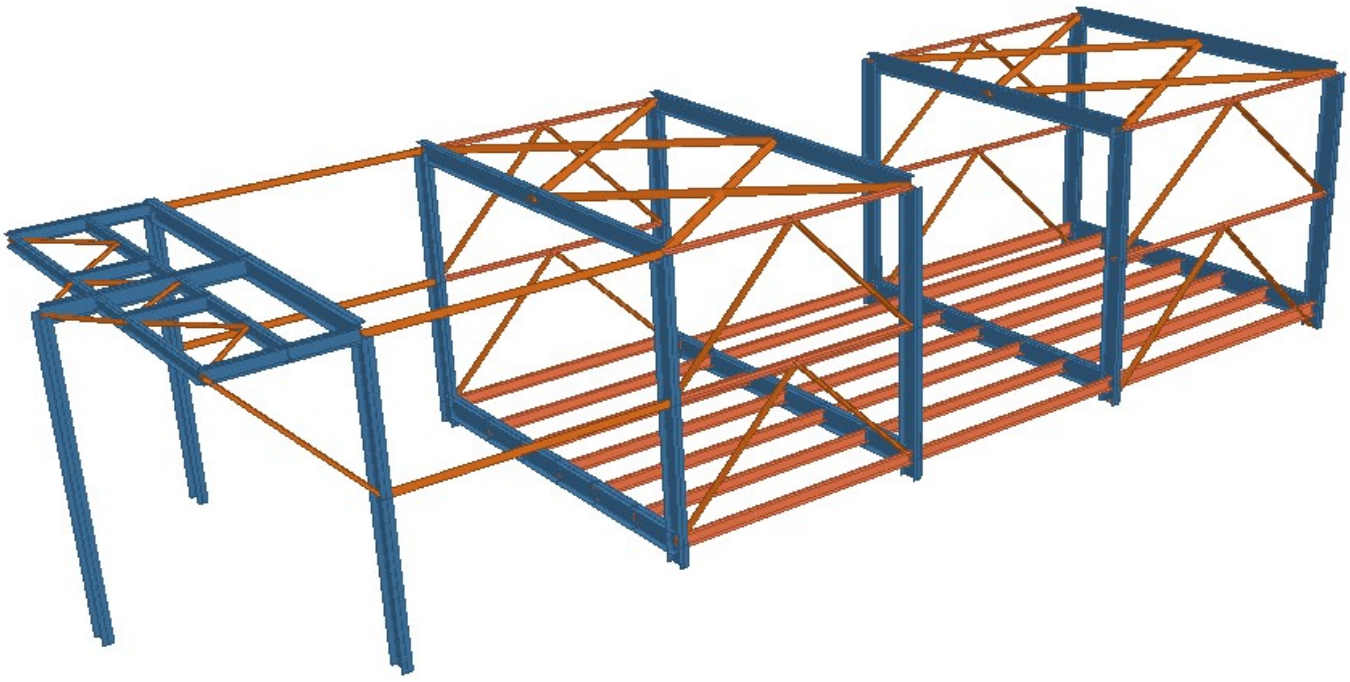
Eq. 2.3.2-3 of SBC 301 - 2007 has not been considered as it would not produce worst load combination in the absence of Roof Live Load & Rain Load

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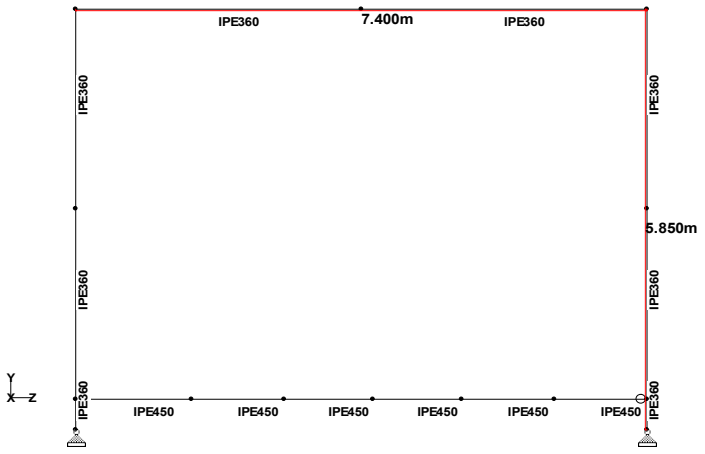
Ma'aden Revision No:	A01		Outotec / L&T Revision No.:	0	
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6 GEOMETRY

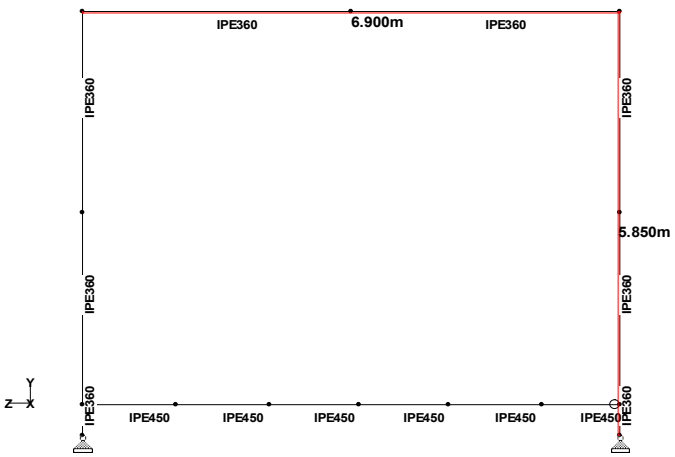
6.1 Geometry of POX Acid wash Monorail supoorting structure



3D-Model of 3650: POX acid wash, Elution and carbon regeneration Monorail supporting structure



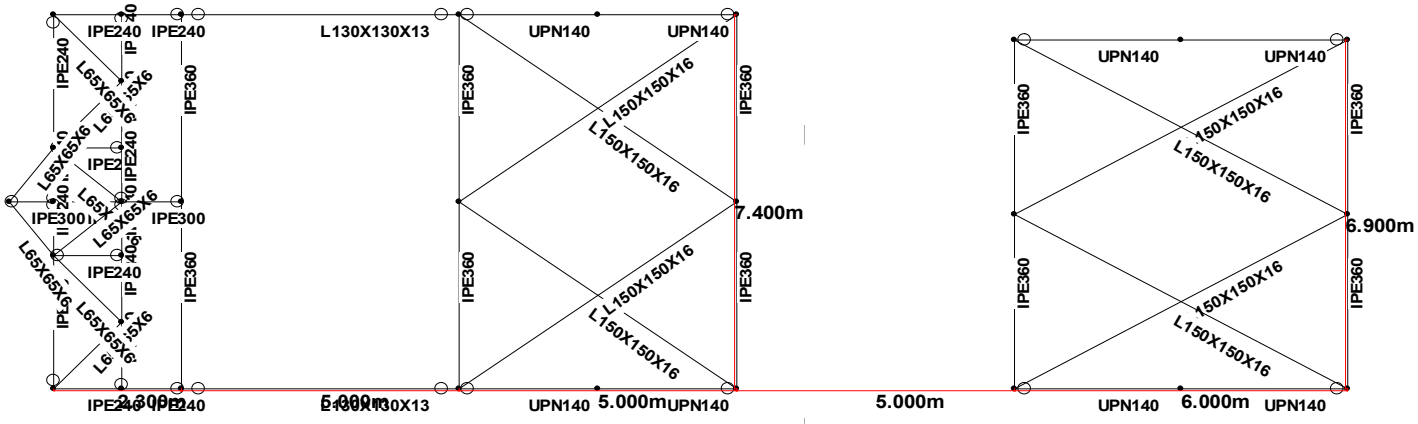
ELEVATION ALONG GRID 3650/D1



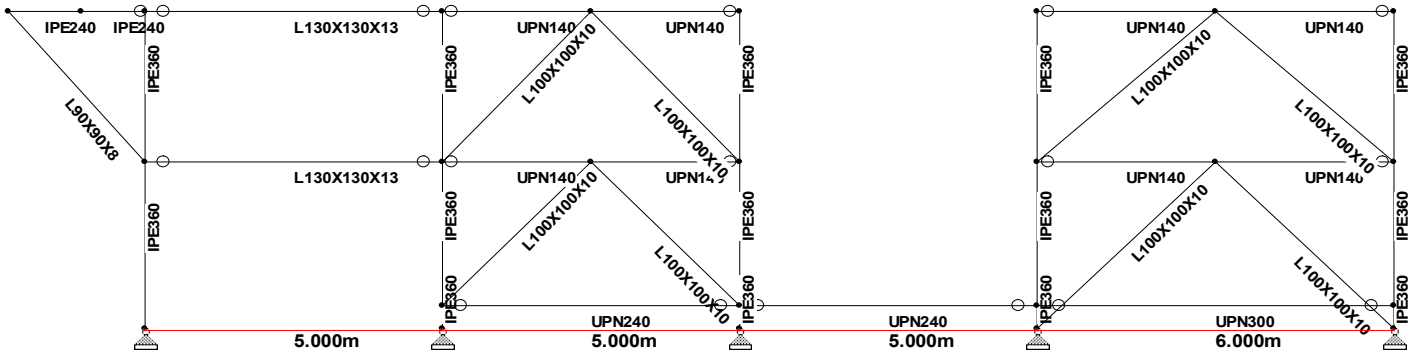
ELEVATION ALONG GRID 3650/E1

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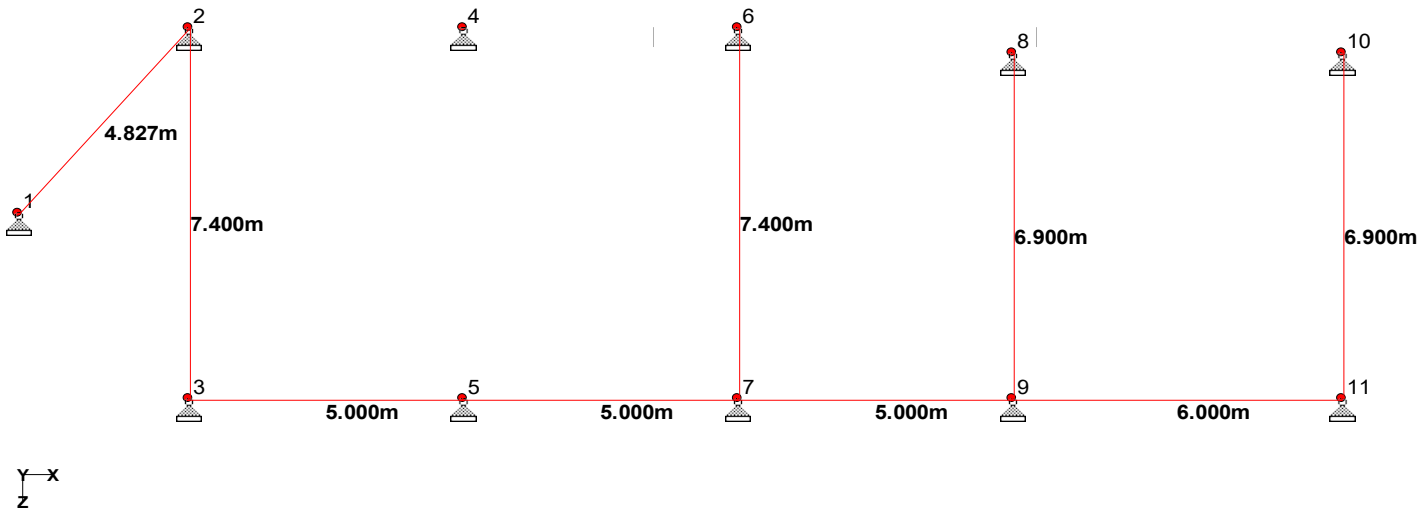


PLAN AT RL. 918.850 (T.O.S.)



ELEVATION ON GRID 3650/4'

6.2 Staad Node no & Layout of Column Support



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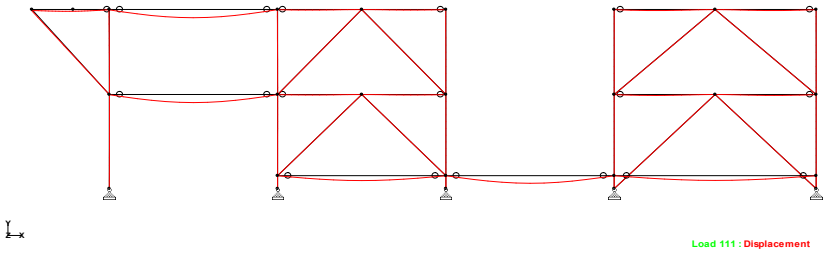
7 ANALYSIS & RESULTS OF POX ACID WASH BUILDING

For 3D analysis and results of secondary crusher house, refer Staad File - **3650_POX acid wash Monorail supporting structure.STD**

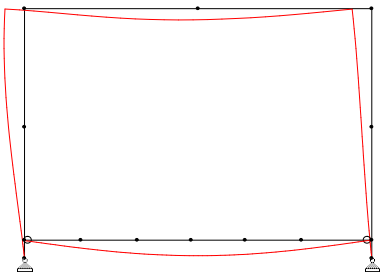
7.1 Deflection of column

Nodal Deflection at Column Top is obtained from Staad-pro and checked as per clause 7.8.1 of Maaden Design criteria Doc. No. MD-101-SMEM-EG-ST-CRT-0001

Column at Row / Axis	Height (m)	Max Deflection in X (mm)	Load Case	Max Deflection in Z (mm)	Load Case	Allowable Deflection H/200 (mm)	Remarks
4" / A1	5.85	1.3	5	17.3	3	29.3	OK
3' / B1	5.85	1.3	5	16.2	3	29.3	OK
4' / B1	5.85	1.0	5	16.2	3	29.3	OK
3' / C1	5.85	1.2	5	11.3	3	29.3	OK
4' / C1	5.85	1.0	5	11.3	3	29.3	OK
3' / D1	5.85	1.2	5	11.2	3	29.3	OK
4' / D1	5.85	1.0	5	11.2	3	29.3	OK
3" / E1	5.85	0.3	5	15.4	3	29.3	OK
4' / E1	5.85	0.3	5	15.4	3	29.3	OK
3" / F1	5.85	0.3	5	15.4	3	29.3	OK
4' / F1	5.85	0.3	5	15.4	3	29.3	OK



Deflection diagram of GRID 3650/4'



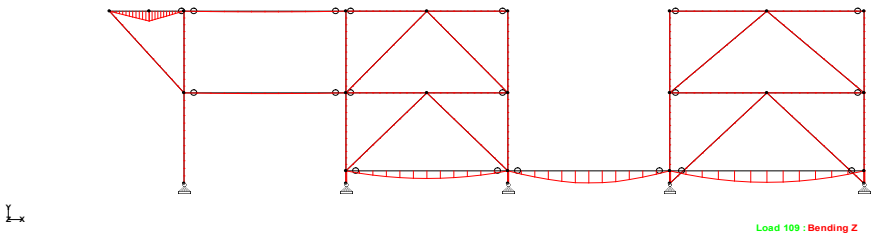
Deflection diagram of GRID 3650/C1

7.2 Deflection of Beam

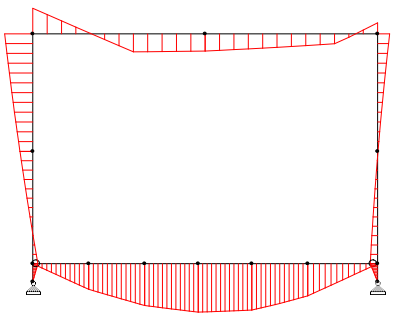
Deflection of all floor has been checked in parameter 2 of Staad-Pro design file considering below deflection limit

1) General Floor Beam - Span/ 360
2) Equipment Supporting Beam - Span/ 500

7.3 Bending Moment Diagram of Frames



Bending Moment diagram on GRID 3650/4'



Bending Moment diagram on GRID 3650/C1

8 DESIGN OF STRUCTURAL MEMBERS

All Major components of structure i.e. Column, Beam & Single angle members have been designed in Staad-Pro by AISC-LRFD Standards.
For Design Results, Refer Attachemnt 3 - Staad Output file.
Base plate design have been performed using MS-Excel as Per SBC-306:2007 & AISC-LRFD.

8.1 Design Summary of Column

Items	Staad Member No.	Pass Ratio
C1 as IPE360	1 To 29	0.62

8.2 Design of Frame Beam, Secondary Beam & Others member

For Beam Design Results, Refer Attachemnt 3 - Staad Output file.

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8.3.1 DESIGN OF BASE PLATE MKD. BP1 FOR COLUMN IPE 360 (STAAD NODE NO. 2 TO 11)

(AS PER SBC - 306 & AISC : 2005 LRFD Specification)

Column Section =	IPE 360		N
d =	360	mm	
t_f =	12.7	mm	
b_f =	170	mm	
t_w =	8	mm	

Force on Column			
Condition	**Load Case**	**Pu (KN)**	
Max. F_y	203	292.0	(comp)
Min. F_y	214	-1.4	(Ten)
Max. F_x	203	133.0	
Max. F_z	204	36.9	

(Refer Staad File for factored Load Combs)

Base Plate Details			Pedestal Details		
Length, N =	450	mm	Length =	600	mm
Width, B =	300	mm	Width =	600	mm
Thick. T =	20	mm	Area A2 =	0.36	m^2
Area A1 =	0.135	m^2			
Bolt Details			**Shear Key (in mm)**		
Dia =	20	mm	bx X tz =	300	32
Spacing, S =	150	mm	bz X tx =	300	32
web stiff thk=	8	mm	Depth =	100	(Below Grout)
Stiff height =	200	mm	Fl. Stiff =	10	mm

1) Check for compression

Grade of Conc. f'_c = 35 N/mm2

Grade of Steel = 275 N/mm2

a) Concrete bearing limit check

Bearing Strength

0.85*fc' = 29.75 N/mm2 (As per Section 10.9 of SBC 306)

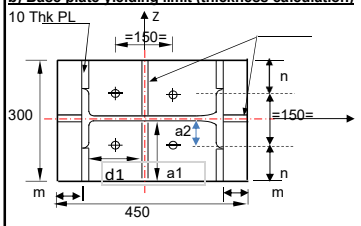
P_p = 0.85*fc' A1*SQRT(A2/A1) = 6559 KN (Eqn. 10.9-2 of SBC 306)

1.7xf'cxA1 = 8033 KN

(Cl.3.1.1 AISC Design Guide-01, Base Plate and Anchor Rod Design)

Design Bearing Load = $f_c P_p$ = 3935.1 KN > 292.012KN - OK (f_c = 0.6)

b) Base plate yielding limit (thickness calculation)



m = (N - 0.95d)/ 2 = 54 mm

n = (B - 0.8bf)/2 = 82 mm

$X = \left\{\frac{4db_f}{(d+b_f)^2}\right\}\frac{P_u}{\phi_c P_p}$ = 0.065

l = 2 √x / (1+ √(1-X)) = 0.26

l n'= l √(db_f)/4 = 15.99 mm

l max = max(m, n, ln') = 82.00 mm

t_{min} = l*sqrt((2*Pu)/(0.9*Fy*B*N))= 10.84 mm **< Tprov OK**

(As per AISC-2005 : LRFD)

c) Flexural Yeilding limit at Bearing Surface

(Cl.3.3.2 AISC Design Guide-01, Base Plate and Anchor Rod Design)

Actual area of base plate = 135000 mm2

Uniform base pressure below base plate, f_{pu} = 2.16 N/mm2

i) For Cantilever Portion :

Moment capacity of Base Plate, M_{pl} = f_{pu} * l^2/2 = 7272 N-mm/mm width

ii) For Plate supported on Three Edge :

a1 = 146.0 mm a1/d1 = 0.894

d1 = 163.3 mm α_3 = 0.106 (As per Mukhanov's Chart)

Now BM per cm = α3 X f_{pu}X $d1^2$ = 0.106 X 2.16 X 163.3^2 = 6137.7 N-mm/mm width

Maximum Bending moment M_{pu} = 7272 N-mm/mm width

Required thk. of Base Plate, $t_{reqd} = \sqrt{\frac{4 * Mpu}{f * F_y}}$ = 10.84 mm **< Tprov OK**

2) Check for Tension

a) Bolt strength check

Considering 4 nos bolt, tension per bolt = 1.35 KN / 4 = 0.3 kN

Ultimate Tensile strength of Bolts, Fu = 400 N/mm2 (For ASTM F1554 Grade 36 Bolts)

Dia of Bolt = 20 mm

Nominal Tensile Capcity of bolt, Rn = 94.23 KN = 0.75*f_u * A_b (Cl. No. 10.3.5- AISC Spec.)

Design Tensile Capcity of bolt, f * Rn = 70.67 KN ***> 0.3375 KN, OK***

b) Base Plate thickness check

Considering 50% of tension will be carried to the web, BM at the face of the web (thickness = 8 mm),

M = 0.5*T*a2 where, a2 = 71 mm

= 11981.25 N-mm

Moment per unit width = 11981.3 / 163.3

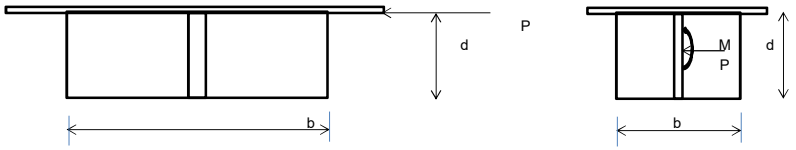
= 73.4 N-mm/mm width

Hence, thickness of baseplate required, treqd = 1.09 mm **< Tprov OK**

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3) Design of Shear Key



		In X Dir^{n}		In Z Dir^{n}	
Max. Factored shear, P_u =	(From STAAD Output)	133.0	KN	36.9	KN
Equating the load to the bearing capacity of concrete,		0		0	
Area reqd. for shear key,	$A_{lreqd.}$ = $P_u/(0.8 * fc')$ =	47.52	cm^2	13.16	cm^2
Shear Key Width ,	bz / bx =	300	mm	300	mm
	d_{reqd} (excluding Grout thk. = 50mm) =	15.84	mm	4.39	mm
		< dprov. OK		< dprov. OK	
Thickness of shear Key,	tx / tz =	32	mm	32	mm
Projected Depth of Shear Key at Pier edge, (d+c)=		384	mm	384	mm
(as per Fig. 4.9.3 of AISC Design Guide-01, Base Plate and anchor rod design)					
Projected area excl. the area of Shear Key, A_v =		200400	mm^2	200400	mm^2
Cap. of conc. In front of shear key = 4*f*sqrt(fc')*Av =		2845.4	KN	2845.4	KN
		> 133.044	KN	> 36.853	KN
Using Cantilever Model for Mid Plate,					
Moment @ shear key base, M = V (G + d/2) =		13.304	KN-m	3.6853	KN-m
Z, $bt^2/4$ =	M/ (f * Fy)			M/ (f * Fy)	
tx_{reqd} / tz_{reqd} =		26.8	mm	14.09	mm
		OK		**OK**	
For Wing Plate, Cantilever Moment =		0	KN-m	0	KN-m
t_{reqd} =		0	mm	0	mm
		< t provided - OK		**< t provided - OK**	

4) Design of Stiffeners

Deisgn Forces		
Max Comp.	292.0	KN
Max Ten.	-1.4	KN

Web Stiffener Plate Thickness, t_s = 8 mm

Web Stiffener Plate height Hs = 300 mm

Weld Size of Stiffener = 6 mm

Zp of Stiffener Plate = $bt^2/4$ = 180000 mm^3

Hs/ts = 37.5 < $1.1\sqrt{(kvE/Fy)}$

Nominal Shear Strength Vn = 0.6*Fy*Aw = 396.0 KN (Ref. Eq. 6.2 of SBC 306 - 2007)

Design Shear Strength = 0.9*Vn = 356.4 KN (Ref. Cl. 6.2.2 of SBC 306-2007)

$1.1\sqrt{(k_vE/F_y)}$ = 66.3

$1.37\sqrt{(k_vE/F_y)}$ = 82.6

where, Kv = 5

Flexural Strength calculation (Ref. Section F11 of AISC 360-2016)

a) Flexure design strength in Yielding = fn*Mp = 0.9*Fy*Zp = 44550 KN-mm

b) Flexure design strength in Lateral-Torsional buckling = fb*Mn = 0.9*Mn

L_b =2*a1 = 292 mm

L_bd/t^2 = 1368.75

> 0.08E/Fy & <1.9E/Fy

So, Mn = Cb(1.52-0.274(Lbd/t2)Fy/E)My

Mn = 26514.1 KN-mm

0.08E/Fy = 58.2

1.9E/Fy = 1381.8

Cb = 1

My = 0.8*Fy*Z = 26400.0 KN-mm

So, Maximum Design Flexure Strength = 26514 KN-mm

	Compression		Tension	
Load on Stiffener Plate ,	P=a1*d1*fpu =	51.57 KN	Ts = T/4 =	0.34 KN
	< 356.4 OK		< 356.4 OK	
Moment at the edge ,	M= P*a1/2=	3764.7 KN-mm	M = Ts*S =	51 KN-mm
	< 26515 OK		< 26515 OK	

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8.3.2 DESIGN OF BASE PLATE MKD. BP1A FOR ONLY COLUMN IPE 360 (ON GRID MKD. 3650/4" & 3650/A1)

(AS PER SBC - 306 & AISC : 2005 LRFD Specification)

Column Section =	IPE 360		N
d =	360	mm	
t_f =	12.7	mm	
b_f =	170	mm	
t_w =	8	mm	

Force on Column			
Condition	**Load Case**	**Pu (KN)**	
Max. F_y	202	74.6	(comp)
Min. F_y	215	8.2	(Ten)
Max. F_x	207	4.7	
Max. F_z	204	5.9	

(Refer Staad File for factored Load Combs)

Base Plate Details			Pedestal Details		
Length, N =	450	mm	Length =	600	mm
Width, B =	300	mm	Width =	600	mm
Thick. T =	20	mm	Area A2 =	0.36	m^2
Area A1 =	0.135	m^2			
Bolt Details			**Shear Key (in mm)**		
Dia =	20	mm	bx X tz =	150	20
Spacing, S =	150	mm	bz X tx =	150	20
web stiff thk=	8	mm	Depth =	100	(Below Grout)
Stiff height =	200	mm	Fl. Stiff =	10	mm

1) Check for compression

Grade of Conc. f'_c = 35 N/mm2

Grade of Steel = 275 N/mm2

a) Concrete bearing limit check

Bearing Strength

0.85*fc' = 29.75 N/mm2 (As per Section 10.9 of SBC 306)

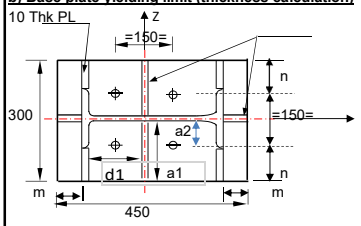
P_p = 0.85*fc' A1*SQRT(A2/A1) = 6559 KN (Eqn. 10.9-2 of SBC 306)

1.7xf'cxA1 = 8033 KN

(Cl.3.1.1 AISC Design Guide-01, Base Plate and Anchor Rod Design)

Design Bearing Load = f_cP_p = 3935.1 KN > 74.629KN - OK (f_c = 0.6)

b) Base plate yielding limit (thickness calculation)



m = (N - 0.95d)/ 2 = 54 mm

n = (B - 0.8bf)/2 = 82 mm

$X = \left\{\frac{4db_f}{(d+b_f)^2}\right\}\frac{P_u}{\phi_c P_p}$ = 0.017

l = 2 √x / (1+ √(1-X)) = 0.13

l n'= l √(db_f)/4 = 7.98 mm

l max = max(m, n, ln') = 82.00 mm

t_{min} = l*sqrt((2*Pu)/(0.9*Fy*B*N))= 5.48 mm < Tprov OK

(As per AISC-2005 : LRFD)

c) Flexural Yeilding limit at Bearing Surface

(Cl.3.3.2 AISC Design Guide-01, Base Plate and Anchor Rod Design)

Actual area of base plate = 135000 mm2

Uniform base pressure below base plate, f_{pu} = 0.55 N/mm2

i) For Cantilever Portion :

Moment capacity of Base Plate, M_{pl} = f_{pu} * l^2/2 = 1859 N-mm/mm width

ii) For Plate supported on Three Edge :

a1 = 146.0 mm a1/d1 = 0.894

d1 = 163.3 mm α_3 = 0.106 (As per Mukhanov's Chart)

Now BM per cm = α3 X f_{pu}X $d1^2$ = 0.106 X 0.55 X 163.3^2 = 1568.6 N-mm/mm width

Maximum Bending moment M_{pu} = 1859 N-mm/mm width

Required thk. of Base Plate, $t_{reqd} = \sqrt{\frac{4 * Mpu}{f * F_y}}$ = 5.48 mm < Tprov OK

2) Check for Tension

a) Bolt strength check

Tension per bolt = 0.0 kN

Ultimate Tensile strength of Bolts, Fu = 400 N/mm2 (For ASTM F1554 Grade 36 Bolts)

Dia of Bolt = 20 mm

Nominal Tensile Capcity of bolt, Rn = 94.23 KN = 0.75*f_u * A_b (Cl. No. 10.3.5- AISC Spec.)

Design Tensile Capcity of bolt, f * Rn = 70.67 KN ***> 0 KN, OK***

b) Base Plate thickness check

M = 0.5*T*a2 where, a2 = 71 mm

= 0 N-mm

Moment per unit width = 0 / 163.3

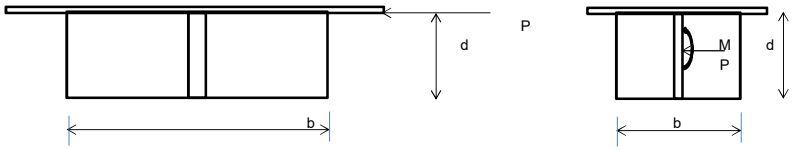
= 0.0 N-mm/mm width

Hence, thickness of baseplate required, treqd = 0.00 mm **< Tprov OK**

Global Project ID	900707
Delivery Type	ENGINEERING PROCUREMENT CONSTRUCTION
Project Name	MANSOURAH MASSARAH GOLD PROJECT
Customer	MA'ADEN GOLD AND BASE METALS (MA'ADEN)
Project Country	KINGDOM OF SAUDI ARABIA

Ma'aden Revision No:	A01		Outotec / L&T Revision No.:	0
Date	06-06-2020		Date	06-06-2020
Revision Text	FIRST ISSUE		Revision Text	FIRST ISSUE
Prepared By	APL		Prepared By	APL
Checked	SYC		Checked	SYC
Approved	PRI		Approved	PRI

3) Design of Shear Key



		In X Dir^{n}		In Z Dir^{n}	
Max. Factored shear, P_u =	(From STAAD Output)	4.7	KN	5.9	KN
Equating the load to the bearing capacity of concrete,		0		0	
Area reqd. for shear key,	$A_{lreqd.}$ = $P_u/(0.8 * fc')$ =	1.67	cm^2	2.10	cm^2
Shear Key Width ,	bz / bx =	150	mm	150	mm
	d_{reqd} (excluding Grout thk. = 50mm) =	1.11	mm	1.40	mm
		< dprov. OK		< dprov. OK	
Thickness of shear Key,	tx / tz =	20	mm	20	mm
Projected Depth of Shear Key at Pier edge, (d+c)=		390	mm	390	mm
(as per Fig. 4.9.3 of AISC Design Guide-01, Base Plate and anchor rod design)					
Projected area excl. the area of Shear Key, A_v =		219000	mm^2	219000	mm^2
Cap. of conc. In front of shear key = 4*f*sqrt(fc')*Av =		3109.5	KN	3109.5	KN
		> 4.68	KN	> 5.883	KN
Using Cantilever Model for Mid Plate,					
Moment @ shear key base, M = V (G + d/2) =		0.468	KN-m	0.5883	KN-m
Z, $bt^2/4$ =	M/ (f * Fy)				
tx_{reqd} = / tz_{reqd} =		7.1	mm	7.96	mm
		OK		OK	
For Wing Plate, Cantilever Moment =		0	KN-m	0	KN-m
t_{reqd} =		0	mm	0	mm
		< t provided - OK		< t provided - OK	

4) Design of Stiffeners

Deisgn Forces		
Max Comp.	74.6	KN
Max Ten.	0.0	KN

Web Stiffener Plate Thickness, t_s = 8 mm

Web Stiffener Plate height Hs = 300 mm

Weld Size of Stiffener = 6 mm

Zp of Stiffener Plate = $bt^2/4$ = 180000 mm^3

Hs/ts = 37.5 < $1.1\sqrt{(kvE/Fy)}$

Nominal Shear Strength Vn = 0.6*Fy*Aw = 396.0 KN (Ref. Eq. 6.2 of SBC 306 - 2007)

Design Shear Strength = 0.9*Vn = 356.4 KN (Ref. Cl. 6.2.2 of SBC 306-2007)

$1.1\sqrt{(k_vE/F_y)}$ = 66.3

$1.37\sqrt{(k_vE/F_y)}$ = 82.6

where, Kv = 5

Flexural Strength calculation (Ref. Section F11 of AISC 360-2016)

a) Flexure design strength in Yielding = fn*Mp = 0.9*Fy*Zp = 44550 KN-mm

b) Flexure design strength in Lateral-Torsional buckling = fb*Mn = 0.9*Mn

L_b =2*a1 = 292 mm

L_bd/t^2 = 1368.75

> 0.08E/Fy & <1.9E/Fy

So, Mn = Cb(1.52-0.274(Lbd/t2)Fy/E)My

Mn = 26514.1 KN-mm

0.08E/Fy = 58.2

1.9E/Fy = 1381.8

Cb = 1

My = 0.8*Fy*Z = 26400.0 KN-mm

So, Maximum Design Flexure Strength = 26514 KN-mm

	Compression		Tension	
Load on Stiffener Plate ,	P=a1*d1*fpu =	13.18 KN	Ts = T/4 =	0.00 KN
		< 356.4 OK		< 356.4 OK
Moment at the edge ,	M= P*a1/2=	962.1 KN-mm	M = Ts*S =	0 KN-mm
		< 26515 OK		< 26515 OK

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Project Country	**KINGDOM OF SAUDI ARABIA**

Ma'aden Revision No:	A01							Outotec / L&T Revision No.:	0					
Date	06-06-2020							Date	06-06-2020					
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Prepared By	APL							Prepared By	APL					
Checked	SYC							Checked	SYC					
Approved	PRI							Approved	PRI					

8.4 Design of Monorail beam (Capacity 3T)

Design parameters:

Capacity =	3.0	T	
Self -wt =	0.5	T	
Wheel load per wheel set =	3.5	T	
Inpact load =	25 % of Wheel load		(Cl. 10.3.2 of SBC 301 : 2007)
Hence, Design wheel load =	4.2	T	
=	4.2	T	(approx)
Effective length , k_zL_z =	6	m	
Factored Maximum Bending moment, M_u =	9.5	T-m	= 1.5 x 4.2 x 6/ 4
Allowable slenderness ratio =	200		
Distance between wheels =	200	mm	

Material Properties :

Grade of steel =	S275 (Ult 450)	
E =	200000	N/mm^2
G =	77200	N/mm^2
F_u =	450	N/mm^2
f_{yf} =	275	N/mm^2
F_{yw} =	275	N/mm^2
Fr=	69	N/mm^2
$F_L(F_{yf}-F_r)$ =	206	N/mm^2

Sectional properties:

Section: IPE 360

D =	360	mm	A = 72.73	cm^2
b_{top} =	170	mm	I_y = 1043	cm^4
t_{top} =	12.7	mm	r_z = 14.95	cm
b_{bot} =	170	mm	r_y = 3.79	cm
t_{bot} =	12.7	mm	Z_{ezT} = 903.6	cm^3
t_w =	8	mm	Z_{ey} = 122.8	cm^3
d =	298.6	mm	Z_{pz} = 1019.1	cm^3
I_z =	16266	cm^4	Z_{py} = 191.1	cm^3
			J = 37.3	cm^4
			Cw = 3.14E+11	cm^6

Unstiffened Element	Flanges	Compact			*(Table 2.5-1)*
		λ=	b/t =	6.7	
Compact Limit		λp=	$0.38(E/F_y)^{1/2}$ =	10.2	
Non Compact Limit		λr=	$0.83(E/F_L)^{1/2}$ =	25.9	
Unstiffened Element	Web	Compact			
		λ=	d/t_w =	37.3	
Compact Limit		λp=	$3.76(E/F_y)^{1/2}$ =	101.4	
Non Compact Limit		λr=	$5.70(E/F_y)^{1/2}$ =	153.7	

Hence the section is compact

Check Bending about the Z-Z Axis *(Section 6.1)*

Actual Unbraced Length		L_b =	6000	mm			
Limit. lat. unbraced length,	L_p, 1.76 *ry sqrt(E/Fyf) =		1799	mm	<	L_b	
Limiting Buckling moment =		$M_r = F_L * Z_{EZ}$ =	186.1	KN-m			
		X_1 =	15902.8	N/mm2			
		X_2 =	118.5783796	mm4/N			
	Lr = (r_y* X_1/F_L)*SQRT(1+SQRT(1+ X_2 F_L^2)) =		138604.42	mm	>	L_b	
Plastic moment of section =		M_p=	280.25	KN-m	<	1.5*M_y =	372.74
		C_b =	1.00			f_b =	0.9
		M_n =	277.36	KN-m	<	M_p	
Flexural Design Strength about the Z-Z Axis = φ_b * M_n =			249.6	KN-m	>	M_u	
Design Flexural Compressive stress = φ_b * M_n/Z_{pz} =			**245.0**	**N/mm2**			
Actual Flexural Compressive stress,	**9.5*10^4/1019.1=**		**93.3**	**N/mm2**	**< 245**	**N/mm2**	**OK**

Check for flange thickness :-

load per wheel =	4.2 /4 =	1.05	T			
Position of wheel from cl of web. =		50.00	mm			
Moment at the flange =	1.05x(50-8/2)/1000x1.5 =	0.08	T-m			
eff. width of the fl.pl.(considering 30 deg dispersion) =		159.35	mm			
thickness of bott. Flg near cl .of web =		12.70	mm			
Calculated tensile stress at bottom flange =	0.08 *4/ (159.35*12.7^2) =	**124.60**	**N/mm2**			
Combined stress =	**93.3 + 124.6 =**	**217.90**	**N/mm2**	**< 245**	**N/mm2**	**OK**

Check for Deflection

Conc. load on beam, P =		42.00	kN	
Actual Deflection,	d_{act} = 1/48 * 42 * 6^3 / (200000 * 16266) =	5.81	mm	
Allowable Deflection,	**d_{allw} = L/ 500 =**	**12.00**	**mm**	
		> Actual deflection- Ok		

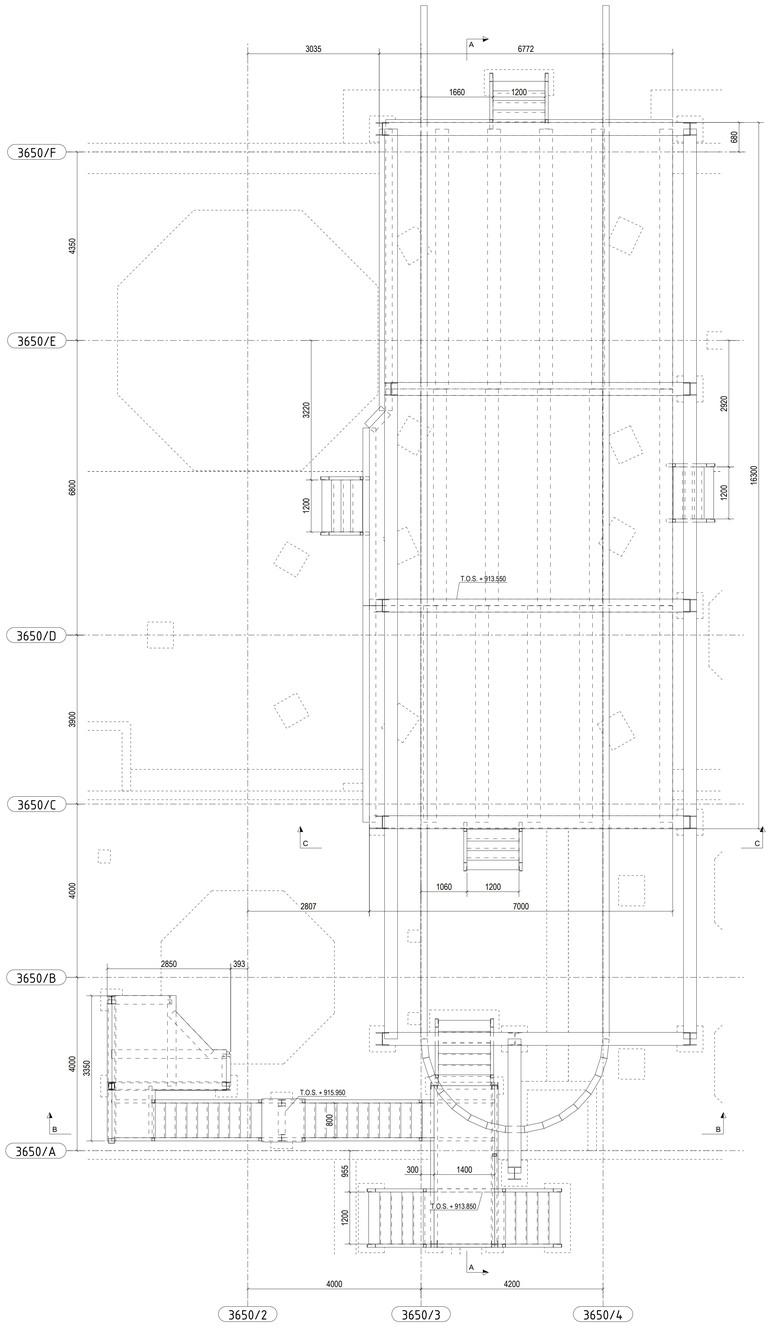
Global Project ID	**900707**
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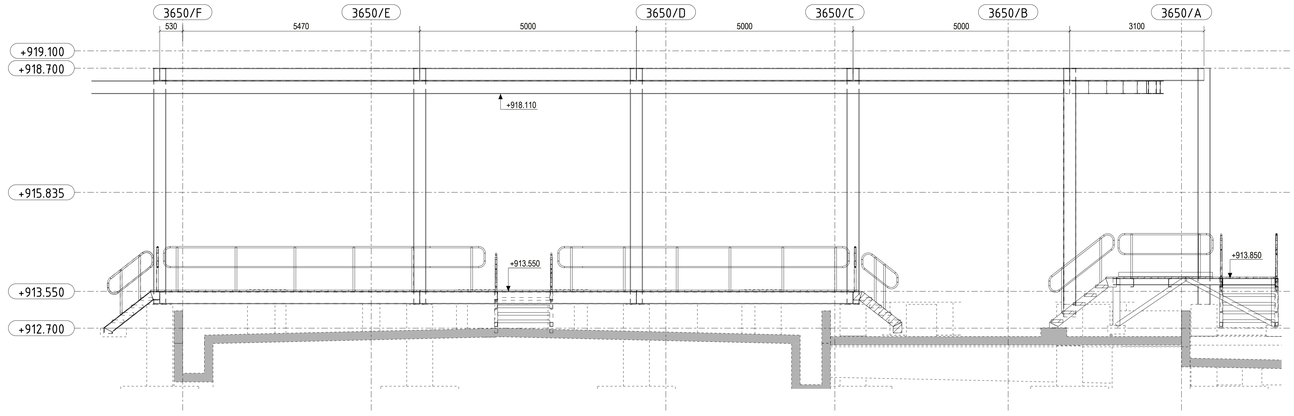
ATTACHMENT 1 **MECHANICAL & STRUCTURAL GUIDE DRAWING**

DRAWING NUMBER: MD-216-3650-EG-ST-GAD-1002
REV: A01

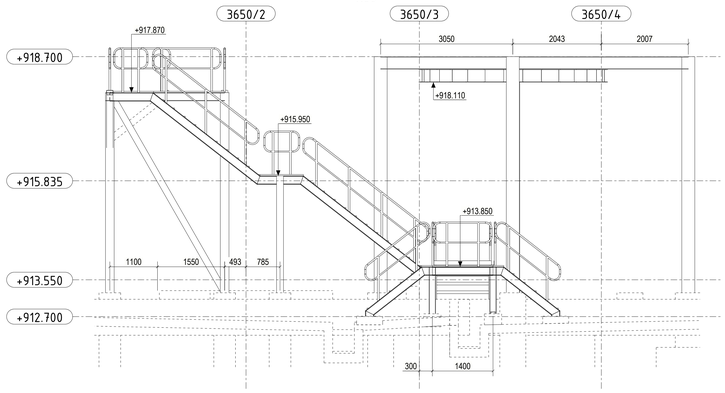
LEVEL +912.700
1:50



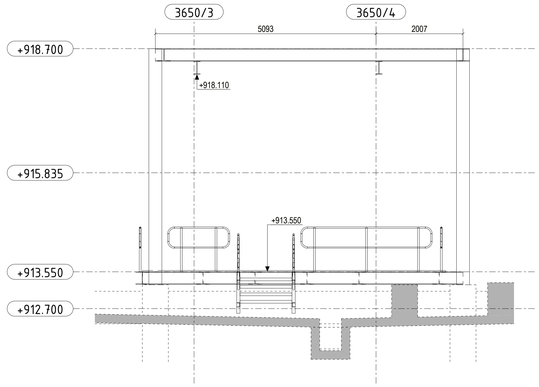
A - A
1:50



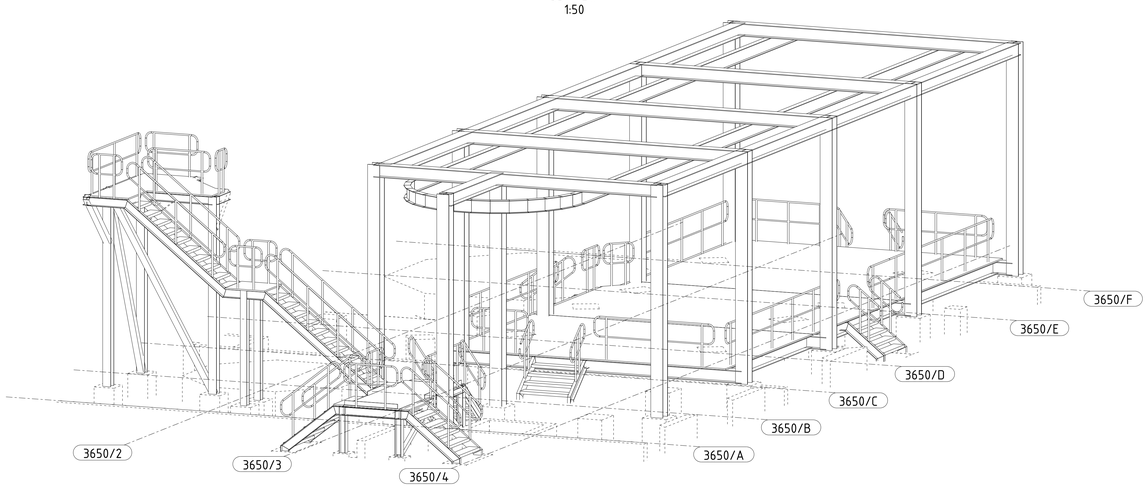
B - B
1:50



C - C
1:50



3D VIEW
1:50



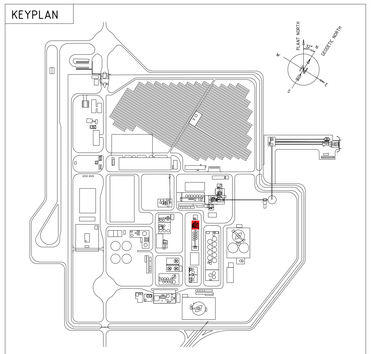
THIS DESIGN IS NOT ISSUED FOR CONSTRUCTION

FINISHED TOP GROUND LEVEL (F.G.L) FOR PROCESS AREA 3650 = RL(+)912.700 m

CHECKERED PLATE
GRATING
GRATING - FRP

1. ALL DIMENSIONS ARE IN MILLIMETER, ELEVATIONS ARE IN METER UNLESS OTHERWISE STATED
2. CO-ORDINATES ARE SHOWN IN GENERAL ARRENGEMENT DRAWINGS
3. GRATING TYPE (FRP OR NORMAL) IS SPECIFIED IN GENERAL ARRENGEMENT DRAWINGS
4. F.F.L, F.G.L, T.O.C = FINISHED FLOOR LEVEL, FINISHED GROUND LEVEL, TOP OF CONCRETE, T.O.S, B.O.S, B.O.B, = TOP OF STEEL, BOTTOM OF STEEL, BOTTOM OF BASE PLATE
5. BUILDING DIMENSIONS AND HEIGHTS ARE TENTATIVE, SHALL BE FINALIZED DURING DETAIL ENGINEERING SUBJECT TO MINMUM AS SHOWN.
6. ALL MAINTENANCE FACILITIES, HANDLING PROVISIONS, ACCESS WAYS, PLATFORMS FOR MAINTENANCE SHALL BE PROVIDED DURING DETAIL ENGINEERING WHICH ARE NOT SHOWN ENTIRELY. THIS DRAWING IS TO BE CONSIDERED AS CONCEPTUAL ARRANGEMENT OF THE FACILITIES.

KEYPLAN



DOCUMENT NUMBER	REFERENCE DOCUMENTS	DOCUMENT NUMBER	REFERENCE DOCUMENTS	REV	DATE
MD-216-1000-EG-ST-CRT-1001	CIVIL, STRUCTURAL AND ARCHITECTUAL DESIGN CRITERIA				
MD-216-3650-EG-ME-GAD-1008-RA01	SECTION C-C	MD-216-3650-EG-ME-GAD-1009	SECTION D-D		
MD-216-3650-EG-ME-GAD-1006	SECTION A-A	MD-216-3650-EG-ME-GAD-1007	SECTION B-B		
MD-216-3650-EG-ME-GAD-1004	PLAN AT EL. +918.700 & EL. +921.300	MD-216-3650-EG-ME-GAD-1005	PLPLAN AT EL. +925.200 & EL. +927.400		
MD-216-3650-EG-ME-GAD-1002	PLAN AT EL. +912.700	MD-216-3650-EG-ME-GAD-1003	PLAN AT EL. +915.400		
MD-216-3650-EG-ST-GAD-1001	PLAN AT EL. +912.700 GRID A-F/6-8	MD-216-3650-EG-ST-GAD-1002	PLAN AT EL. +912.700 GRID A-F/1-5		
MD-216-3650-EG-CV-GAD-1001	PLAN AT EL. +912.700	MD-216-3650-EG-CV-GAD-1002	PLAN AT EL. +912.700 - SECTIONS		

REVISION DESCRIPTION	DRN	CHK	ENG	APP	EMP	REV	DATE
ISSUED FOR APPROVAL						A01	15.04.2020

REVISION DESCRIPTION	DRN	CHK	ENG	APP	EMP
	EliKv	MaLai	LuPiz	SuoCha	

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Saudi Arabian Mining Company شركة معادن العربية السعودية

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OUTOTEC DRAWING NUMBER: 900707_GAU01_EE12_CTA05_36502 — DOCUMENT ID: OU100674884 — REVISION: 0

DRAWN	EliKv	15.04.2020	
CHECKED	MaLai	15.04.2020	
ENGINEER	LuPiz	15.04.2020	
APPROVED	SuoCha	15.04.2020	
EMPLOYER			
SCALE	1:50	A0	

MANSOURAH - MASSARAH GOLD PROJECT
STEEL STRUCTURE GUIDE DRAWING
3650 POX ACID WASH - ELUTION AND CARBON GENERATION
PLAN AT EL. +912.700 GRID A-F/6-8

DRAWING NUMBER: MD-216-3650-EG-ST-GAD-1002 — REV: A01

Tekla Structures

DRAWING NUMBER: MD-216-3650-EG-ME-GAD-1002 | REV: A01

PREVAILING WIND DIRECTION 25° W | PLANT NORTH 32° N GEODETIC NORTH | MECCA | S | E

3650-PMP-008 CARBON TRANSFER WATER PUMP (3650-025 HOLD)
3650-PMP-011 CARBON FINES FILTER FEED PUMP (3650-028 HOLD)
3650-PMP-103 POX REGENERATED CARBON AREA SUMP PUMP (3650-072 HOLD)
E=264212.413 N=2506103.562
TOS EL. +913.550
3650-PMP-012 CARBON FINES FILTER FEED PUMP (3650-029 HOLD)
3650-THK-001 COMBINED CARBON SETTLER (3650-038 HOLD)
TOC EL. +912.540 LP
3650-TNK-006 CARBON PROCESS WATER TANK (3650-044 HOLD)
3650-PMP-007 CARBON TRANSFER WATER PUMP (3650-024 HOLD)
TOC EL. +912.700 HP
3650-PMP-001 POX ELUTION PUMP (3650-018 HOLD)
3650-REA-001 CARBON ATTRITIONING TANK (3650-032 HOLD)
3610-PMP-013 POX CIL WASH SOLUTION PUMP (3610-016 HOLD)
3610-PMP-012 POX CIL WASH SOLUTION PUMP (3610-015 HOLD)
3610-TNK-001 POX CIL WASH SOLUTION TANK
3650-PMP-009 POX REGENERATED CARBON PUMP (3650-026 HOLD)
3650-TNK-002 POX ELUTION TANK (3650-040 HOLD)
3650-TNK-001 POX ACID WASH TANK (3650-039 HOLD)
3650-PMP-102 POX CARBON AREA SUMP PUMP (3650-031 HOLD)
TOC EL. +912.540 LP
3650-PMP-010 POX REGENERATED CARBON PUMP (3650-027 HOLD)
3650-PMP-002 POX ELUTION PUMP (3650-019 HOLD)
TOS EL. +913.350
3620-007 HOLD — 3820-PMP-003 HYDROCHLORIC ACID DOSING PUMP
3820-009 HOLD — 3820-PMP-005 HYDROCHLORIC ACID DOSING PUMP
TOC EL. +912.700
3650-PMP-006 POX ACID WASH PUMP (3650-023 HOLD)
3650-COL-001 POX ACID WASH COLUMN (3650-002 HOLD)
3650-COL-002 POX ELUTION COLUMN (3650-003 HOLD)
3820-012 HOLD — 3820-TNK-001 HYDROCHLORIC ACID STORAGE TANK
3820-008 HOLD — 3820-PMP-004 HYDROCHLORIC ACID DOSING PUMP
TOS EL. +913.850
TOC EL. +912.470 LP
TOC EL. +912.700 HP
E=264197.660 N=2506067.163
3650-PMP-101 POX ACID WASH AREA SUMP PUMP (3650-030 HOLD)
3650-PMP-005 POX ACID WASH PUMP (3650-022 HOLD)
3650-TNK-003 POX NEUTRALIZATION WASH TANK (3650-041 HOLD)

3650/F | 3650/E | 3650/D | 3650/C | 3650/B | 3650/A
3650/1 | 3650/2 | 3650/3 | 3650/4 | 3650/5 | 3650/6 | 3650/7 | 3650/8
4300 | 4000 | 4200 | 4200 | 5200 | 4200 | 5700 — 31800
4350 | 6800 | 2350 | 3900 | 4000 | 4000

CONCRETE ELEMENT FLOOR (PRE-CAST SLAB)
CONCRETE FLOOR
CONCRETE FLOOR WITH ALKALI / ACID RESISTANT COATING
CHECKERED PLATE
GRATING
GRATING - FRP
ACID RESISTANT BRICKLINING
BRICK WALL
CONCRETE WALL

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3. F.F.L, F.G.L, T.O.C = FINISHED FLOOR LEVEL, FINISHED GROUND LEVEL, TOP OF CONCRETE. T.O.S, B.O.S, B.O.B, = TOP OF STEEL, BOTTOM OF STEEL, BOTTOM OF BASE PLATE. T.O.G. = TOP OF GRATING. HP = HIGHT POINT, LP = LOW POINT, T.O.R. = TOP OF RAIL.
4. BUILDING DIMENSIONS AND HEIGHTS ARE TENTATIVE, SHALL BE FINALIZED DURING DETAIL ENGINEERING SUBJECT TO MINIMUM AS SHOWN.
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KEYPLAN

PLAN AT EL.(+)912.700

0 1 10m

DOCUMENT NUMBER	REFERENCE DOCUMENTS
MD-216-3000-EG-GM-LST-1010	HOLD LIST
MD-216-3650-EG-ME-GAD-1008	SECTION C-C
MD-216-3650-EG-ME-GAD-1007	SECTION B-B
MD-216-3650-EG-ME-GAD-1006	SECTION A-A
MD-216-3650-EG-ME-GAD-1005	PLAN AT EL. +925.200 & EL. +927.400
MD-216-3650-EG-ME-GAD-1004	PLAN AT EL. +918.700 & EL. +921.300
MD-216-3650-EG-ME-GAD-1003	PLAN AT EL. +915.400
MD-216-3650-EG-ME-GAD-1015	LOAD PLAN, PLAN AT EL. +912.700

REV	DATE	REVISION DESCRIPTION	DRN	CHK	ENG	APP	EMP
A01	10-04-2020	ISSUED FOR APPROVAL	GK	TV	SC	RL	

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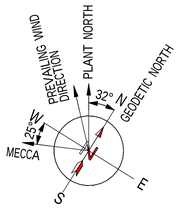
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LARSEN & TOUBRO DRAWING NUMBER	019045_450_M_G_02954_001		REVISION 0
OUTOTEC DRAWING NUMBER	900707 _GAU01 _EE12 _MTL01 _36503	DOCUMENT ID OU100681164	REVISION 0
DRAWN	GK	14-04-2020	MANSOURAH-MASSARAH GOLD PROJECT
CHECKED	TV	14-04-2020	GENERAL ARRANGEMENT DRAWING
ENGINEER	SC	14-04-2020	POX ACID WASH,ELUTION/CARBON REGENERATION
APPROVED	RL	14-04-2020	PLAN AT EL. +912.700
EMPLOYER			
SCALE	1:50	A0	DRAWING NUMBER MD-216-3650-EG-ME-GAD-1002 — REV A01

DRAWING NUMBER: MD-216-3650-EG-ME-GAD-1003 | REV: A01



3650-038 HOLD — 3650-THK-001 COMBINED CARBON SETTLER
3650-071 HOLD — 3650-HST-002 POX ELUTION AREA HOIST
3650-044 HOLD — 3650-TNK-006 CARBON PROCESS WATER TANK
3610-TNK-001 POX CIL WASH SOLUTION TANK
3650-040 HOLD — 3650-TNK-002 POX ELUTION TANK
3820-012 HOLD — 3820-TNK-001 HYDROCHLORIC ACID STORAGE TANK
3820-002 HOLD — 3820-FAN-001 HYDROCHLORIC ACID AREA OFFGAS FAN
3820-011 HOLD — 3820-SCB-001 HYDROCHLORIC ACID AREA OFFGAS DEMISTER
3650-039 HOLD — 3650-TNK-001 POX ACID WASH TANK
3650-041 HOLD — 3650-TNK-003 POX NEUTRALIZATION WASH TANK
3650-003 HOLD — 3650-COL-002 POX ELUTION COLUMN
3650-002 HOLD — 3650-COL-001 POX ACID WASH COLUMN
3650-FLT-001 CARBON FINES FILTER — 3650-009 HOLD
3650-CHT-003 CARBON FINES FILTER CHUTE — 3650-048 HOLD
3650-REA-001 CARBON ATTRITIONING TANK — 3650-032 HOLD
3650-TNK-005 POX QUENCH TANK — 3650-043 HOLD
3650-HEX-005 POX HEATER HEAT EXCHANGER — 3650-070 HOLD
3650-HEX-004 POX HEATER HEAT EXCHANGER — 3650-069 HOLD
3650-HEX-003 POX ELUATE COOLING HEAT EXCHANGER — 3650-013 HOLD
3650-HEX-002 POX RECLAIM HEAT EXCHANGER — 3650-012 HOLD
3650-HEX-001 POX RECLAIM HEAT EXCHANGER — 3650-011 HOLD

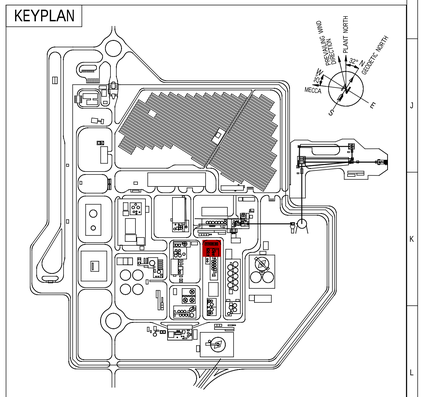
TOS EL. +915.835
TOS EL. +917.870

3650/F 3650/E 3650/D 3650/C 3650/B 3650/A
4350 6800 2350 3900 4000 4000
3650/1 3650/2 3650/3 3650/4 3650/5 3650/6 3650/7 3650/8
4300 4000 4200 4200 5200 4200 5700
31800

- CONCRETE ELEMENT FLOOR (PRE-CAST SLAB)
- CONCRETE FLOOR
- CONCRETE FLOOR WITH ALKALI / ACID RESISTANT COATING
- CHECKERED PLATE
- GRATING
- GRATING - FRP
- ACID RESISTANT BRICKLINING
- BRICK WALL
- CONCRETE WALL

1. ALL DIMENSIONS ARE IN MILLIMETER, CO-ORDINATES AND ELEVATION ARE IN METERS UNLESS OTHERWISE STATED.
2. FINISHED GROUND LEVEL (F.G.L) FOR PROCESS AREA 3650 = RL(+)912.700 m
3. F.F.L, F.G.L, T.O.C = FINISHED FLOOR LEVEL, FINISHED GROUND LEVEL, TOP OF CONCRETE. T.O.S, B.O.S, B.O.B, = TOP OF STEEL, BOTTOM OF STEEL, BOTTOM OF BASE PLATE. T.O.G. = TOP OF GRATING. HP = HIGHT POINT, LP = LOW POINT, T.O.R. = TOP OF RAIL.
4. BUILDING DIMENSIONS AND HEIGHTS ARE TENTATIVE, SHALL BE FINALIZED DURING DETAIL ENGINEERING SUBJECT TO MINIMUM AS SHOWN.
5. ALL MAINTENANCE FACILITIES, HANDLING PROVISIONS, ACCESS WAYS, PLATFORMS FOR MAINTENANCE SHALL BE PROVIDED DURING DETAIL ENGINEERING WHICH ARE NOT SHOWN ENTIRELY. THIS DRAWING IS TO BE CONSIDERED AS CONCEPTUAL ARRANGEMENT OF THE FACILITIES.

KEYPLAN



PLAN AT EL.(+)915.400

0 1 10m

DOCUMENT NUMBER	REFERENCE DOCUMENTS
MD-216-3000-EG-GM-LST-1010	HOLD LIST
MD-216-3650-EG-ME-GAD-1008	SECTION C-C
MD-216-3650-EG-ME-GAD-1007	SECTION B-B
MD-216-3650-EG-ME-GAD-1006	SECTION A-A
MD-216-3650-EG-ME-GAD-1005	PLAN AT EL. +925.200 & EL. +927.400
MD-216-3650-EG-ME-GAD-1004	PLAN AT EL. +918.700 & EL. +921.300
MD-216-3650-EG-ME-GAD-1002	PLAN AT EL. +912.700
MD-216-3650-EG-ME-GAD-1015	LOAD PLAN, PLAN AT EL. +912.700

REV	DATE	REVISION DESCRIPTION	DRN	CHK	ENG	APP	EMP
A01	14-04-2020	ISSUED FOR APPROVAL	GK	TV	SC	RL	

RESERVED FOR VENDOR LOGO AND DRAWING NUMBER

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LARSEN & TOUBRO DRAWING NUMBER			REVISION
O19045_450_M_G_02954_002			0
OUTOTEC DRAWING NUMBER		DOCUMENT ID	REVISION
900707 _GAU01 _EE12 _MTL01 _36504		OU100681165	0
DRAWN	GK	14-04-2020	MANSOURAH-MASSARAH GOLD PROJECT
CHECKED	TV	14-04-2020	GENERAL ARRANGEMENT DRAWING
ENGINEER	SC	14-04-2020	POX ACID WASH,ELUTION/CARBON REGENERATION
APPROVED	RL	14-04-2020	PLAN AT EL. +915.400
EMPLOYER			DRAWING NUMBER: MD-216-3650-EG-ME-GAD-1003
SCALE: 1:50		A0	REV: A01

Global Project ID	**900707**
Delivery Type	**ENGINEERING PROCUREMENT CONSTRUCTION**
Project Name	**MANSOURAH MASSARAH GOLD PROJECT**
Customer	**MA'ADEN GOLD AND BASE METALS (MA'ADEN)**
Project Country	**KINGDOM OF SAUDI ARABIA**

Ma'aden Revision No:	A01						Outotec / L&T Revision No.:	0				
Date	06-06-2020						Date	06-06-2020				
Revision Text	FIRST ISSUE						Revision Text	FIRST ISSUE				
Prepared By	APL						Prepared By	APL				
Checked	SYC						Checked	SYC				
Approved	PRI						Approved	PRI				

ATTACHMENT 2 **STAAD INPUT FILE OF 3650 MONORAIL SUPPORTING STRUCTURE**

```
STAAD SPACE
START JOB INFORMATION
ENGINEER DATE 02-Jun-20
JOB NAME MMGP-SAUDI ARABIA
JOB CLIENT MA'ADEN
JOB NO O19045/900707
JOB REV 0
JOB PART 3650 MONORAIL SUPPORTING STRUCTURE
JOB REF POX ACID WASH, ELUTION AND CARBON REGENERATION
ENGINEER NAME APL
CHECKER NAME SYC
APPROVED NAME PRI
END JOB INFORMATION
INPUT WIDTH 79
UNIT METER KN
JOINT COORDINATES
1 -3.1 0 3.4; 2 0 0 -0.3; 3 0 0 7.1; 4 5 0 -0.3; 5 5 0 7.1; 6 10 0 -0.3;
7 10 0 7.1; 8 15 0 0.2; 9 15 0 7.1; 10 21 0 0.2; 11 21 0 7.1; 12 5 0.425 -0.3;
13 5 0.425 1.2; 14 5 0.425 2.4; 15 5 0.425 3.55; 16 5 0.425 4.7;
17 5 0.425 5.9; 18 5 0.425 7.1; 19 10 0.425 -0.3; 20 10 0.425 1.2;
21 10 0.425 2.4; 22 10 0.425 3.55; 23 10 0.425 4.7; 24 10 0.425 5.9;
25 10 0.425 7.1; 26 15 0.425 0.2; 27 15 0.425 1.2; 28 15 0.425 2.4;
29 15 0.425 3.55; 30 15 0.425 4.7; 31 15 0.425 5.9; 32 15 0.425 7.1;
33 21 0.425 0.2; 34 21 0.425 1.2; 35 21 0.425 2.4; 36 21 0.425 3.55;
37 21 0.425 4.7; 38 21 0.425 5.9; 39 21 0.425 7.1; 40 -3.1 5.85 3.4;
41 0 5.85 -0.3; 42 0 5.85 3.4; 43 0 5.85 7.1; 44 5 5.85 -0.3; 45 5 5.85 7.1;
46 10 5.85 -0.3; 47 10 5.85 7.1; 48 15 5.85 0.2; 49 15 5.85 7.1;
50 21 5.85 0.2; 51 21 5.85 7.1; 52 5 5.85 3.4; 53 10 5.85 3.4; 54 15 5.85 3.65;
55 21 5.85 3.65; 56 5 3.075 -0.3; 57 10 3.075 -0.3; 58 15 3.075 0.2;
59 21 3.075 0.2; 60 5 3.075 7.1; 61 10 3.075 7.1; 62 15 3.075 7.1;
63 21 3.075 7.1; 64 0 3.075 -0.3; 65 0 3.075 7.1; 66 7.5 3.075 -0.3;
67 18 3.075 0.2; 68 7.5 3.075 7.1; 69 18 3.075 7.1; 70 7.5 5.85 -0.3;
71 18 5.85 0.2; 72 7.5 5.85 7.1; 73 18 5.85 7.1; 74 -2.3 5.85 -0.3;
75 -2.3 5.85 3.4; 76 -2.3 5.85 7.1; 77 -1.075 5.85 -0.3; 78 -1.075 5.85 3.4;
79 -1.075 5.85 7.1; 80 -2.3 5.85 4.465; 81 -1.075 5.85 4.465;
82 -2.3 5.85 2.335; 83 -1.075 5.85 2.335; 84 -1.075 5.85 1.0175;
85 -1.075 5.85 5.7825; 86 10 0.425 0.2;
MEMBER INCIDENCES
1 2 64; 2 64 41; 3 4 12; 4 12 56; 5 56 44; 6 6 19; 7 19 57; 8 57 46; 9 8 26;
10 26 58; 11 58 48; 12 10 33; 13 33 59; 14 59 50; 15 1 40; 16 3 65; 17 65 43;
18 5 18; 19 18 60; 20 60 45; 21 7 25; 22 25 61; 23 61 47; 24 9 32; 25 32 62;
26 62 49; 27 11 39; 28 39 63; 29 63 51; 31 41 42; 32 42 43; 33 44 52; 34 52 45;
35 46 53; 36 53 47; 37 48 54; 38 54 49; 39 50 55; 40 55 51; 41 12 13; 42 13 14;
43 14 15; 44 15 16; 45 16 17; 46 17 18; 47 19 86; 48 20 21; 49 21 22; 50 22 23;
51 23 24; 52 24 25; 53 26 27; 54 27 28; 55 28 29; 56 29 30; 57 30 31; 58 31 32;
59 33 34; 60 34 35; 61 35 36; 62 36 37; 63 37 38; 64 38 39; 71 77 74; 72 41 77;
73 77 84; 74 74 82; 75 84 83; 76 82 83; 77 82 75; 78 83 78; 79 40 75; 80 75 78;
81 78 42; 82 75 80; 83 78 81; 84 80 81; 85 81 85; 86 80 76; 87 85 79; 88 79 76;
89 43 79; 91 26 33; 92 32 39; 94 27 34; 95 28 35; 96 29 36; 97 30 37; 98 31 38;
101 12 19; 102 86 26; 103 18 25; 104 25 32; 111 13 20; 112 14 21; 113 15 22;
114 16 23; 115 17 24; 116 20 27; 117 21 28; 118 22 29; 119 23 30; 120 24 31;
125 56 66; 126 44 70; 127 60 68; 128 45 72; 129 66 57; 130 70 46; 131 68 61;
132 72 47; 133 58 67; 134 48 71; 135 62 69; 136 49 73; 137 67 59; 138 71 50;
139 69 63; 140 73 51; 151 64 56; 152 41 44; 153 65 60; 154 43 45; 159 74 64;
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160 76 65; 161 44 53; 162 46 52; 163 52 47; 164 53 45; 165 48 55; 166 50 54;
167 54 51; 168 55 49; 181 12 66; 182 56 70; 183 18 68; 184 60 72; 185 19 66;
186 57 70; 187 25 68; 188 61 72; 189 8 67; 190 58 71; 191 9 69; 192 62 73;
193 10 67; 194 59 71; 195 11 69; 196 63 73; 201 40 82; 202 40 80; 203 74 84;
204 82 84; 205 78 82; 206 78 80; 207 80 85; 208 76 85; 209 86 20;
********************************************************************************
START GROUP DEFINITION
FLOOR
_FL_913.550_1 41 TO 64 91 92 94 TO 98 101 TO 104 111 TO 120 209
END GROUP DEFINITION
********************************************************************************
DEFINE MATERIAL START
ISOTROPIC STEEL
E 2.05e+008
POISSON 0.3
DENSITY 76.8195
ALPHA 1.2e-005
DAMP 0.03
TYPE STEEL
STRENGTH FY 253200 FU 407800 RY 1.5 RT 1.2
END DEFINE MATERIAL
MEMBER PROPERTY EUROPEAN
1 TO 29 TABLE ST IPE360
31 TO 40 TABLE ST IPE360
47 TO 58 209 TABLE ST IPE450
41 TO 46 59 TO 64 TABLE ST IPE360
79 TO 81 TABLE ST IPE300
71 TO 78 82 TO 89 TABLE ST IPE240
*
91 92 TABLE ST UPN300
94 TO 98 TABLE ST UPN300
101 TO 104 TABLE ST UPN240
111 TO 120 TABLE ST UPN240
125 TO 140 TABLE ST UPN140
*
151 TO 154 TABLE ST L130X130X13
159 160 TABLE ST L90X90X8
161 TO 168 TABLE ST L150X150X16
181 TO 196 TABLE ST L100X100X10
201 TO 208 TABLE ST L65X65X6
********************************************************************************
CONSTANTS
BETA 90 MEMB 1 TO 29 125 TO 140
MATERIAL STEEL ALL
********************************************************************************
SUPPORTS
1 TO 11 PINNED
********************************************************************************
MEMBER RELEASE
41 47 53 59 73 74 76 82 TO 84 91 92 94 TO 98 101 TO 104 111 TO 120 -
125 TO 128 133 TO 136 151 TO 154 START MY MZ
46 52 58 64 76 TO 78 84 86 87 91 92 94 TO 98 101 TO 104 111 TO 120 -
129 TO 132 137 TO 140 151 TO 154 END MY MZ
72 79 89 START MP 0.99
```

```
81 END MP 0.99
MEMBER TRUSS
159 TO 168 181 TO 196 201 TO 207
************************************************
DEFINE IBC 2003
SDS 0.066 SD1 0.0133 S1 0.02 IE 1 RX 4 RZ 4 SCLASS 2 PX 0.21 PZ 0.21
SELFWEIGHT 1.1
*SELF WEIGHT OF FRP GRATING @20KG/SQM
ONEWAY LOAD
_FL_913.550_1 ONE 0.2 GY
*Selfweight of 3T CAPACITY MONORAIL 4KN
MEMBER WEIGHT
31 33 35 37 39 73 CON 4 0.893
32 34 36 38 40 CON 4 1.543
85 CON 4 0.973
80 CON 4 0.2
*76 84 CON 4
*Weight of stringer including grating and handrail
*Wt of stringer of approx. 2 M length, 38 thk. FRP grating and H/Rail*
*20kg/sqm x 2m x 1.2m = 48 kg, (25+15)x2=80 kg
102 CON 0.52 1.7
102 CON 0.52 2.9
*
104 CON 0.52 2
104 CON 0.52 3.2
*
42 CON 0.52
42 CON 0.52 1.2
*
64 CON 0.52
64 CON 0.52 1.2
*LOAD FROM HANDRAIL @ 15Kg/m
MEMBER WEIGHT
41 TO 46 59 TO 64 91 92 101 TO 104 UNI 0.15
*Load due to Cable Tray, Ducts, Pipe etc. (Misc. 50 kN per column point)
*JOINT WEIGHT
*40 41 43 TO 51 WEIGHT 50
******25% LIVE LOAD
ONEWAY LOAD
_FL_913.550_1 ONE 1.25 GY
*Live Load due to 1.2m wide stair
*500kg/sqm x 2m x 1.2m = 1200 kg
MEMBER WEIGHT
102 CON 0.75 1.7
102 CON 0.75 2.9
*
104 CON 0.75 2
104 CON 0.75 3.2
*
42 CON 0.75
42 CON 0.75 1.2
*
64 CON 0.75
64 CON 0.75 1.2
```

```
*Load FROM 3T CAPACITY MONORAIL INCLUDING 0.5T HOIST= (30+ 5)*1.2 = 42KN
MEMBER WEIGHT
31 33 35 37 39 73 CON 10.5 0.893
32 34 36 38 40 CON 10.5 1.543
85 CON 10.5 0.973
80 CON 10.5 0.2
*76 84 CON 42
**************************************
LOAD 9 EL (+X)
IBC LOAD X 1
*PERFORM ANALYSIS
*CHANGE
**************************************
LOAD 10 EL (+Z)
IBC LOAD Z 1
*PERFORM ANALYSIS
*CHANGE
*******************************************************************************
LOAD 1 DEAD LOAD
SELFWEIGHT Y -1.1
*SELF WEIGHT OF FRP GRATING @20KG/SQM
ONEWAY LOAD
_FL_913.550_1 ONE -0.2 GY
*Selfweight of 3T CAPACITY MONORAIL 4KN
MEMBER LOAD
31 33 35 37 39 73 CON GY -4 0.893
32 34 36 38 40 CON GY -4 1.543
85 CON GY -4 0.973
80 CON GY -4 0.2
*76 84 CON GY -4
*Weight of stringer including grating and handrail
*Wt of stringer of approx. 2 M length, 38 thk. FRP grating and H/Rail*
*20kg/sqm x 2m x 1.2m = 48 kg, (25+15)x2=80 kg
102 CON GY -0.52 1.7
102 CON GY -0.52 2.9
*
104 CON GY -0.52 2
104 CON GY -0.52 3.2
*
42 CON GY -0.52
42 CON GY -0.52 1.2
*
64 CON GY -0.52
64 CON GY -0.52 1.2
*LOAD FROM HANDRAIL @ 15Kg/m
MEMBER LOAD
41 TO 46 59 TO 64 91 92 101 TO 104 UNI GY -0.15
*Additional support
116 CON GY -4 4.45
118 CON GY 1 4.45
*Load due to Cable Tray, Ducts, Pipe etc. (Misc. 50 kN per column point)
*JOINT LOAD
*40 41 43 TO 51 FY -50
*******************************************************************************
```

```
LOAD 2 LIVE LOAD
*LIVE LOAD ON FLOOR @500KG/SQM
ONEWAY LOAD
_FL_913.550_1 ONE -5 GY
*Live Load due to 1.2m wide stair
*500kg/sqm x 2m x 1.2m = 1200 kg
MEMBER LOAD
102 CON GY -3 1.7
102 CON GY -3 2.9
*
104 CON GY -3 2
104 CON GY -3 3.2
*
42 CON GY -3
42 CON GY -3 1.2
*
64 CON GY -3
64 CON GY -3 1.2
*Load FROM 3T CAPACITY MONORAIL INCLUDING 0.5T HOIST= (30+ 5)*1.2 = 42KN
MEMBER LOAD
31 33 35 37 39 73 CON GY -42 0.893
32 34 36 38 40 CON GY -42 1.543
85 CON GY -42 0.973
80 CON GY -42 0.2
*76 84 CON GY -42
*Additional support
116 CON GY -17 4.45
118 CON GY 7 4.45
*********************************************************************************
LOAD 3 WIND LOAD IN +Z DIRN
**At Grids 3650/A',  3650/B', 3650/D'', 3650/D' & 3650/E'
MEMBER LOAD
1 TO 8 15 TO 23 UNI GZ 1
9 TO 14 24 TO 29 UNI GZ 1.4
*********************************************************************************
LOAD 4 WIND LOAD IN -Z DIRN
**At Grids 3650/A',  3650/B', 3650/D'', 3650/D' & 3650/E'
MEMBER LOAD
1 TO 8 15 TO 23 UNI GZ -1
9 TO 14 24 TO 29 UNI GZ -1.4
*********************************************************************************
LOAD 5 WIND LOAD IN +X DIRN
**At Grids 3650/6',  3650/7', 3650/8'' & 3650/8'
MEMBER LOAD
1 TO 29 UNI GX 1.3
*********************************************************************************
LOAD 6 WIND LOAD IN -X DIRN
**At Grids 3650/6',  3650/7', 3650/8'' & 3650/8'
MEMBER LOAD
1 TO 29 UNI GX -1.3
************************************************
LOAD 7 TL1  (TEMP LOAD +25)
TEMPERATURE LOAD
1 TO 29 31 TO 64 71 TO 89 91 92 94 TO 98 101 TO 104 111 TO 120 125 TO 140 -
```

```
151 TO 154 159 TO 168 181 TO 196 201 TO 208 TEMP 25
*********************************************************************************
LOAD 8 TL2 (TEMP LOAD -25)
TEMPERATURE LOAD
1 TO 29 31 TO 64 71 TO 89 91 92 94 TO 98 101 TO 104 111 TO 120 125 TO 140 -
151 TO 154 159 TO 168 181 TO 196 201 TO 208 TEMP -25
****************** LOAD COMBINATION FOR SERVICEABILITY CHECK
LOAD COMB 101 DL
1 1.0
LOAD COMB 102 (DL + TL1) + LL
1 1.0 7 1.0 2 1.0
LOAD COMB 103 (DL + TL2) + LL+
1 1.0 8 1.0 2 1.0
LOAD COMB 104 DL+ 0.75 WL(+Z) + 0.75 LL
1 1.0 3 0.75 2 0.75
LOAD COMB 105 DL+ 0.75 WL(-Z) + 0.75 LL
1 1.0 4 0.75 2 0.75
LOAD COMB 106 DL+  0.75 WL(+X) + 0.75 LL
1 1.0 5 0.75 2 0.75
LOAD COMB 107 DL+  0.75 WL(-X) + 0.75 LL
1 1.0 6 0.75 2 0.75
LOAD COMB 108 DL + 0.525 EL(+X) + 0.75 LL
1 1.0 9 0.525 2 0.75
LOAD COMB 109 DL + 0.525 EL(-X) + 0.75 LL
1 1.0 9 -0.525 2 0.75
LOAD COMB 110 DL + 0.525 EL(+Z) + 0.75 LL
1 1.0 10 0.525 2 0.75
LOAD COMB 111 DL + 0.525 EL(-Z) + 0.75 LL
1 1.0 10 -0.525 2 0.75
LOAD COMB 112 0.6 DL + WL(+Z)
1 0.6 3 1.0
LOAD COMB 113 0.6 DL +  WL(-Z)
1 0.6 4 1.0
LOAD COMB 114 0.6 DL +  WL(+X)
1 0.6 5 1.0
LOAD COMB 115 0.6 DL + WL(-X)
1 0.6 6 1.0
LOAD COMB 116 0.6 DL + 0.7 EL(+X)
1 0.6 9 0.7
LOAD COMB 117 0.6 DL + 0.7 EL(-X)
1 0.6 9 -0.7
LOAD COMB 118 0.6 DL + 0.7 EL(+Z)
1 0.6 10 0.7
LOAD COMB 119 0.6 DL + 0.7 EL(-Z)
1 0.6 10 -0.7
*********************************************************************************
****LOAD COMBINATION for structural design
LOAD COMB 201 1.4 DL
1 1.4
LOAD COMB 202 1.2 (DL + TL1) + 1.6 LL
1 1.2 7 1.2 2 1.6
LOAD COMB 203 1.2 (DL + TL2) + 1.6 LL
1 1.2 8 1.2 2 1.6
LOAD COMB 204 1.2 DL+ 1.6 WL(+Z) + 0.5 LL
```

```
1 1.2 3 1.6 2 0.5
LOAD COMB 205 1.2 DL+ 1.6 WL(-Z) + 0.5 LL
1 1.2 4 1.6 2 0.5
LOAD COMB 206 1.2 DL+ 1.6 WL(+Z) + 0.5 LL
1 1.2 5 1.6 2 0.5
LOAD COMB 207 1.2 DL+ 1.6 WL(-Z) + 0.5 LL
1 1.2 6 1.6 2 0.5
LOAD COMB 208 1.2 DL + 1.0 EL(+X) + 0.5 LL
1 1.2 9 1.0 2 0.5
LOAD COMB 209 1.2 DL + 1.0 EL(-X) + 0.5 LL
1 1.2 9 -1.0 2 0.5
LOAD COMB 210 1.2 DL + 1.0 EL(+Z) + 0.5 LL
1 1.2 10 1.0 2 0.5
LOAD COMB 211 1.2 DL + 1.0 EL(-Z) + 0.5 LL
1 1.2 10 -1.0 2 0.5
LOAD COMB 212 0.9 DL + 1.6 WL(+Z)
1 0.9 3 1.6
LOAD COMB 213 0.9 DL + 1.6 WL(-Z)
1 0.9 4 1.6
LOAD COMB 214 0.9 DL + 1.6 WL(+X)
1 0.9 5 1.6
LOAD COMB 215 0.9 DL + 1.6 WL(-X)
1 0.9 6 1.6
LOAD COMB 216 0.9 DL + 1.0 EL(+X)
1 0.9 9 1.0
LOAD COMB 217 0.9 DL + 1.0 EL(-X)
1 0.9 9 -1.0
LOAD COMB 218 0.9 DL + 1.0 EL(+Z)
1 0.9 10 1.0
LOAD COMB 219 0.9 DL + 1.0 EL(-Z)
1 0.9 10 -1.0
************************************************
CALCULATE RAYLEIGH FREQUENCY
PERFORM ANALYSIS PRINT STATICS CHECK
**
LOAD LIST 1 TO 10
PRINT SUPPORT REACTION LIST 1 TO 11
************************************************
LOAD LIST 201 TO 219
PARAMETER 1
CODE LRFD
FYLD 275000 ALL
FU 410000 ALL
TRACK 1 ALL
BEAM 1 ALL
TMAIN 300 ALL
MAIN 200 ALL
NSF 0.8 ALL
***BEAM MAJOR AXIS
LZ 7.4 MEMB 41 TO 52 209
LZ 6.9 MEMB 53 TO 64
LZ 7.4 MEMB 31 TO 36
LZ 3.7 MEMB 73 TO 75 77 78 82 83 85 TO 87
UNB 7.4 MEMB 41 TO 52 209
```

```
UNB 6.9 MEMB 53 TO 64
*
LZ 6 MEMB 133 TO 140
LZ 5 MEMB 125 TO 132
***BEAM MINOR AXIS
LY 3.5 MEMB 91 92 94 TO 98
LY 3 MEMB 101 TO 104 111 TO 120
****BRACING
KX 0.85 MEMB 159 TO 164 181 TO 196 201 TO 207
KZ 0.85 MEMB 159 TO 164 181 TO 196 201 TO 207
KY 0.85 MEMB 159 TO 164 181 TO 196 201 TO 207
*
KX 0.8 MEMB 165 TO 168
KZ 0.8 MEMB 165 TO 168
KY 0.8 MEMB 165 TO 168
*
KX 0.95 MEMB 125 TO 140 151 TO 154
KZ 0.95 MEMB 125 TO 140 151 TO 154
KY 0.95 MEMB 125 TO 140 151 TO 154
****COLUMN MAJOR AXIS
LZ 5.75 MEMB 1 TO 14 16 TO 29
****COLUMN MINOR AXIS
LY 3.5 MEMB 1 TO 14 16 TO 29
***
CHECK CODE ALL
STEEL TAKE OFF ALL
************************************************
***////Parameter for deflection check of beams////
***////To be uncommented while deflection check////
***////To be put in comment mode while checking for strength////
*LOAD LIST 101 TO 119
*PARAMETER 2
*CODE LRFD
***FL_913.550
*DJ1 133 MEMB 101 TO 104
*DJ2 134 MEMB 101 TO 104
*DFF 360 MEMB 488 TO 494 621 TO 624
*DFF 500 MEMB 172 TO 188
*CHECK CODE MEMB 172 TO 188 488 TO 494 621 TO 624
FINISH
```

Global Project ID	**900707**
Delivery Type	**ENGINEERING PROCUREMENT CONSTRUCTION**
Project Name	**MANSOURAH MASSARAH GOLD PROJECT**
Customer	**MA'ADEN GOLD AND BASE METALS (MA'ADEN)**
Project Country	**KINGDOM OF SAUDI ARABIA**

Ma'aden Revision No:	A01						Outotec / L&T Revision No.:	0					
Date	06-06-2020						Date	06-06-2020					
Revision Text	FIRST ISSUE						Revision Text	FIRST ISSUE					
Prepared By	APL						Prepared By	APL					
Checked	SYC						Checked	SYC					
Approved	PRI						Approved	PRI					

ATTACHMENT 3 **STAAD OUTPUT FILE OF 3650 MONORAIL SUPPORTING STRUCTURE**

```
**************************************************
*                                                *
*           STAAD.Pro V8i SELECTseries6          *
*           Version  20.07.11.70                 *
*           Proprietary Program of               *
*           Bentley Systems, Inc.                *
*           Date=    JUN  5, 2020                *
*           Time=    12:33: 6                    *
*                                                *
*      USER ID: lnt                              *
**************************************************
```

```
    1. STAAD SPACE
INPUT FILE: D:\APL\Assignments\MMGP-2020\3650_POX MONORAIL SUPPORTING STR\Staad\3650_POX acid wash Mon.. .STD
    2. START JOB INFORMATION
    3. ENGINEER DATE 02-JUN-20
    4. JOB NAME MMGP-SAUDI ARABIA
    5. JOB CLIENT MA'ADEN
    6. JOB NO O19045/900707
    7. JOB REV 0
    8. JOB PART 3650 MONORAIL SUPPORTING STRUCTURE
    9. JOB REF POX ACID WASH, ELUTION AND CARBON REGENERATION
   10. ENGINEER NAME APL
   11. CHECKER NAME SYC
   12. APPROVED NAME PRI
   13. END JOB INFORMATION
   14. INPUT WIDTH 79
   15. UNIT METER KN
   16. JOINT COORDINATES
   17. 1 -3.1 0 3.4; 2 0 0 -0.3; 3 0 0 7.1; 4 5 0 -0.3; 5 5 0 7.1; 6 10 0 -0.3
   18. 7 10 0 7.1; 8 15 0 0.2; 9 15 0 7.1; 10 21 0 0.2; 11 21 0 7.1; 12 5 0.425 -0.3
   19. 13 5 0.425 1.2; 14 5 0.425 2.4; 15 5 0.425 3.55; 16 5 0.425 4.7
   20. 17 5 0.425 5.9; 18 5 0.425 7.1; 19 10 0.425 -0.3; 20 10 0.425 1.2
   21. 21 10 0.425 2.4; 22 10 0.425 3.55; 23 10 0.425 4.7; 24 10 0.425 5.9
   22. 25 10 0.425 7.1; 26 15 0.425 0.2; 27 15 0.425 1.2; 28 15 0.425 2.4
   23. 29 15 0.425 3.55; 30 15 0.425 4.7; 31 15 0.425 5.9; 32 15 0.425 7.1
   24. 33 21 0.425 0.2; 34 21 0.425 1.2; 35 21 0.425 2.4; 36 21 0.425 3.55
   25. 37 21 0.425 4.7; 38 21 0.425 5.9; 39 21 0.425 7.1; 40 -3.1 5.85 3.4
   26. 41 0 5.85 -0.3; 42 0 5.85 3.4; 43 0 5.85 7.1; 44 5 5.85 -0.3; 45 5 5.85 7.1
   27. 46 10 5.85 -0.3; 47 10 5.85 7.1; 48 15 5.85 0.2; 49 15 5.85 7.1
   28. 50 21 5.85 0.2; 51 21 5.85 7.1; 52 5 5.85 3.4; 53 10 5.85 3.4; 54 15 5.85 3.65
   29. 55 21 5.85 3.65; 56 5 3.075 -0.3; 57 10 3.075 -0.3; 58 15 3.075 0.2
   30. 59 21 3.075 0.2; 60 5 3.075 7.1; 61 10 3.075 7.1; 62 15 3.075 7.1
   31. 63 21 3.075 7.1; 64 0 3.075 -0.3; 65 0 3.075 7.1; 66 7.5 3.075 -0.3
   32. 67 18 3.075 0.2; 68 7.5 3.075 7.1; 69 18 3.075 7.1; 70 7.5 5.85 -0.3
   33. 71 18 5.85 0.2; 72 7.5 5.85 7.1; 73 18 5.85 7.1; 74 -2.3 5.85 -0.3
   34. 75 -2.3 5.85 3.4; 76 -2.3 5.85 7.1; 77 -1.075 5.85 -0.3; 78 -1.075 5.85 3.4
   35. 79 -1.075 5.85 7.1; 80 -2.3 5.85 4.465; 81 -1.075 5.85 4.465
   36. 82 -2.3 5.85 2.335; 83 -1.075 5.85 2.335; 84 -1.075 5.85 1.0175
   37. 85 -1.075 5.85 5.7825; 86 10 0.425 0.2
   38. MEMBER INCIDENCES
```

```
39. 1 2 64; 2 64 41; 3 4 12; 4 12 56; 5 56 44; 6 6 19; 7 19 57; 8 57 46; 9 8 26
40. 10 26 58; 11 58 48; 12 10 33; 13 33 59; 14 59 50; 15 1 40; 16 3 65; 17 65 43
41. 18 5 18; 19 18 60; 20 60 45; 21 7 25; 22 25 61; 23 61 47; 24 9 32; 25 32 62
42. 26 62 49; 27 11 39; 28 39 63; 29 63 51; 31 41 42; 32 42 43; 33 44 52; 34 52 45
43. 35 46 53; 36 53 47; 37 48 54; 38 54 49; 39 50 55; 40 55 51; 41 12 13; 42 13 14
44. 43 14 15; 44 15 16; 45 16 17; 46 17 18; 47 19 86; 48 20 21; 49 21 22; 50 22 23
45. 51 23 24; 52 24 25; 53 26 27; 54 27 28; 55 28 29; 56 29 30; 57 30 31; 58 31 32
46. 59 33 34; 60 34 35; 61 35 36; 62 36 37; 63 37 38; 64 38 39; 71 77 74; 72 41 77
47. 73 77 84; 74 74 82; 75 84 83; 76 82 83; 77 82 75; 78 83 78; 79 40 75; 80 75 78
48. 81 78 42; 82 75 80; 83 78 81; 84 80 81; 85 81 85; 86 80 76; 87 85 79; 88 79 76
49. 89 43 79; 91 26 33; 92 32 39; 94 27 34; 95 28 35; 96 29 36; 97 30 37; 98 31 38
50. 101 12 19; 102 86 26; 103 18 25; 104 25 32; 111 13 20; 112 14 21; 113 15 22
51. 114 16 23; 115 17 24; 116 20 27; 117 21 28; 118 22 29; 119 23 30; 120 24 31
52. 125 56 66; 126 44 70; 127 60 68; 128 45 72; 129 66 57; 130 70 46; 131 68 61
53. 132 72 47; 133 58 67; 134 48 71; 135 62 69; 136 49 73; 137 67 59; 138 71 50
54. 139 69 63; 140 73 51; 151 64 56; 152 41 44; 153 65 60; 154 43 45; 159 74 64
55. 160 76 65; 161 44 53; 162 46 52; 163 52 47; 164 53 45; 165 48 55; 166 50 54
56. 167 54 51; 168 55 49; 181 12 66; 182 56 70; 183 18 68; 184 60 72; 185 19 66
57. 186 57 70; 187 25 68; 188 61 72; 189 8 67; 190 58 71; 191 9 69; 192 62 73
58. 193 10 67; 194 59 71; 195 11 69; 196 63 73; 201 40 82; 202 40 80; 203 74 84
59. 204 82 84; 205 78 82; 206 78 80; 207 80 85; 208 76 85; 209 86 20
60. **************************************************************************
61. START GROUP DEFINITION
62. FLOOR
63. _FL_913.550_1 41 TO 64 91 92 94 TO 98 101 TO 104 111 TO 120 209
64. END GROUP DEFINITION
65. **************************************************************************
66. DEFINE MATERIAL START
67. ISOTROPIC STEEL
68. E 2.05E+008
69. POISSON 0.3
70. DENSITY 76.8195
71. ALPHA 1.2E-005
72. DAMP 0.03
73. TYPE STEEL
74. STRENGTH FY 253200 FU 407800 RY 1.5 RT 1.2
75. END DEFINE MATERIAL
76. MEMBER PROPERTY EUROPEAN
77. 1 TO 29 TABLE ST IPE360
78. 31 TO 40 TABLE ST IPE360
79. 47 TO 58 209 TABLE ST IPE450
80. 41 TO 46 59 TO 64 TABLE ST IPE360
81. 79 TO 81 TABLE ST IPE300
82. 71 TO 78 82 TO 89 TABLE ST IPE240
83. *
84. 91 92 TABLE ST UPN300
85. 94 TO 98 TABLE ST UPN300
86. 101 TO 104 TABLE ST UPN240
87. 111 TO 120 TABLE ST UPN240
88. 125 TO 140 TABLE ST UPN140
89. *
90. 151 TO 154 TABLE ST L130X130X13
91. 159 160 TABLE ST L90X90X8
92. 161 TO 168 TABLE ST L150X150X16
93. 181 TO 196 TABLE ST L100X100X10
94. 201 TO 208 TABLE ST L65X65X6
```

```
 95. ********************************************************************************
 96. CONSTANTS
 97. BETA 90 MEMB 1 TO 29 125 TO 140
 98. MATERIAL STEEL ALL
 99. ********************************************************************************
100. SUPPORTS
101. 1 TO 11 PINNED
102. ********************************************************************************
103. MEMBER RELEASE
104. 41 47 53 59 73 74 76 82 TO 84 91 92 94 TO 98 101 TO 104 111 TO 120 -
105. 125 TO 128 133 TO 136 151 TO 154 START MY MZ
106. 46 52 58 64 76 TO 78 84 86 87 91 92 94 TO 98 101 TO 104 111 TO 120 -
107. 129 TO 132 137 TO 140 151 TO 154 END MY MZ
108. 72 79 89 START MP 0.99
109. 81 END MP 0.99
110. MEMBER TRUSS
111. 159 TO 168 181 TO 196 201 TO 207
112. ************************************************
113. DEFINE IBC 2003
114. SDS 0.066 SD1 0.0133 S1 0.02 IE 1 RX 4 RZ 4 SCLASS 2 PX 0.21 PZ 0.21
115. SELFWEIGHT 1.1
116. *SELF WEIGHT OF FRP GRATING @20KG/SQM
117. ONEWAY LOAD
**WARNING** about Floor/OneWay Loads/Weights.
  Please note that depending on the shape of the floor you may
  have to break up the FLOOR/ONEWAY LOAD into multiple commands.
  For details please refer to Technical Reference Manual
  Section 5.32.4.2 Note d and/or "5.32.4.3 Note f.

118. _FL_913.550_1 ONE 0.2 GY
119. *SELFWEIGHT OF 3T CAPACITY MONORAIL 4KN
120. MEMBER WEIGHT
121. 31 33 35 37 39 73 CON 4 0.893
122. 32 34 36 38 40 CON 4 1.543
123. 85 CON 4 0.973
124. 80 CON 4 0.2
125. *76 84 CON 4
126. *WEIGHT OF STRINGER INCLUDING GRATING AND HANDRAIL
127. *WT OF STRINGER OF APPROX. 2 M LENGTH, 38 THK. FRP GRATING AND H/RAIL*
128. *20KG/SQM X 2M X 1.2M = 48 KG, (25+15)X2=80 KG
129. 102 CON 0.52 1.7
130. 102 CON 0.52 2.9
131. *
132. 104 CON 0.52 2
133. 104 CON 0.52 3.2
134. *
135. 42 CON 0.52
136. 42 CON 0.52 1.2
137. *
138. 64 CON 0.52
139. 64 CON 0.52 1.2
140. *LOAD FROM HANDRAIL @ 15KG/M
141. MEMBER WEIGHT
142. 41 TO 46 59 TO 64 91 92 101 TO 104 UNI 0.15
143. *LOAD DUE TO CABLE TRAY, DUCTS, PIPE ETC. (MISC. 50 KN PER COLUMN POINT)
144. *JOINT WEIGHT
```

```
145. *40 41 43 TO 51 WEIGHT 50
146. ******25% LIVE LOAD
147. ONEWAY LOAD
148. _FL_913.550_1 ONE 1.25 GY
149. *LIVE LOAD DUE TO 1.2M WIDE STAIR
150. *500KG/SQM X 2M X 1.2M = 1200 KG
151. MEMBER WEIGHT
152. 102 CON 0.75 1.7
153. 102 CON 0.75 2.9
154. *
155. 104 CON 0.75 2
156. 104 CON 0.75 3.2
157. *
158. 42 CON 0.75
159. 42 CON 0.75 1.2
160. *
161. 64 CON 0.75
162. 64 CON 0.75 1.2
163. *LOAD FROM 3T CAPACITY MONORAIL INCLUDING 0.5T HOIST= (30+ 5)*1.2 = 42KN
164. MEMBER WEIGHT
165. 31 33 35 37 39 73 CON 10.5 0.893
166. 32 34 36 38 40 CON 10.5 1.543
167. 85 CON 10.5 0.973
168. 80 CON 10.5 0.2
169. *76 84 CON 42
170. ****************************************
171. LOAD 9 EL (+X)
172. IBC LOAD X 1
173. *PERFORM ANALYSIS
174. *CHANGE
175. ****************************************
176. LOAD 10 EL (+Z)
177. IBC LOAD Z 1
178. *PERFORM ANALYSIS
179. *CHANGE
180. ********************************************************************************
181. LOAD 1 DEAD LOAD
182. SELFWEIGHT Y -1.1
183. *SELF WEIGHT OF FRP GRATING @20KG/SQM
184. ONEWAY LOAD
185. _FL_913.550_1 ONE -0.2 GY
186. *SELFWEIGHT OF 3T CAPACITY MONORAIL 4KN
187. MEMBER LOAD
188. 31 33 35 37 39 73 CON GY -4 0.893
189. 32 34 36 38 40 CON GY -4 1.543
190. 85 CON GY -4 0.973
191. 80 CON GY -4 0.2
```

```
192. *76 84 CON GY -4
193. *WEIGHT OF STRINGER INCLUDING GRATING AND HANDRAIL
194. *WT OF STRINGER OF APPROX. 2 M LENGTH, 38 THK. FRP GRATING AND H/RAIL*
195. *20KG/SQM X 2M X 1.2M = 48 KG, (25+15)X2=80 KG
196. 102 CON GY -0.52 1.7
197. 102 CON GY -0.52 2.9
198. *
199. 104 CON GY -0.52 2
200. 104 CON GY -0.52 3.2
201. *
202. 42 CON GY -0.52
203. 42 CON GY -0.52 1.2
204. *
205. 64 CON GY -0.52
206. 64 CON GY -0.52 1.2
207. *LOAD FROM HANDRAIL @ 15KG/M
208. MEMBER LOAD
209. 41 TO 46 59 TO 64 91 92 101 TO 104 UNI GY -0.15
210. *ADDITIONAL SUPPORT
211. 116 CON GY -4 4.45
212. 118 CON GY 1 4.45
213. *LOAD DUE TO CABLE TRAY, DUCTS, PIPE ETC. (MISC. 50 KN PER COLUMN POINT)
214. *JOINT LOAD
215. *40 41 43 TO 51 FY -50
216. ********************************************************************************
217. LOAD 2 LIVE LOAD
218. *LIVE LOAD ON FLOOR @500KG/SQM
219. ONEWAY LOAD
220. _FL_913.550_1 ONE -5 GY
221. *LIVE LOAD DUE TO 1.2M WIDE STAIR
222. *500KG/SQM X 2M X 1.2M = 1200 KG
223. MEMBER LOAD
224. 102 CON GY -3 1.7
225. 102 CON GY -3 2.9
226. *
227. 104 CON GY -3 2
228. 104 CON GY -3 3.2
229. *
230. 42 CON GY -3
231. 42 CON GY -3 1.2
232. *
233. 64 CON GY -3
234. 64 CON GY -3 1.2
235. *LOAD FROM 3T CAPACITY MONORAIL INCLUDING 0.5T HOIST= (30+ 5)*1.2 = 42KN
236. MEMBER LOAD
237. 31 33 35 37 39 73 CON GY -42 0.893
238. 32 34 36 38 40 CON GY -42 1.543
239. 85 CON GY -42 0.973
240. 80 CON GY -42 0.2
241. *76 84 CON GY -42
242. *ADDITIONAL SUPPORT
243. 116 CON GY -17 4.45
244. 118 CON GY 7 4.45
```

```
245. ********************************************************************************
246. LOAD 3 WIND LOAD IN +Z DIRN
247. **AT GRIDS 3650/A',  3650/B', 3650/D'', 3650/D' _3650/E'
248. MEMBER LOAD
249. 1 TO 8 15 TO 23 UNI GZ 1
250. 9 TO 14 24 TO 29 UNI GZ 1.4
251. ********************************************************************************
252. LOAD 4 WIND LOAD IN -Z DIRN
253. **AT GRIDS 3650/A',  3650/B', 3650/D'', 3650/D' _3650/E'
254. MEMBER LOAD
255. 1 TO 8 15 TO 23 UNI GZ -1.
256. 9 TO 14 24 TO 29 UNI GZ -1.4
257. ********************************************************************************
258. LOAD 5 WIND LOAD IN +X DIRN
259. **AT GRIDS 3650/6',  3650/7', 3650/8'' _3650/8'
260. MEMBER LOAD
261. 1 TO 29 UNI GX 1.3
262. ********************************************************************************
263. LOAD 6 WIND LOAD IN -X DIRN
264. **AT GRIDS 3650/6',  3650/7', 3650/8'' _3650/8'
265. MEMBER LOAD
266. 1 TO 29 UNI GX -1.3
267. ************************************************
268. LOAD 7 TL1  (TEMP LOAD +25)
269. TEMPERATURE LOAD
270. 1 TO 29 31 TO 64 71 TO 89 91 92 94 TO 98 101 TO 104 111 TO 120 125 TO 140 -
271. 151 TO 154 159 TO 168 181 TO 196 201 TO 208 TEMP 25
272. ********************************************************************************
273. LOAD 8 TL2 (TEMP LOAD -25)
274. TEMPERATURE LOAD
275. 1 TO 29 31 TO 64 71 TO 89 91 92 94 TO 98 101 TO 104 111 TO 120 125 TO 140 -
276. 151 TO 154 159 TO 168 181 TO 196 201 TO 208 TEMP -25
277. ****************** LOAD COMBINATION FOR SERVICEABILITY CHECK
278. LOAD COMB 101 DL
279. 1 1.0
280. LOAD COMB 102 (DL + TL1) + LL
281. 1 1.0 7 1.0 2 1.0
282. LOAD COMB 103 (DL + TL2) + LL+
283. 1 1.0 8 1.0 2 1.0
284. LOAD COMB 104 DL+ 0.75 WL(+Z) + 0.75 LL
285. 1 1.0 3 0.75 2 0.75
286. LOAD COMB 105 DL+ 0.75 WL(-Z) + 0.75 LL
287. 1 1.0 4 0.75 2 0.75
288. LOAD COMB 106 DL+  0.75 WL(+X) + 0.75 LL
289. 1 1.0 5 0.75 2 0.75
290. LOAD COMB 107 DL+  0.75 WL(-X) + 0.75 LL
291. 1 1.0 6 0.75 2 0.75
292. LOAD COMB 108 DL + 0.525 EL(+X) + 0.75 LL
293. 1 1.0 9 0.525 2 0.75
294. LOAD COMB 109 DL + 0.525 EL(-X) + 0.75 LL
295. 1 1.0 9 -0.525 2 0.75
296. LOAD COMB 110 DL + 0.525 EL(+Z) + 0.75 LL
297. 1 1.0 10 0.525 2 0.75
298. LOAD COMB 111 DL + 0.525 EL(-Z) + 0.75 LL
299. 1 1.0 10 -0.525 2 0.75
300. LOAD COMB 112 0.6 DL + WL(+Z)
```

```
301. 1 0.6 3 1.0
302. LOAD COMB 113 0.6 DL +  WL(-Z)
303. 1 0.6 4 1.0
304. LOAD COMB 114 0.6 DL +  WL(+X)
305. 1 0.6 5 1.0
306. LOAD COMB 115 0.6 DL + WL(-X)
307. 1 0.6 6 1.0
308. LOAD COMB 116 0.6 DL + 0.7 EL(+X)
309. 1 0.6 9 0.7
310. LOAD COMB 117 0.6 DL + 0.7 EL(-X)
311. 1 0.6 9 -0.7
312. LOAD COMB 118 0.6 DL + 0.7 EL(+Z)
313. 1 0.6 10 0.7
314. LOAD COMB 119 0.6 DL + 0.7 EL(-Z)
315. 1 0.6 10 -0.7
316. ********************************************************************************
317. ****LOAD COMBINATION FOR STRUCTURAL DESIGN
318. LOAD COMB 201 1.4 DL
319. 1 1.4
320. LOAD COMB 202 1.2 (DL + TL1) + 1.6 LL
321. 1 1.2 7 1.2 2 1.6
322. LOAD COMB 203 1.2 (DL + TL2) + 1.6 LL
323. 1 1.2 8 1.2 2 1.6
324. LOAD COMB 204 1.2 DL+ 1.6 WL(+Z) + 0.5 LL
325. 1 1.2 3 1.6 2 0.5
326. LOAD COMB 205 1.2 DL+ 1.6 WL(-Z) + 0.5 LL
327. 1 1.2 4 1.6 2 0.5
328. LOAD COMB 206 1.2 DL+ 1.6 WL(+Z) + 0.5 LL
329. 1 1.2 5 1.6 2 0.5
330. LOAD COMB 207 1.2 DL+ 1.6 WL(-Z) + 0.5 LL
331. 1 1.2 6 1.6 2 0.5
332. LOAD COMB 208 1.2 DL + 1.0 EL(+X) + 0.5 LL
333. 1 1.2 9 1.0 2 0.5
334. LOAD COMB 209 1.2 DL + 1.0 EL(-X) + 0.5 LL
335. 1 1.2 9 -1.0 2 0.5
336. LOAD COMB 210 1.2 DL + 1.0 EL(+Z) + 0.5 LL
337. 1 1.2 10 1.0 2 0.5
338. LOAD COMB 211 1.2 DL + 1.0 EL(-Z) + 0.5 LL
339. 1 1.2 10 -1.0 2 0.5
340. LOAD COMB 212 0.9 DL + 1.6 WL(+Z)
341. 1 0.9 3 1.6
342. LOAD COMB 213 0.9 DL + 1.6 WL(-Z)
343. 1 0.9 4 1.6
344. LOAD COMB 214 0.9 DL + 1.6 WL(+X)
345. 1 0.9 5 1.6
346. LOAD COMB 215 0.9 DL + 1.6 WL(-X)
347. 1 0.9 6 1.6
348. LOAD COMB 216 0.9 DL + 1.0 EL(+X)
349. 1 0.9 9 1.0
350. LOAD COMB 217 0.9 DL + 1.0 EL(-X)
351. 1 0.9 9 -1.0
352. LOAD COMB 218 0.9 DL + 1.0 EL(+Z)
353. 1 0.9 10 1.0
354. LOAD COMB 219 0.9 DL + 1.0 EL(-Z)
355. 1 0.9 10 -1.0
356. ************************************************
```

357. CALCULATE RAYLEIGH FREQUENCY
358. PERFORM ANALYSIS PRINT STATICS CHECK

P R O B L E M S T A T I S T I C S

NUMBER OF JOINTS	**86**	**NUMBER OF MEMBERS**	**158**
NUMBER OF PLATES	**0**	**NUMBER OF SOLIDS**	**0**
NUMBER OF SURFACES	**0**	**NUMBER OF SUPPORTS**	**11**

SOLVER USED IS THE IN-CORE ADVANCED MATH SOLVER

TOTAL PRIMARY LOAD CASES = 10, TOTAL DEGREES OF FREEDOM = 483
TOTAL LOAD COMBINATION CASES = 38 SO FAR.

**WARNING: IF THIS UBC/IBC ANALYSIS HAS TENSION/COMPRESSION
OR REPEAT LOAD OR RE-ANALYSIS OR SELECT OPTIMIZE, THEN EACH
UBC/IBC CASE SHOULD BE FOLLOWED BY PERFORM ANALYSIS _CHANGE.

```
************************************************************
*  IBC 2003 SEISMIC LOAD ALONG X  :                        *
*     CT =  0.028 Cu =  1.700                              *
*  TIME PERIODS :                                          *
*     Ta =  0.298 T =  0.210 Tuser = 0.210                 *
*  TIME PERIOD USED (T) =  0.210                           *
*  LOAD FACTOR          =  1.000                           *
*  DESIGN BASE SHEAR    =  1.000 X  0.016 X       551.45   *
*                       =         8.73  KN                 *
************************************************************
```

```
************************************************************
*  IBC 2003 SEISMIC LOAD ALONG Z  :                        *
*     CT =  0.028 Cu =  1.700                              *
*  TIME PERIODS :                                          *
*     Ta =  0.298 T =  0.210 Tuser = 0.210                 *
*  TIME PERIOD USED (T) =  0.210                           *
*  LOAD FACTOR          =  1.000                           *
*  DESIGN BASE SHEAR    =  1.000 X  0.016 X       551.45   *
*                       =         8.73  KN                 *
************************************************************
```

*** LOADING 9 : ADDITIONAL APPLIED FORCES AND MOMENTS ABOUT ORIGIN
DUE TO SEISMIC LOADS(KN ,METE)

FORCE-X =	8.73	FORCE-Y =	0.00	FORCE-Z =	0.00
MOM-X =	0.00	MOM-Y =	27.62	MOM-Z =	-46.82

*** LOADING 10 : ADDITIONAL APPLIED FORCES AND MOMENTS ABOUT ORIGIN
DUE TO SEISMIC LOADS(KN ,METE)

FORCE-X =	0.00	FORCE-Y =	0.00	FORCE-Z =	8.73
MOM-X =	46.82	MOM-Y =	-72.86	MOM-Z =	0.00

STATIC LOAD/REACTION/EQUILIBRIUM SUMMARY FOR CASE NO. 9
EL (+X)

CENTER OF FORCE BASED ON X FORCES ONLY (METE).
(FORCES IN NON-GLOBAL DIRECTIONS WILL INVALIDATE RESULTS)

X = 0.834419501E+01
Y = 0.536284328E+01
Z = 0.316293269E+01

***TOTAL APPLIED LOAD (KN METE) SUMMARY (LOADING 9)
SUMMATION FORCE-X = 8.73
SUMMATION FORCE-Y = 0.00
SUMMATION FORCE-Z = 0.00

SUMMATION OF MOMENTS AROUND THE ORIGIN-
MX= 0.00 MY= 27.62 MZ= -46.82

***TOTAL REACTION LOAD(KN METE) SUMMARY (LOADING 9)
SUMMATION FORCE-X = -8.73
SUMMATION FORCE-Y = 0.00
SUMMATION FORCE-Z = 0.00

SUMMATION OF MOMENTS AROUND THE ORIGIN-
MX= 0.00 MY= -27.62 MZ= 46.82

MAXIMUM DISPLACEMENTS (CM /RADIANS) (LOADING 9)

	MAXIMUMS	AT NODE
X =	2.28144E-02	83
Y =	4.99259E-03	74
Z =	-2.11150E-03	76
RX=	1.36234E-05	82
RY=	-4.86881E-05	86
RZ=	-2.20009E-04	4

STATIC LOAD/REACTION/EQUILIBRIUM SUMMARY FOR CASE NO. 10
EL (+Z)

CENTER OF FORCE BASED ON Z FORCES ONLY (METE).
(FORCES IN NON-GLOBAL DIRECTIONS WILL INVALIDATE RESULTS)

X = 0.834419501E+01
Y = 0.536284328E+01
Z = 0.316293269E+01

***TOTAL APPLIED LOAD (KN METE) SUMMARY (LOADING 10)
SUMMATION FORCE-X = 0.00
SUMMATION FORCE-Y = 0.00
SUMMATION FORCE-Z = 8.73

SUMMATION OF MOMENTS AROUND THE ORIGIN-
MX= 46.82 MY= -72.86 MZ= 0.00

***TOTAL REACTION LOAD(KN METE) SUMMARY (LOADING 10)
SUMMATION FORCE-X = -0.00
SUMMATION FORCE-Y = -0.00
SUMMATION FORCE-Z = -8.73

SUMMATION OF MOMENTS AROUND THE ORIGIN-
MX= -46.82 MY= 72.86 MZ= 0.00

MAXIMUM DISPLACEMENTS (CM /RADIANS) (LOADING 10)

	MAXIMUMS	AT NODE
X =	-2.52566E-03	74
Y =	8.59385E-04	41
Z =	5.24003E-01	40
RX=	1.02741E-03	2
RY=	2.93008E-04	77
RZ=	2.74858E-05	4

STATIC LOAD/REACTION/EQUILIBRIUM SUMMARY FOR CASE NO. 1
DEAD LOAD

CENTER OF FORCE BASED ON Y FORCES ONLY (METE).
(FORCES IN NON-GLOBAL DIRECTIONS WILL INVALIDATE RESULTS)

X = 0.104345970E+02
Y = 0.317649407E+01
Z = 0.338839786E+01

***TOTAL APPLIED LOAD (KN METE) SUMMARY (LOADING 1)
SUMMATION FORCE-X = -0.00
SUMMATION FORCE-Y = -270.83
SUMMATION FORCE-Z = 0.00

SUMMATION OF MOMENTS AROUND THE ORIGIN-
MX= 917.67 MY= -0.00 MZ= -2825.96

***TOTAL REACTION LOAD(KN METE) SUMMARY (LOADING 1)
SUMMATION FORCE-X = 0.00
SUMMATION FORCE-Y = 270.83
SUMMATION FORCE-Z = 0.00

SUMMATION OF MOMENTS AROUND THE ORIGIN-
MX= -917.67 MY= 0.00 MZ= 2825.96

MAXIMUM DISPLACEMENTS (CM /RADIANS) (LOADING 1)

	MAXIMUMS	AT NODE
X =	-7.11635E-03	74
Y =	-2.60724E-01	15
Z =	-5.28543E-02	64
RX=	-1.49105E-03	76
RY=	5.14382E-06	56
RZ=	-7.68956E-04	75

STATIC LOAD/REACTION/EQUILIBRIUM SUMMARY FOR CASE NO. 2
LIVE LOAD

CENTER OF FORCE BASED ON Y FORCES ONLY (METE).
(FORCES IN NON-GLOBAL DIRECTIONS WILL INVALIDATE RESULTS)

X = 0.103321541E+02
Y = 0.301307318E+01
Z = 0.325783049E+01

***TOTAL APPLIED LOAD (KN METE) SUMMARY (LOADING 2)
SUMMATION FORCE-X = 0.00
SUMMATION FORCE-Y = -1144.50
SUMMATION FORCE-Z = 0.00

SUMMATION OF MOMENTS AROUND THE ORIGIN-
MX= 3728.59 MY= 0.00 MZ= -11825.15

***TOTAL REACTION LOAD(KN METE) SUMMARY (LOADING 2)
SUMMATION FORCE-X = -0.00
SUMMATION FORCE-Y = 1144.50
SUMMATION FORCE-Z = -0.00

SUMMATION OF MOMENTS AROUND THE ORIGIN-
MX= -3728.59 MY= 0.00 MZ= 11825.15

MAXIMUM DISPLACEMENTS (CM /RADIANS) (LOADING 2)

	MAXIMUMS	AT NODE
X =	-3.19882E-02	74
Y =	-1.59772E+00	15
Z =	-3.39531E-01	64
RX=	-6.13477E-03	76
RY=	2.63950E-05	56
RZ=	-4.85156E-03	75

STATIC LOAD/REACTION/EQUILIBRIUM SUMMARY FOR CASE NO. 3
WIND LOAD IN +Z DIRN

CENTER OF FORCE BASED ON Z FORCES ONLY (METE).
(FORCES IN NON-GLOBAL DIRECTIONS WILL INVALIDATE RESULTS)

X = 0.101349206E+02
Y = 0.292499995E+01
Z = 0.351111106E+01

***TOTAL APPLIED LOAD (KN METE) SUMMARY (LOADING 3)
SUMMATION FORCE-X = 0.00
SUMMATION FORCE-Y = 0.00
SUMMATION FORCE-Z = 73.71

SUMMATION OF MOMENTS AROUND THE ORIGIN-
MX= 215.60 MY= -747.05 MZ= 0.00

***TOTAL REACTION LOAD(KN METE) SUMMARY (LOADING 3)
SUMMATION FORCE-X = -0.00
SUMMATION FORCE-Y = -0.00
SUMMATION FORCE-Z = -73.71

SUMMATION OF MOMENTS AROUND THE ORIGIN-
MX= -215.60 MY= 747.04 MZ= 0.00

MAXIMUM DISPLACEMENTS (CM /RADIANS) (LOADING 3)

	MAXIMUMS	AT NODE
X =	-8.95828E-03	74
Y =	-2.72805E-03	49
Z =	1.72755E+00	40
RX=	3.73776E-03	3
RY=	5.43145E-04	77
RZ=	9.61029E-05	4

STATIC LOAD/REACTION/EQUILIBRIUM SUMMARY FOR CASE NO. 4
WIND LOAD IN -Z DIRN

CENTER OF FORCE BASED ON Z FORCES ONLY (METE).
(FORCES IN NON-GLOBAL DIRECTIONS WILL INVALIDATE RESULTS)

X = 0.101349206E+02
Y = 0.292499995E+01
Z = 0.351111106E+01

***TOTAL APPLIED LOAD (KN METE) SUMMARY (LOADING 4)
SUMMATION FORCE-X = 0.00
SUMMATION FORCE-Y = 0.00
SUMMATION FORCE-Z = -73.71

SUMMATION OF MOMENTS AROUND THE ORIGIN-
MX= -215.60 MY= 747.05 MZ= 0.00

***TOTAL REACTION LOAD(KN METE) SUMMARY (LOADING 4)
SUMMATION FORCE-X = 0.00
SUMMATION FORCE-Y = 0.00
SUMMATION FORCE-Z = 73.71

SUMMATION OF MOMENTS AROUND THE ORIGIN-
MX= 215.60 MY= -747.04 MZ= -0.00

MAXIMUM DISPLACEMENTS (CM /RADIANS) (LOADING 4)

	MAXIMUMS	AT NODE
X =	8.95828E-03	74
Y =	2.72805E-03	49
Z =	-1.72755E+00	40
RX=	-3.73776E-03	3
RY=	-5.43145E-04	77
RZ=	-9.61029E-05	4

STATIC LOAD/REACTION/EQUILIBRIUM SUMMARY FOR CASE NO. 5
WIND LOAD IN +X DIRN

CENTER OF FORCE BASED ON X FORCES ONLY (METE).
(FORCES IN NON-GLOBAL DIRECTIONS WILL INVALIDATE RESULTS)

X = 0.899090938E+01
Y = 0.292499993E+01
Z = 0.349090905E+01

***TOTAL APPLIED LOAD (KN METE) SUMMARY (LOADING 5)
SUMMATION FORCE-X = 83.65
SUMMATION FORCE-Y = 0.00
SUMMATION FORCE-Z = 0.00

SUMMATION OF MOMENTS AROUND THE ORIGIN-
MX= 0.00 MY= 292.03 MZ= -244.69

***TOTAL REACTION LOAD(KN METE) SUMMARY (LOADING 5)
SUMMATION FORCE-X = -83.65
SUMMATION FORCE-Y = 0.00
SUMMATION FORCE-Z = 0.00

SUMMATION OF MOMENTS AROUND THE ORIGIN-
MX= -0.00 MY= -292.03 MZ= 244.69

MAXIMUM DISPLACEMENTS (CM /RADIANS) (LOADING 5)

	MAXIMUMS	AT NODE
X =	1.31764E-01	82
Y =	8.52836E-03	78
Z =	-1.03835E-02	40
RX=	1.84255E-05	6
RY=	-7.04181E-04	86
RZ=	-4.61922E-03	1

STATIC LOAD/REACTION/EQUILIBRIUM SUMMARY FOR CASE NO. 6
WIND LOAD IN -X DIRN

CENTER OF FORCE BASED ON X FORCES ONLY (METE).
(FORCES IN NON-GLOBAL DIRECTIONS WILL INVALIDATE RESULTS)

X = 0.899090938E+01
Y = 0.292499993E+01
Z = 0.349090905E+01

***TOTAL APPLIED LOAD (KN METE) SUMMARY (LOADING 6)
SUMMATION FORCE-X = -83.65
SUMMATION FORCE-Y = 0.00
SUMMATION FORCE-Z = 0.00

SUMMATION OF MOMENTS AROUND THE ORIGIN-
MX= 0.00 MY= -292.03 MZ= 244.69

***TOTAL REACTION LOAD(KN METE) SUMMARY (LOADING 6)
SUMMATION FORCE-X = 83.65
SUMMATION FORCE-Y = -0.00
SUMMATION FORCE-Z = -0.00

SUMMATION OF MOMENTS AROUND THE ORIGIN-
MX= 0.00 MY= 292.03 MZ= -244.69

MAXIMUM DISPLACEMENTS (CM /RADIANS) (LOADING 6)

	MAXIMUMS	AT NODE
X =	-1.31764E-01	82
Y =	-8.52836E-03	78
Z =	1.03835E-02	40
RX=	-1.84255E-05	6
RY=	7.04181E-04	86
RZ=	4.61922E-03	1

STATIC LOAD/REACTION/EQUILIBRIUM SUMMARY FOR CASE NO. 7
TL1 (TEMP LOAD +25)

***TOTAL APPLIED LOAD (KN METE) SUMMARY (LOADING 7)
SUMMATION FORCE-X = 3.1606516E-14
SUMMATION FORCE-Y = -3.1606516E-12
SUMMATION FORCE-Z = 0.0000000E+00

SUMMATION OF MOMENTS AROUND THE ORIGIN-
MX= 7.7069327E-13 MY= -1.0578088E-06 MZ= -7.3468191E-04

***TOTAL REACTION LOAD(KN METE) SUMMARY (LOADING 7)
SUMMATION FORCE-X = 9.4819548E-14
SUMMATION FORCE-Y = -1.2642606E-13
SUMMATION FORCE-Z = 3.4767168E-13

SUMMATION OF MOMENTS AROUND THE ORIGIN-
MX= -3.9461776E-09 MY= 3.1602596E-06 MZ= 4.9429100E-06

MAXIMUM DISPLACEMENTS (CM /RADIANS) (LOADING 7)

	MAXIMUMS	AT NODE
X =	-5.52588E-01	14
Y =	1.99280E-01	42
Z =	-2.71387E-01	56
RX=	-1.93768E-03	4
RY=	-2.53349E-03	86
RZ=	5.70292E-03	5

STATIC LOAD/REACTION/EQUILIBRIUM SUMMARY FOR CASE NO. 8
TL2 (TEMP LOAD -25)

***TOTAL APPLIED LOAD (KN METE) SUMMARY (LOADING 8)
SUMMATION FORCE-X = -3.1606516E-14
SUMMATION FORCE-Y = 3.1606516E-12
SUMMATION FORCE-Z = 0.0000000E+00

SUMMATION OF MOMENTS AROUND THE ORIGIN-
MX= -7.7069327E-13 MY= 1.0578088E-06 MZ= 7.3468191E-04

***TOTAL REACTION LOAD(KN METE) SUMMARY (LOADING 8)
SUMMATION FORCE-X = -9.4819548E-14
SUMMATION FORCE-Y = 1.2642606E-13
SUMMATION FORCE-Z = -3.4767168E-13

SUMMATION OF MOMENTS AROUND THE ORIGIN-
MX= 3.9461776E-09 MY= -3.1602596E-06 MZ= -4.9429100E-06

MAXIMUM DISPLACEMENTS (CM /RADIANS) (LOADING 8)

	MAXIMUMS	AT NODE
X =	5.52588E-01	14
Y =	-1.99280E-01	42
Z =	2.71387E-01	56
RX=	1.93768E-03	4
RY=	2.53349E-03	86
RZ=	-5.70292E-03	5

********** END OF DATA FROM INTERNAL STORAGE **********

359. **
360. LOAD LIST 1 TO 10
361. PRINT SUPPORT REACTION LIST 1 TO 11

SUPPORT REACTIONS -UNIT KN METE STRUCTURE TYPE = SPACE

JOINT	LOAD	FORCE-X	FORCE-Y	FORCE-Z	MOM-X	MOM-Y	MOM Z
1	9	-0.00	-0.00	0.00	0.00	0.00	0.00
	10	0.00	0.00	-0.00	0.00	0.00	0.00
	1	0.05	10.01	0.00	0.00	0.00	0.00
	2	0.30	39.08	0.00	0.00	0.00	0.00
	3	0.00	0.00	-2.93	0.00	0.00	0.00
	4	-0.00	-0.00	2.93	0.00	0.00	0.00
	5	-3.55	0.48	0.00	0.00	0.00	0.00
	6	3.55	-0.48	-0.00	0.00	0.00	0.00
	7	0.04	0.07	0.00	0.00	0.00	0.00
	8	-0.04	-0.07	-0.00	0.00	0.00	0.00
2	9	-0.01	0.00	0.00	0.00	0.00	0.00
	10	0.00	-2.19	-1.41	0.00	0.00	0.00
	1	-0.01	20.28	1.37	0.00	0.00	0.00
	2	-0.05	97.00	8.70	0.00	0.00	0.00
	3	0.00	-6.94	-7.31	0.00	0.00	0.00
	4	-0.00	6.94	7.31	0.00	0.00	0.00
	5	-1.68	-0.24	-0.05	0.00	0.00	0.00
	6	1.68	0.24	0.05	0.00	0.00	0.00
	7	0.36	-0.03	0.18	0.00	0.00	0.00
	8	-0.36	0.03	-0.18	0.00	0.00	0.00
3	9	-0.00	0.00	-0.00	0.00	0.00	0.00
	10	-0.00	2.19	-1.40	0.00	0.00	0.00
	1	-0.01	18.08	-1.37	0.00	0.00	0.00
	2	-0.04	73.92	-8.70	0.00	0.00	0.00
	3	-0.00	6.94	-7.31	0.00	0.00	0.00
	4	0.00	-6.94	7.31	0.00	0.00	0.00
	5	-1.65	-0.24	0.05	0.00	0.00	0.00
	6	1.65	0.24	-0.05	0.00	0.00	0.00
	7	0.47	-0.03	-0.18	0.00	0.00	0.00
	8	-0.47	0.03	0.18	0.00	0.00	0.00
4	9	-1.03	-3.35	0.01	0.00	0.00	0.00
	10	0.13	-0.65	-0.73	0.00	0.00	0.00
	1	0.04	26.58	-4.64	0.00	0.00	0.00
	2	0.10	106.14	-23.35	0.00	0.00	0.00
	3	0.45	-3.26	-5.86	0.00	0.00	0.00
	4	-0.45	3.26	5.86	0.00	0.00	0.00
	5	-8.88	-15.17	0.02	0.00	0.00	0.00
	6	8.88	15.17	-0.02	0.00	0.00	0.00
	7	18.33	1.95	75.10	0.00	0.00	0.00
	8	-18.33	-1.95	-75.10	0.00	0.00	0.00
5	9	-0.64	-2.74	0.00	0.00	0.00	0.00
	10	-0.08	0.66	-0.73	0.00	0.00	0.00
	1	0.05	24.85	4.64	0.00	0.00	0.00
	2	0.09	87.85	23.34	0.00	0.00	0.00
	3	-0.29	3.29	-5.86	0.00	0.00	0.00
	4	0.29	-3.29	5.86	0.00	0.00	0.00

SUPPORT REACTIONS -UNIT KN METE STRUCTURE TYPE = SPACE

JOINT	LOAD	FORCE-X	FORCE-Y	FORCE-Z	MOM-X	MOM-Y	MOM Z
	5	-6.07	-14.82	0.02	0.00	0.00	0.00
	6	6.07	14.82	-0.02	0.00	0.00	0.00
	7	28.43	3.20	-74.92	0.00	0.00	0.00
	8	-28.43	-3.20	74.92	0.00	0.00	0.00
6	9	-0.99	3.35	-0.01	0.00	0.00	0.00
	10	0.13	-1.43	-0.75	0.00	0.00	0.00
	1	-0.03	27.71	-4.96	0.00	0.00	0.00
	2	-0.13	133.12	-24.94	0.00	0.00	0.00
	3	0.44	-5.99	-5.84	0.00	0.00	0.00
	4	-0.44	5.99	5.84	0.00	0.00	0.00
	5	-8.77	15.17	-0.02	0.00	0.00	0.00
	6	8.77	-15.17	0.02	0.00	0.00	0.00
	7	1.15	-1.94	67.42	0.00	0.00	0.00
	8	-1.15	1.94	-67.42	0.00	0.00	0.00
7	9	-0.58	2.74	-0.00	0.00	0.00	0.00
	10	-0.07	1.42	-0.75	0.00	0.00	0.00
	1	-0.02	27.03	4.96	0.00	0.00	0.00
	2	-0.07	125.36	24.95	0.00	0.00	0.00
	3	-0.26	5.96	-5.83	0.00	0.00	0.00
	4	0.26	-5.96	5.83	0.00	0.00	0.00
	5	-5.68	14.82	-0.02	0.00	0.00	0.00
	6	5.68	-14.82	0.02	0.00	0.00	0.00
	7	12.26	-3.20	-67.60	0.00	0.00	0.00
	8	-12.26	3.20	67.60	0.00	0.00	0.00
8	9	-1.42	-1.48	0.00	0.00	0.00	0.00
	10	0.05	-1.10	-0.76	0.00	0.00	0.00
	1	0.63	34.45	-4.61	0.00	0.00	0.00
	2	-0.04	155.53	-23.06	0.00	0.00	0.00
	3	0.17	-6.92	-8.20	0.00	0.00	0.00
	4	-0.17	6.92	8.20	0.00	0.00	0.00
	5	-10.67	-7.84	-0.00	0.00	0.00	0.00
	6	10.67	7.84	0.00	0.00	0.00	0.00
	7	-0.26	-1.52	74.80	0.00	0.00	0.00
	8	0.26	1.52	-74.80	0.00	0.00	0.00
9	9	-1.36	-1.25	-0.00	0.00	0.00	0.00
	10	-0.10	1.09	-0.76	0.00	0.00	0.00
	1	0.62	30.49	4.61	0.00	0.00	0.00
	2	-0.06	132.98	23.06	0.00	0.00	0.00
	3	-0.35	6.90	-8.18	0.00	0.00	0.00
	4	0.35	-6.90	8.18	0.00	0.00	0.00
	5	-13.34	-8.20	-0.00	0.00	0.00	0.00
	6	13.34	8.20	0.00	0.00	0.00	0.00
	7	-9.88	-2.81	-74.81	0.00	0.00	0.00
	8	9.88	2.81	74.81	0.00	0.00	0.00
10	9	-1.37	1.48	-0.00	0.00	0.00	0.00
	10	0.05	-1.12	-0.72	0.00	0.00	0.00

SUPPORT REACTIONS -UNIT KN METE STRUCTURE TYPE = SPACE

JOINT	LOAD	FORCE-X	FORCE-Y	FORCE-Z	MOM-X	MOM-Y	MOM Z
	1	-0.65	25.90	-4.33	0.00	0.00	0.00
	2	-0.04	100.19	-21.67	0.00	0.00	0.00
	3	0.16	-6.96	-8.19	0.00	0.00	0.00
	4	-0.16	6.96	8.19	0.00	0.00	0.00
	5	-10.43	7.84	0.00	0.00	0.00	0.00
	6	10.43	-7.84	-0.00	0.00	0.00	0.00
	7	-20.92	1.52	71.06	0.00	0.00	0.00
	8	20.92	-1.52	-71.06	0.00	0.00	0.00
11	9	-1.32	1.25	0.00	0.00	0.00	0.00
	10	-0.10	1.12	-0.73	0.00	0.00	0.00
	1	-0.66	25.46	4.33	0.00	0.00	0.00
	2	-0.06	93.33	21.67	0.00	0.00	0.00
	3	-0.33	6.99	-8.19	0.00	0.00	0.00
	4	0.33	-6.99	8.19	0.00	0.00	0.00
	5	-12.95	8.20	0.00	0.00	0.00	0.00
	6	12.95	-8.20	-0.00	0.00	0.00	0.00
	7	-29.97	2.81	-71.06	0.00	0.00	0.00
	8	29.97	-2.81	71.06	0.00	0.00	0.00

********** END OF LATEST ANALYSIS RESULT **********

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362. ************************************************
363. LOAD LIST 201 TO 219
364. PARAMETER 1
365. CODE LRFD
366. FYLD 275000 ALL
367. FU 410000 ALL
368. TRACK 1 ALL
369. BEAM 1 ALL
370. TMAIN 300 ALL
371. MAIN 200 ALL
372. NSF 0.8 ALL
373. ***BEAM MAJOR AXIS
374. LZ 7.4 MEMB 41 TO 52 209
375. LZ 6.9 MEMB 53 TO 64
376. LZ 7.4 MEMB 31 TO 36
377. LZ 3.7 MEMB 73 TO 75 77 78 82 83 85 TO 87
378. UNB 7.4 MEMB 41 TO 52 209
379. UNB 6.9 MEMB 53 TO 64
380. *
381. LZ 6 MEMB 133 TO 140
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382. LZ 5 MEMB 125 TO 132
383. ***BEAM MINOR AXIS
384. LY 3.5 MEMB 91 92 94 TO 98
385. LY 3 MEMB 101 TO 104 111 TO 120
386. ****BRACING
387. KX 0.85 MEMB 159 TO 164 181 TO 196 201 TO 207
388. KZ 0.85 MEMB 159 TO 164 181 TO 196 201 TO 207
389. KY 0.85 MEMB 159 TO 164 181 TO 196 201 TO 207
390. *
391. KX 0.8 MEMB 165 TO 168
392. KZ 0.8 MEMB 165 TO 168
393. KY 0.8 MEMB 165 TO 168
394. *
395. KX 0.95 MEMB 125 TO 140 151 TO 154
396. KZ 0.95 MEMB 125 TO 140 151 TO 154
397. KY 0.95 MEMB 125 TO 140 151 TO 154
398. ****COLUMN MAJOR AXIS
399. LZ 5.75 MEMB 1 TO 14 16 TO 29
400. ****COLUMN MINOR AXIS
401. LY 3.5 MEMB 1 TO 14 16 TO 29
402. ***
403. CHECK CODE ALL
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STAAD.Pro CODE CHECKING - (LRFD 3RD EDITION) v1.0
****

ALL UNITS ARE - KN METE (UNLESS OTHERWISE Noted)

MEMBER	TABLE	RESULT/ FX	CRITICAL COND/ MY	RATIO/ MZ	LOADING/ LOCATION
1	ST IPE360	(EUROPEAN SECTIONS)			
		PASS	LRFD-H1-1B-C	0.332	203
		177.30 C	-1.63	-47.18	3.08
DESIGN STRENGTHS FOR MEMBER 1 UNITS - KIP IN					
PNC= 235.04 PNT= 402.05 MNZ= 1978.83 MNY= 403.19 VN= 96.15					
2	ST IPE360	(EUROPEAN SECTIONS)			
		PASS	LRFD-H1-1B-C	0.482	202
		146.75 C	-0.49	-92.35	2.77
DESIGN STRENGTHS FOR MEMBER 2 UNITS - KIP IN					
PNC= 235.04 PNT= 402.05 MNZ= 2039.40 MNY= 403.19 VN= 96.15					
3	ST IPE360	(EUROPEAN SECTIONS)			
		PASS	LRFD-H1-1B-C	0.523	203
		199.06 C	-9.26	56.54	0.43
DESIGN STRENGTHS FOR MEMBER 3 UNITS - KIP IN					
PNC= 235.04 PNT= 402.05 MNZ= 2232.18 MNY= 403.19 VN= 96.15					
4	ST IPE360	(EUROPEAN SECTIONS)			
		PASS	LRFD-H1-1B-C	0.498	203
		109.34 C	-9.27	56.54	0.00
DESIGN STRENGTHS FOR MEMBER 4 UNITS - KIP IN					
PNC= 235.04 PNT= 402.05 MNZ= 2064.64 MNY= 403.19 VN= 96.15					
5	ST IPE360	(EUROPEAN SECTIONS)			
		PASS	LRFD-H1-1B-C	0.336	203
		90.73 C	0.00	-67.48	2.77
DESIGN STRENGTHS FOR MEMBER 5 UNITS - KIP IN					
PNC= 235.04 PNT= 402.05 MNZ= 2039.40 MNY= 403.19 VN= 96.15					
6	ST IPE360	(EUROPEAN SECTIONS)			
		PASS	LRFD-H1-1A-C	0.441	203
		248.26 C	-0.70	53.87	0.43
DESIGN STRENGTHS FOR MEMBER 6 UNITS - KIP IN					
PNC= 235.04 PNT= 402.05 MNZ= 2232.18 MNY= 403.19 VN= 96.15					
7	ST IPE360	(EUROPEAN SECTIONS)			
		PASS	LRFD-H1-1B-C	0.286	203
		82.99 C	-0.71	53.87	0.00

ALL UNITS ARE - KN METE (UNLESS OTHERWISE Noted)

MEMBER	TABLE	RESULT/ FX	CRITICAL COND/ MY	RATIO/ MZ	LOADING/ LOCATION
DESIGN STRENGTHS FOR MEMBER 7	UNITS - KIP IN				
PNC= 235.04 PNT= 402.05 MNZ= 2064.64 MNY= 403.19 VN= 96.15					
8	ST IPE360	(EUROPEAN SECTIONS)			
		PASS	LRFD-H1-1B-C	0.340	203
		90.28 C	0.00	-68.37	2.77
DESIGN STRENGTHS FOR MEMBER 8	UNITS - KIP IN				
PNC= 235.04 PNT= 402.05 MNZ= 2039.40 MNY= 403.19 VN= 96.15					
9	ST IPE360	(EUROPEAN SECTIONS)			
		PASS	LRFD-H1-1A-C	0.490	203
		288.24 C	-0.83	56.18	0.43
DESIGN STRENGTHS FOR MEMBER 9	UNITS - KIP IN				
PNC= 235.04 PNT= 402.05 MNZ= 2232.18 MNY= 403.19 VN= 96.15					
10	ST IPE360	(EUROPEAN SECTIONS)			
		PASS	LRFD-H1-1B-C	0.304	203
		93.55 C	-0.83	56.20	0.00
DESIGN STRENGTHS FOR MEMBER 10	UNITS - KIP IN				
PNC= 235.04 PNT= 402.05 MNZ= 2064.64 MNY= 403.19 VN= 96.15					
11	ST IPE360	(EUROPEAN SECTIONS)			
		PASS	LRFD-H1-1B-C	0.317	203
		88.57 C	-0.00	-63.39	2.77
DESIGN STRENGTHS FOR MEMBER 11	UNITS - KIP IN				
PNC= 235.04 PNT= 402.05 MNZ= 2039.40 MNY= 403.19 VN= 96.15					
12	ST IPE360	(EUROPEAN SECTIONS)			
		PASS	LRFD-H1-1B-C	0.515	203
		190.28 C	9.71	53.19	0.43
DESIGN STRENGTHS FOR MEMBER 12	UNITS - KIP IN				
PNC= 235.04 PNT= 402.05 MNZ= 2232.18 MNY= 403.19 VN= 96.15					
13	ST IPE360	(EUROPEAN SECTIONS)			
		PASS	LRFD-H1-1B-C	0.486	203
		94.43 C	9.71	53.19	0.00

ALL UNITS ARE - KN METE (UNLESS OTHERWISE Noted)

MEMBER	TABLE	RESULT/ FX	CRITICAL COND/ MY	RATIO/ MZ	LOADING/ LOCATION
DESIGN STRENGTHS FOR MEMBER 13		UNITS - KIP IN			
PNC= 235.04 PNT= 402.05 MNZ= 2064.64 MNY= 403.19 VN= 96.15					
14	ST IPE360	(EUROPEAN SECTIONS)			
	PASS	LRFD-H1-1B-C	0.315	203	
	88.55 C	-0.00	-62.75	2.77	
DESIGN STRENGTHS FOR MEMBER 14		UNITS - KIP IN			
PNC= 235.04 PNT= 402.05 MNZ= 2039.40 MNY= 403.19 VN= 96.15					
15	ST IPE360	(EUROPEAN SECTIONS)			
	PASS	LRFD-H1-1B-C	0.214	207	
	28.63 C	8.31	-0.00	2.92	
DESIGN STRENGTHS FOR MEMBER 15		UNITS - KIP IN			
PNC= 103.34 PNT= 402.05 MNZ= 1370.48 MNY= 403.19 VN= 96.15					
16	ST IPE360	(EUROPEAN SECTIONS)			
	PASS	LRFD-H1-1B-C	0.320	203	
	137.74 C	-1.95	47.18	3.08	
DESIGN STRENGTHS FOR MEMBER 16		UNITS - KIP IN			
PNC= 235.04 PNT= 402.05 MNZ= 1978.83 MNY= 403.19 VN= 96.15					
17	ST IPE360	(EUROPEAN SECTIONS)			
	PASS	LRFD-H1-1B-C	0.464	202	
	114.14 C	-0.39	92.35	2.77	
DESIGN STRENGTHS FOR MEMBER 17		UNITS - KIP IN			
PNC= 235.04 PNT= 402.05 MNZ= 2039.40 MNY= 403.19 VN= 96.15					
18	ST IPE360	(EUROPEAN SECTIONS)			
	PASS	LRFD-H1-1B-C	0.620	203	
	166.23 C	-14.41	-56.44	0.43	
DESIGN STRENGTHS FOR MEMBER 18		UNITS - KIP IN			
PNC= 235.04 PNT= 402.05 MNZ= 2232.18 MNY= 403.19 VN= 96.15					
19	ST IPE360	(EUROPEAN SECTIONS)			
	PASS	LRFD-H1-1B-C	0.598	203	
	82.53 C	-14.40	-56.44	0.00	

ALL UNITS ARE - KN METE (UNLESS OTHERWISE Noted)

MEMBER	TABLE	RESULT/ FX	CRITICAL COND/ MY	RATIO/ MZ	LOADING/ LOCATION
DESIGN STRENGTHS FOR MEMBER 19		UNITS - KIP IN			
PNC= 235.04	PNT= 402.05	MNZ= 2064.64	MNY= 403.19	VN= 96.15	
20	ST IPE360	(EUROPEAN SECTIONS)			
	PASS	LRFD-H1-1B-C	0.331	203	
	66.51 C	-0.00	68.85	2.77	
DESIGN STRENGTHS FOR MEMBER 20		UNITS - KIP IN			
PNC= 235.04	PNT= 402.05	MNZ= 2039.40	MNY= 403.19	VN= 96.15	
21	ST IPE360	(EUROPEAN SECTIONS)			
	PASS	LRFD-H1-1A-C	0.539	203	
	236.55 C	-6.31	-53.97	0.43	
DESIGN STRENGTHS FOR MEMBER 21		UNITS - KIP IN			
PNC= 235.04	PNT= 402.05	MNZ= 2232.18	MNY= 403.19	VN= 96.15	
22	ST IPE360	(EUROPEAN SECTIONS)			
	PASS	LRFD-H1-1B-C	0.399	203	
	60.60 C	-6.29	-53.97	0.00	
DESIGN STRENGTHS FOR MEMBER 22		UNITS - KIP IN			
PNC= 235.04	PNT= 402.05	MNZ= 2064.64	MNY= 403.19	VN= 96.15	
23	ST IPE360	(EUROPEAN SECTIONS)			
	PASS	LRFD-H1-1B-C	0.322	203	
	65.32 C	-0.00	66.99	2.77	
DESIGN STRENGTHS FOR MEMBER 23		UNITS - KIP IN			
PNC= 235.04	PNT= 402.05	MNZ= 2039.40	MNY= 403.19	VN= 96.15	
24	ST IPE360	(EUROPEAN SECTIONS)			
	PASS	LRFD-H1-1A-C	0.497	203	
	246.99 C	3.24	-56.18	0.43	
DESIGN STRENGTHS FOR MEMBER 24		UNITS - KIP IN			
PNC= 235.04	PNT= 402.05	MNZ= 2232.18	MNY= 403.19	VN= 96.15	
25	ST IPE360	(EUROPEAN SECTIONS)			
	PASS	LRFD-H1-1B-C	0.346	203	
	71.97 C	3.23	-56.18	0.00	

ALL UNITS ARE - KN METE (UNLESS OTHERWISE Noted)

MEMBER	TABLE	RESULT/ FX	CRITICAL COND/ MY	RATIO/ MZ	LOADING/ LOCATION
DESIGN STRENGTHS FOR MEMBER 25	UNITS - KIP IN				
PNC= 235.04 PNT= 402.05 MNZ= 2064.64 MNY= 403.19 VN= 96.15					
26	ST IPE360	(EUROPEAN SECTIONS)			
	PASS	LRFD-H1-1B-C	0.307	203	
	67.40 C	0.00	63.35	2.77	
DESIGN STRENGTHS FOR MEMBER 26	UNITS - KIP IN				
PNC= 235.04 PNT= 402.05 MNZ= 2039.40 MNY= 403.19 VN= 96.15					
27	ST IPE360	(EUROPEAN SECTIONS)			
	PASS	LRFD-H1-1B-C	0.593	203	
	179.19 C	13.49	-53.18	0.43	
DESIGN STRENGTHS FOR MEMBER 27	UNITS - KIP IN				
PNC= 235.04 PNT= 402.05 MNZ= 2232.18 MNY= 403.19 VN= 96.15					
28	ST IPE360	(EUROPEAN SECTIONS)			
	PASS	LRFD-H1-1B-C	0.559	203	
	73.69 C	13.48	-53.18	0.00	
DESIGN STRENGTHS FOR MEMBER 28	UNITS - KIP IN				
PNC= 235.04 PNT= 402.05 MNZ= 2064.64 MNY= 403.19 VN= 96.15					
29	ST IPE360	(EUROPEAN SECTIONS)			
	PASS	LRFD-H1-1B-C	0.305	203	
	67.41 C	0.00	62.82	2.77	
DESIGN STRENGTHS FOR MEMBER 29	UNITS - KIP IN				
PNC= 235.04 PNT= 402.05 MNZ= 2039.40 MNY= 403.19 VN= 96.15					
31	ST IPE360	(EUROPEAN SECTIONS)			
	PASS	LRFD-H1-1B-C	0.636	203	
	15.32 C	-0.02	-131.37	3.70	
DESIGN STRENGTHS FOR MEMBER 31	UNITS - KIP IN				
PNC= 221.99 PNT= 402.05 MNZ= 1852.65 MNY= 403.19 VN= 96.15					
32	ST IPE360	(EUROPEAN SECTIONS)			
	PASS	LRFD-H1-1B-C	0.636	203	
	15.32 C	-0.02	-131.37	0.00	

ALL UNITS ARE - KN METE (UNLESS OTHERWISE Noted)

MEMBER	TABLE	RESULT/ FX	CRITICAL COND/ MY	RATIO/ MZ	LOADING/ LOCATION
DESIGN STRENGTHS FOR MEMBER 32		UNITS - KIP IN			
PNC= 221.99	PNT= 402.05	MNZ= 1852.65	MNY= 403.19	VN= 96.15	
33	ST IPE360	(EUROPEAN SECTIONS)			
	PASS	LRFD-H1-1B-C	0.334	203	
	23.30 C	0.00	67.47	0.00	
DESIGN STRENGTHS FOR MEMBER 33		UNITS - KIP IN			
PNC= 221.99	PNT= 402.05	MNZ= 1852.65	MNY= 403.19	VN= 96.15	
34	ST IPE360	(EUROPEAN SECTIONS)			
	PASS	LRFD-H1-1B-C	0.384	202	
	6.36 C	0.03	-79.66	1.54	
DESIGN STRENGTHS FOR MEMBER 34		UNITS - KIP IN			
PNC= 221.99	PNT= 402.05	MNZ= 1852.65	MNY= 403.19	VN= 96.15	
35	ST IPE360	(EUROPEAN SECTIONS)			
	PASS	LRFD-H1-1B-C	0.338	203	
	22.85 C	0.00	68.37	0.00	
DESIGN STRENGTHS FOR MEMBER 35		UNITS - KIP IN			
PNC= 221.99	PNT= 402.05	MNZ= 1852.65	MNY= 403.19	VN= 96.15	
36	ST IPE360	(EUROPEAN SECTIONS)			
	PASS	LRFD-H1-1B-C	0.378	202	
	7.57 C	-0.03	-78.24	1.54	
DESIGN STRENGTHS FOR MEMBER 36		UNITS - KIP IN			
PNC= 221.99	PNT= 402.05	MNZ= 1852.65	MNY= 403.19	VN= 96.15	
37	ST IPE360	(EUROPEAN SECTIONS)			
	PASS	LRFD-H1-1B-C	0.305	203	
	22.04 C	0.00	63.39	0.00	
DESIGN STRENGTHS FOR MEMBER 37		UNITS - KIP IN			
PNC= 238.30	PNT= 402.05	MNZ= 1903.12	MNY= 403.19	VN= 96.15	
38	ST IPE360	(EUROPEAN SECTIONS)			
	PASS	LRFD-H1-1B-C	0.336	202	
	5.73 C	-0.00	-71.70	1.44	

ALL UNITS ARE - KN METE (UNLESS OTHERWISE Noted)

MEMBER	TABLE	RESULT/ FX	CRITICAL COND/ MY	RATIO/ MZ	LOADING/ LOCATION
DESIGN STRENGTHS FOR MEMBER 38		UNITS - KIP IN			
PNC= 238.30 PNT= 402.05 MNZ= 1903.12 MNY= 403.19 VN= 96.15					
39	ST IPE360	(EUROPEAN SECTIONS)			
	PASS	LRFD-H1-1B-C	0.302	203	
	21.37 C	0.00	62.75	0.00	
DESIGN STRENGTHS FOR MEMBER 39		UNITS - KIP IN			
PNC= 238.30 PNT= 402.05 MNZ= 1903.12 MNY= 403.19 VN= 96.15					
40	ST IPE360	(EUROPEAN SECTIONS)			
	PASS	LRFD-H1-1B-C	0.336	202	
	5.90 C	0.00	-71.58	1.44	
DESIGN STRENGTHS FOR MEMBER 40		UNITS - KIP IN			
PNC= 238.30 PNT= 402.05 MNZ= 1903.12 MNY= 403.19 VN= 96.15					
41	ST IPE360	(EUROPEAN SECTIONS)			
	PASS	LRFD-H1-1B-T	0.559	203	
	155.90 T	3.62	-110.07	1.50	
DESIGN STRENGTHS FOR MEMBER 41		UNITS - KIP IN			
PNC= 332.39 PNT= 402.05 MNZ= 2232.18 MNY= 403.19 VN= 96.15					
42	ST IPE360	(EUROPEAN SECTIONS)			
	PASS	LRFD-H1-1B-T	0.699	203	
	155.90 T	1.20	-158.72	1.20	
DESIGN STRENGTHS FOR MEMBER 42		UNITS - KIP IN			
PNC= 332.39 PNT= 402.05 MNZ= 2232.18 MNY= 403.19 VN= 96.15					
43	ST IPE360	(EUROPEAN SECTIONS)			
	PASS	LRFD-H1-1B-T	0.727	203	
	155.90 T	1.21	-165.68	1.15	
DESIGN STRENGTHS FOR MEMBER 43		UNITS - KIP IN			
PNC= 332.39 PNT= 402.05 MNZ= 2232.18 MNY= 403.19 VN= 96.15					
44	ST IPE360	(EUROPEAN SECTIONS)			
	PASS	LRFD-H1-1B-T	0.727	203	
	155.90 T	1.21	-165.68	0.00	

ALL UNITS ARE - KN METE (UNLESS OTHERWISE Noted)

MEMBER	TABLE	RESULT/ FX	CRITICAL COND/ MY	RATIO/ MZ	LOADING/ LOCATION
DESIGN STRENGTHS FOR MEMBER 44		UNITS - KIP IN			
PNC= 332.39 PNT= 402.05 MNZ= 2232.18 MNY= 403.19 VN= 96.15					
45	ST IPE360	(EUROPEAN SECTIONS)			
	PASS	LRFD-H1-1B-T	0.625	203	
	155.90 T	0.68	-142.95	0.00	
DESIGN STRENGTHS FOR MEMBER 45		UNITS - KIP IN			
PNC= 332.39 PNT= 402.05 MNZ= 2232.18 MNY= 403.19 VN= 96.15					
46	ST IPE360	(EUROPEAN SECTIONS)			
	PASS	LRFD-H1-1B-T	0.415	203	
	155.90 T	1.09	-87.61	0.00	
DESIGN STRENGTHS FOR MEMBER 46		UNITS - KIP IN			
PNC= 332.39 PNT= 402.05 MNZ= 2232.18 MNY= 403.19 VN= 96.15					
47	ST IPE450	(EUROPEAN SECTIONS)			
	PASS	LRFD-H1-1B-T	0.364	203	
	149.29 T	10.57	-72.34	0.50	
DESIGN STRENGTHS FOR MEMBER 47		UNITS - KIP IN			
PNC= 473.92 PNT= 546.39 MNZ= 3728.33 MNY= 579.69 VN= 141.21					
48	ST IPE450	(EUROPEAN SECTIONS)			
	PASS	LRFD-H1-1B-T	0.740	203	
	149.29 T	2.70	-281.36	1.20	
DESIGN STRENGTHS FOR MEMBER 48		UNITS - KIP IN			
PNC= 473.92 PNT= 546.39 MNZ= 3728.33 MNY= 579.69 VN= 141.21					
49	ST IPE450	(EUROPEAN SECTIONS)			
	PASS	LRFD-H1-1B-T	0.773	203	
	149.29 T	1.81	-301.23	1.15	
DESIGN STRENGTHS FOR MEMBER 49		UNITS - KIP IN			
PNC= 473.92 PNT= 546.39 MNZ= 3728.33 MNY= 579.69 VN= 141.21					
50	ST IPE450	(EUROPEAN SECTIONS)			
	PASS	LRFD-H1-1B-T	0.773	203	
	149.29 T	1.81	-301.24	0.00	

ALL UNITS ARE - KN METE (UNLESS OTHERWISE Noted)

MEMBER	TABLE	RESULT/ FX	CRITICAL COND/ MY	RATIO/ MZ	LOADING/ LOCATION
DESIGN STRENGTHS FOR MEMBER 50		UNITS - KIP IN			
PNC= 473.92 PNT= 546.39 MNZ= 3728.33 MNY= 579.69 VN= 141.21					
51	ST IPE450		(EUROPEAN SECTIONS)		
	PASS	LRFD-H1-1B-T	0.677	203	
	149.29 T	1.22	-264.39	0.00	
DESIGN STRENGTHS FOR MEMBER 51		UNITS - KIP IN			
PNC= 473.92 PNT= 546.39 MNZ= 3728.33 MNY= 579.69 VN= 141.21					
52	ST IPE450		(EUROPEAN SECTIONS)		
	PASS	LRFD-H1-1B-T	0.436	203	
	149.29 T	1.06	-163.87	0.00	
DESIGN STRENGTHS FOR MEMBER 52		UNITS - KIP IN			
PNC= 473.92 PNT= 546.39 MNZ= 3728.33 MNY= 579.69 VN= 141.21					
53	ST IPE450		(EUROPEAN SECTIONS)		
	PASS	LRFD-H1-1B-T	0.465	203	
	154.22 T	3.10	-162.69	1.00	
DESIGN STRENGTHS FOR MEMBER 53		UNITS - KIP IN			
PNC= 479.60 PNT= 546.39 MNZ= 3728.33 MNY= 579.69 VN= 141.21					
54	ST IPE450		(EUROPEAN SECTIONS)		
	PASS	LRFD-H1-1B-T	0.685	203	
	154.22 T	2.47	-259.16	1.20	
DESIGN STRENGTHS FOR MEMBER 54		UNITS - KIP IN			
PNC= 479.60 PNT= 546.39 MNZ= 3728.33 MNY= 579.69 VN= 141.21					
55	ST IPE450		(EUROPEAN SECTIONS)		
	PASS	LRFD-H1-1B-T	0.739	203	
	154.22 T	1.89	-285.73	1.15	
DESIGN STRENGTHS FOR MEMBER 55		UNITS - KIP IN			
PNC= 479.60 PNT= 546.39 MNZ= 3728.33 MNY= 579.69 VN= 141.21					
56	ST IPE450		(EUROPEAN SECTIONS)		
	PASS	LRFD-H1-1B-T	0.739	203	
	154.22 T	1.89	-285.73	0.00	

ALL UNITS ARE - KN METE (UNLESS OTHERWISE Noted)

MEMBER	TABLE	RESULT/ FX	CRITICAL COND/ MY	RATIO/ MZ	LOADING/ LOCATION
DESIGN STRENGTHS FOR MEMBER 56		UNITS - KIP IN			
PNC= 479.60 PNT= 546.39 MNZ= 3728.33 MNY= 579.69 VN= 141.21					
57	ST IPE450		(EUROPEAN SECTIONS)		
		PASS	LRFD-H1-1B-T	0.670	203
		154.22 T	1.31	-260.43	0.00
DESIGN STRENGTHS FOR MEMBER 57		UNITS - KIP IN			
PNC= 479.60 PNT= 546.39 MNZ= 3728.33 MNY= 579.69 VN= 141.21					
58	ST IPE450		(EUROPEAN SECTIONS)		
		PASS	LRFD-H1-1B-T	0.427	203
		154.22 T	0.18	-165.28	0.00
DESIGN STRENGTHS FOR MEMBER 58		UNITS - KIP IN			
PNC= 479.60 PNT= 546.39 MNZ= 3728.33 MNY= 579.69 VN= 141.21					
59	ST IPE360		(EUROPEAN SECTIONS)		
		PASS	LRFD-H1-1B-T	0.393	203
		146.52 T	1.47	-80.70	1.00
DESIGN STRENGTHS FOR MEMBER 59		UNITS - KIP IN			
PNC= 338.49 PNT= 402.05 MNZ= 2232.18 MNY= 403.19 VN= 96.15					
60	ST IPE360		(EUROPEAN SECTIONS)		
		PASS	LRFD-H1-1B-T	0.639	203
		146.52 T	1.65	-141.55	1.20
DESIGN STRENGTHS FOR MEMBER 60		UNITS - KIP IN			
PNC= 338.49 PNT= 402.05 MNZ= 2232.18 MNY= 403.19 VN= 96.15					
61	ST IPE360		(EUROPEAN SECTIONS)		
		PASS	LRFD-H1-1B-T	0.713	203
		146.52 T	1.15	-163.17	1.15
DESIGN STRENGTHS FOR MEMBER 61		UNITS - KIP IN			
PNC= 338.49 PNT= 402.05 MNZ= 2232.18 MNY= 403.19 VN= 96.15					
62	ST IPE360		(EUROPEAN SECTIONS)		
		PASS	LRFD-H1-1B-T	0.713	203
		146.52 T	1.15	-163.17	0.00

ALL UNITS ARE - KN METE (UNLESS OTHERWISE Noted)

MEMBER	TABLE	RESULT/ FX	CRITICAL COND/ MY	RATIO/ MZ	LOADING/ LOCATION
DESIGN STRENGTHS FOR MEMBER 62		UNITS - KIP IN			
PNC= 338.49	PNT= 402.05	MNZ= 2232.18	MNY= 403.19	VN= 96.15	
63	ST IPE360	(EUROPEAN SECTIONS)			
	PASS	LRFD-H1-1B-T	0.651	203	
	146.52 T	0.89	-148.83	0.00	
DESIGN STRENGTHS FOR MEMBER 63		UNITS - KIP IN			
PNC= 338.49	PNT= 402.05	MNZ= 2232.18	MNY= 403.19	VN= 96.15	
64	ST IPE360	(EUROPEAN SECTIONS)			
	PASS	LRFD-H1-1B-T	0.426	203	
	146.52 T	-0.28	-95.58	0.00	
DESIGN STRENGTHS FOR MEMBER 64		UNITS - KIP IN			
PNC= 338.49	PNT= 402.05	MNZ= 2232.18	MNY= 403.19	VN= 96.15	
71	ST IPE240	(EUROPEAN SECTIONS)			
	PASS	LRFD-H1-1B-T	0.364	203	
	22.44 T	-0.03	-31.83	0.00	
DESIGN STRENGTHS FOR MEMBER 71		UNITS - KIP IN			
PNC= 182.68	PNT= 216.23	MNZ= 803.94	MNY= 155.53	VN= 49.68	
72	ST IPE240	(EUROPEAN SECTIONS)			
	PASS	LRFD-H1-1B-T	0.364	203	
	22.45 T	-0.03	-31.84	1.07	
DESIGN STRENGTHS FOR MEMBER 72		UNITS - KIP IN			
PNC= 187.69	PNT= 216.23	MNZ= 803.94	MNY= 155.53	VN= 49.68	
73	ST IPE240	(EUROPEAN SECTIONS)			
	PASS	LRFD-H1-1B-C	0.537	202	
	0.05 C	-0.01	-48.60	0.88	
DESIGN STRENGTHS FOR MEMBER 73		UNITS - KIP IN			
PNC= 179.35	PNT= 216.23	MNZ= 801.90	MNY= 155.53	VN= 49.68	
74	ST IPE240	(EUROPEAN SECTIONS)			
	PASS	LRFD-H1-1B-C	0.017	206	
	3.52 C	0.02	-0.93	2.20	

ALL UNITS ARE - KN METE (UNLESS OTHERWISE Noted)

MEMBER	TABLE	RESULT/ FX	CRITICAL COND/ MY	RATIO/ MZ	LOADING/ LOCATION
DESIGN STRENGTHS FOR MEMBER 74		UNITS - KIP IN			
PNC= 119.28 PNT= 216.23 MNZ= 682.44 MNY= 155.53 VN= 49.68					
75	ST IPE240	(EUROPEAN SECTIONS)			
	PASS	LRFD-H1-1B-T	0.467	202	
	0.56 T	-0.02	-42.23	0.00	
DESIGN STRENGTHS FOR MEMBER 75		UNITS - KIP IN			
PNC= 179.35 PNT= 216.23 MNZ= 801.90 MNY= 155.53 VN= 49.68					
76	ST IPE240	(EUROPEAN SECTIONS)			
	PASS	SHEAR-Y	0.001	201	
	0.00 C	0.00	0.00	0.00	
DESIGN STRENGTHS FOR MEMBER 76		UNITS - KIP IN			
PNC= 182.68 PNT= 216.23 MNZ= 803.94 MNY= 155.53 VN= 49.68					
77	ST IPE240	(EUROPEAN SECTIONS)			
	PASS	LRFD-H1-1B-C	0.015	206	
	5.61 C	0.03	-0.88	0.00	
DESIGN STRENGTHS FOR MEMBER 77		UNITS - KIP IN			
PNC= 188.00 PNT= 216.23 MNZ= 803.94 MNY= 155.53 VN= 49.68					
78	ST IPE240	(EUROPEAN SECTIONS)			
	PASS	LRFD-H1-1B-T	0.213	202	
	0.56 T	0.01	-19.30	0.00	
DESIGN STRENGTHS FOR MEMBER 78		UNITS - KIP IN			
PNC= 188.00 PNT= 216.23 MNZ= 803.94 MNY= 155.53 VN= 49.68					
79	ST IPE300	(EUROPEAN SECTIONS)			
	PASS	LRFD-H1-1B-C	0.341	203	
	0.19 C	-0.00	-53.01	0.80	
DESIGN STRENGTHS FOR MEMBER 79		UNITS - KIP IN			
PNC= 273.69 PNT= 297.53 MNZ= 1375.67 MNY= 264.62 VN= 71.11					
80	ST IPE300	(EUROPEAN SECTIONS)			
	PASS	LRFD-H1-1B-C	0.428	203	
	0.20 C	0.00	-66.45	0.20	

ALL UNITS ARE - KN METE (UNLESS OTHERWISE Noted)

MEMBER	TABLE	RESULT/ FX	CRITICAL COND/ MY	RATIO/ MZ	LOADING/ LOCATION
DESIGN STRENGTHS FOR MEMBER 80		UNITS - KIP IN			
PNC= 262.01 PNT= 297.53 MNZ= 1375.67 MNY= 264.62 VN= 71.11					
81	ST IPE300	(EUROPEAN SECTIONS)			
		PASS	LRFD-H1-1B-T	0.396	203
		0.01 T	0.00	-61.50	0.00
DESIGN STRENGTHS FOR MEMBER 81		UNITS - KIP IN			
PNC= 266.63 PNT= 297.53 MNZ= 1375.67 MNY= 264.62 VN= 71.11					
82	ST IPE240	(EUROPEAN SECTIONS)			
		PASS	LRFD-H1-1B-C	0.015	206
		5.61 C	0.03	-0.88	1.06
DESIGN STRENGTHS FOR MEMBER 82		UNITS - KIP IN			
PNC= 188.00 PNT= 216.23 MNZ= 803.94 MNY= 155.53 VN= 49.68					
83	ST IPE240	(EUROPEAN SECTIONS)			
		PASS	LRFD-H1-1B-T	0.389	202
		0.57 T	0.01	-35.24	1.06
DESIGN STRENGTHS FOR MEMBER 83		UNITS - KIP IN			
PNC= 188.00 PNT= 216.23 MNZ= 803.94 MNY= 155.53 VN= 49.68					
84	ST IPE240	(EUROPEAN SECTIONS)			
		PASS	SHEAR-Y	0.001	201
		0.00 C	0.00	0.00	0.00
DESIGN STRENGTHS FOR MEMBER 84		UNITS - KIP IN			
PNC= 182.68 PNT= 216.23 MNZ= 803.94 MNY= 155.53 VN= 49.68					
85	ST IPE240	(EUROPEAN SECTIONS)			
		PASS	LRFD-H1-1B-T	0.731	202
		0.57 T	-0.01	-66.19	0.99
DESIGN STRENGTHS FOR MEMBER 85		UNITS - KIP IN			
PNC= 179.35 PNT= 216.23 MNZ= 801.90 MNY= 155.53 VN= 49.68					
86	ST IPE240	(EUROPEAN SECTIONS)			
		PASS	LRFD-H1-1B-C	0.017	206
		3.52 C	0.02	-0.93	0.44

ALL UNITS ARE - KN METE (UNLESS OTHERWISE Noted)

MEMBER	TABLE	RESULT/ FX	CRITICAL COND/ MY	RATIO/ MZ	LOADING/ LOCATION
DESIGN STRENGTHS FOR MEMBER 86		UNITS - KIP IN			
PNC= 119.28 PNT= 216.23 MNZ= 682.44 MNY= 155.53 VN= 49.68					
87	ST IPE240	(EUROPEAN SECTIONS)			
	PASS	LRFD-H1-1B-C	0.586	203	
	0.05 C	-0.01	-53.05	0.00	
DESIGN STRENGTHS FOR MEMBER 87		UNITS - KIP IN			
PNC= 179.35 PNT= 216.23 MNZ= 801.90 MNY= 155.53 VN= 49.68					
88	ST IPE240	(EUROPEAN SECTIONS)			
	PASS	LRFD-H1-1B-T	0.267	203	
	16.70 T	0.03	-23.32	0.00	
DESIGN STRENGTHS FOR MEMBER 88		UNITS - KIP IN			
PNC= 182.68 PNT= 216.23 MNZ= 803.94 MNY= 155.53 VN= 49.68					
89	ST IPE240	(EUROPEAN SECTIONS)			
	PASS	LRFD-H1-1B-T	0.267	203	
	16.71 T	0.03	-23.34	1.07	
DESIGN STRENGTHS FOR MEMBER 89		UNITS - KIP IN			
PNC= 187.69 PNT= 216.23 MNZ= 803.94 MNY= 155.53 VN= 49.68					
91	ST UPN300	(EUROPEAN SECTIONS)			
	PASS	LRFD-H1-1B-C	0.320	202	
	29.07 C	0.00	-22.03	3.00	
DESIGN STRENGTHS FOR MEMBER 91		UNITS - KIP IN			
PNC= 45.14 PNT= 325.18 MNZ= 786.62 MNY= 222.81 VN= 100.15					
92	ST UPN300	(EUROPEAN SECTIONS)			
	PASS	LRFD-H1-1B-C	0.385	202	
	38.19 C	0.00	-25.74	3.00	
DESIGN STRENGTHS FOR MEMBER 92		UNITS - KIP IN			
PNC= 45.14 PNT= 325.18 MNZ= 786.62 MNY= 222.81 VN= 100.15					
94	ST UPN300	(EUROPEAN SECTIONS)			
	PASS	LRFD-H1-1B-C	0.492	203	
	1.32 C	0.00	-43.47	3.00	

ALL UNITS ARE - KN METE (UNLESS OTHERWISE Noted)

MEMBER	TABLE	RESULT/ FX	CRITICAL COND/ MY	RATIO/ MZ	LOADING/ LOCATION
DESIGN STRENGTHS FOR MEMBER 94		UNITS - KIP IN			
PNC= 45.14 PNT= 325.18 MNZ= 786.62 MNY= 222.81 VN= 100.15					
95	ST UPN300	(EUROPEAN SECTIONS)			
	PASS	LRFD-H1-1B-C	0.522	203	
	0.59 C	0.00	-46.25	3.00	
DESIGN STRENGTHS FOR MEMBER 95		UNITS - KIP IN			
PNC= 45.14 PNT= 325.18 MNZ= 786.62 MNY= 222.81 VN= 100.15					
96	ST UPN300	(EUROPEAN SECTIONS)			
	PASS	LRFD-H1-1B-C	0.510	202	
	0.20 C	0.00	-45.33	3.00	
DESIGN STRENGTHS FOR MEMBER 96		UNITS - KIP IN			
PNC= 45.14 PNT= 325.18 MNZ= 786.62 MNY= 222.81 VN= 100.15					
97	ST UPN300	(EUROPEAN SECTIONS)			
	PASS	LRFD-H1-1B-C	0.522	203	
	0.74 C	0.00	-46.25	3.00	
DESIGN STRENGTHS FOR MEMBER 97		UNITS - KIP IN			
PNC= 45.14 PNT= 325.18 MNZ= 786.62 MNY= 222.81 VN= 100.15					
98	ST UPN300	(EUROPEAN SECTIONS)			
	PASS	LRFD-H1-1B-C	0.534	202	
	1.21 C	0.00	-47.18	3.00	
DESIGN STRENGTHS FOR MEMBER 98		UNITS - KIP IN			
PNC= 45.14 PNT= 325.18 MNZ= 786.62 MNY= 222.81 VN= 100.15					
101	ST UPN240	(EUROPEAN SECTIONS)			
	PASS	LRFD-H1-1B-C	0.469	202	
	21.48 C	0.00	-21.22	2.50	
DESIGN STRENGTHS FOR MEMBER 101		UNITS - KIP IN			
PNC= 32.63 PNT= 233.93 MNZ= 475.90 MNY= 129.35 VN= 76.12					
102	ST UPN240	(EUROPEAN SECTIONS)			
	PASS	LRFD-H1-1A-C	0.622	202	
	30.18 C	0.00	-25.08	2.50	

ALL UNITS ARE - KN METE (UNLESS OTHERWISE Noted)

MEMBER	TABLE	RESULT/ FX	CRITICAL COND/ MY	RATIO/ MZ	LOADING/ LOCATION
DESIGN STRENGTHS FOR MEMBER 102		UNITS - KIP IN			
PNC= 32.63	PNT= 233.93	MNZ= 475.90	MNY= 129.35	VN= 76.12	
103 ST	UPN240	(EUROPEAN SECTIONS)			
	PASS	LRFD-H1-1A-C	0.529	202	
	35.19 C	0.00	-17.35	2.50	
DESIGN STRENGTHS FOR MEMBER 103		UNITS - KIP IN			
PNC= 32.63	PNT= 233.93	MNZ= 475.90	MNY= 129.35	VN= 76.12	
104 ST	UPN240	(EUROPEAN SECTIONS)			
	PASS	LRFD-H1-1A-C	0.787	202	
	48.00 C	0.00	-27.66	2.50	
DESIGN STRENGTHS FOR MEMBER 104		UNITS - KIP IN			
PNC= 32.63	PNT= 233.93	MNZ= 475.90	MNY= 129.35	VN= 76.12	
111 ST	UPN240	(EUROPEAN SECTIONS)			
	PASS	LRFD-H1-1B-C	0.687	202	
	4.47 C	0.00	-36.10	2.50	
DESIGN STRENGTHS FOR MEMBER 111		UNITS - KIP IN			
PNC= 32.63	PNT= 233.93	MNZ= 475.90	MNY= 129.35	VN= 76.12	
112 ST	UPN240	(EUROPEAN SECTIONS)			
	PASS	LRFD-H1-1B-C	0.595	203	
	2.03 C	0.00	-31.60	2.50	
DESIGN STRENGTHS FOR MEMBER 112		UNITS - KIP IN			
PNC= 32.63	PNT= 233.93	MNZ= 475.90	MNY= 129.35	VN= 76.12	
113 ST	UPN240	(EUROPEAN SECTIONS)			
	PASS	LRFD-H1-1B-C	0.577	202	
	0.47 C	0.00	-30.95	2.50	
DESIGN STRENGTHS FOR MEMBER 113		UNITS - KIP IN			
PNC= 32.63	PNT= 233.93	MNZ= 475.90	MNY= 129.35	VN= 76.12	
114 ST	UPN240	(EUROPEAN SECTIONS)			
	PASS	LRFD-H1-1B-C	0.590	203	
	0.80 C	0.00	-31.60	2.50	

ALL UNITS ARE - KN METE (UNLESS OTHERWISE Noted)

MEMBER	TABLE	RESULT/ FX	CRITICAL COND/ MY	RATIO/ MZ	LOADING/ LOCATION
DESIGN STRENGTHS FOR MEMBER 114		UNITS - KIP IN			
PNC= 32.63 PNT= 233.93 MNZ= 475.90 MNY= 129.35 VN= 76.12					
115 ST	UPN240	(EUROPEAN SECTIONS)			
	PASS	LRFD-H1-1B-C	0.604	202	
	1.24 C	0.00	-32.24	2.50	
DESIGN STRENGTHS FOR MEMBER 115		UNITS - KIP IN			
PNC= 32.63 PNT= 233.93 MNZ= 475.90 MNY= 129.35 VN= 76.12					
116 ST	UPN240	(EUROPEAN SECTIONS)			
	PASS	LRFD-H1-1B-C	0.744	203	
	4.94 C	0.00	-39.11	2.92	
DESIGN STRENGTHS FOR MEMBER 116		UNITS - KIP IN			
PNC= 32.63 PNT= 233.93 MNZ= 475.90 MNY= 129.35 VN= 76.12					
117 ST	UPN240	(EUROPEAN SECTIONS)			
	PASS	LRFD-H1-1B-C	0.590	203	
	0.57 C	0.00	-31.60	2.50	
DESIGN STRENGTHS FOR MEMBER 117		UNITS - KIP IN			
PNC= 32.63 PNT= 233.93 MNZ= 475.90 MNY= 129.35 VN= 76.12					
118 ST	UPN240	(EUROPEAN SECTIONS)			
	PASS	LRFD-H1-1B-C	0.513	202	
	0.21 C	0.00	-27.54	2.50	
DESIGN STRENGTHS FOR MEMBER 118		UNITS - KIP IN			
PNC= 32.63 PNT= 233.93 MNZ= 475.90 MNY= 129.35 VN= 76.12					
119 ST	UPN240	(EUROPEAN SECTIONS)			
	PASS	LRFD-H1-1B-C	0.592	203	
	1.18 C	0.00	-31.60	2.50	
DESIGN STRENGTHS FOR MEMBER 119		UNITS - KIP IN			
PNC= 32.63 PNT= 233.93 MNZ= 475.90 MNY= 129.35 VN= 76.12					
120 ST	UPN240	(EUROPEAN SECTIONS)			
	PASS	LRFD-H1-1B-C	0.607	202	
	2.00 C	0.00	-32.24	2.50	

ALL UNITS ARE - KN METE (UNLESS OTHERWISE Noted)

MEMBER	TABLE	RESULT/ FX	CRITICAL COND/ MY	RATIO/ MZ	LOADING/ LOCATION
DESIGN STRENGTHS FOR MEMBER 120		UNITS - KIP IN			
PNC= 32.63 PNT= 233.93 MNZ= 475.90 MNY= 129.35 VN= 76.12					
125	ST UPN140	(EUROPEAN SECTIONS)			
		PASS	LRFD-H1-1B-C	0.112	206
		25.79 C	0.17	0.00	2.50
DESIGN STRENGTHS FOR MEMBER 125		UNITS - KIP IN			
PNC= 35.22 PNT= 112.82 MNZ= 186.00 MNY= 49.05 VN= 32.72					
126	ST UPN140	(EUROPEAN SECTIONS)			
		PASS	LRFD-H1-1B-T	0.054	203
		22.11 T	0.18	0.00	2.50
DESIGN STRENGTHS FOR MEMBER 126		UNITS - KIP IN			
PNC= 35.22 PNT= 112.82 MNZ= 186.00 MNY= 49.05 VN= 32.72					
127	ST UPN140	(EUROPEAN SECTIONS)			
		PASS	LRFD-H1-1B-C	0.107	206
		24.02 C	0.17	0.00	2.50
DESIGN STRENGTHS FOR MEMBER 127		UNITS - KIP IN			
PNC= 35.22 PNT= 112.82 MNZ= 186.00 MNY= 49.05 VN= 32.72					
128	ST UPN140	(EUROPEAN SECTIONS)			
		PASS	LRFD-H1-1B-T	0.049	203
		17.82 T	0.17	0.00	2.50
DESIGN STRENGTHS FOR MEMBER 128		UNITS - KIP IN			
PNC= 35.22 PNT= 112.82 MNZ= 186.00 MNY= 49.05 VN= 32.72					
129	ST UPN140	(EUROPEAN SECTIONS)			
		PASS	LRFD-H1-1B-C	0.089	207
		18.56 C	0.16	0.00	0.00
DESIGN STRENGTHS FOR MEMBER 129		UNITS - KIP IN			
PNC= 35.22 PNT= 112.82 MNZ= 186.00 MNY= 49.05 VN= 32.72					
130	ST UPN140	(EUROPEAN SECTIONS)			
		PASS	LRFD-H1-1B-C	0.036	207
		1.91 C	0.17	0.00	0.00

ALL UNITS ARE - KN METE (UNLESS OTHERWISE Noted)

MEMBER	TABLE	RESULT/ FX	CRITICAL COND/ MY	RATIO/ MZ	LOADING/ LOCATION
DESIGN STRENGTHS FOR MEMBER 130		UNITS - KIP IN			
PNC= 35.22 PNT= 112.82 MNZ= 186.00 MNY= 49.05 VN= 32.72					
131	ST UPN140	(EUROPEAN SECTIONS)			
	PASS	LRFD-H1-1B-C	0.082	207	
	16.59 C	0.16	0.00	0.00	
DESIGN STRENGTHS FOR MEMBER 131		UNITS - KIP IN			
PNC= 35.22 PNT= 112.82 MNZ= 186.00 MNY= 49.05 VN= 32.72					
132	ST UPN140	(EUROPEAN SECTIONS)			
	PASS	LRFD-H1-1B-C	0.035	207	
	1.79 C	0.16	0.00	0.00	
DESIGN STRENGTHS FOR MEMBER 132		UNITS - KIP IN			
PNC= 35.22 PNT= 112.82 MNZ= 186.00 MNY= 49.05 VN= 32.72					
133	ST UPN140	(EUROPEAN SECTIONS)			
	PASS	LRFD-H1-1B-C	0.086	206	
	9.70 C	0.24	0.00	3.00	
DESIGN STRENGTHS FOR MEMBER 133		UNITS - KIP IN			
PNC= 24.94 PNT= 112.82 MNZ= 173.92 MNY= 49.05 VN= 32.72					
134	ST UPN140	(EUROPEAN SECTIONS)			
	PASS	LRFD-H1-1B-C	0.051	206	
	1.91 C	0.23	0.00	3.00	
DESIGN STRENGTHS FOR MEMBER 134		UNITS - KIP IN			
PNC= 24.94 PNT= 112.82 MNZ= 173.92 MNY= 49.05 VN= 32.72					
135	ST UPN140	(EUROPEAN SECTIONS)			
	PASS	LRFD-H1-1B-C	0.090	206	
	10.45 C	0.24	0.00	3.00	
DESIGN STRENGTHS FOR MEMBER 135		UNITS - KIP IN			
PNC= 24.94 PNT= 112.82 MNZ= 173.92 MNY= 49.05 VN= 32.72					
136	ST UPN140	(EUROPEAN SECTIONS)			
	PASS	LRFD-H1-1B-C	0.050	206	
	1.75 C	0.23	0.00	3.00	

ALL UNITS ARE - KN METE (UNLESS OTHERWISE Noted)

MEMBER	TABLE	RESULT/ FX	CRITICAL COND/ MY	RATIO/ MZ	LOADING/ LOCATION
DESIGN STRENGTHS FOR MEMBER 136		UNITS - KIP IN			
PNC= 24.94 PNT= 112.82 MNZ= 173.92 MNY= 49.05 VN= 32.72					
137	ST UPN140		(EUROPEAN SECTIONS)		
		PASS	LRFD-H1-1B-C	0.086	207
		9.59 C	0.24	0.00	0.00
DESIGN STRENGTHS FOR MEMBER 137		UNITS - KIP IN			
PNC= 24.94 PNT= 112.82 MNZ= 173.92 MNY= 49.05 VN= 32.72					
138	ST UPN140		(EUROPEAN SECTIONS)		
		PASS	LRFD-H1-1B-C	0.051	207
		1.93 C	0.23	0.00	0.00
DESIGN STRENGTHS FOR MEMBER 138		UNITS - KIP IN			
PNC= 24.94 PNT= 112.82 MNZ= 173.92 MNY= 49.05 VN= 32.72					
139	ST UPN140		(EUROPEAN SECTIONS)		
		PASS	LRFD-H1-1B-C	0.089	207
		10.27 C	0.24	0.00	0.00
DESIGN STRENGTHS FOR MEMBER 139		UNITS - KIP IN			
PNC= 24.94 PNT= 112.82 MNZ= 173.92 MNY= 49.05 VN= 32.72					
140	ST UPN140		(EUROPEAN SECTIONS)		
		PASS	LRFD-H1-1B-C	0.050	207
		1.78 C	0.23	0.00	0.00
DESIGN STRENGTHS FOR MEMBER 140		UNITS - KIP IN			
PNC= 24.94 PNT= 112.82 MNZ= 173.92 MNY= 49.05 VN= 32.72					
151	ST L130X130X13		(EUROPEAN SECTIONS)		
		PASS	LRFD-H1-1B-C	0.150	202
		23.55 C	0.00	-1.02	2.50
DESIGN STRENGTHS FOR MEMBER 151		UNITS - KIP IN			
PNC= 32.24 PNT= 178.63 MNZ= 133.10 MNY= 220.18 VN= 37.61					
152	ST L130X130X13		(EUROPEAN SECTIONS)		
		PASS	LRFD-H1-1B-T	0.082	202
		22.93 T	0.00	-1.02	2.50

ALL UNITS ARE - KN METE (UNLESS OTHERWISE Noted)

MEMBER	TABLE	RESULT/ FX	CRITICAL COND/ MY	RATIO/ MZ	LOADING/ LOCATION
DESIGN STRENGTHS FOR MEMBER 152		UNITS - KIP IN			
PNC= 32.24 PNT= 178.63 MNZ= 133.10 MNY= 220.18 VN= 37.61					
153	ST L130X130X13		(EUROPEAN SECTIONS)		
	PASS	LRFD-H1-1B-C	0.131	202	
	18.08 C	0.00	-1.02	2.50	
DESIGN STRENGTHS FOR MEMBER 153		UNITS - KIP IN			
PNC= 32.24 PNT= 178.63 MNZ= 133.10 MNY= 220.18 VN= 37.61					
154	ST L130X130X13		(EUROPEAN SECTIONS)		
	PASS	LRFD-H1-1B-T	0.081	201	
	3.13 T	0.00	-1.19	2.50	
DESIGN STRENGTHS FOR MEMBER 154		UNITS - KIP IN			
PNC= 32.24 PNT= 178.63 MNZ= 133.10 MNY= 220.18 VN= 37.61					
159	ST L90X90X8		(EUROPEAN SECTIONS)		
	PASS	COMPRESSION	0.493	203	
	35.77 C	0.00	0.00	3.60	
DESIGN STRENGTHS FOR MEMBER 159		UNITS - KIP IN			
PNC= 16.30 PNT= 76.87 MNZ= 0.00 MNY= 0.00 VN= 0.00					
160	ST L90X90X8		(EUROPEAN SECTIONS)		
	PASS	COMPRESSION	0.369	203	
	26.77 C	0.00	0.00	3.60	
DESIGN STRENGTHS FOR MEMBER 160		UNITS - KIP IN			
PNC= 16.30 PNT= 76.87 MNZ= 0.00 MNY= 0.00 VN= 0.00					
161	ST L150X150X16		(EUROPEAN SECTIONS)		
	PASS	COMPRESSION	0.002	213	
	0.40 C	0.00	0.00	0.00	
DESIGN STRENGTHS FOR MEMBER 161		UNITS - KIP IN			
PNC= 48.54 PNT= 252.73 MNZ= 0.00 MNY= 0.00 VN= 0.00					
162	ST L150X150X16		(EUROPEAN SECTIONS)		
	PASS	COMPRESSION	0.002	213	
	0.36 C	0.00	0.00	0.00	

ALL UNITS ARE - KN METE (UNLESS OTHERWISE Noted)

MEMBER	TABLE	RESULT/ FX	CRITICAL COND/ MY	RATIO/ MZ	LOADING/ LOCATION
DESIGN STRENGTHS FOR MEMBER 162		UNITS - KIP IN			
PNC= 48.54 PNT= 252.73 MNZ= 0.00 MNY= 0.00 VN= 0.00					
163	ST L150X150X16	(EUROPEAN SECTIONS)			
	PASS	COMPRESSION	0.003		202
	0.65 C	0.00	0.00		0.00
DESIGN STRENGTHS FOR MEMBER 163		UNITS - KIP IN			
PNC= 48.54 PNT= 252.73 MNZ= 0.00 MNY= 0.00 VN= 0.00					
164	ST L150X150X16	(EUROPEAN SECTIONS)			
	PASS	COMPRESSION	0.003		203
	0.71 C	0.00	0.00		0.00
DESIGN STRENGTHS FOR MEMBER 164		UNITS - KIP IN			
PNC= 48.54 PNT= 252.73 MNZ= 0.00 MNY= 0.00 VN= 0.00					
165	ST L150X150X16	(EUROPEAN SECTIONS)			
	PASS	COMPRESSION	0.001		209
	0.28 C	0.00	0.00		0.00
DESIGN STRENGTHS FOR MEMBER 165		UNITS - KIP IN			
PNC= 44.14 PNT= 252.73 MNZ= 0.00 MNY= 0.00 VN= 0.00					
166	ST L150X150X16	(EUROPEAN SECTIONS)			
	PASS	COMPRESSION	0.001		208
	0.28 C	0.00	0.00		0.00
DESIGN STRENGTHS FOR MEMBER 166		UNITS - KIP IN			
PNC= 44.14 PNT= 252.73 MNZ= 0.00 MNY= 0.00 VN= 0.00					
167	ST L150X150X16	(EUROPEAN SECTIONS)			
	PASS	COMPRESSION	0.001		208
	0.29 C	0.00	0.00		0.00
DESIGN STRENGTHS FOR MEMBER 167		UNITS - KIP IN			
PNC= 44.14 PNT= 252.73 MNZ= 0.00 MNY= 0.00 VN= 0.00					
168	ST L150X150X16	(EUROPEAN SECTIONS)			
	PASS	COMPRESSION	0.001		209
	0.29 C	0.00	0.00		0.00

ALL UNITS ARE - KN METE (UNLESS OTHERWISE Noted)

MEMBER	TABLE	RESULT/ FX	CRITICAL COND/ MY	RATIO/ MZ	LOADING/ LOCATION
DESIGN STRENGTHS FOR MEMBER 168		UNITS - KIP IN			
PNC= 44.14 PNT= 252.73 MNZ= 0.00 MNY= 0.00 VN= 0.00					
181	ST L100X100X10		(EUROPEAN SECTIONS)		
	PASS	COMPRESSION	0.231	207	
	27.77 C	0.00	0.00	0.00	
DESIGN STRENGTHS FOR MEMBER 181		UNITS - KIP IN			
PNC= 26.97 PNT= 106.18 MNZ= 0.00 MNY= 0.00 VN= 0.00					
182	ST L100X100X10		(EUROPEAN SECTIONS)		
	PASS	COMPRESSION	0.159	203	
	18.08 C	0.00	0.00	0.00	
DESIGN STRENGTHS FOR MEMBER 182		UNITS - KIP IN			
PNC= 25.58 PNT= 106.18 MNZ= 0.00 MNY= 0.00 VN= 0.00					
183	ST L100X100X10		(EUROPEAN SECTIONS)		
	PASS	COMPRESSION	0.221	207	
	26.53 C	0.00	0.00	0.00	
DESIGN STRENGTHS FOR MEMBER 183		UNITS - KIP IN			
PNC= 26.97 PNT= 106.18 MNZ= 0.00 MNY= 0.00 VN= 0.00					
184	ST L100X100X10		(EUROPEAN SECTIONS)		
	PASS	COMPRESSION	0.128	203	
	14.60 C	0.00	0.00	0.00	
DESIGN STRENGTHS FOR MEMBER 184		UNITS - KIP IN			
PNC= 25.58 PNT= 106.18 MNZ= 0.00 MNY= 0.00 VN= 0.00					
185	ST L100X100X10		(EUROPEAN SECTIONS)		
	PASS	COMPRESSION	0.232	206	
	27.81 C	0.00	0.00	0.00	
DESIGN STRENGTHS FOR MEMBER 185		UNITS - KIP IN			
PNC= 26.97 PNT= 106.18 MNZ= 0.00 MNY= 0.00 VN= 0.00					
186	ST L100X100X10		(EUROPEAN SECTIONS)		
	PASS	COMPRESSION	0.052	214	
	5.91 C	0.00	0.00	0.00	

ALL UNITS ARE - KN METE (UNLESS OTHERWISE Noted)

MEMBER	TABLE	RESULT/ FX	CRITICAL COND/ MY	RATIO/ MZ	LOADING/ LOCATION
DESIGN STRENGTHS FOR MEMBER 186		UNITS - KIP IN			
PNC= 25.58 PNT= 106.18 MNZ= 0.00 MNY= 0.00 VN= 0.00					
187	ST L100X100X10		(EUROPEAN SECTIONS)		
	PASS	COMPRESSION		0.221	206
	26.56 C	0.00		0.00	0.00
DESIGN STRENGTHS FOR MEMBER 187		UNITS - KIP IN			
PNC= 26.97 PNT= 106.18 MNZ= 0.00 MNY= 0.00 VN= 0.00					
188	ST L100X100X10		(EUROPEAN SECTIONS)		
	PASS	COMPRESSION		0.056	214
	6.42 C	0.00		0.00	0.00
DESIGN STRENGTHS FOR MEMBER 188		UNITS - KIP IN			
PNC= 25.58 PNT= 106.18 MNZ= 0.00 MNY= 0.00 VN= 0.00					
189	ST L100X100X10		(EUROPEAN SECTIONS)		
	PASS	COMPRESSION		0.193	207
	16.44 C	0.00		0.00	0.00
DESIGN STRENGTHS FOR MEMBER 189		UNITS - KIP IN			
PNC= 19.11 PNT= 106.18 MNZ= 0.00 MNY= 0.00 VN= 0.00					
190	ST L100X100X10		(EUROPEAN SECTIONS)		
	PASS	COMPRESSION		0.043	207
	4.06 C	0.00		0.00	0.00
DESIGN STRENGTHS FOR MEMBER 190		UNITS - KIP IN			
PNC= 21.20 PNT= 106.18 MNZ= 0.00 MNY= 0.00 VN= 0.00					
191	ST L100X100X10		(EUROPEAN SECTIONS)		
	PASS	COMPRESSION		0.205	207
	17.46 C	0.00		0.00	0.00
DESIGN STRENGTHS FOR MEMBER 191		UNITS - KIP IN			
PNC= 19.11 PNT= 106.18 MNZ= 0.00 MNY= 0.00 VN= 0.00					
192	ST L100X100X10		(EUROPEAN SECTIONS)		
	PASS	COMPRESSION		0.041	207
	3.83 C	0.00		0.00	0.00

ALL UNITS ARE - KN METE (UNLESS OTHERWISE Noted)

MEMBER	TABLE	RESULT/ FX	CRITICAL COND/ MY	RATIO/ MZ	LOADING/ LOCATION
DESIGN STRENGTHS FOR MEMBER 192		UNITS - KIP IN			
PNC= 21.20 PNT= 106.18 MNZ= 0.00 MNY= 0.00 VN= 0.00					
193 ST	L100X100X10	(EUROPEAN SECTIONS)			
	PASS	COMPRESSION	0.194	206	
	16.45 C	0.00	0.00	0.00	
DESIGN STRENGTHS FOR MEMBER 193		UNITS - KIP IN			
PNC= 19.11 PNT= 106.18 MNZ= 0.00 MNY= 0.00 VN= 0.00					
194 ST	L100X100X10	(EUROPEAN SECTIONS)			
	PASS	COMPRESSION	0.043	206	
	4.06 C	0.00	0.00	0.00	
DESIGN STRENGTHS FOR MEMBER 194		UNITS - KIP IN			
PNC= 21.20 PNT= 106.18 MNZ= 0.00 MNY= 0.00 VN= 0.00					
195 ST	L100X100X10	(EUROPEAN SECTIONS)			
	PASS	COMPRESSION	0.206	206	
	17.48 C	0.00	0.00	0.00	
DESIGN STRENGTHS FOR MEMBER 195		UNITS - KIP IN			
PNC= 19.11 PNT= 106.18 MNZ= 0.00 MNY= 0.00 VN= 0.00					
196 ST	L100X100X10	(EUROPEAN SECTIONS)			
	PASS	COMPRESSION	0.041	206	
	3.83 C	0.00	0.00	0.00	
DESIGN STRENGTHS FOR MEMBER 196		UNITS - KIP IN			
PNC= 21.20 PNT= 106.18 MNZ= 0.00 MNY= 0.00 VN= 0.00					
201 ST	L65X65X6	(EUROPEAN SECTIONS)			
	PASS	COMPRESSION	0.028	206	
	3.29 C	0.00	0.00	0.00	
DESIGN STRENGTHS FOR MEMBER 201		UNITS - KIP IN			
PNC= 26.47 PNT= 41.64 MNZ= 0.00 MNY= 0.00 VN= 0.00					
202 ST	L65X65X6	(EUROPEAN SECTIONS)			
	PASS	COMPRESSION	0.028	206	
	3.29 C	0.00	0.00	0.00	

ALL UNITS ARE - KN METE (UNLESS OTHERWISE Noted)

MEMBER	TABLE	RESULT/ FX	CRITICAL COND/ MY	RATIO/ MZ	LOADING/ LOCATION
DESIGN STRENGTHS FOR MEMBER 202		UNITS - KIP IN			
PNC= 26.47 PNT= 41.64 MNZ= 0.00 MNY= 0.00 VN= 0.00					
203	ST L65X65X6	(EUROPEAN SECTIONS)			
	PASS	COMPRESSION	0.058	215	
	4.67 C	0.00	0.00	0.00	
DESIGN STRENGTHS FOR MEMBER 203		UNITS - KIP IN			
PNC= 18.13 PNT= 41.64 MNZ= 0.00 MNY= 0.00 VN= 0.00					
204	ST L65X65X6	(EUROPEAN SECTIONS)			
	PASS	COMPRESSION	0.059	206	
	4.80 C	0.00	0.00	0.00	
DESIGN STRENGTHS FOR MEMBER 204		UNITS - KIP IN			
PNC= 18.13 PNT= 41.64 MNZ= 0.00 MNY= 0.00 VN= 0.00					
205	ST L65X65X6	(EUROPEAN SECTIONS)			
	PASS	COMPRESSION	0.019	215	
	1.78 C	0.00	0.00	0.00	
DESIGN STRENGTHS FOR MEMBER 205		UNITS - KIP IN			
PNC= 21.18 PNT= 41.64 MNZ= 0.00 MNY= 0.00 VN= 0.00					
206	ST L65X65X6	(EUROPEAN SECTIONS)			
	PASS	COMPRESSION	0.019	215	
	1.78 C	0.00	0.00	0.00	
DESIGN STRENGTHS FOR MEMBER 206		UNITS - KIP IN			
PNC= 21.18 PNT= 41.64 MNZ= 0.00 MNY= 0.00 VN= 0.00					
207	ST L65X65X6	(EUROPEAN SECTIONS)			
	PASS	COMPRESSION	0.059	206	
	4.80 C	0.00	0.00	0.00	
DESIGN STRENGTHS FOR MEMBER 207		UNITS - KIP IN			
PNC= 18.13 PNT= 41.64 MNZ= 0.00 MNY= 0.00 VN= 0.00					
208	ST L65X65X6	(EUROPEAN SECTIONS)			
	PASS	LRFD-H1-1B-C	0.057	207	
	4.54 C	-0.00	-0.03	0.90	

ALL UNITS ARE - KN METE (UNLESS OTHERWISE Noted)

MEMBER	TABLE	RESULT/ FX	CRITICAL COND/ MY	RATIO/ MZ	LOADING/ LOCATION
DESIGN STRENGTHS FOR MEMBER 208		UNITS - KIP IN			
PNC= 13.26	PNT= 41.64	MNZ= 15.92	MNY= 27.64	VN= 8.68	
209	ST IPE450	(EUROPEAN SECTIONS)			
	PASS	LRFD-H1-1B-T	0.531	203	
	149.29 T	1.88	-198.55	1.00	
DESIGN STRENGTHS FOR MEMBER 209		UNITS - KIP IN			
PNC= 473.92	PNT= 546.39	MNZ= 3728.33	MNY= 579.69	VN= 141.21	

********** END OF TABULATED RESULT OF DESIGN **********

404. STEEL TAKE OFF ALL

STEEL TAKE-OFF

	PROFILE	**LENGTH(METE)**	**WEIGHT(KN)**
ST	IPE360	114.65	64.029
ST	IPE450	14.30	10.853
ST	IPE240	21.85	6.563
ST	IPE300	3.10	1.281
ST	UPN300	42.00	18.971
ST	UPN240	70.00	22.746
ST	UPN140	44.00	6.895
ST	L130X130X13	20.00	4.963
ST	L90X90X8	7.21	0.770
ST	L150X150X16	52.57	18.454
ST	L100X100X10	63.04	9.298
ST	L65X65X6	13.11	0.758
		TOTAL =	165.583

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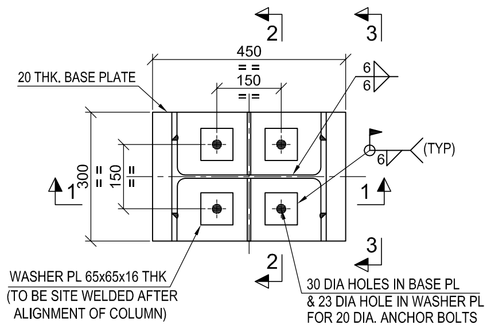
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405. ************************************************
406. ***////PARAMETER FOR DEFLECTION CHECK OF BEAMS////
407. ***////TO BE UNCOMMENTED WHILE DEFLECTION CHECK////
408. ***////TO BE PUT IN COMMENT MODE WHILE CHECKING FOR STRENGTH////
409. *LOAD LIST 101 TO 119
410. *PARAMETER 2
411. *CODE LRFD
412. ***FL_913.550
413. *DJ1 133 MEMB 101 TO 104
414. *DJ2 134 MEMB 101 TO 104
415. *DFF 360 MEMB 488 TO 494 621 TO 624
416. *DFF 500 MEMB 172 TO 188
417. *CHECK CODE MEMB 172 TO 188 488 TO 494 621 TO 624
418. FINISH
```

********** END OF THE STAAD.Pro RUN **********

**** DATE= JUN 5,2020 TIME= 12:33: 9 ****

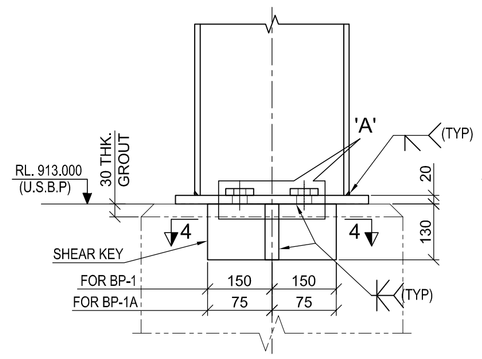
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* For technical assistance on STAAD.Pro, please visit *
* http://selectservices.bentley.com/en-US/ *
* *
* Details about additional assistance from *
* Bentley and Partners can be found at program menu *
* Help->Technical Support *
* *
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****

DRAWING NUMBER MD-216-3650-EG-ST-GAD-2290 REV D01



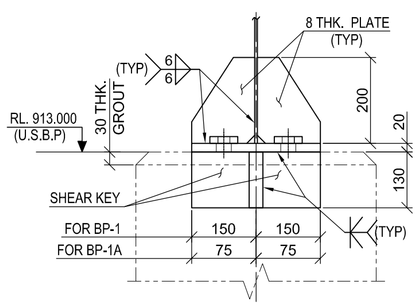
DETAIL OF BASE PLATE MKD. BP-1 & BP-1A

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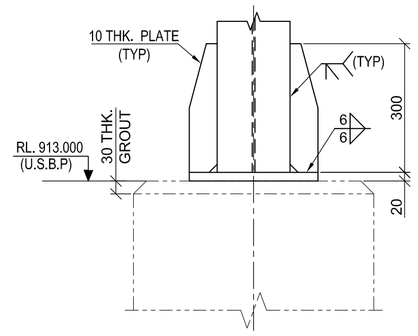
SECTION 1-1

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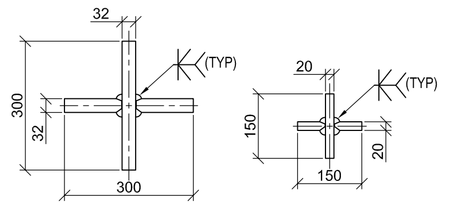
SECTION 2-2

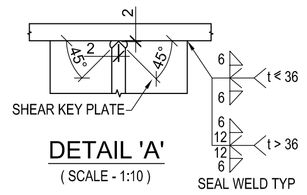
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SECTION 3-3

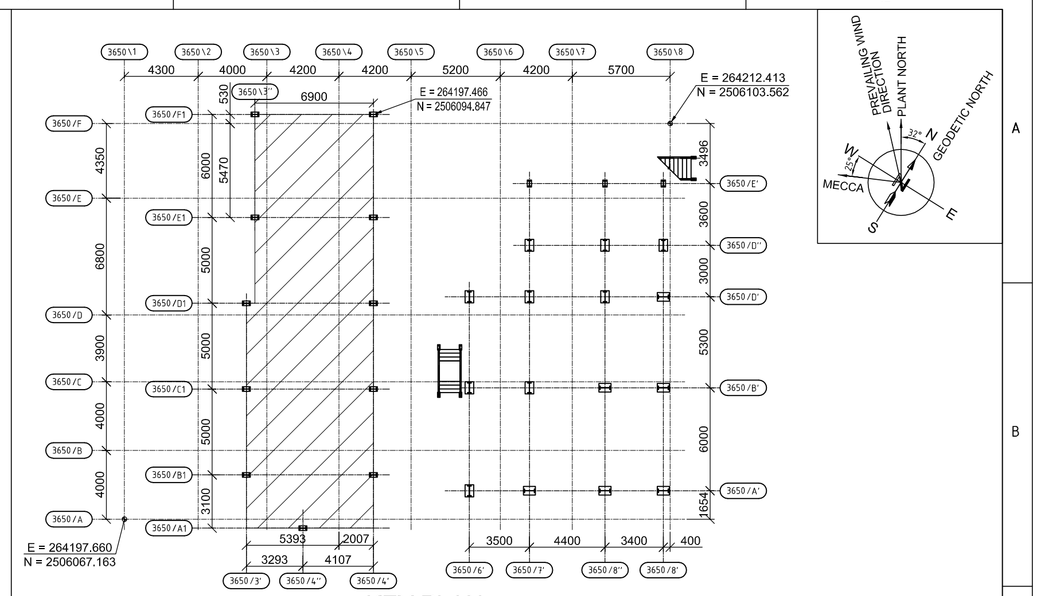
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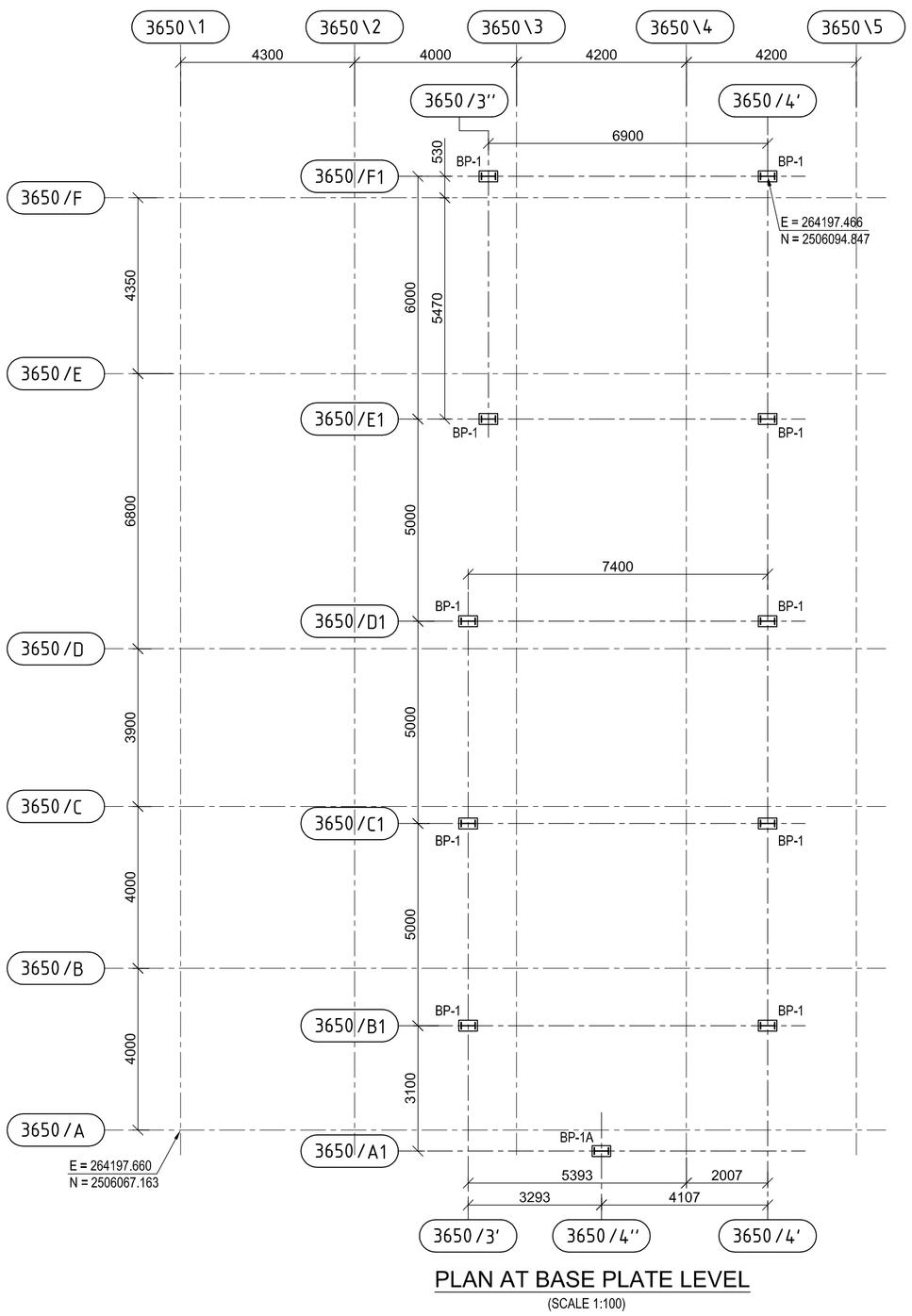
DETAIL 'A'

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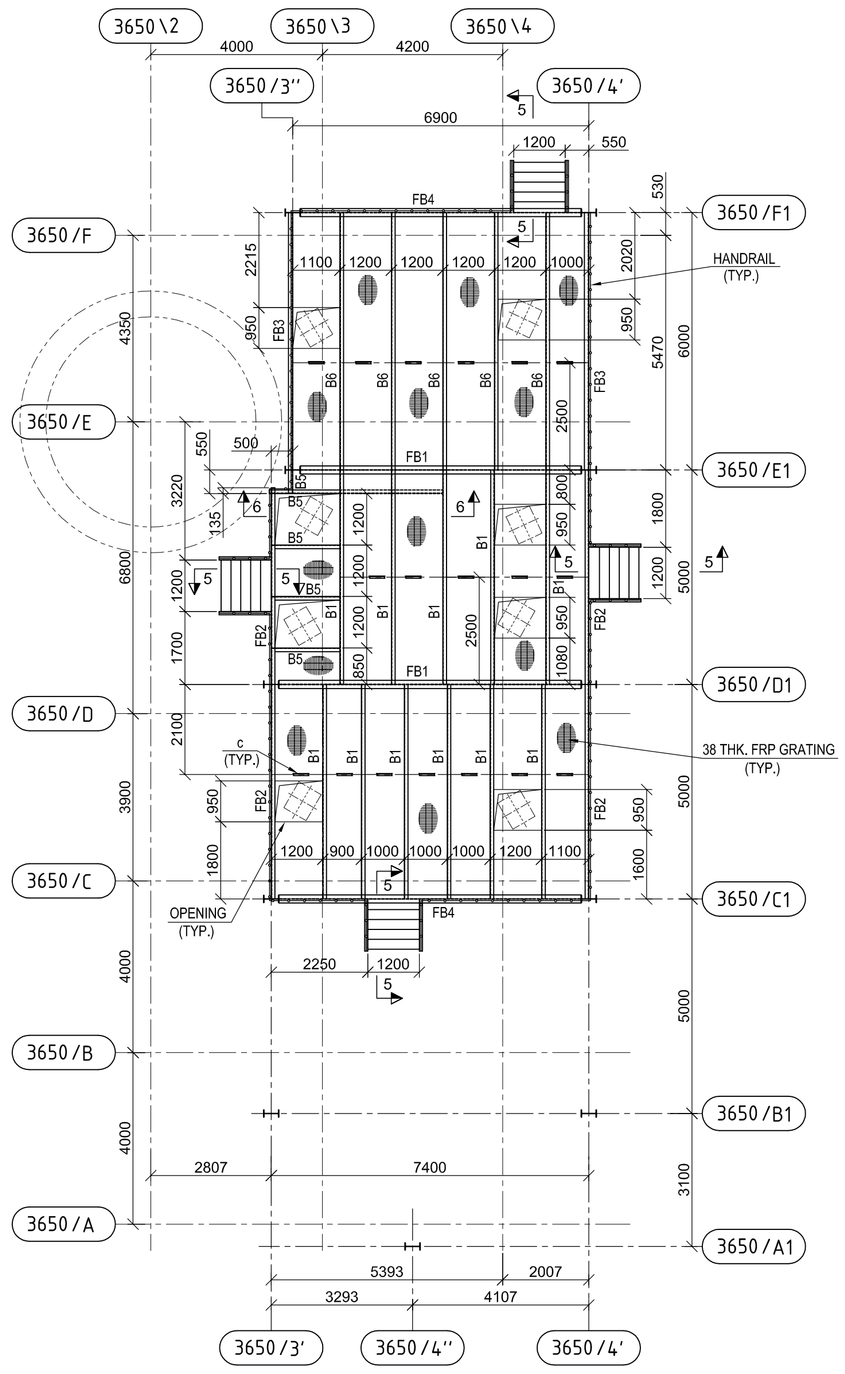
KEY PLAN

(SCALE-1:250)



PLAN AT BASE PLATE LEVEL

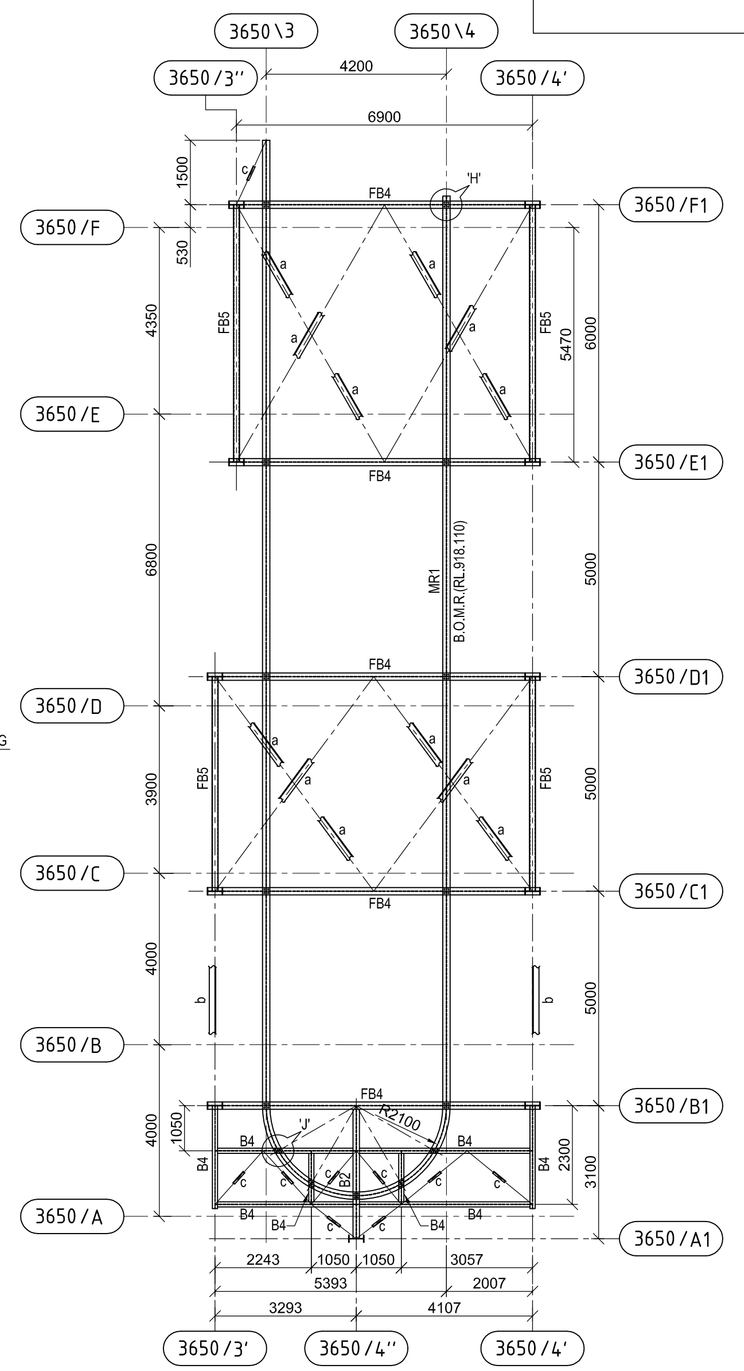
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PLAN AT RL.913.550(T.O.S.)

(TOP OF FLOOR BRACING SHOULD BE KEPT 50mm BELOW T.O.S. U.N.O.)

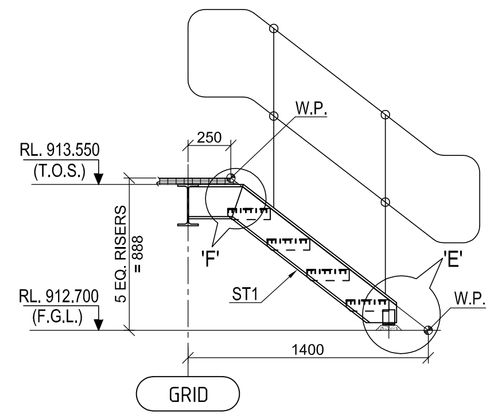
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PLAN AT RL.918.850(T.O.S.)

(TOP OF BRACING SHOULD BE KEPT 180mm BELOW T.O.S. U.N.O.)

(SCALE 1:100)



SECTION 5-5

(SCALE-1:25)

LEGENDS :-

R.L. - REDUCED LEVEL
℄ - CENTRE LINE
T.O.S. - TOP OF STEEL.
F.G.L. - FINISHED GROUND LEVEL.
TYP. - TYPICAL.
B.O.M.R. - BOTTOM OF MONORAIL.
MKD. - MARKED.
MC - MOMENT CONNECTION
THK. - THICK
W.P.-WORKING POINT
U.S.B.P. - UNDER SIDE OF BASE PLATE
FRP. - FIBERGLASS REINFORCED PLASTIC
REF. - REFER
DRG. NO. - DRAWING NUMBER

NOTES:

1. ALL DIMENSIONS ARE IN MILLIMETRE AND LEVELS & COORDINATES ARE IN METRE UNO.
2. ANCHOR BOLT MATERIAL SHALL BE AS PER ASTM F 1554 Gr.36.
3. SIZE OF WELD SHALL BE DECIDED BASED ON LESSER THICKNESS OF THE JOINED MEMBER UNO.
4. ALL WELDS ARE MINIMUM 6 mm LENGTH SIZE CONTINUOUS FILLET UNO.
5. ALL PERMANENT HOLES SHALL BE 21.5 Ø FOR M20 BOLT AS PER (ASTM - A325M) UNO.
6. THREAD TO BE PERMANENTLY DAMAGED AFTER TIGHTENING OF NUT FOR ALL PERMANENT BOLTS BELOW GRIP LENGTH.
7. PROPER ERECTION SEQUENCE TO BE MAINTAINED TO ENSURE STABILITY AND SAFETY OF THE STRUCTURE.
8. GROUTING TO BE DONE AFTER ERECTION AND ALIGNMENT OF STEEL COLUMN.
9. GRADE OF STEEL USED IS S275R AS PER EN10025.
10. ALL STRUCTURAL WORK SHALL CONFORM TO CIVIL STRUCTURAL DESIGN CRITERIA NO MD - 216 - 1000 -EG -ST - CRT - 1001
11. THIS DRAWING IS TO BE READ IN CONJUNCTION WITH DRAWING NO. MD - 216 - 3650 -EG -ST - GAD - 2291.

DOCUMENT NUMBER	REFERENCE DOCUMENTS	REV	DATE	REVISION DESCRIPTION	DRN	CHK	ENG	APP	EMP
MD-216-3650-EG-ST-GAD-1002	PLAN AT EL. +912.700 GRID A-F/6-8	D01	24/11/2023	AS BUILT	SUS	SYC	APL	PRI	
MD-216-3650-EG-ME-GAD-1003	PLAN AT EL. +915.400	C01	10/07/2020	ISSUED FOR CONSTRUCTION	SUS	SYC	APL	PRI	
MD-216-3650-EG-ME-GAD-1002	PLAN AT EL. +912.700	A01	06/06/2020	ISSUED FOR REFERENCE	SUS	SYC	APL	PRI	

RESERVED FOR VENDOR LOGO AND DRAWING NUMBER

LARSEN & TOUBRO DRAWING NUMBER: 019045_450_S_G_26059_004 — REVISION: 2

OUTOTEC DRAWING NUMBER: 900707 __GAU01 __EE12 __MTL01 __36516 — DOCUMENT ID: OU100686003 — REVISION: 2

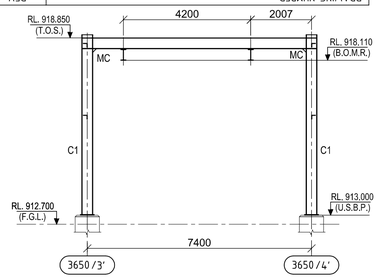
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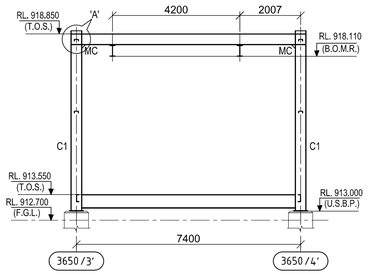
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DRAWN	SUS	03/06/2020	MANSOURAH - MASSARAH GOLD PROJECT
CHECKED	SYC	05/06/2020	GENERAL ARRANGEMENT DRAWING
ENGINEER	APL	04/06/2020	POX ACID WASH, ELUTN. AND CARBON REGN.
APPROVED	PRI	06/06/2020	PLAN, ELEVATIONS & DETAILS
EMPLOYER			
SCALE 1:250,100,10	A1		DRAWING NUMBER: MD - 216 - 3650 - EG - ST - GAD - 2290 — REV: D01

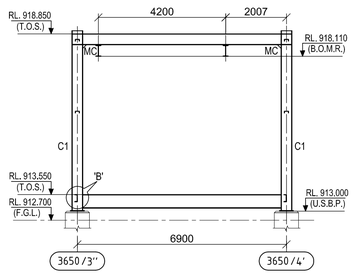
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REV: D 01



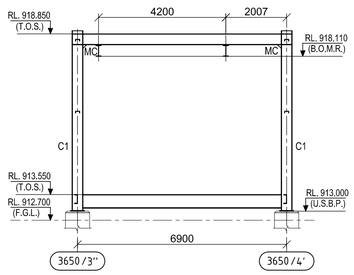
ELEVATION ALONG GRID 3650/B1
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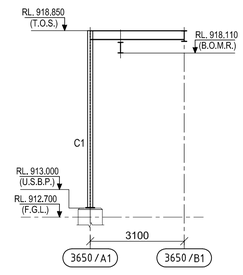
ELEVATION ALONG GRID 3650/C1 & 3650/D1
(SCALE 1:100)



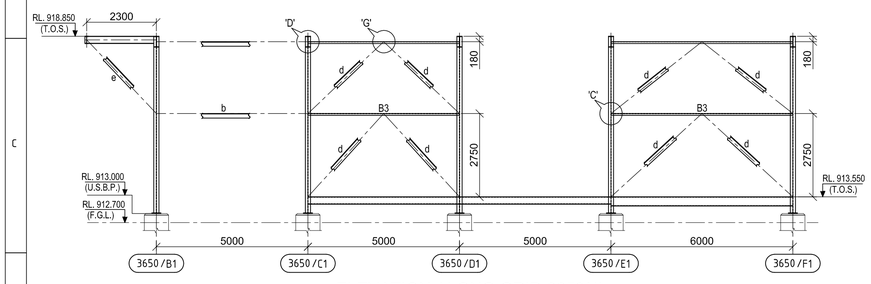
ELEVATION ALONG GRID 3650/E1
(SCALE 1:100)



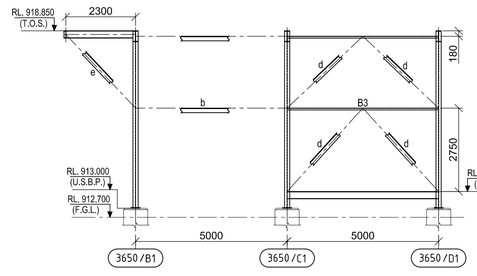
ELEVATION ALONG GRID 3650/F1
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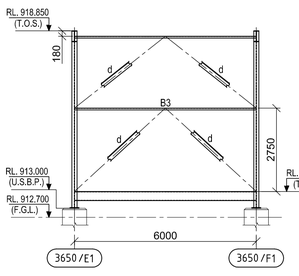
ELEVATION ALONG GRID 3650/4''
(SCALE 1:100)



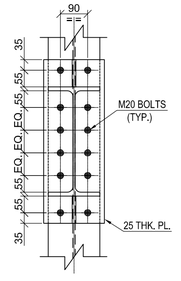
ELEVATION ALONG GRID 3650/4'
(SCALE 1:100)



ELEVATION ALONG GRID 3650/3'
(SCALE 1:100)



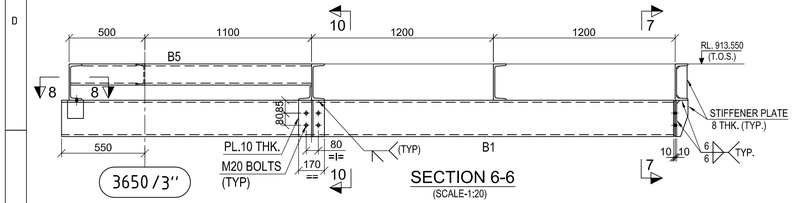
ELEVATION ALONG GRID 3650/3''
(SCALE 1:100)



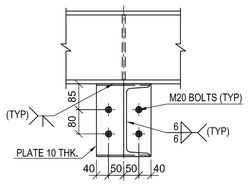
SECTION A1-A1
(SCALE - 1:10)

SCHEDULE OF MEMBERS					
MEMBER MARKED	MEMBER	SECTON	FACTORED AXIAL FORCE(KN)	FACTORED SHEAR FORCE(KN)	REMARKS
C1	IPE 360	I	300	-	GRADE OF MATERIAL IS S275 JR
FB1	IPE 450	I	170	170	
FB2	UPN 240	[	50	25	
FB3	UPN 300	[	40	25	
FB4	IPE 360	I	170	120	
FB5	UPN 140	⌐	-	-	
B1	UPN 240	[	-	60	
B2	IPE 300	I	-	80	
B3	UPN 140	⌐	-	-	
B4	IPE 240	I	30	60	
B5	UPN 140	[	30	-	
B6	UPN 300	[	-	35	
a	L 150X150X16	L	-	-	
b	L 130X130X13	L	30	-	
c	L 65X65X6	L	10	-	
d	L 100X100X10	L	30	-	
e	L 90X90X8	L	40	-	
ST1	UPN 200	[	-	-	
MR1	IPE 360	I	-	-	

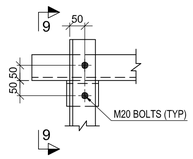
ALL GUSSET PLATE SHOULD BE 8mm THICK U.N.O.
FOR DEVELOPING CONNECTIONS MEMBER END FORCES MENTIONED IN SCHEDULE OF MEMBERS SHALL BE FOLLOWED.
ALL MEMBER CONNECTIONS SHOULD BE DESIGN FOR A MINIMUM FORCE OF 50KN.
MINIMUM 6mm FILLET WELD OF 150 mm LENGTH SHALL BE PROVIDED FOR WELDED CONNECTION. FOR BOLTED CONNECTION, MINIMUM 2 Nos.
M20 BOLTS SHALL BE PROVIDED.



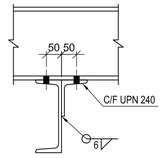
SECTION 6-6
(SCALE-1:20)



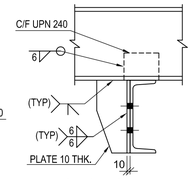
SECTION 7-7
(SCALE 1:10)



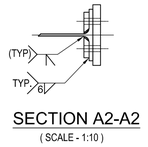
SECTION 8-8
(SCALE 1:10)



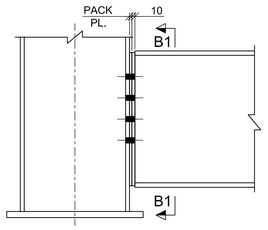
SECTION 9-9
(SCALE 1:10)



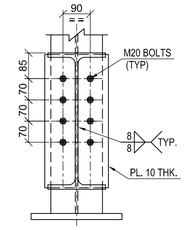
SECTION 10-10
(SCALE 1:10)



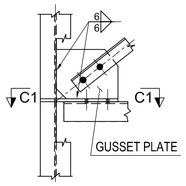
SECTION A2-A2
(SCALE - 1:10)



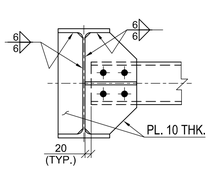
DETAIL 'B'
(SCALE 1:10)



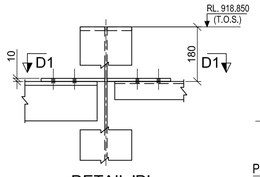
SECTION B1-B1
(SCALE - 1:10)



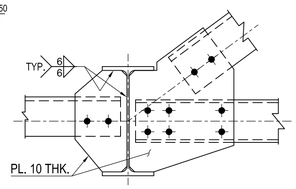
DETAIL 'C'
(SCALE 1:10)



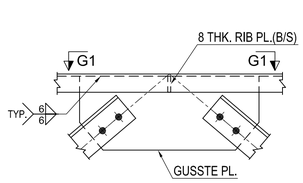
SECTION C1-C1
(SCALE - 1:10)



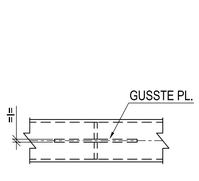
DETAIL 'D'
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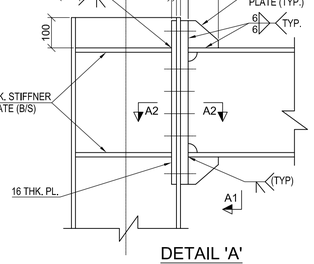
SECTION D1-D1
(SCALE - 1:10)



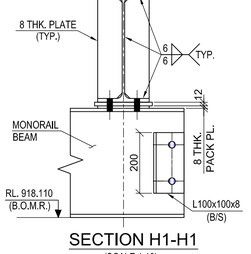
DETAIL 'G'
(SCALE 1:10)



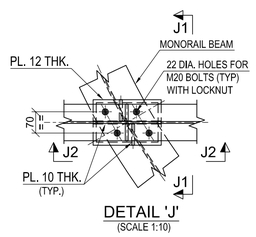
SECTION G1-G1
(SCALE - 1:10)



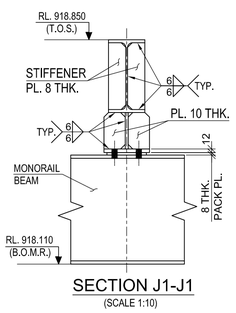
DETAIL 'A'
(SCALE - 1:10)



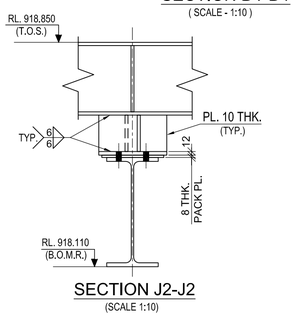
SECTION H1-H1
(SCALE 1:10)



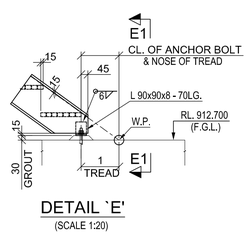
DETAIL 'J'
(SCALE 1:10)



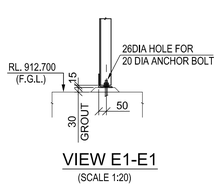
SECTION J1-J1
(SCALE 1:10)



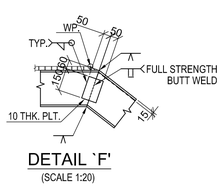
SECTION J2-J2
(SCALE 1:10)



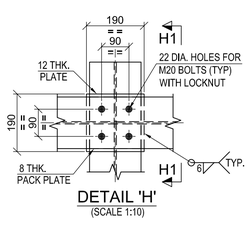
DETAIL `E'
(SCALE 1:20)



VIEW E1-E1
(SCALE 1:20)



DETAIL `F'
(SCALE 1:20)



DETAIL 'H'
(SCALE 1:10)

LEGENDS :-

R.L. - REDUCED LEVEL
℄ - CENTRE LINE
T.O.S. - TOP OF STEEL.
F.G.L. - FINISHED GROUND LEVEL.
TYP. - TYPICAL.
B.O.M.R. - BOTTOM OF MONORAIL.
MKD. - MARKED.
MC - MOMENT CONNECTION
THK. - THICK
W.P.-WORKING POINT
U.S.B.P. - UNDER SIDE OF BASE PLATE
FRP. - FIBERGLASS REINFORCED PLASTIC
REF. - REFER
DRG. NO. - DRAWING NUMBER

NOTES:

1. ALL DIMENSIONS ARE IN MILLIMETRE AND LEVELS & COORDINATES ARE IN METRE UNO.
2. ANCHOR BOLT MATERIAL SHALL BE AS PER ASTM F 1554 Gr.36,
3. SIZE OF WELD SHALL BE DECIDED BASED ON LESSER THICKNESS OF THE JOINED MEMBER UNO.
4. ALL WELDS ARE MINIMUM 6 mm LENGTH SIZE CONTINUOUS FILLET UNO.
5. ALL PERMANENT HOLES SHALL BE 21.5 Ø FOR M20 BOLT AS PER (ASTM - A325M) UNO.
6. THREAD TO BE PERMANENTLY DAMAGED AFTER TIGHTENING OF NUT FOR ALL PERMANENT BOLTS BELOW GRIP LENGTH.
7. PROPER ERECTION SEQUENCE TO BE MAINTAINED TO ENSURE STABILITY AND SAFETY OF THE STRUCTURE.
8. GROUTING TO BE DONE AFTER ERECTION AND ALIGNMENT OF STEEL COLUMN.
9. GRADE OF STEEL USED IS S275R AS PER EN10025.
10. ALL STRUCTURAL WORK SHALL CONFORM TO CIVIL STRUCTURAL DESIGN CRITERIA NO MD - 216 - 1000 -EG -ST - CRT - 1001
11. THIS DRAWING IS TO BE READ IN CONJUNCTION WITH DRAWING NO. MD - 216 - 3650 -EG -ST - GAD - 2290.

DOCUMENT NUMBER	REFERENCE DOCUMENTS	REV	DATE	REVISION DESCRIPTION	DRN	CHK	ENG	APP	EMP
		D01	24/11/2023	AS BUILT	SUS	SYC	APL	PRI	
MD-216-3650-EG-ST-GAD-1002	PLAN AT EL. +912.700 GRID A-F/6-8	C01	10/07/2020	ISSUED FOR CONSTRUCTION, REVISED AS PER EMPLOYER	SUS	SYC	APL	PRI	
MD-216-3650-EG-ME-GAD-1003	PLAN AT EL. +915.400			COMMENTS DATED 02/07/2020 & REVISED AS MARKED C01					
MD-216-3650-EG-ME-GAD-1002	PLAN AT EL. +912.700	A01	06/06/2020	ISSUED FOR REFERENCE	SUS	SYC	APL	PRI	

RESERVED FOR VENDOR LOGO AND DRAWING NUMBER

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Saudi Arabian Mining Company شركة معادن العربية السعودية

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DRAWN	SUS	03/06/2020
CHECKED	SYC	05/06/2020
ENGINEER	APL	04/06/2020
APPROVED	PRI	06/06/2020
EMPLOYER		
SCALE	1:100,25,10	A1

LARSEN & TOUBRO DRAWING NUMBER: 019045_450_S_G_26059_005 — REVISION: 2
OUTOTEC DRAWING NUMBER: 900707 __GAU01 __EE12 __MTL01 __36517 — DOCUMENT ID: OU100686004 — REVISION: 2

MANSOURAH - MASSARAH GOLD PROJECT
GENERAL ARRANGEMENT DRAWING
POX ACID WASH, ELUTN. AND CARBON REGN.
PLAN, ELEVATIONS & DETAILS

DRAWING NUMBER: MD - 216 - 3650 - EG - ST - GAD - 2291 — REV: D01

MA'ADEN معادن
Saudi Arabian Mining Company شركة التعدين العربية السعودية

BECHTEL

Mansourah Massarah Gold Project
Comment Resolution Log Sheet / Compliance Report

Outotec
LARSEN & TOUBRO

PROJECT: MANSOURAH MASSARAH GOLD PROJECT **PROJECT NO.:** 216

STAGE: 4 - PROJECT EXECUTION. **REVIEW LOG DOC. NO. & REV.:**

ACONEX REF. NO. & DATE: BECHTEL-WTRAN-003435, 03 July 2020

STATUS CODE: A - APPROVED, C - APPROVED WITH COMMENTS, R - REVISE & RESUBMIT & O - OTHERS (INFORMATION, REFERENCE, REVIEWED)

DOCUMENT NUMBER	Area / Title	Comment No.	PMC/PMT COMMENT DATE: 03 July 2020	Comment by	EPC Response DATE: 10 July 2020	Status	Remarks
			General Comments (cc)				
MD-216-3650-EG-ST-GAD-2291-A01	3650	1	With this shop welded connection you will need to transport on a truck a frame with dimensions of aprox 6 x 7 mtrs . Otherwise you need to change this design.	Mr. Doron Golan Bechtel	We have changed the detail from shop welded type to bolted type moment connection.	Closed	

ATTACHMENT 2 STAAD INPUT – FOUNDATION F1

PROPRIETARY & CONFIDENTIAL

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```
STAAD SPACE
START JOB INFORMATION
ENGINEER DATE 08/05/2020
JOB NAME FDN. MKD. F1
JOB CLIENT MA'ADEN
JOB NO O19045
JOB REV 0
ENGINEER NAME SOR
END JOB INFORMATION
INPUT WIDTH 79
SET NL 200
SET ITERLIM 15
SET PRINT 1
UNIT METER KN
JOINT COORDINATES
456 -0.75 911.5 0.42438; 457 -0.5 911.5 0.42438; 458 -0.5 911.5 0.17538;
459 -0.75 911.5 0.17538; 460 -0.25 911.5 0.42438; 461 -0.25 911.5 0.17538;
462 0 911.5 0.42438; 463 0 911.5 0.17538; 464 0.25 911.5 0.42438;
465 0.25 911.5 0.17538; 466 0.5 911.5 0.42438; 467 0.5 911.5 0.17538;
468 0.75 911.5 0.42438; 469 0.75 911.5 0.17538; 474 -0.75 911.5 0.67438;
475 -0.5 911.5 0.67438; 476 -0.25 911.5 0.67438; 477 0 911.5 0.67438;
478 0.25 911.5 0.67438; 479 0.5 911.5 0.67438; 480 0.75 911.5 0.67438;
483 -0.75 911.5 0.87538; 484 -0.5 911.5 0.87538; 485 -0.25 911.5 0.87538;
486 0 911.5 0.87538; 487 0.25 911.5 0.87538; 488 0.5 911.5 0.87538;
489 0.75 911.5 0.87538; 492 -0.75 911.5 -0.62362; 493 -0.5 911.5 -0.62362;
494 -0.5 911.5 -0.87362; 495 -0.75 911.5 -0.87362; 496 -0.25 911.5 -0.62362;
497 -0.25 911.5 -0.87362; 498 0 911.5 -0.62362; 499 0 911.5 -0.87362;
500 0.25 911.5 -0.62362; 501 0.25 911.5 -0.87362; 502 0.5 911.5 -0.62362;
503 0.5 911.5 -0.87362; 504 0.75 911.5 -0.62362; 505 0.75 911.5 -0.87362;
510 -0.75 911.5 -0.37362; 511 -0.5 911.5 -0.37362; 512 -0.25 911.5 -0.37362;
513 0 911.5 -0.37362; 514 0.25 911.5 -0.37362; 515 0.5 911.5 -0.37362;
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797 -0.75 911.5 -1.12462; 798 -0.5 911.5 -1.12462; 801 -0.25 911.5 -1.12462;
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809 0.75 911.5 -1.12462; 810 1 911.5 0.42438; 811 1 911.5 0.17538;
812 1 911.5 0.67438; 813 1 911.5 0.87538; 814 1 911.5 -0.62362;
815 1 911.5 -0.87362; 816 1 911.5 -0.37362; 817 1 911.5 -0.12462;
818 1 911.5 -1.12462; 819 -1 911.5 0.42438; 820 -1 911.5 0.17538;
821 -1 911.5 0.67438; 822 -1 911.5 0.87538; 823 -1 911.5 -0.62362;
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827 -1 911.5 -1.12462;
MEMBER INCIDENCES
529 522 528;
ELEMENT INCIDENCES SHELL
465 456 457 458 459; 466 457 460 461 458; 467 460 462 463 461;
468 462 464 465 463; 469 464 466 467 465; 470 466 468 469 467;
473 474 475 457 456; 474 475 476 460 457; 475 476 477 462 460;
476 477 478 464 462; 477 478 479 466 464; 478 479 480 468 466;
481 483 484 475 474; 482 484 485 476 475; 483 485 486 477 476;
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521 459 458 520 519; 522 458 461 521 520; 523 461 463 522 521;
524 463 465 523 522; 525 465 467 524 523; 526 467 469 525 524;
785 495 494 798 797; 786 494 497 801 798; 787 497 499 803 801;
788 499 501 805 803; 789 501 503 807 805; 790 503 505 809 807;
791 468 810 811 469; 792 480 812 810 468; 793 489 813 812 480;
794 504 814 815 505; 795 516 816 814 504; 796 525 817 816 516;
797 469 811 817 525; 798 505 815 818 809; 799 819 456 459 820;
800 821 474 456 819; 801 822 483 474 821; 802 823 492 495 824;
803 825 510 492 823; 804 826 519 510 825; 805 820 459 519 826;
806 824 495 797 827;
START GROUP DEFINITION
ELEMENT
_RAFT 465 TO 470 473 TO 478 481 TO 486 497 TO 502 505 TO 510 513 TO 518 521 -
522 TO 526 785 TO 806
_RAFT_MOM 465 TO 470 473 TO 478 481 TO 486 497 TO 502 505 TO 510 513 514 517 -
518 521 522 525 526 785 TO 806
_RAFT_SHEAR 465 470 473 TO 478 481 TO 486 497 TO 502 505 510 513 518 521 526 -
785 TO 806
MEMBER
END GROUP DEFINITION
ELEMENT PROPERTY
```



```
ELEMENT PROPERTY
465 TO 470 473 TO 478 481 TO 486 497 TO 502 505 TO 510 513 TO 518 521 TO 526 -
785 TO 806 THICKNESS 0.5
DEFINE MATERIAL START
ISOTROPIC CONCRETE
E 2.17185e+007
POISSON 0.17
DENSITY 25
ALPHA 1e-005
DAMP 0.05
TYPE CONCRETE
STRENGTH FCU 27579
END DEFINE MATERIAL
MEMBER PROPERTY AMERICAN
529 PRIS YD 0.6 ZD 0.6
CONSTANTS
MATERIAL CONCRETE ALL
SUPPORTS
465 TO 470 473 TO 478 481 TO 486 497 TO 502 505 TO 510 513 TO 518 521 TO 526 -
785 TO 806 PLATE MAT DIRECT Y SUBGRADE 57600 PRINT
*************************************************************************************************
LOAD 1 LOADTYPE None  TITLE DEAD LOAD ( DL )
SELFWEIGHT Y -1
ELEMENT LOAD
* LOAD DUE TO BACKFILL
465 TO 470 473 TO 478 481 TO 486 497 TO 502 505 TO 510 513 TO 518 521 TO 526 -
785 TO 806 PR GY -15
* LOAD DUE TO SURCHARGE
465 TO 470 473 TO 478 481 TO 486 497 TO 502 505 TO 510 513 TO 518 521 TO 526 -
785 TO 806 PR GY -20
JOINT LOAD
528 FX -0.034 FY -27.7 FZ 5
*************************************************************************************************
LOAD 2 LOADTYPE None  TITLE LIVE LOAD ( LL )
JOINT LOAD
528 FX 0.13 FY -133.12 FZ 25
*************************************************************************************************
LOAD 3 LOADTYPE None  TITLE WIND LOAD +X  ( WLX1 )
JOINT LOAD
528 FX 8.7 FY -15.2 FZ 0.02
*************************************************************************************************
LOAD 4 LOADTYPE None  TITLE WIND LOAD -X  ( WLX2 )
JOINT LOAD
528 FX -8.7 FY 15.2 FZ -0.02
*************************************************************************************************
LOAD 5 LOADTYPE None  TITLE WIND LOAD +Z  ( WLZ1 )
JOINT LOAD
528 FX -0.44 FY 6 FZ 5.8
*************************************************************************************************
LOAD 6 LOADTYPE None  TITLE WIND LOAD -Z  ( WLZ2 )
JOINT LOAD
528 FX 0.44 FY -6 FZ -5.8
*************************************************************************************************
LOAD 7 LOADTYPE None  TITLE TEMPERATURE LOAD +25  ( TL1 )
JOINT LOAD
528 FX -1.1 FY 2 FZ -67.4
*************************************************************************************************
LOAD 8 LOADTYPE None  TITLE TEMPERATURE LOAD -25  ( TL2 )
JOINT LOAD
528 FX 1.1 FY -2 FZ 67.4
*************************************************************************************************
LOAD 9 LOADTYPE None  TITLE SEISMIC LOAD  ( ELX )
JOINT LOAD
528 FX 1 FY -3.35 FZ 0.008
*************************************************************************************************
LOAD 10 LOADTYPE None  TITLE SEISMIC LOAD  ( ELZ )
JOINT LOAD
528 FX -0.127 FY 1.428 FZ 0.75
*************************************************************************************************
PERFORM ANALYSIS
CHANGE
SUPPORTS
465 TO 470 473 TO 478 481 TO 486 497 TO 502 505 TO 510 513 TO 518 521 TO 526 -
785 TO 806 PLATE MAT DIRECT Y SUBGRADE 57600 COMPRESSION
*************************************************************************************************
***FACTORED LOAD COMBINATIONS FOR STRENGTH DESIGN
LOAD 101 1.4DL
REPEAT LOAD
1 1.4
```

```
1 1.4
PERFORM ANALYSIS
CHANGE
LOAD 102 1.4DL + 1.7LL
REPEAT LOAD
1 1.4 2 1.7
PERFORM ANALYSIS
CHANGE
LOAD 103 1.2DL + 1.6WLX1 + 1.0 LL
REPEAT LOAD
1 1.2 3 1.6 2 1.0
PERFORM ANALYSIS
CHANGE
LOAD 104 1.2DL + 1.6WLX2 + 1.0 LL
REPEAT LOAD
1 1.2 4 1.6 2 1.0
PERFORM ANALYSIS
CHANGE
LOAD 105 1.2DL + 1.6WLZ1 + 1.0 LL
REPEAT LOAD
1 1.2 5 1.6 2 1.0
PERFORM ANALYSIS
CHANGE
LOAD 106 1.2DL + 1.6WLZ2 + 1.0 LL
REPEAT LOAD
1 1.2 6 1.6 2 1.0
PERFORM ANALYSIS
CHANGE
LOAD 107 0.9DL + 1.6WLX1
REPEAT LOAD
1 0.9 3 1.6
PERFORM ANALYSIS
CHANGE
LOAD 108 0.9DL + 1.6WLX2
REPEAT LOAD
1 0.9 4 1.6
PERFORM ANALYSIS
CHANGE
LOAD 109 0.9DL + 1.6WLZ1
REPEAT LOAD
1 0.9 5 1.6
PERFORM ANALYSIS
CHANGE
LOAD 110 0.9DL + 1.6WLZ2
REPEAT LOAD
1 0.9 6 1.6
PERFORM ANALYSIS
CHANGE
LOAD 111 1.2DL + 1.0ELX + 1.0 LL
REPEAT LOAD
1 1.2 2 1.0 9 1.0
PERFORM ANALYSIS
CHANGE
LOAD 112 1.2DL + 1.0ELZ + 1.0 LL
REPEAT LOAD
1 1.2 2 1.0 10 1.0
PERFORM ANALYSIS
CHANGE
LOAD 113 0.9DL + 1.0ELX
REPEAT LOAD
1 0.9 9 1.0
PERFORM ANALYSIS
CHANGE
LOAD 114 0.9DL + 1.0ELZ
REPEAT LOAD
1 0.9 10 1.0
PERFORM ANALYSIS
CHANGE
LOAD 115 1.4DL + 1.7LL + 1.0 TL1
REPEAT LOAD
1 1.4 2 1.7 7 1.0
PERFORM ANALYSIS
CHANGE
LOAD 116 1.4DL + 1.7LL + 1.0 TL2
REPEAT LOAD
1 1.4 2 1.7 8 1.0
PERFORM ANALYSIS
CHANGE
*********************************************************************************************
```



```
*********************************************************************************************
* UNFACTORED LOAD COMBINATIONS FOR STABILITY CHECK
LOAD 201 1.0DL
REPEAT LOAD
1 1.0
PERFORM ANALYSIS
CHANGE
LOAD 202 1.0DL + 1.0LL
REPEAT LOAD
1 1.0 2 1.0
PERFORM ANALYSIS
CHANGE
LOAD 203 1.0DL + 1.0WLX1
REPEAT LOAD
1 1.0 3 1.0
PERFORM ANALYSIS
CHANGE
LOAD 204 1.0DL + 1.0WLX2
REPEAT LOAD
1 1.0 4 1.0
PERFORM ANALYSIS
CHANGE
LOAD 205 1.0DL + 1.0WLZ1
REPEAT LOAD
1 1.0 5 1.0
PERFORM ANALYSIS
CHANGE
LOAD 206 1.0DL + 1.0WLZ2
REPEAT LOAD
1 1.0 6 1.0
PERFORM ANALYSIS
CHANGE
LOAD 207 1.0DL + 0.75WLX1 + 0.75LL
REPEAT LOAD
1 1.0 3 0.75 2 0.75
PERFORM ANALYSIS
CHANGE
LOAD 208 1.0DL + 0.75WLX2 + 0.75LL
REPEAT LOAD
1 1.0 4 0.75 2 0.75
PERFORM ANALYSIS
CHANGE
LOAD 209 1.0DL + 0.75WLZ1 + 0.75LL
REPEAT LOAD
1 1.0 5 0.75 2 0.75
PERFORM ANALYSIS
CHANGE
LOAD 210 1.0DL + 0.75WLZ2 + 0.75LL
REPEAT LOAD
1 1.0 6 0.75 2 0.75
PERFORM ANALYSIS
CHANGE
LOAD 211 0.6DL + 1.0WLX1
REPEAT LOAD
1 0.6 3 1.0
PERFORM ANALYSIS
CHANGE
LOAD 212 0.6DL + 1.0WLX2
REPEAT LOAD
1 0.6 4 1.0
PERFORM ANALYSIS
CHANGE
LOAD 213 0.6DL + 1.0WLZ1
REPEAT LOAD
1 0.6 5 1.0
PERFORM ANALYSIS
CHANGE
LOAD 214 0.6DL + 1.0WLZ2
REPEAT LOAD
1 0.6 6 1.0
PERFORM ANALYSIS
CHANGE
LOAD 215 1.0DL + 1.0LL + 1.0TL1
REPEAT LOAD
1 1.0 2 1.0 7 1.0
PERFORM ANALYSIS
CHANGE
LOAD 216 1.0DL + 1.0LL + 1.0TL2
REPEAT LOAD
```

```
REPEAT LOAD
1 1.0 2 1.0 8 1.0
PERFORM ANALYSIS
CHANGE
LOAD 217 0.6DL + 0.7ELX
REPEAT LOAD
1 0.6 9 0.7
PERFORM ANALYSIS
CHANGE
LOAD 218 0.6DL + 0.7ELZ
REPEAT LOAD
1 0.6 10 0.7
PERFORM ANALYSIS
CHANGE
*********************************************************************************************
PERFORM ANALYSIS
LOAD LIST 101 TO 116
START CONCRETE DESIGN
CODE ACI
TRACK 2 ALL
FC 35000 MEMB 529
CLB 0.05 MEMB 529
CLS 0.05 ALL
MAXMAIN 25 ALL
MINMAIN 25 ALL
MINSEC 10 ALL
DESIGN COLUMN 529
END CONCRETE DESIGN
FINISH
```

ATTACHMENT 3 STAAD INPUT – FOUNDATION F2

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```
STAAD SPACE
START JOB INFORMATION
ENGINEER DATE 08/05/2020
JOB NAME FDN. MKD. F1
JOB CLIENT MA'ADEN
JOB NO O19045
JOB REV 0
ENGINEER NAME SOR
END JOB INFORMATION
INPUT WIDTH 79
SET NL 200
SET ITERLIM 15
SET PRINT 1
UNIT METER KN
JOINT COORDINATES
456 -0.75 911.5 0.42438; 457 -0.5 911.5 0.42438; 458 -0.5 911.5 0.17538;
459 -0.75 911.5 0.17538; 460 -0.25 911.5 0.42438; 461 -0.25 911.5 0.17538;
462 0 911.5 0.42438; 463 0 911.5 0.17538; 464 0.25 911.5 0.42438;
465 0.25 911.5 0.17538; 466 0.5 911.5 0.42438; 467 0.5 911.5 0.17538;
468 0.75 911.5 0.42438; 469 0.75 911.5 0.17538; 474 -0.75 911.5 0.67438;
475 -0.5 911.5 0.67438; 476 -0.25 911.5 0.67438; 477 0 911.5 0.67438;
478 0.25 911.5 0.67438; 479 0.5 911.5 0.67438; 480 0.75 911.5 0.67438;
483 -0.75 911.5 0.87538; 484 -0.5 911.5 0.87538; 485 -0.25 911.5 0.87538;
486 0 911.5 0.87538; 487 0.25 911.5 0.87538; 488 0.5 911.5 0.87538;
489 0.75 911.5 0.87538; 492 -0.75 911.5 -0.62362; 493 -0.5 911.5 -0.62362;
494 -0.5 911.5 -0.87362; 495 -0.75 911.5 -0.87362; 496 -0.25 911.5 -0.62362;
497 -0.25 911.5 -0.87362; 498 0 911.5 -0.62362; 499 0 911.5 -0.87362;
500 0.25 911.5 -0.62362; 501 0.25 911.5 -0.87362; 502 0.5 911.5 -0.62362;
503 0.5 911.5 -0.87362; 504 0.75 911.5 -0.62362; 505 0.75 911.5 -0.87362;
510 -0.75 911.5 -0.37362; 511 -0.5 911.5 -0.37362; 512 -0.25 911.5 -0.37362;
513 0 911.5 -0.37362; 514 0.25 911.5 -0.37362; 515 0.5 911.5 -0.37362;
516 0.75 911.5 -0.37362; 519 -0.75 911.5 -0.12462; 520 -0.5 911.5 -0.12462;
521 -0.25 911.5 -0.12462; 522 0 911.5 -0.12462; 523 0.25 911.5 -0.12462;
524 0.5 911.5 -0.12462; 525 0.75 911.5 -0.12462; 528 0 913 -0.125;
797 -0.75 911.5 -1.12462; 798 -0.5 911.5 -1.12462; 801 -0.25 911.5 -1.12462;
803 0 911.5 -1.12462; 805 0.25 911.5 -1.12462; 807 0.5 911.5 -1.12462;
809 0.75 911.5 -1.12462; 819 -1 911.5 0.42438; 820 -1 911.5 0.17538;
821 -1 911.5 0.67438; 822 -1 911.5 0.87538; 823 -1 911.5 -0.62362;
824 -1 911.5 -0.87362; 825 -1 911.5 -0.37362; 826 -1 911.5 -0.12462;
827 -1 911.5 -1.12462; 828 -1.25 911.5 0.42438; 829 -1.25 911.5 0.17538;
830 -1.25 911.5 0.67438; 831 -1.25 911.5 0.87538; 832 -1.25 911.5 -0.62362;
833 -1.25 911.5 -0.87362; 834 -1.25 911.5 -0.37362; 835 -1.25 911.5 -0.12462;
836 -1.25 911.5 -1.12462;
MEMBER INCIDENCES
529 522 528;
ELEMENT INCIDENCES SHELL
465 456 457 458 459; 466 457 460 461 458; 467 460 462 463 461;
468 462 464 465 463; 469 464 466 467 465; 470 466 468 469 467;
473 474 475 457 456; 474 475 476 460 457; 475 476 477 462 460;
476 477 478 464 462; 477 478 479 466 464; 478 479 480 468 466;
481 483 484 475 474; 482 484 485 476 475; 483 485 486 477 476;
484 486 487 478 477; 485 487 488 479 478; 486 488 489 480 479;
497 492 493 494 495; 498 493 496 497 494; 499 496 498 499 497;
500 498 500 501 499; 501 500 502 503 501; 502 502 504 505 503;
505 510 511 493 492; 506 511 512 496 493; 507 512 513 498 496;
508 513 514 500 498; 509 514 515 502 500; 510 515 516 504 502;
513 519 520 511 510; 514 520 521 512 511; 515 521 522 513 512;
516 522 523 514 513; 517 523 524 515 514; 518 524 525 516 515;
521 459 458 520 519; 522 458 461 521 520; 523 461 463 522 521;
524 463 465 523 522; 525 465 467 524 523; 526 467 469 525 524;
785 495 494 798 797; 786 494 497 801 798; 787 497 499 803 801;
788 499 501 805 803; 789 501 503 807 805; 790 503 505 809 807;
799 819 456 459 820; 800 821 474 456 819; 801 822 483 474 821;
802 823 492 495 824; 803 825 510 492 823; 804 826 519 510 825;
805 820 459 519 826; 806 824 495 797 827; 807 828 819 820 829;
808 830 821 819 828; 809 831 822 821 830; 810 832 823 824 833;
811 834 825 823 832; 812 835 826 825 834; 813 829 820 826 835;
814 833 824 827 836;
START GROUP DEFINITION
ELEMENT
_RAFT 465 TO 470 473 TO 478 481 TO 486 497 TO 502 505 TO 510 513 TO 518 521 -
522 TO 526 785 TO 790 799 TO 814
_RAFT_MOM 465 TO 470 473 TO 478 481 TO 486 497 TO 502 505 TO 510 513 514 517 -
518 521 522 525 526 785 TO 790 799 TO 814
_RAFT_SHEAR 465 470 473 TO 478 481 TO 486 497 TO 502 505 510 513 518 521 526 -
785 TO 790 799 TO 814
MEMBER
END GROUP DEFINITION
ELEMENT PROPERTY
```



```
ELEMENT PROPERTY
465 TO 470 473 TO 478 481 TO 486 497 TO 502 505 TO 510 513 TO 518 521 TO 526 -
785 TO 790 799 TO 814 THICKNESS 0.5
DEFINE MATERIAL START
ISOTROPIC CONCRETE
E 2.17185e+007
POISSON 0.17
DENSITY 25
ALPHA 1e-005
DAMP 0.05
TYPE CONCRETE
STRENGTH FCU 27579
END DEFINE MATERIAL
MEMBER PROPERTY AMERICAN
529 PRIS YD 0.6 ZD 0.6
CONSTANTS
MATERIAL CONCRETE ALL
SUPPORTS
465 TO 470 473 TO 478 481 TO 486 497 TO 502 505 TO 510 513 TO 518 521 TO 526 -
785 TO 790 799 TO 814 PLATE MAT DIRECT Y SUBGRADE 57600 PRINT
**************************************************************************************************
LOAD 1 LOADTYPE None  TITLE DEAD LOAD ( DL )
SELFWEIGHT Y -1
ELEMENT LOAD
* LOAD DUE TO BACKFILL
465 TO 470 473 TO 478 481 TO 486 497 TO 502 505 TO 510 513 TO 518 521 TO 526 -
785 TO 790 799 TO 814 PR GY -15
* LOAD DUE TO SURCHARGE
465 TO 470 473 TO 478 481 TO 486 497 TO 502 505 TO 510 513 TO 518 521 TO 526 -
785 TO 790 799 TO 814 PR GY -20
JOINT LOAD
528 FX -0.034 FY -27.7 FZ 5
**************************************************************************************************
LOAD 2 LOADTYPE None  TITLE LIVE LOAD ( LL )
JOINT LOAD
528 FX 0.13 FY -133.12 FZ 25
**************************************************************************************************
LOAD 3 LOADTYPE None  TITLE WIND LOAD +X  ( WLX1 )
JOINT LOAD
528 FX 8.7 FY -15.2 FZ 0.02
**************************************************************************************************
LOAD 4 LOADTYPE None  TITLE WIND LOAD -X  ( WLX2 )
JOINT LOAD
528 FX -8.7 FY 15.2 FZ -0.02
**************************************************************************************************
LOAD 5 LOADTYPE None  TITLE WIND LOAD +Z  ( WLZ1 )
JOINT LOAD
528 FX -0.44 FY 6 FZ 5.8
**************************************************************************************************
LOAD 6 LOADTYPE None  TITLE WIND LOAD -Z  ( WLZ2 )
JOINT LOAD
528 FX 0.44 FY -6 FZ -5.8
**************************************************************************************************
LOAD 7 LOADTYPE None  TITLE TEMPERATURE LOAD +25  ( TL1 )
JOINT LOAD
528 FX -1.1 FY 2 FZ -67.4
**************************************************************************************************
LOAD 8 LOADTYPE None  TITLE TEMPERATURE LOAD -25  ( TL2 )
JOINT LOAD
528 FX 1.1 FY -2 FZ 67.4
**************************************************************************************************
LOAD 9 LOADTYPE None  TITLE SEISMIC LOAD  ( ELX )
JOINT LOAD
528 FX 1 FY -3.35 FZ 0.008
**************************************************************************************************
LOAD 10 LOADTYPE None  TITLE SEISMIC LOAD  ( ELZ )
JOINT LOAD
528 FX -0.127 FY 1.428 FZ 0.75
**************************************************************************************************
PERFORM ANALYSIS
CHANGE
SUPPORTS
465 TO 470 473 TO 478 481 TO 486 497 TO 502 505 TO 510 513 TO 518 521 TO 526 -
785 TO 790 799 TO 814 PLATE MAT DIRECT Y SUBGRADE 57600 COMPRESSION
**************************************************************************************************
***FACTORED LOAD COMBINATIONS FOR STRENGTH DESIGN
LOAD 101 1.4DL
REPEAT LOAD
1 1.4
```

```
1 1.4
PERFORM ANALYSIS
CHANGE
LOAD 102 1.4DL + 1.7LL
REPEAT LOAD
1 1.4 2 1.7
PERFORM ANALYSIS
CHANGE
LOAD 103 1.2DL + 1.6WLX1 + 1.0 LL
REPEAT LOAD
1 1.2 3 1.6 2 1.0
PERFORM ANALYSIS
CHANGE
LOAD 104 1.2DL + 1.6WLX2 + 1.0 LL
REPEAT LOAD
1 1.2 4 1.6 2 1.0
PERFORM ANALYSIS
CHANGE
LOAD 105 1.2DL + 1.6WLZ1 + 1.0 LL
REPEAT LOAD
1 1.2 5 1.6 2 1.0
PERFORM ANALYSIS
CHANGE
LOAD 106 1.2DL + 1.6WLZ2 + 1.0 LL
REPEAT LOAD
1 1.2 6 1.6 2 1.0
PERFORM ANALYSIS
CHANGE
LOAD 107 0.9DL + 1.6WLX1
REPEAT LOAD
1 0.9 3 1.6
PERFORM ANALYSIS
CHANGE
LOAD 108 0.9DL + 1.6WLX2
REPEAT LOAD
1 0.9 4 1.6
PERFORM ANALYSIS
CHANGE
LOAD 109 0.9DL + 1.6WLZ1
REPEAT LOAD
1 0.9 5 1.6
PERFORM ANALYSIS
CHANGE
LOAD 110 0.9DL + 1.6WLZ2
REPEAT LOAD
1 0.9 6 1.6
PERFORM ANALYSIS
CHANGE
LOAD 111 1.2DL + 1.0ELX + 1.0 LL
REPEAT LOAD
1 1.2 2 1.0 9 1.0
PERFORM ANALYSIS
CHANGE
LOAD 112 1.2DL + 1.0ELZ + 1.0 LL
REPEAT LOAD
1 1.2 2 1.0 10 1.0
PERFORM ANALYSIS
CHANGE
LOAD 113 0.9DL + 1.0ELX
REPEAT LOAD
1 0.9 9 1.0
PERFORM ANALYSIS
CHANGE
LOAD 114 0.9DL + 1.0ELZ
REPEAT LOAD
1 0.9 10 1.0
PERFORM ANALYSIS
CHANGE
LOAD 115 1.4DL + 1.7LL + 1.0 TL1
REPEAT LOAD
1 1.4 2 1.7 7 1.0
PERFORM ANALYSIS
CHANGE
LOAD 116 1.4DL + 1.7LL + 1.0 TL2
REPEAT LOAD
1 1.4 2 1.7 8 1.0
PERFORM ANALYSIS
CHANGE
*********************************************************************************************
```



```
*********************************************************************************************
* UNFACTORED LOAD COMBINATIONS FOR STABILITY CHECK
LOAD 201 1.0DL
REPEAT LOAD
1 1.0
PERFORM ANALYSIS
CHANGE
LOAD 202 1.0DL + 1.0LL
REPEAT LOAD
1 1.0 2 1.0
PERFORM ANALYSIS
CHANGE
LOAD 203 1.0DL + 1.0WLX1
REPEAT LOAD
1 1.0 3 1.0
PERFORM ANALYSIS
CHANGE
LOAD 204 1.0DL + 1.0WLX2
REPEAT LOAD
1 1.0 4 1.0
PERFORM ANALYSIS
CHANGE
LOAD 205 1.0DL + 1.0WLZ1
REPEAT LOAD
1 1.0 5 1.0
PERFORM ANALYSIS
CHANGE
LOAD 206 1.0DL + 1.0WLZ2
REPEAT LOAD
1 1.0 6 1.0
PERFORM ANALYSIS
CHANGE
LOAD 207 1.0DL + 0.75WLX1 + 0.75LL
REPEAT LOAD
1 1.0 3 0.75 2 0.75
PERFORM ANALYSIS
CHANGE
LOAD 208 1.0DL + 0.75WLX2 + 0.75LL
REPEAT LOAD
1 1.0 4 0.75 2 0.75
PERFORM ANALYSIS
CHANGE
LOAD 209 1.0DL + 0.75WLZ1 + 0.75LL
REPEAT LOAD
1 1.0 5 0.75 2 0.75
PERFORM ANALYSIS
CHANGE
LOAD 210 1.0DL + 0.75WLZ2 + 0.75LL
REPEAT LOAD
1 1.0 6 0.75 2 0.75
PERFORM ANALYSIS
CHANGE
LOAD 211 0.6DL + 1.0WLX1
REPEAT LOAD
1 0.6 3 1.0
PERFORM ANALYSIS
CHANGE
LOAD 212 0.6DL + 1.0WLX2
REPEAT LOAD
1 0.6 4 1.0
PERFORM ANALYSIS
CHANGE
LOAD 213 0.6DL + 1.0WLZ1
REPEAT LOAD
1 0.6 5 1.0
PERFORM ANALYSIS
CHANGE
LOAD 214 0.6DL + 1.0WLZ2
REPEAT LOAD
1 0.6 6 1.0
PERFORM ANALYSIS
CHANGE
LOAD 215 1.0DL + 1.0LL + 1.0TL1
REPEAT LOAD
1 1.0 2 1.0 7 1.0
PERFORM ANALYSIS
CHANGE
LOAD 216 1.0DL + 1.0LL + 1.0TL2
REPEAT LOAD
```

```
REPEAT LOAD
1 1.0 2 1.0 8 1.0
PERFORM ANALYSIS
CHANGE
LOAD 217 0.6DL + 0.7ELX
REPEAT LOAD
1 0.6 9 0.7
PERFORM ANALYSIS
CHANGE
LOAD 218 0.6DL + 0.7ELZ
REPEAT LOAD
1 0.6 10 0.7
PERFORM ANALYSIS
CHANGE
*********************************************************************************************
PERFORM ANALYSIS
LOAD LIST 101 TO 116
START CONCRETE DESIGN
CODE ACI
TRACK 2 ALL
FC 35000 MEMB 529
CLB 0.05 MEMB 529
CLS 0.05 ALL
MAXMAIN 25 ALL
MINMAIN 25 ALL
MINSEC 10 ALL
DESIGN COLUMN 529
END CONCRETE DESIGN
FINISH
```

ATTACHMENT 4 STAAD INPUT – FOUNDATION F3

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```
STAAD SPACE
START JOB INFORMATION
ENGINEER DATE 08/05/2020
JOB NAME FDN. MKD. F3
JOB CLIENT MA'ADEN
JOB NO O19045
JOB REV 0
ENGINEER NAME SOR
END JOB INFORMATION
INPUT WIDTH 79
SET NL 200
SET ITERLIM 15
SET PRINT 1
UNIT METER KN
JOINT COORDINATES
457 -0.5 911.5 0.42438; 460 -0.25 911.5 0.42438; 461 -0.25 911.5 0.17538;
462 0 911.5 0.42438; 463 0 911.5 0.17538; 464 0.25 911.5 0.42438;
465 0.25 911.5 0.17538; 466 0.5 911.5 0.42438; 467 0.5 911.5 0.17538;
468 0.75 911.5 0.42438; 469 0.75 911.5 0.17538; 475 -0.5 911.5 0.67438;
476 -0.25 911.5 0.67438; 477 0 911.5 0.67438; 478 0.25 911.5 0.67438;
479 0.5 911.5 0.67438; 480 0.75 911.5 0.67438; 483 -0.7 911.5 0.87538;
484 -0.5 911.5 0.87538; 485 -0.25 911.5 0.87538; 486 0 911.5 0.87538;
487 0.25 911.5 0.87538; 488 0.5 911.5 0.87538; 489 0.75 911.5 0.87538;
498 0 911.5 -0.62362; 499 0 911.5 -0.87362; 500 0.25 911.5 -0.62362;
501 0.25 911.5 -0.87362; 502 0.5 911.5 -0.62362; 503 0.5 911.5 -0.87362;
504 0.75 911.5 -0.62362; 505 0.75 911.5 -0.87362; 512 -0.25 911.5 -0.37362;
513 0 911.5 -0.37362; 514 0.25 911.5 -0.37362; 515 0.5 911.5 -0.37362;
516 0.75 911.5 -0.37362; 521 -0.25 911.5 -0.12462; 522 0 911.5 -0.12462;
523 0.25 911.5 -0.12462; 524 0.5 911.5 -0.12462; 525 0.75 911.5 -0.12462;
528 0 913 -0.125; 803 0 911.5 -1.12462; 805 0.25 911.5 -1.12462;
807 0.5 911.5 -1.12462; 809 0.75 911.5 -1.12462; 837 1 911.5 0.42438;
838 1 911.5 0.17538; 839 1.25 911.5 0.42438; 840 1.25 911.5 0.17538;
841 1.5 911.5 0.42438; 842 1.5 911.5 0.17538; 843 1 911.5 0.67438;
844 1.25 911.5 0.67438; 845 1.5 911.5 0.67438; 846 1 911.5 0.87538;
847 1.25 911.5 0.87538; 848 1.5 911.5 0.87538; 849 1 911.5 -0.62362;
850 1 911.5 -0.87362; 851 1.25 911.5 -0.62362; 852 1.25 911.5 -0.87362;
853 1.5 911.5 -0.62362; 854 1.5 911.5 -0.87362; 855 1 911.5 -0.37362;
856 1.25 911.5 -0.37362; 857 1.5 911.5 -0.37362; 858 1 911.5 -0.12462;
859 1.25 911.5 -0.12462; 860 1.5 911.5 -0.12462; 861 1 911.5 -1.12462;
862 1.25 911.5 -1.12462; 863 1.5 911.5 -1.12462; 864 1.8 911.5 0.42438;
865 1.8 911.5 0.17538; 866 1.8 911.5 0.67438; 867 1.8 911.5 0.87538;
868 1.8 911.5 -0.62362; 869 1.8 911.5 -0.87362; 870 1.8 911.5 -0.37362;
871 1.8 911.5 -0.12462; 872 1.8 911.5 -1.12462; 873 -0.075 911.5 -1.12462;
874 -0.139786 911.5 -0.917303; 875 -0.210961 911.5 -0.689545;
876 -0.304377 911.5 -0.390613; 877 -0.156584 911.5 -0.672552;
878 -0.375267 911.5 -0.163766; 879 -0.460676 911.5 0.109544;
880 -0.553808 911.5 0.407565; 881 -0.406868 911.5 0.126359;
882 -0.624982 911.5 0.635323;
MEMBER INCIDENCES
529 522 528;
ELEMENT INCIDENCES SHELL
467 460 462 463 461; 468 462 464 465 463; 469 464 466 467 465;
470 466 468 469 467; 474 475 476 460 457; 475 476 477 462 460;
476 477 478 464 462; 477 478 479 466 464; 478 479 480 468 466;
482 484 485 476 475; 483 485 486 477 476; 484 486 487 478 477;
485 487 488 479 478; 486 488 489 480 479; 500 498 500 501 499;
501 500 502 503 501; 502 502 504 505 503; 508 513 514 500 498;
509 514 515 502 500; 510 515 516 504 502; 515 521 522 513 512;
516 522 523 514 513; 517 523 524 515 514; 518 524 525 516 515;
523 461 463 522 521; 524 463 465 523 522; 525 465 467 524 523;
526 467 469 525 524; 788 499 501 805 803; 789 501 503 807 805;
790 503 505 809 807; 815 468 837 838 469; 816 837 839 840 838;
817 839 841 842 840; 818 480 843 837 468; 819 843 844 839 837;
820 844 845 841 839; 821 489 846 843 480; 822 846 847 844 843;
823 847 848 845 844; 824 504 849 850 505; 825 849 851 852 850;
826 851 853 854 852; 827 516 855 849 504; 828 855 856 851 849;
829 856 857 853 851; 830 525 858 855 516; 831 858 859 856 855;
832 859 860 857 856; 833 469 838 858 525; 834 838 840 859 858;
835 840 842 860 859; 836 505 850 861 809; 837 850 852 862 861;
838 852 854 863 862; 839 841 864 865 842; 840 845 866 864 841;
841 848 867 866 845; 842 853 868 869 854; 843 857 870 868 853;
844 860 871 870 857; 845 842 865 871 860; 846 854 869 872 863;
867 483 484 475 882; 868 882 475 457 880; 869 457 460 461 881;
870 880 457 881 879; 871 881 461 521; 872 879 881 521 878; 873 878 521 512 876;
874 876 512 877 875; 875 512 513 498 877; 876 877 498 499; 877 875 877 499 874;
878 874 499 803 873;
START GROUP DEFINITION
ELEMENT
_RAFT 467 TO 470 474 TO 478 482 TO 486 500 TO 502 508 TO 510 515 TO 518 523 -
```



```
_RAFT 467 TO 470 474 TO 478 482 TO 486 500 TO 502 508 TO 510 515 TO 518 523 -
524 TO 526 788 TO 790 815 TO 846
_RAFT_MOM 467 TO 470 474 TO 478 482 TO 486 500 TO 502 508 TO 510 517 518 525 -
526 788 TO 790 815 TO 829 831 832 834 TO 846 867 TO 878
_RAFT_SHEAR 470 474 TO 478 482 TO 486 500 TO 502 510 518 526 788 TO 790 815 -
816 TO 846 867 868 876 TO 878
MEMBER
END GROUP DEFINITION
ELEMENT PROPERTY
467 TO 470 474 TO 478 482 TO 486 500 TO 502 508 TO 510 515 TO 518 523 TO 526 -
788 TO 790 815 TO 846 867 TO 878 THICKNESS 0.5
DEFINE MATERIAL START
ISOTROPIC CONCRETE
E 2.17185e+007
POISSON 0.17
DENSITY 25
ALPHA 1e-005
DAMP 0.05
TYPE CONCRETE
STRENGTH FCU 27579
END DEFINE MATERIAL
MEMBER PROPERTY AMERICAN
529 PRIS YD 0.6 ZD 0.6
CONSTANTS
MATERIAL CONCRETE ALL
SUPPORTS
467 TO 470 474 TO 478 482 TO 486 500 TO 502 508 TO 510 515 TO 518 523 TO 526 -
788 TO 790 815 TO 846 867 TO 878 PLATE MAT DIRECT Y SUBGRADE 57600 PRINT
*********************************************************************************************
LOAD 1 LOADTYPE None  TITLE DEAD LOAD ( DL )
SELFWEIGHT Y -1
ELEMENT LOAD
* LOAD DUE TO BACKFILL
467 TO 470 474 TO 478 482 TO 486 500 TO 502 508 TO 510 515 TO 518 523 TO 526 -
788 TO 790 815 TO 846 867 TO 878 PR GY -15
* LOAD DUE TO SURCHARGE
467 TO 470 474 TO 478 482 TO 486 500 TO 502 508 TO 510 515 TO 518 523 TO 526 -
788 TO 790 815 TO 846 867 TO 878 PR GY -20
JOINT LOAD
528 FX -0.034 FY -27.7 FZ 5
*********************************************************************************************
LOAD 2 LOADTYPE None  TITLE LIVE LOAD ( LL )
JOINT LOAD
528 FX 0.13 FY -133.12 FZ 25
*********************************************************************************************
LOAD 3 LOADTYPE None  TITLE WIND LOAD +X  ( WLX1 )
JOINT LOAD
528 FX 8.7 FY -15.2 FZ 0.02
*********************************************************************************************
LOAD 4 LOADTYPE None  TITLE WIND LOAD -X  ( WLX2 )
JOINT LOAD
528 FX -8.7 FY 15.2 FZ -0.02
*********************************************************************************************
LOAD 5 LOADTYPE None  TITLE WIND LOAD +Z  ( WLZ1 )
JOINT LOAD
528 FX -0.44 FY 6 FZ 5.8
*********************************************************************************************
LOAD 6 LOADTYPE None  TITLE WIND LOAD -Z  ( WLZ2 )
JOINT LOAD
528 FX 0.44 FY -6 FZ -5.8
*********************************************************************************************
LOAD 7 LOADTYPE None  TITLE TEMPERATURE LOAD +25  ( TL1 )
JOINT LOAD
528 FX -1.1 FY 2 FZ -67.4
*********************************************************************************************
LOAD 8 LOADTYPE None  TITLE TEMPERATURE LOAD -25  ( TL2 )
JOINT LOAD
528 FX 1.1 FY -2 FZ 67.4
*********************************************************************************************
LOAD 9 LOADTYPE None  TITLE SEISMIC LOAD  ( ELX )
JOINT LOAD
528 FX 1 FY -3.35 FZ 0.008
*********************************************************************************************
LOAD 10 LOADTYPE None  TITLE SEISMIC LOAD  ( ELZ )
JOINT LOAD
528 FX -0.127 FY 1.428 FZ 0.75
*********************************************************************************************
PERFORM ANALYSIS
CHANGE
```

```
CHANGE
SUPPORTS
467 TO 470 474 TO 478 482 TO 486 500 TO 502 508 TO 510 515 TO 518 523 TO 526 -
788 TO 790 815 TO 846 867 TO 877 -
878 PLATE MAT DIRECT Y SUBGRADE 57600 COMPRESSION
**********************************************************************************************
***FACTORED LOAD COMBINATIONS FOR STRENGTH DESIGN
LOAD 101 1.4DL
REPEAT LOAD
1 1.4
PERFORM ANALYSIS
CHANGE
LOAD 102 1.4DL + 1.7LL
REPEAT LOAD
1 1.4 2 1.7
PERFORM ANALYSIS
CHANGE
LOAD 103 1.2DL + 1.6WLX1 + 1.0 LL
REPEAT LOAD
1 1.2 3 1.6 2 1.0
PERFORM ANALYSIS
CHANGE
LOAD 104 1.2DL + 1.6WLX2 + 1.0 LL
REPEAT LOAD
1 1.2 4 1.6 2 1.0
PERFORM ANALYSIS
CHANGE
LOAD 105 1.2DL + 1.6WLZ1 + 1.0 LL
REPEAT LOAD
1 1.2 5 1.6 2 1.0
PERFORM ANALYSIS
CHANGE
LOAD 106 1.2DL + 1.6WLZ2 + 1.0 LL
REPEAT LOAD
1 1.2 6 1.6 2 1.0
PERFORM ANALYSIS
CHANGE
LOAD 107 0.9DL + 1.6WLX1
REPEAT LOAD
1 0.9 3 1.6
PERFORM ANALYSIS
CHANGE
LOAD 108 0.9DL + 1.6WLX2
REPEAT LOAD
1 0.9 4 1.6
PERFORM ANALYSIS
CHANGE
LOAD 109 0.9DL + 1.6WLZ1
REPEAT LOAD
1 0.9 5 1.6
PERFORM ANALYSIS
CHANGE
LOAD 110 0.9DL + 1.6WLZ2
REPEAT LOAD
1 0.9 6 1.6
PERFORM ANALYSIS
CHANGE
LOAD 111 1.2DL + 1.0ELX + 1.0 LL
REPEAT LOAD
1 1.2 2 1.0 9 1.0
PERFORM ANALYSIS
CHANGE
LOAD 112 1.2DL + 1.0ELZ + 1.0 LL
REPEAT LOAD
1 1.2 2 1.0 10 1.0
PERFORM ANALYSIS
CHANGE
LOAD 113 0.9DL + 1.0ELX
REPEAT LOAD
1 0.9 9 1.0
PERFORM ANALYSIS
CHANGE
LOAD 114 0.9DL + 1.0ELZ
REPEAT LOAD
1 0.9 10 1.0
PERFORM ANALYSIS
CHANGE
LOAD 115 1.4DL + 1.7LL + 1.0 TL1
REPEAT LOAD
```



```
REPEAT LOAD
1 1.4 2 1.7 7 1.0
PERFORM ANALYSIS
CHANGE
LOAD 116 1.4DL + 1.7LL + 1.0 TL2
REPEAT LOAD
1 1.4 2 1.7 8 1.0
PERFORM ANALYSIS
CHANGE
**********************************************************************************************
* UNFACTORED LOAD COMBINATIONS FOR STABILITY CHECK
LOAD 201 1.0DL
REPEAT LOAD
1 1.0
PERFORM ANALYSIS
CHANGE
LOAD 202 1.0DL + 1.0LL
REPEAT LOAD
1 1.0 2 1.0
PERFORM ANALYSIS
CHANGE
LOAD 203 1.0DL + 1.0WLX1
REPEAT LOAD
1 1.0 3 1.0
PERFORM ANALYSIS
CHANGE
LOAD 204 1.0DL + 1.0WLX2
REPEAT LOAD
1 1.0 4 1.0
PERFORM ANALYSIS
CHANGE
LOAD 205 1.0DL + 1.0WLZ1
REPEAT LOAD
1 1.0 5 1.0
PERFORM ANALYSIS
CHANGE
LOAD 206 1.0DL + 1.0WLZ2
REPEAT LOAD
1 1.0 6 1.0
PERFORM ANALYSIS
CHANGE
LOAD 207 1.0DL + 0.75WLX1 + 0.75LL
REPEAT LOAD
1 1.0 3 0.75 2 0.75
PERFORM ANALYSIS
CHANGE
LOAD 208 1.0DL + 0.75WLX2 + 0.75LL
REPEAT LOAD
1 1.0 4 0.75 2 0.75
PERFORM ANALYSIS
CHANGE
LOAD 209 1.0DL + 0.75WLZ1 + 0.75LL
REPEAT LOAD
1 1.0 5 0.75 2 0.75
PERFORM ANALYSIS
CHANGE
LOAD 210 1.0DL + 0.75WLZ2 + 0.75LL
REPEAT LOAD
1 1.0 6 0.75 2 0.75
PERFORM ANALYSIS
CHANGE
LOAD 211 0.6DL + 1.0WLX1
REPEAT LOAD
1 0.6 3 1.0
PERFORM ANALYSIS
CHANGE
LOAD 212 0.6DL + 1.0WLX2
REPEAT LOAD
1 0.6 4 1.0
PERFORM ANALYSIS
CHANGE
LOAD 213 0.6DL + 1.0WLZ1
REPEAT LOAD
1 0.6 5 1.0
PERFORM ANALYSIS
CHANGE
LOAD 214 0.6DL + 1.0WLZ2
REPEAT LOAD
1 0.6 6 1.0
```

```
1 0.6 6 1.0
PERFORM ANALYSIS
CHANGE
LOAD 215 1.0DL + 1.0LL + 1.0TL1
REPEAT LOAD
1 1.0 2 1.0 7 1.0
PERFORM ANALYSIS
CHANGE
LOAD 216 1.0DL + 1.0LL + 1.0TL2
REPEAT LOAD
1 1.0 2 1.0 8 1.0
PERFORM ANALYSIS
CHANGE
LOAD 217 0.6DL + 0.7ELX
REPEAT LOAD
1 0.6 9 0.7
PERFORM ANALYSIS
CHANGE
LOAD 218 0.6DL + 0.7ELZ
REPEAT LOAD
1 0.6 10 0.7
PERFORM ANALYSIS
CHANGE
*********************************************************************************************
PERFORM ANALYSIS
LOAD LIST 101 TO 116
START CONCRETE DESIGN
CODE ACI
TRACK 2 ALL
FC 35000 MEMB 529
CLB 0.05 MEMB 529
CLS 0.05 ALL
MAXMAIN 25 ALL
MINMAIN 25 ALL
MINSEC 10 ALL
DESIGN COLUMN 529
END CONCRETE DESIGN
FINISH
```

ATTACHMENT 5 STAAD OUTPUT – FOUNDATION F1

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****
* *
* STAAD.Pro V8i SELECTseries5 *
* Version 20.07.10.64 *
* Proprietary Program of *
* Bentley Systems, Inc. *
* Date= OCT 19, 2020 *
* Time= 18:25:38 *
* *
* USER ID: LNT ECC DIV *
****

1. STAAD SPACE
INPUT FILE: F1.STD
2. START JOB INFORMATION
3. ENGINEER DATE 08/05/2020
4. JOB NAME FDN. MKD. F1
5. JOB CLIENT MA'ADEN
6. JOB NO O19045
7. JOB REV 0
8. ENGINEER NAME SOR
9. END JOB INFORMATION
10. INPUT WIDTH 79
11. SET NL 200
12. SET ITERLIM 15
13. SET PRINT 1
14. UNIT METER KN
15. JOINT COORDINATES
16. 456 -0.75 911.5 0.42438; 457 -0.5 911.5 0.42438; 458 -0.5 911.5 0.17538
17. 459 -0.75 911.5 0.17538; 460 -0.25 911.5 0.42438; 461 -0.25 911.5 0.17538
18. 462 0 911.5 0.42438; 463 0 911.5 0.17538; 464 0.25 911.5 0.42438
19. 465 0.25 911.5 0.17538; 466 0.5 911.5 0.42438; 467 0.5 911.5 0.17538
20. 468 0.75 911.5 0.42438; 469 0.75 911.5 0.17538; 474 -0.75 911.5 0.67438
21. 475 -0.5 911.5 0.67438; 476 -0.25 911.5 0.67438; 477 0 911.5 0.67438
22. 478 0.25 911.5 0.67438; 479 0.5 911.5 0.67438; 480 0.75 911.5 0.67438
23. 483 -0.75 911.5 0.87538; 484 -0.5 911.5 0.87538; 485 -0.25 911.5 0.87538
24. 486 0 911.5 0.87538; 487 0.25 911.5 0.87538; 488 0.5 911.5 0.87538
25. 489 0.75 911.5 0.87538; 492 -0.75 911.5 -0.62362; 493 -0.5 911.5 -0.62362
26. 494 -0.5 911.5 -0.87362; 495 -0.75 911.5 -0.87362; 496 -0.25 911.5 -0.62362
27. 497 -0.25 911.5 -0.87362; 498 0 911.5 -0.62362; 499 0 911.5 -0.87362
28. 500 0.25 911.5 -0.62362; 501 0.25 911.5 -0.87362; 502 0.5 911.5 -0.62362
29. 503 0.5 911.5 -0.87362; 504 0.75 911.5 -0.62362; 505 0.75 911.5 -0.87362
30. 510 -0.75 911.5 -0.37362; 511 -0.5 911.5 -0.37362; 512 -0.25 911.5 -0.37362
31. 513 0 911.5 -0.37362; 514 0.25 911.5 -0.37362; 515 0.5 911.5 -0.37362
32. 516 0.75 911.5 -0.37362; 519 -0.75 911.5 -0.12462; 520 -0.5 911.5 -0.12462
33. 521 -0.25 911.5 -0.12462; 522 0 911.5 -0.12462; 523 0.25 911.5 -0.12462
34. 524 0.5 911.5 -0.12462; 525 0.75 911.5 -0.12462; 528 0 913 -0.125
35. 797 -0.75 911.5 -1.12462; 798 -0.5 911.5 -1.12462; 801 -0.25 911.5 -1.12462
36. 803 0 911.5 -1.12462; 805 0.25 911.5 -1.12462; 807 0.5 911.5 -1.12462
37. 809 0.75 911.5 -1.12462; 810 1 911.5 0.42438; 811 1 911.5 0.17538
38. 812 1 911.5 0.67438; 813 1 911.5 0.87538; 814 1 911.5 -0.62362

39. 815 1 911.5 -0.87362; 816 1 911.5 -0.37362; 817 1 911.5 -0.12462
40. 818 1 911.5 -1.12462; 819 -1 911.5 0.42438; 820 -1 911.5 0.17538
41. 821 -1 911.5 0.67438; 822 -1 911.5 0.87538; 823 -1 911.5 -0.62362
42. 824 -1 911.5 -0.87362; 825 -1 911.5 -0.37362; 826 -1 911.5 -0.12462
43. 827 -1 911.5 -1.12462
44. MEMBER INCIDENCES
45. 529 522 528
46. ELEMENT INCIDENCES SHELL
47. 465 456 457 458 459; 466 457 460 461 458; 467 460 462 463 461
48. 468 462 464 465 463; 469 464 466 467 465; 470 466 468 469 467
49. 473 474 475 457 456; 474 475 476 460 457; 475 476 477 462 460
50. 476 477 478 464 462; 477 478 479 466 464; 478 479 480 468 466
51. 481 483 484 475 474; 482 484 485 476 475; 483 485 486 477 476
52. 484 486 487 478 477; 485 487 488 479 478; 486 488 489 480 479
53. 497 492 493 494 495; 498 493 496 497 494; 499 496 498 499 497
54. 500 498 500 501 499; 501 500 502 503 501; 502 502 504 505 503
55. 505 510 511 493 492; 506 511 512 496 493; 507 512 513 498 496
56. 508 513 514 500 498; 509 514 515 502 500; 510 515 516 504 502
57. 513 519 520 511 510; 514 520 521 512 511; 515 521 522 513 512
58. 516 522 523 514 513; 517 523 524 515 514; 518 524 525 516 515
59. 521 459 458 520 519; 522 458 461 521 520; 523 461 463 522 521
60. 524 463 465 523 522; 525 465 467 524 523; 526 467 469 525 524
61. 785 495 494 798 797; 786 494 497 801 798; 787 497 499 803 801
62. 788 499 501 805 803; 789 501 503 807 805; 790 503 505 809 807
63. 791 468 810 811 469; 792 480 812 810 468; 793 489 813 812 480
64. 794 504 814 815 505; 795 516 816 814 504; 796 525 817 816 516
65. 797 469 811 817 525; 798 505 815 818 809; 799 819 456 459 820
66. 800 821 474 456 819; 801 822 483 474 821; 802 823 492 495 824
67. 803 825 510 492 823; 804 826 519 510 825; 805 820 459 519 826
68. 806 824 495 797 827
69. START GROUP DEFINITION
70. ELEMENT
71. _RAFT 465 TO 470 473 TO 478 481 TO 486 497 TO 502 505 TO 510 513 TO 518 521 -
72. 522 TO 526 785 TO 806
73. _RAFT_MOM 465 TO 470 473 TO 478 481 TO 486 497 TO 502 505 TO 510 513 514 517 -
74. 518 521 522 525 526 785 TO 806
75. _RAFT_SHEAR 465 470 473 TO 478 481 TO 486 497 TO 502 505 510 513 518 521 526 -
76. 785 TO 806
77. MEMBER
78. END GROUP DEFINITION
79. ELEMENT PROPERTY
80. 465 TO 470 473 TO 478 481 TO 486 497 TO 502 505 TO 510 513 TO 518 521 TO 526 -
81. 785 TO 806 THICKNESS 0.5
82. DEFINE MATERIAL START
83. ISOTROPIC CONCRETE
84. E 2.17185E+007
85. POISSON 0.17
86. DENSITY 25
87. ALPHA 1E-005
88. DAMP 0.05
89. TYPE CONCRETE
90. STRENGTH FCU 27579
91. END DEFINE MATERIAL
92. MEMBER PROPERTY AMERICAN
93. 529 PRIS YD 0.6 ZD 0.6
94. CONSTANTS

```
95. MATERIAL CONCRETE ALL
96. SUPPORTS
97. 465 TO 470 473 TO 478 481 TO 486 497 TO 502 505 TO 510 513 TO 518 521 TO 526 -
98. 785 TO 806 PLATE MAT DIRECT Y SUBGRADE 57600 PRINT
```

JOINT	INFLUENCE AREA
456	0.062375
457	0.062375
458	0.068625
459	0.068625
460	0.062375
461	0.068625
462	0.062375
463	0.068625
464	0.062375
465	0.068625
466	0.062375
467	0.068625
468	0.062375
469	0.068625
474	0.056375
475	0.056375
476	0.056375
477	0.056375
478	0.056375
479	0.056375
480	0.056375
483	0.025125
484	0.025125
485	0.025125
486	0.025125
487	0.025125
488	0.025125
489	0.025125
492	0.062500
493	0.062500
494	0.062625
495	0.062625
496	0.062500
497	0.062625
498	0.062500
499	0.062625
500	0.062500
501	0.062625
502	0.062500
503	0.062625
504	0.062500
505	0.062625
510	0.062375
511	0.062375
512	0.062375
513	0.062375
514	0.062375
515	0.062375

516	0.062375
519	0.068625
520	0.068625
521	0.068625
522	0.068625
523	0.068625
524	0.068625
525	0.068625
797	0.031375
798	0.031375
801	0.031375
803	0.031375
805	0.031375
807	0.031375
809	0.031375
810	0.031188
811	0.034313
812	0.028187
813	0.012562
814	0.031250
815	0.031312
816	0.031188
817	0.034313
818	0.015687
819	0.031188
820	0.034313
821	0.028187
822	0.012562
823	0.031250
824	0.031312
825	0.031188
826	0.034313
827	0.015687
TOTAL	4.000000

```
 99. ********************************************************************************
100. LOAD 1 LOADTYPE NONE  TITLE DEAD LOAD ( DL )
101. SELFWEIGHT Y -1
102. ELEMENT LOAD
103. * LOAD DUE TO BACKFILL
104. 465 TO 470 473 TO 478 481 TO 486 497 TO 502 505 TO 510 513 TO 518 521 TO 526 -
105. 785 TO 806 PR GY -15
106. * LOAD DUE TO SURCHARGE
107. 465 TO 470 473 TO 478 481 TO 486 497 TO 502 505 TO 510 513 TO 518 521 TO 526 -
108. 785 TO 806 PR GY -20
109. JOINT LOAD
110. 528 FX -0.034 FY -27.7 FZ 5
111. ********************************************************************************
112. LOAD 2 LOADTYPE NONE  TITLE LIVE LOAD ( LL )
113. JOINT LOAD
114. 528 FX 0.13 FY -133.12 FZ 25
115. ********************************************************************************
116. LOAD 3 LOADTYPE NONE  TITLE WIND LOAD +X  ( WLX1 )
117. JOINT LOAD
118. 528 FX 8.7 FY -15.2 FZ 0.02
119. ********************************************************************************
120. LOAD 4 LOADTYPE NONE  TITLE WIND LOAD -X  ( WLX2 )
```

```
121. JOINT LOAD
122. 528 FX -8.7 FY 15.2 FZ -0.02
123. ******************************************************************************
124. LOAD 5 LOADTYPE NONE  TITLE WIND LOAD +Z  ( WLZ1 )
125. JOINT LOAD
126. 528 FX -0.44 FY 6 FZ 5.8
127. ******************************************************************************
128. LOAD 6 LOADTYPE NONE  TITLE WIND LOAD -Z  ( WLZ2 )
129. JOINT LOAD
130. 528 FX 0.44 FY -6 FZ -5.8
131. ******************************************************************************
132. LOAD 7 LOADTYPE NONE  TITLE TEMPERATURE LOAD +25  ( TL1 )
133. JOINT LOAD
134. 528 FX -1.1 FY 2 FZ -67.4
135. ******************************************************************************
136. LOAD 8 LOADTYPE NONE  TITLE TEMPERATURE LOAD -25  ( TL2 )
137. JOINT LOAD
138. 528 FX 1.1 FY -2 FZ 67.4
139. ******************************************************************************
140. LOAD 9 LOADTYPE NONE  TITLE SEISMIC LOAD  ( ELX )
141. JOINT LOAD
142. 528 FX 1 FY -3.35 FZ 0.008
143. ******************************************************************************
144. LOAD 10 LOADTYPE NONE  TITLE SEISMIC LOAD  ( ELZ )
145. JOINT LOAD
146. 528 FX -0.127 FY 1.428 FZ 0.75
147. ******************************************************************************
148. PERFORM ANALYSIS
```

P R O B L E M S T A T I S T I C S

NUMBER OF JOINTS 82 NUMBER OF MEMBERS 1
NUMBER OF PLATES 64 NUMBER OF SOLIDS 0
NUMBER OF SURFACES 0 NUMBER OF SUPPORTS 81

SOLVER USED IS THE OUT-OF-CORE BASIC SOLVER

ORIGINAL/FINAL BAND-WIDTH= 77/ 16/ 54 DOF
TOTAL PRIMARY LOAD CASES = 10, TOTAL DEGREES OF FREEDOM = 249
TOTAL LOAD COMBINATION CASES = 0 SO FAR.
SIZE OF STIFFNESS MATRIX = 14 DOUBLE KILO-WORDS
REQRD/AVAIL. DISK SPACE = 12.5/ 142652.2 MB

```
149. CHANGE
150. SUPPORTS
151. 465 TO 470 473 TO 478 481 TO 486 497 TO 502 505 TO 510 513 TO 518 521 TO 526 -
152. 785 TO 806 PLATE MAT DIRECT Y SUBGRADE 57600 COMPRESSION
```



```
153. ******************************************************************************
154. ***FACTORED LOAD COMBINATIONS FOR STRENGTH DESIGN
155. LOAD 101 1.4DL
156. REPEAT LOAD
157. 1 1.4
158. PERFORM ANALYSIS

**NOTE-Tension/Compression converged after  1 iterations, Case=  101

159. CHANGE
160. LOAD 102 1.4DL + 1.7LL
161. REPEAT LOAD
162. 1 1.4 2 1.7
163. PERFORM ANALYSIS

**NOTE-Tension/Compression converged after  1 iterations, Case=  102

164. CHANGE
165. LOAD 103 1.2DL + 1.6WLX1 + 1.0 LL
166. REPEAT LOAD
167. 1 1.2 3 1.6 2 1.0
168. PERFORM ANALYSIS

**NOTE-Tension/Compression converged after  1 iterations, Case=  103

169. CHANGE
170. LOAD 104 1.2DL + 1.6WLX2 + 1.0 LL
171. REPEAT LOAD
172. 1 1.2 4 1.6 2 1.0
173. PERFORM ANALYSIS

**NOTE-Tension/Compression converged after  1 iterations, Case=  104

174. CHANGE
175. LOAD 105 1.2DL + 1.6WLZ1 + 1.0 LL
176. REPEAT LOAD
```

```
177. 1 1.2 5 1.6 2 1.0
178. PERFORM ANALYSIS

**NOTE-Tension/Compression converged after  1 iterations, Case=  105


179. CHANGE
180. LOAD 106 1.2DL + 1.6WLZ2 + 1.0 LL
181. REPEAT LOAD
182. 1 1.2 6 1.6 2 1.0
183. PERFORM ANALYSIS

**NOTE-Tension/Compression converged after  1 iterations, Case=  106


184. CHANGE
185. LOAD 107 0.9DL + 1.6WLX1
186. REPEAT LOAD
187. 1 0.9 3 1.6
188. PERFORM ANALYSIS

**NOTE-Tension/Compression converged after  1 iterations, Case=  107


189. CHANGE
190. LOAD 108 0.9DL + 1.6WLX2
191. REPEAT LOAD
192. 1 0.9 4 1.6
193. PERFORM ANALYSIS

**NOTE-Tension/Compression converged after  1 iterations, Case=  108


194. CHANGE
195. LOAD 109 0.9DL + 1.6WLZ1
196. REPEAT LOAD
197. 1 0.9 5 1.6
198. PERFORM ANALYSIS
```



```
**NOTE-Tension/Compression converged after  1 iterations, Case=  109


199. CHANGE
200. LOAD 110 0.9DL + 1.6WLZ2
201. REPEAT LOAD
202. 1 0.9 6 1.6
203. PERFORM ANALYSIS

**NOTE-Tension/Compression converged after  1 iterations, Case=  110


204. CHANGE
205. LOAD 111 1.2DL + 1.0ELX + 1.0 LL
206. REPEAT LOAD
207. 1 1.2 2 1.0 9 1.0
208. PERFORM ANALYSIS

**NOTE-Tension/Compression converged after  1 iterations, Case=  111


209. CHANGE
210. LOAD 112 1.2DL + 1.0ELZ + 1.0 LL
211. REPEAT LOAD
212. 1 1.2 2 1.0 10 1.0
213. PERFORM ANALYSIS

**NOTE-Tension/Compression converged after  1 iterations, Case=  112


214. CHANGE
215. LOAD 113 0.9DL + 1.0ELX
216. REPEAT LOAD
217. 1 0.9 9 1.0
218. PERFORM ANALYSIS
```

**NOTE-Tension/Compression converged after 1 iterations, Case= 113

```
219. CHANGE
220. LOAD 114 0.9DL + 1.0ELZ
221. REPEAT LOAD
222. 1 0.9 10 1.0
223. PERFORM ANALYSIS
```

**NOTE-Tension/Compression converged after 1 iterations, Case= 114

```
224. CHANGE
225. LOAD 115 1.4DL + 1.7LL + 1.0 TL1
226. REPEAT LOAD
227. 1 1.4 2 1.7 7 1.0
228. PERFORM ANALYSIS
```

**NOTE-Tension/Compression converged after 1 iterations, Case= 115

```
229. CHANGE
230. LOAD 116 1.4DL + 1.7LL + 1.0 TL2
231. REPEAT LOAD
232. 1 1.4 2 1.7 8 1.0
233. PERFORM ANALYSIS
```

**NOTE-Tension/Compression converged after 1 iterations, Case= 116

```
234. CHANGE
235. ******************************************************************************
236. * UNFACTORED LOAD COMBINATIONS FOR STABILITY CHECK
237. LOAD 201 1.0DL
238. REPEAT LOAD
239. 1 1.0
240. PERFORM ANALYSIS
```


**NOTE-Tension/Compression converged after 1 iterations, Case= 201

```
241. CHANGE
242. LOAD 202 1.0DL + 1.0LL
243. REPEAT LOAD
244. 1 1.0 2 1.0
245. PERFORM ANALYSIS
```

**NOTE-Tension/Compression converged after 1 iterations, Case= 202

```
246. CHANGE
247. LOAD 203 1.0DL + 1.0WLX1
248. REPEAT LOAD
249. 1 1.0 3 1.0
250. PERFORM ANALYSIS
```

**NOTE-Tension/Compression converged after 1 iterations, Case= 203

```
251. CHANGE
252. LOAD 204 1.0DL + 1.0WLX2
253. REPEAT LOAD
254. 1 1.0 4 1.0
255. PERFORM ANALYSIS
```

**NOTE-Tension/Compression converged after 1 iterations, Case= 204

```
256. CHANGE
257. LOAD 205 1.0DL + 1.0WLZ1
258. REPEAT LOAD
259. 1 1.0 5 1.0
260. PERFORM ANALYSIS
```

**NOTE-Tension/Compression converged after 1 iterations, Case= 205

```
261. CHANGE
262. LOAD 206 1.0DL + 1.0WLZ2
263. REPEAT LOAD
264. 1 1.0 6 1.0
265. PERFORM ANALYSIS

**NOTE-Tension/Compression converged after  1 iterations, Case=  206



266. CHANGE
267. LOAD 207 1.0DL + 0.75WLX1 + 0.75LL
268. REPEAT LOAD
269. 1 1.0 3 0.75 2 0.75
270. PERFORM ANALYSIS

**NOTE-Tension/Compression converged after  1 iterations, Case=  207



271. CHANGE
272. LOAD 208 1.0DL + 0.75WLX2 + 0.75LL
273. REPEAT LOAD
274. 1 1.0 4 0.75 2 0.75
275. PERFORM ANALYSIS

**NOTE-Tension/Compression converged after  1 iterations, Case=  208



276. CHANGE
277. LOAD 209 1.0DL + 0.75WLZ1 + 0.75LL
278. REPEAT LOAD
279. 1 1.0 5 0.75 2 0.75
280. PERFORM ANALYSIS

**NOTE-Tension/Compression converged after  1 iterations, Case=  209



281. CHANGE
282. LOAD 210 1.0DL + 0.75WLZ2 + 0.75LL
283. REPEAT LOAD
284. 1 1.0 6 0.75 2 0.75
```



```
285. PERFORM ANALYSIS

**NOTE-Tension/Compression converged after  1 iterations, Case=  210



286. CHANGE
287. LOAD 211 0.6DL + 1.0WLX1
288. REPEAT LOAD
289. 1 0.6 3 1.0
290. PERFORM ANALYSIS

**NOTE-Tension/Compression converged after  1 iterations, Case=  211



291. CHANGE
292. LOAD 212 0.6DL + 1.0WLX2
293. REPEAT LOAD
294. 1 0.6 4 1.0
295. PERFORM ANALYSIS

**NOTE-Tension/Compression converged after  1 iterations, Case=  212



296. CHANGE
297. LOAD 213 0.6DL + 1.0WLZ1
298. REPEAT LOAD
299. 1 0.6 5 1.0
300. PERFORM ANALYSIS

**NOTE-Tension/Compression converged after  1 iterations, Case=  213



301. CHANGE
302. LOAD 214 0.6DL + 1.0WLZ2
303. REPEAT LOAD
304. 1 0.6 6 1.0
305. PERFORM ANALYSIS
```

```
**NOTE-Tension/Compression converged after  1 iterations, Case=  214


 306. CHANGE
 307. LOAD 215 1.0DL + 1.0LL + 1.0TL1
 308. REPEAT LOAD
 309. 1 1.0 2 1.0 7 1.0
 310. PERFORM ANALYSIS

**NOTE-Tension/Compression converged after  1 iterations, Case=  215


 311. CHANGE
 312. LOAD 216 1.0DL + 1.0LL + 1.0TL2
 313. REPEAT LOAD
 314. 1 1.0 2 1.0 8 1.0
 315. PERFORM ANALYSIS
**START ITERATION NO.           2

**NOTE-Tension/Compression converged after  2 iterations, Case=  216


 316. CHANGE
 317. LOAD 217 0.6DL + 0.7ELX
 318. REPEAT LOAD
 319. 1 0.6 9 0.7
 320. PERFORM ANALYSIS

**NOTE-Tension/Compression converged after  1 iterations, Case=  217


 321. CHANGE
 322. LOAD 218 0.6DL + 0.7ELZ
 323. REPEAT LOAD
 324. 1 0.6 10 0.7
 325. PERFORM ANALYSIS
```



```
**NOTE-Tension/Compression converged after  1 iterations, Case=  218


 326. CHANGE
 327. ******************************************************************************
 328. PERFORM ANALYSIS

  ** ALL CASES BEING MADE ACTIVE BEFORE RE-ANALYSIS. **

**NOTE-Tension/Compression converged after  1 iterations, Case=    1



**NOTE-Tension/Compression converged after  1 iterations, Case=    2



**START ITERATION NO.           2
**START ITERATION NO.           3
**START ITERATION NO.           4

**NOTE-Tension/Compression converged after  4 iterations, Case=    3



**START ITERATION NO.           2
**START ITERATION NO.           3
**START ITERATION NO.           4

***WARNING -  INSTABILITY AT JOINT   827   DIRECTION = MZ
 PROBABLE CAUSE SINGULAR-ADDING WEAK SPRING
 K-MATRIX DIAG=  1.1642613E+06 L-MATRIX DIAG= -5.3108670E-07 EQN NO     249
 ***NOTE - VERY WEAK SPRING ADDED FOR STABILITY

   **NOTE** STAAD DETECTS INSTABILITIES AS EXCESSIVE LOSS OF SIGNIFICANT DIGITS
            DURING DECOMPOSITION. WHEN A DECOMPOSED DIAGONAL IS LESS THAN THE
            BUILT-IN REDUCTION FACTOR TIMES THE ORIGINAL STIFFNESS MATRIX DIAGONAL,
            STAAD PRINTS A SINGULARITY NOTICE. THE BUILT-IN REDUCTION FACTOR
            IS   1.000E-09

            THE ABOVE CONDITIONS COULD ALSO BE CAUSED BY VERY STIFF OR VERY WEAK
            ELEMENTS AS WELL AS TRUE SINGULARITIES.

 **START ITERATION NO.           5

***WARNING -  INSTABILITY AT JOINT   827   DIRECTION = FY
 PROBABLE CAUSE SINGULAR-ADDING WEAK SPRING
 K-MATRIX DIAG=  1.6396673E+04 L-MATRIX DIAG= -2.2919266E-10 EQN NO     247
 ***NOTE - VERY WEAK SPRING ADDED FOR STABILITY
```

NOTE STAAD DETECTS INSTABILITIES AS EXCESSIVE LOSS OF SIGNIFICANT DIGITS
DURING DECOMPOSITION. WHEN A DECOMPOSED DIAGONAL IS LESS THAN THE
BUILT-IN REDUCTION FACTOR TIMES THE ORIGINAL STIFFNESS MATRIX DIAGONAL,
STAAD PRINTS A SINGULARITY NOTICE. THE BUILT-IN REDUCTION FACTOR
IS 1.000E-09

THE ABOVE CONDITIONS COULD ALSO BE CAUSED BY VERY STIFF OR VERY WEAK
ELEMENTS AS WELL AS TRUE SINGULARITIES.

***WARNING - INSTABILITY AT JOINT 827 DIRECTION = MX
PROBABLE CAUSE SINGULAR-ADDING WEAK SPRING
K-MATRIX DIAG= 1.1614363E+06 L-MATRIX DIAG= -6.1956234E-07 EQN NO 248
***NOTE - VERY WEAK SPRING ADDED FOR STABILITY

***WARNING - INSTABILITY AT JOINT 827 DIRECTION = MZ
PROBABLE CAUSE SINGULAR-ADDING WEAK SPRING
K-MATRIX DIAG= 1.1642613E+06 L-MATRIX DIAG= -4.5495108E-07 EQN NO 249
***NOTE - VERY WEAK SPRING ADDED FOR STABILITY

**NOTE-Tension/Compression converged after 5 iterations, Case= 4

**START ITERATION NO. 2
**START ITERATION NO. 3
**START ITERATION NO. 4

***WARNING - INSTABILITY AT JOINT 827 DIRECTION = MX
PROBABLE CAUSE SINGULAR-ADDING WEAK SPRING
K-MATRIX DIAG= 1.1614363E+06 L-MATRIX DIAG= -1.7345883E-07 EQN NO 248
***NOTE - VERY WEAK SPRING ADDED FOR STABILITY

NOTE STAAD DETECTS INSTABILITIES AS EXCESSIVE LOSS OF SIGNIFICANT DIGITS
DURING DECOMPOSITION. WHEN A DECOMPOSED DIAGONAL IS LESS THAN THE
BUILT-IN REDUCTION FACTOR TIMES THE ORIGINAL STIFFNESS MATRIX DIAGONAL,
STAAD PRINTS A SINGULARITY NOTICE. THE BUILT-IN REDUCTION FACTOR
IS 1.000E-09

THE ABOVE CONDITIONS COULD ALSO BE CAUSED BY VERY STIFF OR VERY WEAK
ELEMENTS AS WELL AS TRUE SINGULARITIES.

**START ITERATION NO. 5

***WARNING - INSTABILITY AT JOINT 827 DIRECTION = FY
PROBABLE CAUSE SINGULAR-ADDING WEAK SPRING
K-MATRIX DIAG= 1.6396673E+04 L-MATRIX DIAG= -2.2919266E-10 EQN NO 247
***NOTE - VERY WEAK SPRING ADDED FOR STABILITY

NOTE STAAD DETECTS INSTABILITIES AS EXCESSIVE LOSS OF SIGNIFICANT DIGITS
DURING DECOMPOSITION. WHEN A DECOMPOSED DIAGONAL IS LESS THAN THE
BUILT-IN REDUCTION FACTOR TIMES THE ORIGINAL STIFFNESS MATRIX DIAGONAL,
STAAD PRINTS A SINGULARITY NOTICE. THE BUILT-IN REDUCTION FACTOR
IS 1.000E-09

THE ABOVE CONDITIONS COULD ALSO BE CAUSED BY VERY STIFF OR VERY WEAK
ELEMENTS AS WELL AS TRUE SINGULARITIES.

***WARNING - INSTABILITY AT JOINT 827 DIRECTION = MX
PROBABLE CAUSE SINGULAR-ADDING WEAK SPRING
K-MATRIX DIAG= 1.1614363E+06 L-MATRIX DIAG= -6.1956234E-07 EQN NO 248
***NOTE - VERY WEAK SPRING ADDED FOR STABILITY

***WARNING - INSTABILITY AT JOINT 827 DIRECTION = MZ
PROBABLE CAUSE SINGULAR-ADDING WEAK SPRING
K-MATRIX DIAG= 1.1642613E+06 L-MATRIX DIAG= -4.5495108E-07 EQN NO 249
***NOTE - VERY WEAK SPRING ADDED FOR STABILITY
**START ITERATION NO. 6
**START ITERATION NO. 7
**START ITERATION NO. 8
**START ITERATION NO. 9

***WARNING - INSTABILITY AT JOINT 827 DIRECTION = MZ
PROBABLE CAUSE SINGULAR-ADDING WEAK SPRING
K-MATRIX DIAG= 1.1642613E+06 L-MATRIX DIAG= -3.9138831E-07 EQN NO 249
***NOTE - VERY WEAK SPRING ADDED FOR STABILITY

NOTE STAAD DETECTS INSTABILITIES AS EXCESSIVE LOSS OF SIGNIFICANT DIGITS
DURING DECOMPOSITION. WHEN A DECOMPOSED DIAGONAL IS LESS THAN THE
BUILT-IN REDUCTION FACTOR TIMES THE ORIGINAL STIFFNESS MATRIX DIAGONAL,
STAAD PRINTS A SINGULARITY NOTICE. THE BUILT-IN REDUCTION FACTOR
IS 1.000E-09

THE ABOVE CONDITIONS COULD ALSO BE CAUSED BY VERY STIFF OR VERY WEAK
ELEMENTS AS WELL AS TRUE SINGULARITIES.

**START ITERATION NO. 10

***WARNING - INSTABILITY AT JOINT 827 DIRECTION = MZ
PROBABLE CAUSE SINGULAR-ADDING WEAK SPRING
K-MATRIX DIAG= 1.1642613E+06 L-MATRIX DIAG= -4.1979365E-07 EQN NO 249
***NOTE - VERY WEAK SPRING ADDED FOR STABILITY

NOTE STAAD DETECTS INSTABILITIES AS EXCESSIVE LOSS OF SIGNIFICANT DIGITS
DURING DECOMPOSITION. WHEN A DECOMPOSED DIAGONAL IS LESS THAN THE
BUILT-IN REDUCTION FACTOR TIMES THE ORIGINAL STIFFNESS MATRIX DIAGONAL,
STAAD PRINTS A SINGULARITY NOTICE. THE BUILT-IN REDUCTION FACTOR
IS 1.000E-09

THE ABOVE CONDITIONS COULD ALSO BE CAUSED BY VERY STIFF OR VERY WEAK
ELEMENTS AS WELL AS TRUE SINGULARITIES.

**START ITERATION NO. 11

***WARNING - INSTABILITY AT JOINT 827 DIRECTION = MX
PROBABLE CAUSE SINGULAR-ADDING WEAK SPRING
K-MATRIX DIAG= 1.1614363E+06 L-MATRIX DIAG= -5.2200630E-07 EQN NO 248
***NOTE - VERY WEAK SPRING ADDED FOR STABILITY

NOTE STAAD DETECTS INSTABILITIES AS EXCESSIVE LOSS OF SIGNIFICANT DIGITS
DURING DECOMPOSITION. WHEN A DECOMPOSED DIAGONAL IS LESS THAN THE
BUILT-IN REDUCTION FACTOR TIMES THE ORIGINAL STIFFNESS MATRIX DIAGONAL,
STAAD PRINTS A SINGULARITY NOTICE. THE BUILT-IN REDUCTION FACTOR
IS 1.000E-09

THE ABOVE CONDITIONS COULD ALSO BE CAUSED BY VERY STIFF OR VERY WEAK
ELEMENTS AS WELL AS TRUE SINGULARITIES.

***WARNING - INSTABILITY AT JOINT 827 DIRECTION = MZ
PROBABLE CAUSE SINGULAR-ADDING WEAK SPRING
K-MATRIX DIAG= 1.1642613E+06 L-MATRIX DIAG= -4.2957254E-07 EQN NO 249
***NOTE - VERY WEAK SPRING ADDED FOR STABILITY
**START ITERATION NO. 12

***WARNING - INSTABILITY AT JOINT 827 DIRECTION = FY
PROBABLE CAUSE SINGULAR-ADDING WEAK SPRING
K-MATRIX DIAG= 1.6396673E+04 L-MATRIX DIAG= -2.2919266E-10 EQN NO 247
***NOTE - VERY WEAK SPRING ADDED FOR STABILITY

NOTE STAAD DETECTS INSTABILITIES AS EXCESSIVE LOSS OF SIGNIFICANT DIGITS
DURING DECOMPOSITION. WHEN A DECOMPOSED DIAGONAL IS LESS THAN THE
BUILT-IN REDUCTION FACTOR TIMES THE ORIGINAL STIFFNESS MATRIX DIAGONAL,
STAAD PRINTS A SINGULARITY NOTICE. THE BUILT-IN REDUCTION FACTOR
IS 1.000E-09

THE ABOVE CONDITIONS COULD ALSO BE CAUSED BY VERY STIFF OR VERY WEAK
ELEMENTS AS WELL AS TRUE SINGULARITIES.

***WARNING - INSTABILITY AT JOINT 827 DIRECTION = MX
PROBABLE CAUSE SINGULAR-ADDING WEAK SPRING
K-MATRIX DIAG= 1.1614363E+06 L-MATRIX DIAG= -6.1956234E-07 EQN NO 248
***NOTE - VERY WEAK SPRING ADDED FOR STABILITY

***WARNING - INSTABILITY AT JOINT 827 DIRECTION = MZ
PROBABLE CAUSE SINGULAR-ADDING WEAK SPRING
K-MATRIX DIAG= 1.1642613E+06 L-MATRIX DIAG= -4.5495108E-07 EQN NO 249
***NOTE - VERY WEAK SPRING ADDED FOR STABILITY
**START ITERATION NO. 13
**START ITERATION NO. 14
**START ITERATION NO. 15

**WARNING-Tension/Compression did not converge after 15 iterations, Case= 5
** If results are not satisfactory then enter a
** SET ITERLIM i command (i>10) before loads to get more iterations.

**START ITERATION NO. 2

**START ITERATION NO. 3
**START ITERATION NO. 4
**START ITERATION NO. 5

***WARNING - INSTABILITY AT JOINT 827 DIRECTION = MX
PROBABLE CAUSE SINGULAR-ADDING WEAK SPRING
K-MATRIX DIAG= 1.1614363E+06 L-MATRIX DIAG= -6.1932951E-07 EQN NO 248
***NOTE - VERY WEAK SPRING ADDED FOR STABILITY

NOTE STAAD DETECTS INSTABILITIES AS EXCESSIVE LOSS OF SIGNIFICANT DIGITS
DURING DECOMPOSITION. WHEN A DECOMPOSED DIAGONAL IS LESS THAN THE
BUILT-IN REDUCTION FACTOR TIMES THE ORIGINAL STIFFNESS MATRIX DIAGONAL,
STAAD PRINTS A SINGULARITY NOTICE. THE BUILT-IN REDUCTION FACTOR
IS 1.000E-09

THE ABOVE CONDITIONS COULD ALSO BE CAUSED BY VERY STIFF OR VERY WEAK
ELEMENTS AS WELL AS TRUE SINGULARITIES.

**NOTE-Tension/Compression converged after 5 iterations, Case= 6

**START ITERATION NO. 2
**START ITERATION NO. 3
**START ITERATION NO. 4
**START ITERATION NO. 5

***WARNING - INSTABILITY AT JOINT 827 DIRECTION = MX
PROBABLE CAUSE SINGULAR-ADDING WEAK SPRING
K-MATRIX DIAG= 1.1614363E+06 L-MATRIX DIAG= -6.1956234E-07 EQN NO 248
***NOTE - VERY WEAK SPRING ADDED FOR STABILITY

NOTE STAAD DETECTS INSTABILITIES AS EXCESSIVE LOSS OF SIGNIFICANT DIGITS
DURING DECOMPOSITION. WHEN A DECOMPOSED DIAGONAL IS LESS THAN THE
BUILT-IN REDUCTION FACTOR TIMES THE ORIGINAL STIFFNESS MATRIX DIAGONAL,
STAAD PRINTS A SINGULARITY NOTICE. THE BUILT-IN REDUCTION FACTOR
IS 1.000E-09

THE ABOVE CONDITIONS COULD ALSO BE CAUSED BY VERY STIFF OR VERY WEAK
ELEMENTS AS WELL AS TRUE SINGULARITIES.

**START ITERATION NO. 6

***WARNING - INSTABILITY AT JOINT 827 DIRECTION = MX
PROBABLE CAUSE SINGULAR-ADDING WEAK SPRING
K-MATRIX DIAG= 1.1614363E+06 L-MATRIX DIAG= -6.1979517E-07 EQN NO 248
***NOTE - VERY WEAK SPRING ADDED FOR STABILITY

NOTE STAAD DETECTS INSTABILITIES AS EXCESSIVE LOSS OF SIGNIFICANT DIGITS
DURING DECOMPOSITION. WHEN A DECOMPOSED DIAGONAL IS LESS THAN THE
BUILT-IN REDUCTION FACTOR TIMES THE ORIGINAL STIFFNESS MATRIX DIAGONAL,
STAAD PRINTS A SINGULARITY NOTICE. THE BUILT-IN REDUCTION FACTOR
IS 1.000E-09

THE ABOVE CONDITIONS COULD ALSO BE CAUSED BY VERY STIFF OR VERY WEAK
ELEMENTS AS WELL AS TRUE SINGULARITIES.

**START ITERATION NO. 7

***WARNING - INSTABILITY AT JOINT 827 DIRECTION = MX
PROBABLE CAUSE SINGULAR-ADDING WEAK SPRING
K-MATRIX DIAG= 1.1614363E+06 L-MATRIX DIAG= -6.1956234E-07 EQN NO 248
***NOTE - VERY WEAK SPRING ADDED FOR STABILITY

NOTE STAAD DETECTS INSTABILITIES AS EXCESSIVE LOSS OF SIGNIFICANT DIGITS
DURING DECOMPOSITION. WHEN A DECOMPOSED DIAGONAL IS LESS THAN THE
BUILT-IN REDUCTION FACTOR TIMES THE ORIGINAL STIFFNESS MATRIX DIAGONAL,
STAAD PRINTS A SINGULARITY NOTICE. THE BUILT-IN REDUCTION FACTOR
IS 1.000E-09

THE ABOVE CONDITIONS COULD ALSO BE CAUSED BY VERY STIFF OR VERY WEAK
ELEMENTS AS WELL AS TRUE SINGULARITIES.

**START ITERATION NO. 8

***WARNING - INSTABILITY AT JOINT 827 DIRECTION = MX
PROBABLE CAUSE SINGULAR-ADDING WEAK SPRING
K-MATRIX DIAG= 1.1614363E+06 L-MATRIX DIAG= -6.1956234E-07 EQN NO 248
***NOTE - VERY WEAK SPRING ADDED FOR STABILITY

NOTE STAAD DETECTS INSTABILITIES AS EXCESSIVE LOSS OF SIGNIFICANT DIGITS
DURING DECOMPOSITION. WHEN A DECOMPOSED DIAGONAL IS LESS THAN THE
BUILT-IN REDUCTION FACTOR TIMES THE ORIGINAL STIFFNESS MATRIX DIAGONAL,
STAAD PRINTS A SINGULARITY NOTICE. THE BUILT-IN REDUCTION FACTOR
IS 1.000E-09

THE ABOVE CONDITIONS COULD ALSO BE CAUSED BY VERY STIFF OR VERY WEAK
ELEMENTS AS WELL AS TRUE SINGULARITIES.

***WARNING - INSTABILITY AT JOINT 827 DIRECTION = MZ
PROBABLE CAUSE SINGULAR-ADDING WEAK SPRING
K-MATRIX DIAG= 1.1642613E+06 L-MATRIX DIAG= -4.5495108E-07 EQN NO 249
***NOTE - VERY WEAK SPRING ADDED FOR STABILITY
**START ITERATION NO. 9

***WARNING - INSTABILITY AT JOINT 827 DIRECTION = FY
PROBABLE CAUSE SINGULAR-ADDING WEAK SPRING
K-MATRIX DIAG= 1.6396673E+04 L-MATRIX DIAG= -2.2919266E-10 EQN NO 247
***NOTE - VERY WEAK SPRING ADDED FOR STABILITY

NOTE STAAD DETECTS INSTABILITIES AS EXCESSIVE LOSS OF SIGNIFICANT DIGITS
DURING DECOMPOSITION. WHEN A DECOMPOSED DIAGONAL IS LESS THAN THE
BUILT-IN REDUCTION FACTOR TIMES THE ORIGINAL STIFFNESS MATRIX DIAGONAL,
STAAD PRINTS A SINGULARITY NOTICE. THE BUILT-IN REDUCTION FACTOR
IS 1.000E-09

THE ABOVE CONDITIONS COULD ALSO BE CAUSED BY VERY STIFF OR VERY WEAK
ELEMENTS AS WELL AS TRUE SINGULARITIES.

***WARNING - INSTABILITY AT JOINT 827 DIRECTION = MX
PROBABLE CAUSE SINGULAR-ADDING WEAK SPRING
K-MATRIX DIAG= 1.1614363E+06 L-MATRIX DIAG= -6.1956234E-07 EQN NO 248
***NOTE - VERY WEAK SPRING ADDED FOR STABILITY

***WARNING - INSTABILITY AT JOINT 827 DIRECTION = MZ
PROBABLE CAUSE SINGULAR-ADDING WEAK SPRING
K-MATRIX DIAG= 1.1642613E+06 L-MATRIX DIAG= -4.5495108E-07 EQN NO 249
***NOTE - VERY WEAK SPRING ADDED FOR STABILITY

**NOTE-Tension/Compression converged after 9 iterations, Case= 7

**START ITERATION NO. 2
**START ITERATION NO. 3
**START ITERATION NO. 4
**START ITERATION NO. 5

***WARNING - INSTABILITY AT JOINT 827 DIRECTION = MX
PROBABLE CAUSE SINGULAR-ADDING WEAK SPRING
K-MATRIX DIAG= 1.1614363E+06 L-MATRIX DIAG= -1.7345883E-07 EQN NO 248
***NOTE - VERY WEAK SPRING ADDED FOR STABILITY

NOTE STAAD DETECTS INSTABILITIES AS EXCESSIVE LOSS OF SIGNIFICANT DIGITS
DURING DECOMPOSITION. WHEN A DECOMPOSED DIAGONAL IS LESS THAN THE
BUILT-IN REDUCTION FACTOR TIMES THE ORIGINAL STIFFNESS MATRIX DIAGONAL,
STAAD PRINTS A SINGULARITY NOTICE. THE BUILT-IN REDUCTION FACTOR
IS 1.000E-09

THE ABOVE CONDITIONS COULD ALSO BE CAUSED BY VERY STIFF OR VERY WEAK
ELEMENTS AS WELL AS TRUE SINGULARITIES.

**START ITERATION NO. 6

***WARNING - INSTABILITY AT JOINT 827 DIRECTION = MX
PROBABLE CAUSE SINGULAR-ADDING WEAK SPRING
K-MATRIX DIAG= 1.1614363E+06 L-MATRIX DIAG= -3.5227276E-07 EQN NO 248
***NOTE - VERY WEAK SPRING ADDED FOR STABILITY

NOTE STAAD DETECTS INSTABILITIES AS EXCESSIVE LOSS OF SIGNIFICANT DIGITS
DURING DECOMPOSITION. WHEN A DECOMPOSED DIAGONAL IS LESS THAN THE
BUILT-IN REDUCTION FACTOR TIMES THE ORIGINAL STIFFNESS MATRIX DIAGONAL,
STAAD PRINTS A SINGULARITY NOTICE. THE BUILT-IN REDUCTION FACTOR
IS 1.000E-09

THE ABOVE CONDITIONS COULD ALSO BE CAUSED BY VERY STIFF OR VERY WEAK
ELEMENTS AS WELL AS TRUE SINGULARITIES.

**START ITERATION NO. 7

***WARNING - INSTABILITY AT JOINT 827 DIRECTION = MX
PROBABLE CAUSE SINGULAR-ADDING WEAK SPRING
K-MATRIX DIAG= 1.1614363E+06 L-MATRIX DIAG= -3.4761615E-07 EQN NO 248
***NOTE - VERY WEAK SPRING ADDED FOR STABILITY

NOTE STAAD DETECTS INSTABILITIES AS EXCESSIVE LOSS OF SIGNIFICANT DIGITS
DURING DECOMPOSITION. WHEN A DECOMPOSED DIAGONAL IS LESS THAN THE
BUILT-IN REDUCTION FACTOR TIMES THE ORIGINAL STIFFNESS MATRIX DIAGONAL,
STAAD PRINTS A SINGULARITY NOTICE. THE BUILT-IN REDUCTION FACTOR
IS 1.000E-09

THE ABOVE CONDITIONS COULD ALSO BE CAUSED BY VERY STIFF OR VERY WEAK
ELEMENTS AS WELL AS TRUE SINGULARITIES.

**START ITERATION NO. 8

***WARNING - INSTABILITY AT JOINT 827 DIRECTION = MX
PROBABLE CAUSE SINGULAR-ADDING WEAK SPRING
K-MATRIX DIAG= 1.1614363E+06 L-MATRIX DIAG= -3.6880374E-07 EQN NO 248
***NOTE - VERY WEAK SPRING ADDED FOR STABILITY

NOTE STAAD DETECTS INSTABILITIES AS EXCESSIVE LOSS OF SIGNIFICANT DIGITS
DURING DECOMPOSITION. WHEN A DECOMPOSED DIAGONAL IS LESS THAN THE
BUILT-IN REDUCTION FACTOR TIMES THE ORIGINAL STIFFNESS MATRIX DIAGONAL,
STAAD PRINTS A SINGULARITY NOTICE. THE BUILT-IN REDUCTION FACTOR
IS 1.000E-09

THE ABOVE CONDITIONS COULD ALSO BE CAUSED BY VERY STIFF OR VERY WEAK
ELEMENTS AS WELL AS TRUE SINGULARITIES.

**NOTE-Tension/Compression converged after 8 iterations, Case= 8

**START ITERATION NO. 2
**START ITERATION NO. 3

**NOTE-Tension/Compression converged after 3 iterations, Case= 9

**START ITERATION NO. 2
**START ITERATION NO. 3
**START ITERATION NO. 4

***WARNING - INSTABILITY AT JOINT 827 DIRECTION = MX
PROBABLE CAUSE SINGULAR-ADDING WEAK SPRING
K-MATRIX DIAG= 1.1614363E+06 L-MATRIX DIAG= -1.7345883E-07 EQN NO 248
***NOTE - VERY WEAK SPRING ADDED FOR STABILITY

NOTE STAAD DETECTS INSTABILITIES AS EXCESSIVE LOSS OF SIGNIFICANT DIGITS
DURING DECOMPOSITION. WHEN A DECOMPOSED DIAGONAL IS LESS THAN THE
BUILT-IN REDUCTION FACTOR TIMES THE ORIGINAL STIFFNESS MATRIX DIAGONAL,
STAAD PRINTS A SINGULARITY NOTICE. THE BUILT-IN REDUCTION FACTOR
IS 1.000E-09

THE ABOVE CONDITIONS COULD ALSO BE CAUSED BY VERY STIFF OR VERY WEAK
ELEMENTS AS WELL AS TRUE SINGULARITIES.

**START ITERATION NO. 5

***WARNING - INSTABILITY AT JOINT 827 DIRECTION = FY
PROBABLE CAUSE SINGULAR-ADDING WEAK SPRING
K-MATRIX DIAG= 1.6396673E+04 L-MATRIX DIAG= -2.2919266E-10 EQN NO 247
***NOTE - VERY WEAK SPRING ADDED FOR STABILITY

NOTE STAAD DETECTS INSTABILITIES AS EXCESSIVE LOSS OF SIGNIFICANT DIGITS
DURING DECOMPOSITION. WHEN A DECOMPOSED DIAGONAL IS LESS THAN THE
BUILT-IN REDUCTION FACTOR TIMES THE ORIGINAL STIFFNESS MATRIX DIAGONAL,
STAAD PRINTS A SINGULARITY NOTICE. THE BUILT-IN REDUCTION FACTOR
IS 1.000E-09

THE ABOVE CONDITIONS COULD ALSO BE CAUSED BY VERY STIFF OR VERY WEAK
ELEMENTS AS WELL AS TRUE SINGULARITIES.

***WARNING - INSTABILITY AT JOINT 827 DIRECTION = MX
PROBABLE CAUSE SINGULAR-ADDING WEAK SPRING
K-MATRIX DIAG= 1.1614363E+06 L-MATRIX DIAG= -6.1956234E-07 EQN NO 248
***NOTE - VERY WEAK SPRING ADDED FOR STABILITY

***WARNING - INSTABILITY AT JOINT 827 DIRECTION = MZ
PROBABLE CAUSE SINGULAR-ADDING WEAK SPRING
K-MATRIX DIAG= 1.1642613E+06 L-MATRIX DIAG= -4.5495108E-07 EQN NO 249
***NOTE - VERY WEAK SPRING ADDED FOR STABILITY

**NOTE-Tension/Compression converged after 5 iterations, Case= 10

**NOTE-Tension/Compression converged after 1 iterations, Case= 101

**NOTE-Tension/Compression converged after 1 iterations, Case= 102

**NOTE-Tension/Compression converged after 1 iterations, Case= 103

**NOTE-Tension/Compression converged after 1 iterations, Case= 104

**NOTE-Tension/Compression converged after 1 iterations, Case= 105

**NOTE-Tension/Compression converged after 1 iterations, Case= 106

**NOTE-Tension/Compression converged after 1 iterations, Case= 107

**NOTE-Tension/Compression converged after 1 iterations, Case= 108

**NOTE-Tension/Compression converged after 1 iterations, Case= 109

**NOTE-Tension/Compression converged after 1 iterations, Case= 110

**NOTE-Tension/Compression converged after 1 iterations, Case= 111

**NOTE-Tension/Compression converged after 1 iterations, Case= 112

**NOTE-Tension/Compression converged after 1 iterations, Case= 113

**NOTE-Tension/Compression converged after 1 iterations, Case= 114

**NOTE-Tension/Compression converged after 1 iterations, Case= 115

**NOTE-Tension/Compression converged after 1 iterations, Case= 116

**NOTE-Tension/Compression converged after 1 iterations, Case= 202

**NOTE-Tension/Compression converged after 1 iterations, Case= 203

**NOTE-Tension/Compression converged after 1 iterations, Case= 204

**NOTE-Tension/Compression converged after 1 iterations, Case= 205

**NOTE-Tension/Compression converged after 1 iterations, Case= 206

**NOTE-Tension/Compression converged after 1 iterations, Case= 207

**NOTE-Tension/Compression converged after 1 iterations, Case= 208

**NOTE-Tension/Compression converged after 1 iterations, Case= 209

**NOTE-Tension/Compression converged after 1 iterations, Case= 210

**NOTE-Tension/Compression converged after 1 iterations, Case= 211

**NOTE-Tension/Compression converged after 1 iterations, Case= 212

**NOTE-Tension/Compression converged after 1 iterations, Case= 213

**NOTE-Tension/Compression converged after 1 iterations, Case= 214

**START ITERATION NO. 2

**NOTE-Tension/Compression converged after 2 iterations, Case= 216

**NOTE-Tension/Compression converged after 1 iterations, Case= 217

**NOTE-Tension/Compression converged after 1 iterations, Case= 218

329. LOAD LIST 101 TO 116
330. START CONCRETE DESIGN

331. CODE ACI
332. TRACK 2 ALL
333. FC 35000 MEMB 529
334. CLB 0.05 MEMB 529
335. CLS 0.05 ALL
336. MAXMAIN 25 ALL
337. MINMAIN 25 ALL
338. MINSEC 10 ALL
339. DESIGN COLUMN 529

==========

COLUMN NO. 529 DESIGN PER ACI 318-08 - AXIAL + BENDING

FY - 413.7 FC - 35.0 MPA, SQRE SIZE - 600.0 X 600.0 MMS, TIED
ONLY MINIMUM STEEL IS REQUIRED.
AREA OF STEEL REQUIRED = 3600.0 SQ. MM

BAR CONFIGURATION	REINF PCT.	LOAD	LOCATION	PHI
8 - 25 MM	1.091	101	END	0.650

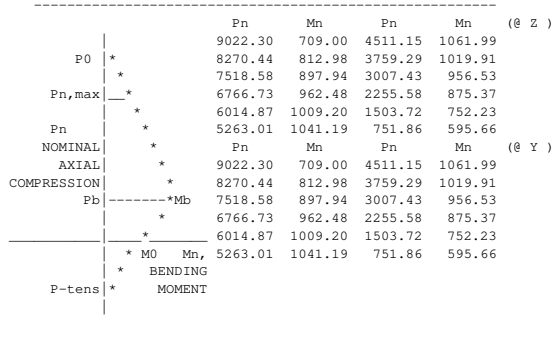
(PROVIDE EQUAL NUMBER OF BARS ON EACH FACE)
TIE BAR NUMBER 10 SPACING 400.00 MM

COLUMN INTERACTION: MOMENT ABOUT Z -AXIS (KN-MET)

P0	Pn max	P-bal.	M-bal.	e-bal. (MM)
12217.70	9774.16	4481.49	1062.73	237.1
M0	P-tens.	Des.Pn	Des.Mn	e/h
420.14	-1624.53	59.66	0.00	0.00000

COLUMN INTERACTION: MOMENT ABOUT Y -AXIS (KN-MET)

P0	Pn max	P-bal.	M-bal.	e-bal. (MM)
12217.70	9774.16	4481.49	1062.73	237.1
M0	P-tens.	Des.Pn	Des.Mn	e/h
420.14	-1624.53	59.66	0.00	0.00000



**********END OF COLUMN DESIGN RESULTS**********


```
340. END CONCRETE DESIGN
341. FINISH
```

********** END OF THE STAAD.Pro RUN **********

**** DATE= OCT 19,2020 TIME= 18:28: 7 ****

****
* For technical assistance on STAAD.Pro, please visit *
* http://selectservices.bentley.com/en-US/ *
* *
* Details about additional assistance from *
* Bentley and Partners can be found at program menu *
* Help->Technical Support *
* *
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* http://www.bentley.com *
****

ATTACHMENT 6 STAAD OUTPUT – FOUNDATION F2

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```
**************************************************
*                                                *
*          STAAD.Pro V8i SELECTseries5           *
*          Version  20.07.10.64                  *
*          Proprietary Program of                *
*          Bentley Systems, Inc.                 *
*          Date=    OCT 19, 2020                 *
*          Time=    18:44:24                     *
*                                                *
*     USER ID: LNT ECC DIV                       *
**************************************************
```

```
    1. STAAD SPACE
INPUT FILE: F2.STD
    2. START JOB INFORMATION
    3. ENGINEER DATE 08/05/2020
    4. JOB NAME FDN. MKD. F1
    5. JOB CLIENT MA'ADEN
    6. JOB NO O19045
    7. JOB REV 0
    8. ENGINEER NAME SOR
    9. END JOB INFORMATION
   10. INPUT WIDTH 79
   11. SET NL 200
   12. SET ITERLIM 15
   13. SET PRINT 1
   14. UNIT METER KN
   15. JOINT COORDINATES
   16. 456 -0.75 911.5 0.42438; 457 -0.5 911.5 0.42438; 458 -0.5 911.5 0.17538
   17. 459 -0.75 911.5 0.17538; 460 -0.25 911.5 0.42438; 461 -0.25 911.5 0.17538
   18. 462 0 911.5 0.42438; 463 0 911.5 0.17538; 464 0.25 911.5 0.42438
   19. 465 0.25 911.5 0.17538; 466 0.5 911.5 0.42438; 467 0.5 911.5 0.17538
   20. 468 0.75 911.5 0.42438; 469 0.75 911.5 0.17538; 474 -0.75 911.5 0.67438
   21. 475 -0.5 911.5 0.67438; 476 -0.25 911.5 0.67438; 477 0 911.5 0.67438
   22. 478 0.25 911.5 0.67438; 479 0.5 911.5 0.67438; 480 0.75 911.5 0.67438
   23. 483 -0.75 911.5 0.87538; 484 -0.5 911.5 0.87538; 485 -0.25 911.5 0.87538
   24. 486 0 911.5 0.87538; 487 0.25 911.5 0.87538; 488 0.5 911.5 0.87538
   25. 489 0.75 911.5 0.87538; 492 -0.75 911.5 -0.62362; 493 -0.5 911.5 -0.62362
   26. 494 -0.5 911.5 -0.87362; 495 -0.75 911.5 -0.87362; 496 -0.25 911.5 -0.62362
   27. 497 -0.25 911.5 -0.87362; 498 0 911.5 -0.62362; 499 0 911.5 -0.87362
   28. 500 0.25 911.5 -0.62362; 501 0.25 911.5 -0.87362; 502 0.5 911.5 -0.62362
   29. 503 0.5 911.5 -0.87362; 504 0.75 911.5 -0.62362; 505 0.75 911.5 -0.87362
   30. 510 -0.75 911.5 -0.37362; 511 -0.5 911.5 -0.37362; 512 -0.25 911.5 -0.37362
   31. 513 0 911.5 -0.37362; 514 0.25 911.5 -0.37362; 515 0.5 911.5 -0.37362
   32. 516 0.75 911.5 -0.37362; 519 -0.75 911.5 -0.12462; 520 -0.5 911.5 -0.12462
   33. 521 -0.25 911.5 -0.12462; 522 0 911.5 -0.12462; 523 0.25 911.5 -0.12462
   34. 524 0.5 911.5 -0.12462; 525 0.75 911.5 -0.12462; 528 0 913 -0.125
   35. 797 -0.75 911.5 -1.12462; 798 -0.5 911.5 -1.12462; 801 -0.25 911.5 -1.12462
   36. 803 0 911.5 -1.12462; 805 0.25 911.5 -1.12462; 807 0.5 911.5 -1.12462
   37. 809 0.75 911.5 -1.12462; 819 -1 911.5 0.42438; 820 -1 911.5 0.17538
   38. 821 -1 911.5 0.67438; 822 -1 911.5 0.87538; 823 -1 911.5 -0.62362
```



```
   39. 824 -1 911.5 -0.87362; 825 -1 911.5 -0.37362; 826 -1 911.5 -0.12462
   40. 827 -1 911.5 -1.12462; 828 -1.25 911.5 0.42438; 829 -1.25 911.5 0.17538
   41. 830 -1.25 911.5 0.67438; 831 -1.25 911.5 0.87538; 832 -1.25 911.5 -0.62362
   42. 833 -1.25 911.5 -0.87362; 834 -1.25 911.5 -0.37362; 835 -1.25 911.5 -0.12462
   43. 836 -1.25 911.5 -1.12462
   44. MEMBER INCIDENCES
   45. 529 522 528
   46. ELEMENT INCIDENCES SHELL
   47. 465 456 457 458 459; 466 457 460 461 458; 467 460 462 463 461
   48. 468 462 464 465 463; 469 464 466 467 465; 470 466 468 469 467
   49. 473 474 475 457 456; 474 475 476 460 457; 475 476 477 462 460
   50. 476 477 478 464 462; 477 478 479 466 464; 478 479 480 468 466
   51. 481 483 484 475 474; 482 484 485 476 475; 483 485 486 477 476
   52. 484 486 487 478 477; 485 487 488 479 478; 486 488 489 480 479
   53. 497 492 493 494 495; 498 493 496 497 494; 499 496 498 499 497
   54. 500 498 500 501 499; 501 500 502 503 501; 502 502 504 505 503
   55. 505 510 511 493 492; 506 511 512 496 493; 507 512 513 498 496
   56. 508 513 514 500 498; 509 514 515 502 500; 510 515 516 504 502
   57. 513 519 520 511 510; 514 520 521 512 511; 515 521 522 513 512
   58. 516 522 523 514 513; 517 523 524 515 514; 518 524 525 516 515
   59. 521 459 458 520 519; 522 458 461 521 520; 523 461 463 522 521
   60. 524 463 465 523 522; 525 465 467 524 523; 526 467 469 525 524
   61. 785 495 494 798 797; 786 494 497 801 798; 787 497 499 803 801
   62. 788 499 501 805 803; 789 501 503 807 805; 790 503 505 809 807
   63. 799 819 456 459 820; 800 821 474 456 819; 801 822 483 474 821
   64. 802 823 492 495 824; 803 825 510 492 823; 804 826 519 510 825
   65. 805 820 459 519 826; 806 824 495 797 827; 807 828 819 820 829
   66. 808 830 821 819 828; 809 831 822 821 830; 810 832 823 824 833
   67. 811 834 825 823 832; 812 835 826 825 834; 813 829 820 826 835
   68. 814 833 824 827 836
   69. START GROUP DEFINITION
   70. ELEMENT
   71. _RAFT 465 TO 470 473 TO 478 481 TO 486 497 TO 502 505 TO 510 513 TO 518 521 -
   72. 522 TO 526 785 TO 790 799 TO 814
   73. _RAFT_MOM 465 TO 470 473 TO 478 481 TO 486 497 TO 502 505 TO 510 513 514 517 -
   74. 518 521 522 525 526 785 TO 790 799 TO 814
   75. _RAFT_SHEAR 465 470 473 TO 478 481 TO 486 497 TO 502 505 510 513 518 521 526 -
   76. 785 TO 790 799 TO 814
   77. MEMBER
   78. END GROUP DEFINITION
   79. ELEMENT PROPERTY
   80. 465 TO 470 473 TO 478 481 TO 486 497 TO 502 505 TO 510 513 TO 518 521 TO 526 -
   81. 785 TO 790 799 TO 814 THICKNESS 0.5
   82. DEFINE MATERIAL START
   83. ISOTROPIC CONCRETE
   84. E 2.17185E+007
   85. POISSON 0.17
   86. DENSITY 25
   87. ALPHA 1E-005
   88. DAMP 0.05
   89. TYPE CONCRETE
   90. STRENGTH FCU 27579
   91. END DEFINE MATERIAL
   92. MEMBER PROPERTY AMERICAN
   93. 529 PRIS YD 0.6 ZD 0.6
   94. CONSTANTS
```

```
95. MATERIAL CONCRETE ALL
96. SUPPORTS
97. 465 TO 470 473 TO 478 481 TO 486 497 TO 502 505 TO 510 513 TO 518 521 TO 526 -
98. 785 TO 790 799 TO 814 PLATE MAT DIRECT Y SUBGRADE 57600 PRINT
```

JOINT	INFLUENCE AREA
456	0.062375
457	0.062375
458	0.068625
459	0.068625
460	0.062375
461	0.068625
462	0.062375
463	0.068625
464	0.062375
465	0.068625
466	0.062375
467	0.068625
468	0.031188
469	0.034313
474	0.056375
475	0.056375
476	0.056375
477	0.056375
478	0.056375
479	0.056375
480	0.028187
483	0.025125
484	0.025125
485	0.025125
486	0.025125
487	0.025125
488	0.025125
489	0.012562
492	0.062500
493	0.062500
494	0.062625
495	0.062625
496	0.062500
497	0.062625
498	0.062500
499	0.062625
500	0.062500
501	0.062625
502	0.062500
503	0.062625
504	0.031250
505	0.031312
510	0.062375
511	0.062375
512	0.062375
513	0.062375
514	0.062375
515	0.062375

516	0.031188
519	0.068625
520	0.068625
521	0.068625
522	0.068625
523	0.068625
524	0.068625
525	0.034313
797	0.031375
798	0.031375
801	0.031375
803	0.031375
805	0.031375
807	0.031375
809	0.015687
819	0.062375
820	0.068625
821	0.056375
822	0.025125
823	0.062500
824	0.062625
825	0.062375
826	0.068625
827	0.031375
828	0.031188
829	0.034313
830	0.028188
831	0.012563
832	0.031250
833	0.031313
834	0.031188
835	0.034313
836	0.015688
TOTAL	4.000000

```
 99. ********************************************************************************
100. LOAD 1 LOADTYPE NONE  TITLE DEAD LOAD ( DL )
101. SELFWEIGHT Y -1
102. ELEMENT LOAD
103. * LOAD DUE TO BACKFILL
104. 465 TO 470 473 TO 478 481 TO 486 497 TO 502 505 TO 510 513 TO 518 521 TO 526 -
105. 785 TO 790 799 TO 814 PR GY -15
106. * LOAD DUE TO SURCHARGE
107. 465 TO 470 473 TO 478 481 TO 486 497 TO 502 505 TO 510 513 TO 518 521 TO 526 -
108. 785 TO 790 799 TO 814 PR GY -20
109. JOINT LOAD
110. 528 FX -0.034 FY -27.7 FZ 5
111. ********************************************************************************
112. LOAD 2 LOADTYPE NONE  TITLE LIVE LOAD ( LL )
113. JOINT LOAD
114. 528 FX 0.13 FY -133.12 FZ 25
115. ********************************************************************************
116. LOAD 3 LOADTYPE NONE  TITLE WIND LOAD +X  ( WLX1 )
117. JOINT LOAD
118. 528 FX 8.7 FY -15.2 FZ 0.02
119. ********************************************************************************
120. LOAD 4 LOADTYPE NONE  TITLE WIND LOAD -X  ( WLX2 )
```

```
121. JOINT LOAD
122. 528 FX -8.7 FY 15.2 FZ -0.02
123. ********************************************************************************
124. LOAD 5 LOADTYPE NONE  TITLE WIND LOAD +Z  ( WLZ1 )
125. JOINT LOAD
126. 528 FX -0.44 FY 6 FZ 5.8
127. ********************************************************************************
128. LOAD 6 LOADTYPE NONE  TITLE WIND LOAD -Z  ( WLZ2 )
129. JOINT LOAD
130. 528 FX 0.44 FY -6 FZ -5.8
131. ********************************************************************************
132. LOAD 7 LOADTYPE NONE  TITLE TEMPERATURE LOAD +25  ( TL1 )
133. JOINT LOAD
134. 528 FX -1.1 FY 2 FZ -67.4
135. ********************************************************************************
136. LOAD 8 LOADTYPE NONE  TITLE TEMPERATURE LOAD -25  ( TL2 )
137. JOINT LOAD
138. 528 FX 1.1 FY -2 FZ 67.4
139. ********************************************************************************
140. LOAD 9 LOADTYPE NONE  TITLE SEISMIC LOAD  ( ELX )
141. JOINT LOAD
142. 528 FX 1 FY -3.35 FZ 0.008
143. ********************************************************************************
144. LOAD 10 LOADTYPE NONE  TITLE SEISMIC LOAD  ( ELZ )
145. JOINT LOAD
146. 528 FX -0.127 FY 1.428 FZ 0.75
147. ********************************************************************************
148. PERFORM ANALYSIS
```

P R O B L E M S T A T I S T I C S

NUMBER OF JOINTS 82 NUMBER OF MEMBERS 1
NUMBER OF PLATES 64 NUMBER OF SOLIDS 0
NUMBER OF SURFACES 0 NUMBER OF SUPPORTS 81

SOLVER USED IS THE OUT-OF-CORE BASIC SOLVER

ORIGINAL/FINAL BAND-WIDTH= 68/ 16/ 54 DOF
TOTAL PRIMARY LOAD CASES = 10, TOTAL DEGREES OF FREEDOM = 249
TOTAL LOAD COMBINATION CASES = 0 SO FAR.
SIZE OF STIFFNESS MATRIX = 14 DOUBLE KILO-WORDS
REQRD/AVAIL. DISK SPACE = 12.5/ 142650.0 MB

```
149. CHANGE
150. SUPPORTS
151. 465 TO 470 473 TO 478 481 TO 486 497 TO 502 505 TO 510 513 TO 518 521 TO 526 -
152. 785 TO 790 799 TO 814 PLATE MAT DIRECT Y SUBGRADE 57600 COMPRESSION
```



```
153. ********************************************************************************
154. ***FACTORED LOAD COMBINATIONS FOR STRENGTH DESIGN
155. LOAD 101 1.4DL
156. REPEAT LOAD
157. 1 1.4
158. PERFORM ANALYSIS

**NOTE-Tension/Compression converged after  1 iterations, Case=  101

159. CHANGE
160. LOAD 102 1.4DL + 1.7LL
161. REPEAT LOAD
162. 1 1.4 2 1.7
163. PERFORM ANALYSIS

**NOTE-Tension/Compression converged after  1 iterations, Case=  102

164. CHANGE
165. LOAD 103 1.2DL + 1.6WLX1 + 1.0 LL
166. REPEAT LOAD
167. 1 1.2 3 1.6 2 1.0
168. PERFORM ANALYSIS

**NOTE-Tension/Compression converged after  1 iterations, Case=  103

169. CHANGE
170. LOAD 104 1.2DL + 1.6WLX2 + 1.0 LL
171. REPEAT LOAD
172. 1 1.2 4 1.6 2 1.0
173. PERFORM ANALYSIS

**NOTE-Tension/Compression converged after  1 iterations, Case=  104

174. CHANGE
175. LOAD 105 1.2DL + 1.6WLZ1 + 1.0 LL
176. REPEAT LOAD
```

```
177. 1 1.2 5 1.6 2 1.0
178. PERFORM ANALYSIS

**NOTE-Tension/Compression converged after  1 iterations, Case=  105


179. CHANGE
180. LOAD 106 1.2DL + 1.6WLZ2 + 1.0 LL
181. REPEAT LOAD
182. 1 1.2 6 1.6 2 1.0
183. PERFORM ANALYSIS

**NOTE-Tension/Compression converged after  1 iterations, Case=  106


184. CHANGE
185. LOAD 107 0.9DL + 1.6WLX1
186. REPEAT LOAD
187. 1 0.9 3 1.6
188. PERFORM ANALYSIS

**NOTE-Tension/Compression converged after  1 iterations, Case=  107


189. CHANGE
190. LOAD 108 0.9DL + 1.6WLX2
191. REPEAT LOAD
192. 1 0.9 4 1.6
193. PERFORM ANALYSIS

**NOTE-Tension/Compression converged after  1 iterations, Case=  108


194. CHANGE
195. LOAD 109 0.9DL + 1.6WLZ1
196. REPEAT LOAD
197. 1 0.9 5 1.6
198. PERFORM ANALYSIS
```



```
**NOTE-Tension/Compression converged after  1 iterations, Case=  109


199. CHANGE
200. LOAD 110 0.9DL + 1.6WLZ2
201. REPEAT LOAD
202. 1 0.9 6 1.6
203. PERFORM ANALYSIS

**NOTE-Tension/Compression converged after  1 iterations, Case=  110


204. CHANGE
205. LOAD 111 1.2DL + 1.0ELX + 1.0 LL
206. REPEAT LOAD
207. 1 1.2 2 1.0 9 1.0
208. PERFORM ANALYSIS

**NOTE-Tension/Compression converged after  1 iterations, Case=  111


209. CHANGE
210. LOAD 112 1.2DL + 1.0ELZ + 1.0 LL
211. REPEAT LOAD
212. 1 1.2 2 1.0 10 1.0
213. PERFORM ANALYSIS

**NOTE-Tension/Compression converged after  1 iterations, Case=  112


214. CHANGE
215. LOAD 113 0.9DL + 1.0ELX
216. REPEAT LOAD
217. 1 0.9 9 1.0
218. PERFORM ANALYSIS
```

**NOTE-Tension/Compression converged after 1 iterations, Case= 113

219. CHANGE
220. LOAD 114 0.9DL + 1.0ELZ
221. REPEAT LOAD
222. 1 0.9 10 1.0
223. PERFORM ANALYSIS

**NOTE-Tension/Compression converged after 1 iterations, Case= 114

224. CHANGE
225. LOAD 115 1.4DL + 1.7LL + 1.0 TL1
226. REPEAT LOAD
227. 1 1.4 2 1.7 7 1.0
228. PERFORM ANALYSIS

**NOTE-Tension/Compression converged after 1 iterations, Case= 115

229. CHANGE
230. LOAD 116 1.4DL + 1.7LL + 1.0 TL2
231. REPEAT LOAD
232. 1 1.4 2 1.7 8 1.0
233. PERFORM ANALYSIS
**START ITERATION NO. 2
**START ITERATION NO. 3

**NOTE-Tension/Compression converged after 3 iterations, Case= 116

234. CHANGE
235. **
236. * UNFACTORED LOAD COMBINATIONS FOR STABILITY CHECK
237. LOAD 201 1.0DL
238. REPEAT LOAD
239. 1 1.0
240. PERFORM ANALYSIS

**NOTE-Tension/Compression converged after 1 iterations, Case= 201

241. CHANGE
242. LOAD 202 1.0DL + 1.0LL
243. REPEAT LOAD
244. 1 1.0 2 1.0
245. PERFORM ANALYSIS

**NOTE-Tension/Compression converged after 1 iterations, Case= 202

246. CHANGE
247. LOAD 203 1.0DL + 1.0WLX1
248. REPEAT LOAD
249. 1 1.0 3 1.0
250. PERFORM ANALYSIS

**NOTE-Tension/Compression converged after 1 iterations, Case= 203

251. CHANGE
252. LOAD 204 1.0DL + 1.0WLX2
253. REPEAT LOAD
254. 1 1.0 4 1.0
255. PERFORM ANALYSIS

**NOTE-Tension/Compression converged after 1 iterations, Case= 204

256. CHANGE
257. LOAD 205 1.0DL + 1.0WLZ1
258. REPEAT LOAD
259. 1 1.0 5 1.0
260. PERFORM ANALYSIS

**NOTE-Tension/Compression converged after 1 iterations, Case= 205

```
261. CHANGE
262. LOAD 206 1.0DL + 1.0WLZ2
263. REPEAT LOAD
264. 1 1.0 6 1.0
265. PERFORM ANALYSIS

**NOTE-Tension/Compression converged after  1 iterations, Case=  206



266. CHANGE
267. LOAD 207 1.0DL + 0.75WLX1 + 0.75LL
268. REPEAT LOAD
269. 1 1.0 3 0.75 2 0.75
270. PERFORM ANALYSIS

**NOTE-Tension/Compression converged after  1 iterations, Case=  207



271. CHANGE
272. LOAD 208 1.0DL + 0.75WLX2 + 0.75LL
273. REPEAT LOAD
274. 1 1.0 4 0.75 2 0.75
275. PERFORM ANALYSIS

**NOTE-Tension/Compression converged after  1 iterations, Case=  208



276. CHANGE
277. LOAD 209 1.0DL + 0.75WLZ1 + 0.75LL
278. REPEAT LOAD
279. 1 1.0 5 0.75 2 0.75
280. PERFORM ANALYSIS

**NOTE-Tension/Compression converged after  1 iterations, Case=  209



281. CHANGE
282. LOAD 210 1.0DL + 0.75WLZ2 + 0.75LL
283. REPEAT LOAD
284. 1 1.0 6 0.75 2 0.75
```



```
285. PERFORM ANALYSIS

**NOTE-Tension/Compression converged after  1 iterations, Case=  210



286. CHANGE
287. LOAD 211 0.6DL + 1.0WLX1
288. REPEAT LOAD
289. 1 0.6 3 1.0
290. PERFORM ANALYSIS

**NOTE-Tension/Compression converged after  1 iterations, Case=  211



291. CHANGE
292. LOAD 212 0.6DL + 1.0WLX2
293. REPEAT LOAD
294. 1 0.6 4 1.0
295. PERFORM ANALYSIS

**NOTE-Tension/Compression converged after  1 iterations, Case=  212



296. CHANGE
297. LOAD 213 0.6DL + 1.0WLZ1
298. REPEAT LOAD
299. 1 0.6 5 1.0
300. PERFORM ANALYSIS

**NOTE-Tension/Compression converged after  1 iterations, Case=  213



301. CHANGE
302. LOAD 214 0.6DL + 1.0WLZ2
303. REPEAT LOAD
304. 1 0.6 6 1.0
305. PERFORM ANALYSIS
```

**NOTE-Tension/Compression converged after 1 iterations, Case= 214

306. CHANGE
307. LOAD 215 1.0DL + 1.0LL + 1.0TL1
308. REPEAT LOAD
309. 1 1.0 2 1.0 7 1.0
310. PERFORM ANALYSIS

**NOTE-Tension/Compression converged after 1 iterations, Case= 215

311. CHANGE
312. LOAD 216 1.0DL + 1.0LL + 1.0TL2
313. REPEAT LOAD
314. 1 1.0 2 1.0 8 1.0
315. PERFORM ANALYSIS
**START ITERATION NO. 2
**START ITERATION NO. 3

**NOTE-Tension/Compression converged after 3 iterations, Case= 216

316. CHANGE
317. LOAD 217 0.6DL + 0.7ELX
318. REPEAT LOAD
319. 1 0.6 9 0.7
320. PERFORM ANALYSIS

**NOTE-Tension/Compression converged after 1 iterations, Case= 217

321. CHANGE
322. LOAD 218 0.6DL + 0.7ELZ
323. REPEAT LOAD
324. 1 0.6 10 0.7
325. PERFORM ANALYSIS

**NOTE-Tension/Compression converged after 1 iterations, Case= 218

326. CHANGE
327. **
328. PERFORM ANALYSIS

** ALL CASES BEING MADE ACTIVE BEFORE RE-ANALYSIS. **

**NOTE-Tension/Compression converged after 1 iterations, Case= 1

**START ITERATION NO. 2
**START ITERATION NO. 3

**NOTE-Tension/Compression converged after 3 iterations, Case= 2

**START ITERATION NO. 2
**START ITERATION NO. 3
**START ITERATION NO. 4
**START ITERATION NO. 5
**START ITERATION NO. 6

***WARNING - INSTABILITY AT JOINT 836 DIRECTION = MZ
PROBABLE CAUSE SINGULAR-ADDING WEAK SPRING
K-MATRIX DIAG= 1.1642613E+06 L-MATRIX DIAG= -2.0186417E-07 EQN NO 249
***NOTE - VERY WEAK SPRING ADDED FOR STABILITY

NOTE STAAD DETECTS INSTABILITIES AS EXCESSIVE LOSS OF SIGNIFICANT DIGITS
DURING DECOMPOSITION. WHEN A DECOMPOSED DIAGONAL IS LESS THAN THE
BUILT-IN REDUCTION FACTOR TIMES THE ORIGINAL STIFFNESS MATRIX DIAGONAL,
STAAD PRINTS A SINGULARITY NOTICE. THE BUILT-IN REDUCTION FACTOR
IS 1.000E-09

THE ABOVE CONDITIONS COULD ALSO BE CAUSED BY VERY STIFF OR VERY WEAK
ELEMENTS AS WELL AS TRUE SINGULARITIES.

**NOTE-Tension/Compression converged after 6 iterations, Case= 3

**START ITERATION NO. 2
**START ITERATION NO. 3
**START ITERATION NO. 4

***WARNING - INSTABILITY AT JOINT 836 DIRECTION = MZ
PROBABLE CAUSE SINGULAR-ADDING WEAK SPRING
K-MATRIX DIAG= 1.1642613E+06 L-MATRIX DIAG= -7.4296258E-07 EQN NO 249
***NOTE - VERY WEAK SPRING ADDED FOR STABILITY

NOTE STAAD DETECTS INSTABILITIES AS EXCESSIVE LOSS OF SIGNIFICANT DIGITS
DURING DECOMPOSITION. WHEN A DECOMPOSED DIAGONAL IS LESS THAN THE
BUILT-IN REDUCTION FACTOR TIMES THE ORIGINAL STIFFNESS MATRIX DIAGONAL,
STAAD PRINTS A SINGULARITY NOTICE. THE BUILT-IN REDUCTION FACTOR
IS 1.000E-09

THE ABOVE CONDITIONS COULD ALSO BE CAUSED BY VERY STIFF OR VERY WEAK
ELEMENTS AS WELL AS TRUE SINGULARITIES.

**START ITERATION NO. 5

***WARNING - INSTABILITY AT JOINT 836 DIRECTION = FY
PROBABLE CAUSE SINGULAR-ADDING WEAK SPRING
K-MATRIX DIAG= 1.6396673E+04 L-MATRIX DIAG= -2.8740033E-10 EQN NO 247
***NOTE - VERY WEAK SPRING ADDED FOR STABILITY

NOTE STAAD DETECTS INSTABILITIES AS EXCESSIVE LOSS OF SIGNIFICANT DIGITS
DURING DECOMPOSITION. WHEN A DECOMPOSED DIAGONAL IS LESS THAN THE
BUILT-IN REDUCTION FACTOR TIMES THE ORIGINAL STIFFNESS MATRIX DIAGONAL,
STAAD PRINTS A SINGULARITY NOTICE. THE BUILT-IN REDUCTION FACTOR
IS 1.000E-09

THE ABOVE CONDITIONS COULD ALSO BE CAUSED BY VERY STIFF OR VERY WEAK
ELEMENTS AS WELL AS TRUE SINGULARITIES.

***WARNING - INSTABILITY AT JOINT 836 DIRECTION = MX
PROBABLE CAUSE SINGULAR-ADDING WEAK SPRING
K-MATRIX DIAG= 1.1614363E+06 L-MATRIX DIAG= -6.0233288E-07 EQN NO 248
***NOTE - VERY WEAK SPRING ADDED FOR STABILITY

***WARNING - INSTABILITY AT JOINT 836 DIRECTION = MZ
PROBABLE CAUSE SINGULAR-ADDING WEAK SPRING
K-MATRIX DIAG= 1.1642613E+06 L-MATRIX DIAG= -7.4319541E-07 EQN NO 249
***NOTE - VERY WEAK SPRING ADDED FOR STABILITY

**NOTE-Tension/Compression converged after 5 iterations, Case= 4

**START ITERATION NO. 2
**START ITERATION NO. 3
**START ITERATION NO. 4
**START ITERATION NO. 5

***WARNING - INSTABILITY AT JOINT 836 DIRECTION = MX
PROBABLE CAUSE SINGULAR-ADDING WEAK SPRING
K-MATRIX DIAG= 1.1614363E+06 L-MATRIX DIAG= -6.8196096E-07 EQN NO 248
***NOTE - VERY WEAK SPRING ADDED FOR STABILITY

NOTE STAAD DETECTS INSTABILITIES AS EXCESSIVE LOSS OF SIGNIFICANT DIGITS
DURING DECOMPOSITION. WHEN A DECOMPOSED DIAGONAL IS LESS THAN THE
BUILT-IN REDUCTION FACTOR TIMES THE ORIGINAL STIFFNESS MATRIX DIAGONAL,
STAAD PRINTS A SINGULARITY NOTICE. THE BUILT-IN REDUCTION FACTOR
IS 1.000E-09

THE ABOVE CONDITIONS COULD ALSO BE CAUSED BY VERY STIFF OR VERY WEAK
ELEMENTS AS WELL AS TRUE SINGULARITIES.

**START ITERATION NO. 6

***WARNING - INSTABILITY AT JOINT 836 DIRECTION = MX
PROBABLE CAUSE SINGULAR-ADDING WEAK SPRING
K-MATRIX DIAG= 1.1614363E+06 L-MATRIX DIAG= -6.3236803E-07 EQN NO 248
***NOTE - VERY WEAK SPRING ADDED FOR STABILITY

NOTE STAAD DETECTS INSTABILITIES AS EXCESSIVE LOSS OF SIGNIFICANT DIGITS
DURING DECOMPOSITION. WHEN A DECOMPOSED DIAGONAL IS LESS THAN THE
BUILT-IN REDUCTION FACTOR TIMES THE ORIGINAL STIFFNESS MATRIX DIAGONAL,
STAAD PRINTS A SINGULARITY NOTICE. THE BUILT-IN REDUCTION FACTOR
IS 1.000E-09

THE ABOVE CONDITIONS COULD ALSO BE CAUSED BY VERY STIFF OR VERY WEAK
ELEMENTS AS WELL AS TRUE SINGULARITIES.

**START ITERATION NO. 7

***WARNING - INSTABILITY AT JOINT 836 DIRECTION = MX
PROBABLE CAUSE SINGULAR-ADDING WEAK SPRING
K-MATRIX DIAG= 1.1614363E+06 L-MATRIX DIAG= -5.6088902E-07 EQN NO 248
***NOTE - VERY WEAK SPRING ADDED FOR STABILITY

NOTE STAAD DETECTS INSTABILITIES AS EXCESSIVE LOSS OF SIGNIFICANT DIGITS
DURING DECOMPOSITION. WHEN A DECOMPOSED DIAGONAL IS LESS THAN THE
BUILT-IN REDUCTION FACTOR TIMES THE ORIGINAL STIFFNESS MATRIX DIAGONAL,
STAAD PRINTS A SINGULARITY NOTICE. THE BUILT-IN REDUCTION FACTOR
IS 1.000E-09

THE ABOVE CONDITIONS COULD ALSO BE CAUSED BY VERY STIFF OR VERY WEAK
ELEMENTS AS WELL AS TRUE SINGULARITIES.

***WARNING - INSTABILITY AT JOINT 836 DIRECTION = MZ
PROBABLE CAUSE SINGULAR-ADDING WEAK SPRING
K-MATRIX DIAG= 1.1642613E+06 L-MATRIX DIAG= -7.4319541E-07 EQN NO 249
***NOTE - VERY WEAK SPRING ADDED FOR STABILITY
**START ITERATION NO. 8

***WARNING - INSTABILITY AT JOINT 836 DIRECTION = FY
PROBABLE CAUSE SINGULAR-ADDING WEAK SPRING
K-MATRIX DIAG= 1.6396673E+04 L-MATRIX DIAG= -2.8740033E-10 EQN NO 247
***NOTE - VERY WEAK SPRING ADDED FOR STABILITY

NOTE STAAD DETECTS INSTABILITIES AS EXCESSIVE LOSS OF SIGNIFICANT DIGITS
DURING DECOMPOSITION. WHEN A DECOMPOSED DIAGONAL IS LESS THAN THE
BUILT-IN REDUCTION FACTOR TIMES THE ORIGINAL STIFFNESS MATRIX DIAGONAL,
STAAD PRINTS A SINGULARITY NOTICE. THE BUILT-IN REDUCTION FACTOR
IS 1.000E-09

THE ABOVE CONDITIONS COULD ALSO BE CAUSED BY VERY STIFF OR VERY WEAK
ELEMENTS AS WELL AS TRUE SINGULARITIES.

***WARNING - INSTABILITY AT JOINT 836 DIRECTION = MX
PROBABLE CAUSE SINGULAR-ADDING WEAK SPRING
K-MATRIX DIAG= 1.1614363E+06 L-MATRIX DIAG= -6.0233288E-07 EQN NO 248
***NOTE - VERY WEAK SPRING ADDED FOR STABILITY

***WARNING - INSTABILITY AT JOINT 836 DIRECTION = MZ
PROBABLE CAUSE SINGULAR-ADDING WEAK SPRING
K-MATRIX DIAG= 1.1642613E+06 L-MATRIX DIAG= -7.4319541E-07 EQN NO 249
***NOTE - VERY WEAK SPRING ADDED FOR STABILITY
**START ITERATION NO. 9
**START ITERATION NO. 10
**START ITERATION NO. 11

***WARNING - INSTABILITY AT JOINT 836 DIRECTION = MZ
PROBABLE CAUSE SINGULAR-ADDING WEAK SPRING
K-MATRIX DIAG= 1.1642613E+06 L-MATRIX DIAG= -7.4272975E-07 EQN NO 249
***NOTE - VERY WEAK SPRING ADDED FOR STABILITY

NOTE STAAD DETECTS INSTABILITIES AS EXCESSIVE LOSS OF SIGNIFICANT DIGITS
DURING DECOMPOSITION. WHEN A DECOMPOSED DIAGONAL IS LESS THAN THE
BUILT-IN REDUCTION FACTOR TIMES THE ORIGINAL STIFFNESS MATRIX DIAGONAL,
STAAD PRINTS A SINGULARITY NOTICE. THE BUILT-IN REDUCTION FACTOR
IS 1.000E-09

THE ABOVE CONDITIONS COULD ALSO BE CAUSED BY VERY STIFF OR VERY WEAK
ELEMENTS AS WELL AS TRUE SINGULARITIES.

**START ITERATION NO. 12

***WARNING - INSTABILITY AT JOINT 836 DIRECTION = MZ
PROBABLE CAUSE SINGULAR-ADDING WEAK SPRING
K-MATRIX DIAG= 1.1642613E+06 L-MATRIX DIAG= -7.4342825E-07 EQN NO 249
***NOTE - VERY WEAK SPRING ADDED FOR STABILITY

NOTE STAAD DETECTS INSTABILITIES AS EXCESSIVE LOSS OF SIGNIFICANT DIGITS
DURING DECOMPOSITION. WHEN A DECOMPOSED DIAGONAL IS LESS THAN THE
BUILT-IN REDUCTION FACTOR TIMES THE ORIGINAL STIFFNESS MATRIX DIAGONAL,
STAAD PRINTS A SINGULARITY NOTICE. THE BUILT-IN REDUCTION FACTOR
IS 1.000E-09

THE ABOVE CONDITIONS COULD ALSO BE CAUSED BY VERY STIFF OR VERY WEAK
ELEMENTS AS WELL AS TRUE SINGULARITIES.

**START ITERATION NO. 13

***WARNING - INSTABILITY AT JOINT 836 DIRECTION = MX
PROBABLE CAUSE SINGULAR-ADDING WEAK SPRING
K-MATRIX DIAG= 1.1614363E+06 L-MATRIX DIAG= -5.6088902E-07 EQN NO 248
***NOTE - VERY WEAK SPRING ADDED FOR STABILITY

NOTE STAAD DETECTS INSTABILITIES AS EXCESSIVE LOSS OF SIGNIFICANT DIGITS
DURING DECOMPOSITION. WHEN A DECOMPOSED DIAGONAL IS LESS THAN THE
BUILT-IN REDUCTION FACTOR TIMES THE ORIGINAL STIFFNESS MATRIX DIAGONAL,
STAAD PRINTS A SINGULARITY NOTICE. THE BUILT-IN REDUCTION FACTOR
IS 1.000E-09

THE ABOVE CONDITIONS COULD ALSO BE CAUSED BY VERY STIFF OR VERY WEAK
ELEMENTS AS WELL AS TRUE SINGULARITIES.

***WARNING - INSTABILITY AT JOINT 836 DIRECTION = MZ
PROBABLE CAUSE SINGULAR-ADDING WEAK SPRING
K-MATRIX DIAG= 1.1642613E+06 L-MATRIX DIAG= -7.4319541E-07 EQN NO 249
***NOTE - VERY WEAK SPRING ADDED FOR STABILITY
**START ITERATION NO. 14

***WARNING - INSTABILITY AT JOINT 836 DIRECTION = FY
PROBABLE CAUSE SINGULAR-ADDING WEAK SPRING
K-MATRIX DIAG= 1.6396673E+04 L-MATRIX DIAG= -2.8740033E-10 EQN NO 247
***NOTE - VERY WEAK SPRING ADDED FOR STABILITY

NOTE STAAD DETECTS INSTABILITIES AS EXCESSIVE LOSS OF SIGNIFICANT DIGITS
DURING DECOMPOSITION. WHEN A DECOMPOSED DIAGONAL IS LESS THAN THE
BUILT-IN REDUCTION FACTOR TIMES THE ORIGINAL STIFFNESS MATRIX DIAGONAL,
STAAD PRINTS A SINGULARITY NOTICE. THE BUILT-IN REDUCTION FACTOR
IS 1.000E-09

THE ABOVE CONDITIONS COULD ALSO BE CAUSED BY VERY STIFF OR VERY WEAK
ELEMENTS AS WELL AS TRUE SINGULARITIES.

***WARNING - INSTABILITY AT JOINT 836 DIRECTION = MX
PROBABLE CAUSE SINGULAR-ADDING WEAK SPRING
K-MATRIX DIAG= 1.1614363E+06 L-MATRIX DIAG= -6.0233288E-07 EQN NO 248
***NOTE - VERY WEAK SPRING ADDED FOR STABILITY

***WARNING - INSTABILITY AT JOINT 836 DIRECTION = MZ
PROBABLE CAUSE SINGULAR-ADDING WEAK SPRING
K-MATRIX DIAG= 1.1642613E+06 L-MATRIX DIAG= -7.4319541E-07 EQN NO 249
***NOTE - VERY WEAK SPRING ADDED FOR STABILITY
**START ITERATION NO. 15

**WARNING-Tension/Compression did not converge after 15 iterations, Case= 5
** If results are not satisfactory then enter a
** SET ITERLIM i command (i>10) before loads to get more iterations.

**START ITERATION NO. 2
**START ITERATION NO. 3
**START ITERATION NO. 4

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 **START ITERATION NO.           5
 **START ITERATION NO.           6

***WARNING -  INSTABILITY AT JOINT   836   DIRECTION = MX
 PROBABLE CAUSE SINGULAR-ADDING WEAK SPRING
 K-MATRIX DIAG=  1.1614363E+06 L-MATRIX DIAG= -4.6426430E-07 EQN NO     248
 ***NOTE - VERY WEAK SPRING ADDED FOR STABILITY

   **NOTE** STAAD DETECTS INSTABILITIES AS EXCESSIVE LOSS OF SIGNIFICANT DIGITS
            DURING DECOMPOSITION. WHEN A DECOMPOSED DIAGONAL IS LESS THAN THE
            BUILT-IN REDUCTION FACTOR TIMES THE ORIGINAL STIFFNESS MATRIX DIAGONAL,
            STAAD PRINTS A SINGULARITY NOTICE. THE BUILT-IN REDUCTION FACTOR
            IS   1.000E-09

            THE ABOVE CONDITIONS COULD ALSO BE CAUSED BY VERY STIFF OR VERY WEAK
            ELEMENTS AS WELL AS TRUE SINGULARITIES.

 **START ITERATION NO.           7

***WARNING -  INSTABILITY AT JOINT   836   DIRECTION = MX
 PROBABLE CAUSE SINGULAR-ADDING WEAK SPRING
 K-MATRIX DIAG=  1.1614363E+06 L-MATRIX DIAG= -4.5564957E-07 EQN NO     248
 ***NOTE - VERY WEAK SPRING ADDED FOR STABILITY

   **NOTE** STAAD DETECTS INSTABILITIES AS EXCESSIVE LOSS OF SIGNIFICANT DIGITS
            DURING DECOMPOSITION. WHEN A DECOMPOSED DIAGONAL IS LESS THAN THE
            BUILT-IN REDUCTION FACTOR TIMES THE ORIGINAL STIFFNESS MATRIX DIAGONAL,
            STAAD PRINTS A SINGULARITY NOTICE. THE BUILT-IN REDUCTION FACTOR
            IS   1.000E-09

            THE ABOVE CONDITIONS COULD ALSO BE CAUSED BY VERY STIFF OR VERY WEAK
            ELEMENTS AS WELL AS TRUE SINGULARITIES.

 **START ITERATION NO.           8

***WARNING -  INSTABILITY AT JOINT   836   DIRECTION = MX
 PROBABLE CAUSE SINGULAR-ADDING WEAK SPRING
 K-MATRIX DIAG=  1.1614363E+06 L-MATRIX DIAG= -4.5564957E-07 EQN NO     248
 ***NOTE - VERY WEAK SPRING ADDED FOR STABILITY

   **NOTE** STAAD DETECTS INSTABILITIES AS EXCESSIVE LOSS OF SIGNIFICANT DIGITS
            DURING DECOMPOSITION. WHEN A DECOMPOSED DIAGONAL IS LESS THAN THE
            BUILT-IN REDUCTION FACTOR TIMES THE ORIGINAL STIFFNESS MATRIX DIAGONAL,
            STAAD PRINTS A SINGULARITY NOTICE. THE BUILT-IN REDUCTION FACTOR
            IS   1.000E-09

            THE ABOVE CONDITIONS COULD ALSO BE CAUSED BY VERY STIFF OR VERY WEAK
            ELEMENTS AS WELL AS TRUE SINGULARITIES.


 **NOTE-Tension/Compression converged after  8 iterations, Case=    6



 **START ITERATION NO.           2
 **START ITERATION NO.           3
 **START ITERATION NO.           4
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```
 **START ITERATION NO.           5

***WARNING -  INSTABILITY AT JOINT   836   DIRECTION = MX
 PROBABLE CAUSE SINGULAR-ADDING WEAK SPRING
 K-MATRIX DIAG=  1.1614363E+06 L-MATRIX DIAG= -5.7346188E-07 EQN NO     248
 ***NOTE - VERY WEAK SPRING ADDED FOR STABILITY

   **NOTE** STAAD DETECTS INSTABILITIES AS EXCESSIVE LOSS OF SIGNIFICANT DIGITS
            DURING DECOMPOSITION. WHEN A DECOMPOSED DIAGONAL IS LESS THAN THE
            BUILT-IN REDUCTION FACTOR TIMES THE ORIGINAL STIFFNESS MATRIX DIAGONAL,
            STAAD PRINTS A SINGULARITY NOTICE. THE BUILT-IN REDUCTION FACTOR
            IS   1.000E-09

            THE ABOVE CONDITIONS COULD ALSO BE CAUSED BY VERY STIFF OR VERY WEAK
            ELEMENTS AS WELL AS TRUE SINGULARITIES.

 **START ITERATION NO.           6

***WARNING -  INSTABILITY AT JOINT   836   DIRECTION = MX
 PROBABLE CAUSE SINGULAR-ADDING WEAK SPRING
 K-MATRIX DIAG=  1.1614363E+06 L-MATRIX DIAG= -5.6461431E-07 EQN NO     248
 ***NOTE - VERY WEAK SPRING ADDED FOR STABILITY

   **NOTE** STAAD DETECTS INSTABILITIES AS EXCESSIVE LOSS OF SIGNIFICANT DIGITS
            DURING DECOMPOSITION. WHEN A DECOMPOSED DIAGONAL IS LESS THAN THE
            BUILT-IN REDUCTION FACTOR TIMES THE ORIGINAL STIFFNESS MATRIX DIAGONAL,
            STAAD PRINTS A SINGULARITY NOTICE. THE BUILT-IN REDUCTION FACTOR
            IS   1.000E-09

            THE ABOVE CONDITIONS COULD ALSO BE CAUSED BY VERY STIFF OR VERY WEAK
            ELEMENTS AS WELL AS TRUE SINGULARITIES.

 **START ITERATION NO.           7

***WARNING -  INSTABILITY AT JOINT   836   DIRECTION = MX
 PROBABLE CAUSE SINGULAR-ADDING WEAK SPRING
 K-MATRIX DIAG=  1.1614363E+06 L-MATRIX DIAG= -5.9558079E-07 EQN NO     248
 ***NOTE - VERY WEAK SPRING ADDED FOR STABILITY

   **NOTE** STAAD DETECTS INSTABILITIES AS EXCESSIVE LOSS OF SIGNIFICANT DIGITS
            DURING DECOMPOSITION. WHEN A DECOMPOSED DIAGONAL IS LESS THAN THE
            BUILT-IN REDUCTION FACTOR TIMES THE ORIGINAL STIFFNESS MATRIX DIAGONAL,
            STAAD PRINTS A SINGULARITY NOTICE. THE BUILT-IN REDUCTION FACTOR
            IS   1.000E-09

            THE ABOVE CONDITIONS COULD ALSO BE CAUSED BY VERY STIFF OR VERY WEAK
            ELEMENTS AS WELL AS TRUE SINGULARITIES.

 **START ITERATION NO.           8

***WARNING -  INSTABILITY AT JOINT   836   DIRECTION = MX
 PROBABLE CAUSE SINGULAR-ADDING WEAK SPRING
 K-MATRIX DIAG=  1.1614363E+06 L-MATRIX DIAG= -6.0233288E-07 EQN NO     248
 ***NOTE - VERY WEAK SPRING ADDED FOR STABILITY
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NOTE STAAD DETECTS INSTABILITIES AS EXCESSIVE LOSS OF SIGNIFICANT DIGITS
DURING DECOMPOSITION. WHEN A DECOMPOSED DIAGONAL IS LESS THAN THE
BUILT-IN REDUCTION FACTOR TIMES THE ORIGINAL STIFFNESS MATRIX DIAGONAL,
STAAD PRINTS A SINGULARITY NOTICE. THE BUILT-IN REDUCTION FACTOR
IS 1.000E-09

THE ABOVE CONDITIONS COULD ALSO BE CAUSED BY VERY STIFF OR VERY WEAK
ELEMENTS AS WELL AS TRUE SINGULARITIES.

***WARNING - INSTABILITY AT JOINT 836 DIRECTION = MZ
PROBABLE CAUSE SINGULAR-ADDING WEAK SPRING
K-MATRIX DIAG= 1.1642613E+06 L-MATRIX DIAG= -7.4319541E-07 EQN NO 249
***NOTE - VERY WEAK SPRING ADDED FOR STABILITY
**START ITERATION NO. 9

***WARNING - INSTABILITY AT JOINT 836 DIRECTION = FY
PROBABLE CAUSE SINGULAR-ADDING WEAK SPRING
K-MATRIX DIAG= 1.6396673E+04 L-MATRIX DIAG= -2.8740033E-10 EQN NO 247
***NOTE - VERY WEAK SPRING ADDED FOR STABILITY

NOTE STAAD DETECTS INSTABILITIES AS EXCESSIVE LOSS OF SIGNIFICANT DIGITS
DURING DECOMPOSITION. WHEN A DECOMPOSED DIAGONAL IS LESS THAN THE
BUILT-IN REDUCTION FACTOR TIMES THE ORIGINAL STIFFNESS MATRIX DIAGONAL,
STAAD PRINTS A SINGULARITY NOTICE. THE BUILT-IN REDUCTION FACTOR
IS 1.000E-09

THE ABOVE CONDITIONS COULD ALSO BE CAUSED BY VERY STIFF OR VERY WEAK
ELEMENTS AS WELL AS TRUE SINGULARITIES.

***WARNING - INSTABILITY AT JOINT 836 DIRECTION = MX
PROBABLE CAUSE SINGULAR-ADDING WEAK SPRING
K-MATRIX DIAG= 1.1614363E+06 L-MATRIX DIAG= -6.0233288E-07 EQN NO 248
***NOTE - VERY WEAK SPRING ADDED FOR STABILITY

***WARNING - INSTABILITY AT JOINT 836 DIRECTION = MZ
PROBABLE CAUSE SINGULAR-ADDING WEAK SPRING
K-MATRIX DIAG= 1.1642613E+06 L-MATRIX DIAG= -7.4319541E-07 EQN NO 249
***NOTE - VERY WEAK SPRING ADDED FOR STABILITY

**NOTE-Tension/Compression converged after 9 iterations, Case= 7

**START ITERATION NO. 2
**START ITERATION NO. 3
**START ITERATION NO. 4
**START ITERATION NO. 5

***WARNING - INSTABILITY AT JOINT 836 DIRECTION = MX
PROBABLE CAUSE SINGULAR-ADDING WEAK SPRING
K-MATRIX DIAG= 1.1614363E+06 L-MATRIX DIAG= -4.8219226E-07 EQN NO 248
***NOTE - VERY WEAK SPRING ADDED FOR STABILITY

NOTE STAAD DETECTS INSTABILITIES AS EXCESSIVE LOSS OF SIGNIFICANT DIGITS
DURING DECOMPOSITION. WHEN A DECOMPOSED DIAGONAL IS LESS THAN THE
BUILT-IN REDUCTION FACTOR TIMES THE ORIGINAL STIFFNESS MATRIX DIAGONAL,
STAAD PRINTS A SINGULARITY NOTICE. THE BUILT-IN REDUCTION FACTOR
IS 1.000E-09

THE ABOVE CONDITIONS COULD ALSO BE CAUSED BY VERY STIFF OR VERY WEAK
ELEMENTS AS WELL AS TRUE SINGULARITIES.

**START ITERATION NO. 6

***WARNING - INSTABILITY AT JOINT 836 DIRECTION = MX
PROBABLE CAUSE SINGULAR-ADDING WEAK SPRING
K-MATRIX DIAG= 1.1614363E+06 L-MATRIX DIAG= -4.8754737E-07 EQN NO 248
***NOTE - VERY WEAK SPRING ADDED FOR STABILITY

NOTE STAAD DETECTS INSTABILITIES AS EXCESSIVE LOSS OF SIGNIFICANT DIGITS
DURING DECOMPOSITION. WHEN A DECOMPOSED DIAGONAL IS LESS THAN THE
BUILT-IN REDUCTION FACTOR TIMES THE ORIGINAL STIFFNESS MATRIX DIAGONAL,
STAAD PRINTS A SINGULARITY NOTICE. THE BUILT-IN REDUCTION FACTOR
IS 1.000E-09

THE ABOVE CONDITIONS COULD ALSO BE CAUSED BY VERY STIFF OR VERY WEAK
ELEMENTS AS WELL AS TRUE SINGULARITIES.

**START ITERATION NO. 7

***WARNING - INSTABILITY AT JOINT 836 DIRECTION = MX
PROBABLE CAUSE SINGULAR-ADDING WEAK SPRING
K-MATRIX DIAG= 1.1614363E+06 L-MATRIX DIAG= -5.7951547E-07 EQN NO 248
***NOTE - VERY WEAK SPRING ADDED FOR STABILITY

NOTE STAAD DETECTS INSTABILITIES AS EXCESSIVE LOSS OF SIGNIFICANT DIGITS
DURING DECOMPOSITION. WHEN A DECOMPOSED DIAGONAL IS LESS THAN THE
BUILT-IN REDUCTION FACTOR TIMES THE ORIGINAL STIFFNESS MATRIX DIAGONAL,
STAAD PRINTS A SINGULARITY NOTICE. THE BUILT-IN REDUCTION FACTOR
IS 1.000E-09

THE ABOVE CONDITIONS COULD ALSO BE CAUSED BY VERY STIFF OR VERY WEAK
ELEMENTS AS WELL AS TRUE SINGULARITIES.

**START ITERATION NO. 8

***WARNING - INSTABILITY AT JOINT 836 DIRECTION = MX
PROBABLE CAUSE SINGULAR-ADDING WEAK SPRING
K-MATRIX DIAG= 1.1614363E+06 L-MATRIX DIAG= -1.8044375E-07 EQN NO 248
***NOTE - VERY WEAK SPRING ADDED FOR STABILITY

NOTE STAAD DETECTS INSTABILITIES AS EXCESSIVE LOSS OF SIGNIFICANT DIGITS
DURING DECOMPOSITION. WHEN A DECOMPOSED DIAGONAL IS LESS THAN THE
BUILT-IN REDUCTION FACTOR TIMES THE ORIGINAL STIFFNESS MATRIX DIAGONAL,
STAAD PRINTS A SINGULARITY NOTICE. THE BUILT-IN REDUCTION FACTOR
IS 1.000E-09

THE ABOVE CONDITIONS COULD ALSO BE CAUSED BY VERY STIFF OR VERY WEAK
ELEMENTS AS WELL AS TRUE SINGULARITIES.

**START ITERATION NO. 9

***WARNING - INSTABILITY AT JOINT 836 DIRECTION = MX
PROBABLE CAUSE SINGULAR-ADDING WEAK SPRING
K-MATRIX DIAG= 1.1614363E+06 L-MATRIX DIAG= -2.3003668E-07 EQN NO 248
***NOTE - VERY WEAK SPRING ADDED FOR STABILITY

NOTE STAAD DETECTS INSTABILITIES AS EXCESSIVE LOSS OF SIGNIFICANT DIGITS
DURING DECOMPOSITION. WHEN A DECOMPOSED DIAGONAL IS LESS THAN THE
BUILT-IN REDUCTION FACTOR TIMES THE ORIGINAL STIFFNESS MATRIX DIAGONAL,
STAAD PRINTS A SINGULARITY NOTICE. THE BUILT-IN REDUCTION FACTOR
IS 1.000E-09

THE ABOVE CONDITIONS COULD ALSO BE CAUSED BY VERY STIFF OR VERY WEAK
ELEMENTS AS WELL AS TRUE SINGULARITIES.

**START ITERATION NO. 10

***WARNING - INSTABILITY AT JOINT 836 DIRECTION = MX
PROBABLE CAUSE SINGULAR-ADDING WEAK SPRING
K-MATRIX DIAG= 1.1614363E+06 L-MATRIX DIAG= -5.0151721E-07 EQN NO 248
***NOTE - VERY WEAK SPRING ADDED FOR STABILITY

NOTE STAAD DETECTS INSTABILITIES AS EXCESSIVE LOSS OF SIGNIFICANT DIGITS
DURING DECOMPOSITION. WHEN A DECOMPOSED DIAGONAL IS LESS THAN THE
BUILT-IN REDUCTION FACTOR TIMES THE ORIGINAL STIFFNESS MATRIX DIAGONAL,
STAAD PRINTS A SINGULARITY NOTICE. THE BUILT-IN REDUCTION FACTOR
IS 1.000E-09

THE ABOVE CONDITIONS COULD ALSO BE CAUSED BY VERY STIFF OR VERY WEAK
ELEMENTS AS WELL AS TRUE SINGULARITIES.

***WARNING - INSTABILITY AT JOINT 836 DIRECTION = MZ
PROBABLE CAUSE SINGULAR-ADDING WEAK SPRING
K-MATRIX DIAG= 1.1642613E+06 L-MATRIX DIAG= -3.7439167E-07 EQN NO 249
***NOTE - VERY WEAK SPRING ADDED FOR STABILITY

**NOTE-Tension/Compression converged after 10 iterations, Case= 8

**START ITERATION NO. 2
**START ITERATION NO. 3
**START ITERATION NO. 4

**NOTE-Tension/Compression converged after 4 iterations, Case= 9

**START ITERATION NO. 2
**START ITERATION NO. 3
**START ITERATION NO. 4
**START ITERATION NO. 5

***WARNING - INSTABILITY AT JOINT 836 DIRECTION = MX
PROBABLE CAUSE SINGULAR-ADDING WEAK SPRING
K-MATRIX DIAG= 1.1614363E+06 L-MATRIX DIAG= -6.2282197E-07 EQN NO 248
***NOTE - VERY WEAK SPRING ADDED FOR STABILITY

NOTE STAAD DETECTS INSTABILITIES AS EXCESSIVE LOSS OF SIGNIFICANT DIGITS
DURING DECOMPOSITION. WHEN A DECOMPOSED DIAGONAL IS LESS THAN THE
BUILT-IN REDUCTION FACTOR TIMES THE ORIGINAL STIFFNESS MATRIX DIAGONAL,
STAAD PRINTS A SINGULARITY NOTICE. THE BUILT-IN REDUCTION FACTOR
IS 1.000E-09

THE ABOVE CONDITIONS COULD ALSO BE CAUSED BY VERY STIFF OR VERY WEAK
ELEMENTS AS WELL AS TRUE SINGULARITIES.

**START ITERATION NO. 6

***WARNING - INSTABILITY AT JOINT 836 DIRECTION = MX
PROBABLE CAUSE SINGULAR-ADDING WEAK SPRING
K-MATRIX DIAG= 1.1614363E+06 L-MATRIX DIAG= -6.3236803E-07 EQN NO 248
***NOTE - VERY WEAK SPRING ADDED FOR STABILITY

NOTE STAAD DETECTS INSTABILITIES AS EXCESSIVE LOSS OF SIGNIFICANT DIGITS
DURING DECOMPOSITION. WHEN A DECOMPOSED DIAGONAL IS LESS THAN THE
BUILT-IN REDUCTION FACTOR TIMES THE ORIGINAL STIFFNESS MATRIX DIAGONAL,
STAAD PRINTS A SINGULARITY NOTICE. THE BUILT-IN REDUCTION FACTOR
IS 1.000E-09

THE ABOVE CONDITIONS COULD ALSO BE CAUSED BY VERY STIFF OR VERY WEAK
ELEMENTS AS WELL AS TRUE SINGULARITIES.

**START ITERATION NO. 7

***WARNING - INSTABILITY AT JOINT 836 DIRECTION = MX
PROBABLE CAUSE SINGULAR-ADDING WEAK SPRING
K-MATRIX DIAG= 1.1614363E+06 L-MATRIX DIAG= -5.6088902E-07 EQN NO 248
***NOTE - VERY WEAK SPRING ADDED FOR STABILITY

NOTE STAAD DETECTS INSTABILITIES AS EXCESSIVE LOSS OF SIGNIFICANT DIGITS
DURING DECOMPOSITION. WHEN A DECOMPOSED DIAGONAL IS LESS THAN THE
BUILT-IN REDUCTION FACTOR TIMES THE ORIGINAL STIFFNESS MATRIX DIAGONAL,
STAAD PRINTS A SINGULARITY NOTICE. THE BUILT-IN REDUCTION FACTOR
IS 1.000E-09

THE ABOVE CONDITIONS COULD ALSO BE CAUSED BY VERY STIFF OR VERY WEAK
ELEMENTS AS WELL AS TRUE SINGULARITIES.

***WARNING - INSTABILITY AT JOINT 836 DIRECTION = MZ
PROBABLE CAUSE SINGULAR-ADDING WEAK SPRING
K-MATRIX DIAG= 1.1642613E+06 L-MATRIX DIAG= -7.4319541E-07 EQN NO 249
***NOTE - VERY WEAK SPRING ADDED FOR STABILITY
**START ITERATION NO. 8

***WARNING - INSTABILITY AT JOINT 836 DIRECTION = FY
PROBABLE CAUSE SINGULAR-ADDING WEAK SPRING
K-MATRIX DIAG= 1.6396673E+04 L-MATRIX DIAG= -2.8740033E-10 EQN NO 247
***NOTE - VERY WEAK SPRING ADDED FOR STABILITY

NOTE STAAD DETECTS INSTABILITIES AS EXCESSIVE LOSS OF SIGNIFICANT DIGITS
DURING DECOMPOSITION. WHEN A DECOMPOSED DIAGONAL IS LESS THAN THE
BUILT-IN REDUCTION FACTOR TIMES THE ORIGINAL STIFFNESS MATRIX DIAGONAL,
STAAD PRINTS A SINGULARITY NOTICE. THE BUILT-IN REDUCTION FACTOR
IS 1.000E-09

THE ABOVE CONDITIONS COULD ALSO BE CAUSED BY VERY STIFF OR VERY WEAK
ELEMENTS AS WELL AS TRUE SINGULARITIES.

***WARNING - INSTABILITY AT JOINT 836 DIRECTION = MX
PROBABLE CAUSE SINGULAR-ADDING WEAK SPRING
K-MATRIX DIAG= 1.1614363E+06 L-MATRIX DIAG= -6.0233288E-07 EQN NO 248
***NOTE - VERY WEAK SPRING ADDED FOR STABILITY

***WARNING - INSTABILITY AT JOINT 836 DIRECTION = MZ
PROBABLE CAUSE SINGULAR-ADDING WEAK SPRING
K-MATRIX DIAG= 1.1642613E+06 L-MATRIX DIAG= -7.4319541E-07 EQN NO 249
***NOTE - VERY WEAK SPRING ADDED FOR STABILITY

**NOTE-Tension/Compression converged after 8 iterations, Case= 10

**NOTE-Tension/Compression converged after 1 iterations, Case= 101

**NOTE-Tension/Compression converged after 1 iterations, Case= 102

**NOTE-Tension/Compression converged after 1 iterations, Case= 103

**NOTE-Tension/Compression converged after 1 iterations, Case= 104

**NOTE-Tension/Compression converged after 1 iterations, Case= 105

**NOTE-Tension/Compression converged after 1 iterations, Case= 106

**NOTE-Tension/Compression converged after 1 iterations, Case= 107

**NOTE-Tension/Compression converged after 1 iterations, Case= 108

**NOTE-Tension/Compression converged after 1 iterations, Case= 109

**NOTE-Tension/Compression converged after 1 iterations, Case= 110

**NOTE-Tension/Compression converged after 1 iterations, Case= 111

**NOTE-Tension/Compression converged after 1 iterations, Case= 112

**NOTE-Tension/Compression converged after 1 iterations, Case= 113

**NOTE-Tension/Compression converged after 1 iterations, Case= 114

**NOTE-Tension/Compression converged after 1 iterations, Case= 115

**START ITERATION NO. 2
**START ITERATION NO. 3

**NOTE-Tension/Compression converged after 3 iterations, Case= 116

**NOTE-Tension/Compression converged after 1 iterations, Case= 201

**NOTE-Tension/Compression converged after 1 iterations, Case= 202

**NOTE-Tension/Compression converged after 1 iterations, Case= 203

**NOTE-Tension/Compression converged after 1 iterations, Case= 204

**NOTE-Tension/Compression converged after 1 iterations, Case= 205

**NOTE-Tension/Compression converged after 1 iterations, Case= 206

**NOTE-Tension/Compression converged after 1 iterations, Case= 207

**NOTE-Tension/Compression converged after 1 iterations, Case= 208

**NOTE-Tension/Compression converged after 1 iterations, Case= 209

**NOTE-Tension/Compression converged after 1 iterations, Case= 210

**NOTE-Tension/Compression converged after 1 iterations, Case= 211

**NOTE-Tension/Compression converged after 1 iterations, Case= 212

**NOTE-Tension/Compression converged after 1 iterations, Case= 213

**NOTE-Tension/Compression converged after 1 iterations, Case= 214

**NOTE-Tension/Compression converged after 1 iterations, Case= 215

**START ITERATION NO. 2

**NOTE-Tension/Compression converged after 3 iterations, Case= 216

**NOTE-Tension/Compression converged after 1 iterations, Case= 217

**NOTE-Tension/Compression converged after 1 iterations, Case= 218

329. LOAD LIST 101 TO 116
330. START CONCRETE DESIGN

```
331. CODE ACI
332. TRACK 2 ALL
333. FC 35000 MEMB 529
334. CLB 0.05 MEMB 529
335. CLS 0.05 ALL
336. MAXMAIN 25 ALL
337. MINMAIN 25 ALL
338. MINSEC 10 ALL
339. DESIGN COLUMN 529
```



```
================================================================

     COLUMN  NO.   529  DESIGN PER ACI 318-08 - AXIAL + BENDING


FY - 413.7  FC -  35.0 MPA,  SQRE SIZE - 600.0 X 600.0 MMS, TIED
        ONLY MINIMUM STEEL IS REQUIRED.
        AREA OF STEEL REQUIRED =  3600.0  SQ. MM

BAR CONFIGURATION       REINF PCT.   LOAD   LOCATION   PHI
----------------------------------------------------------

  8 - 25 MM               1.091      101      END     0.650
(PROVIDE EQUAL NUMBER OF BARS ON EACH FACE)
TIE BAR NUMBER   10 SPACING 400.00 MM

    COLUMN INTERACTION: MOMENT ABOUT Z -AXIS (KN-MET)

 ----------------------------------------------------------
   P0        Pn max    P-bal.    M-bal.     e-bal. (MM)
12217.70   9774.16   4481.49   1062.73     237.1
   M0       P-tens.   Des.Pn    Des.Mn     e/h
  420.14  -1624.53     59.66      0.00    0.00000
 ----------------------------------------------------------

    COLUMN INTERACTION: MOMENT ABOUT Y -AXIS (KN-MET)

 ----------------------------------------------------------
   P0        Pn max    P-bal.    M-bal.     e-bal. (MM)
12217.70   9774.16   4481.49   1062.73     237.1
   M0       P-tens.   Des.Pn    Des.Mn     e/h
  420.14  -1624.53     59.66      0.00    0.00000
 ----------------------------------------------------------
                          Pn       Mn       Pn       Mn    (@ Z )
          |              9022.30   709.00  4511.15  1061.99
       P0 |*             8270.44   812.98  3759.29  1019.91
          | *            7518.58   897.94  3007.43   956.53
   Pn,max|__*           6766.73   962.48  2255.58   875.37
          |   *          6014.87  1009.20  1503.72   752.23
   Pn     |    *         5263.01  1041.19   751.86   595.66
  NOMINAL|     *          Pn       Mn       Pn       Mn    (@ Y )
    AXIAL|      *       9022.30   709.00  4511.15  1061.99
COMPRESSION|       *     8270.44   812.98  3759.29  1019.91
        Pb|-------*Mb    7518.58   897.94  3007.43   956.53
          |      *       6766.73   962.48  2255.58   875.37
__________|____*_______  6014.87  1009.20  1503.72   752.23
          |  * M0   Mn,  5263.01  1041.19   751.86   595.66
          | *   BENDING
    P-tens|*     MOMENT
          |


********************END OF COLUMN DESIGN RESULTS********************
```

```
340. END CONCRETE DESIGN
341. FINISH


          ********** END OF THE STAAD.Pro RUN **********

            **** DATE= OCT 19,2020   TIME= 18:47:15 ****

      ************************************************************
      *   For technical assistance on STAAD.Pro, please visit    *
      *   http://selectservices.bentley.com/en-US/               *
      *                                                          *
      *   Details about additional assistance from               *
      *   Bentley and Partners can be found at program menu      *
      *   Help->Technical Support                                *
      *                                                          *
      *      Copyright (c) 1997-2014 Bentley Systems, Inc.       *
      *                http://www.bentley.com                    *
      ************************************************************
```

ATTACHMENT 7 STAAD OUTPUT – FOUNDATION F3

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```
**************************************************
*                                                *
*          STAAD.Pro V8i SELECTseries5           *
*          Version  20.07.10.64                  *
*          Proprietary Program of                *
*          Bentley Systems, Inc.                 *
*          Date=    OCT 19, 2020                 *
*          Time=    19:12:48                     *
*                                                *
*     USER ID: LNT ECC DIV                       *
**************************************************
```

```
    1. STAAD SPACE
INPUT FILE: F3.STD
    2. START JOB INFORMATION
    3. ENGINEER DATE 08/05/2020
    4. JOB NAME FDN. MKD. F3
    5. JOB CLIENT MA'ADEN
    6. JOB NO O19045
    7. JOB REV 0
    8. ENGINEER NAME SOR
    9. END JOB INFORMATION
   10. INPUT WIDTH 79
   11. SET NL 200
   12. SET ITERLIM 15
   13. SET PRINT 1
   14. UNIT METER KN
   15. JOINT COORDINATES
   16. 457 -0.5 911.5 0.42438; 460 -0.25 911.5 0.42438; 461 -0.25 911.5 0.17538
   17. 462 0 911.5 0.42438; 463 0 911.5 0.17538; 464 0.25 911.5 0.42438
   18. 465 0.25 911.5 0.17538; 466 0.5 911.5 0.42438; 467 0.5 911.5 0.17538
   19. 468 0.75 911.5 0.42438; 469 0.75 911.5 0.17538; 475 -0.5 911.5 0.67438
   20. 476 -0.25 911.5 0.67438; 477 0 911.5 0.67438; 478 0.25 911.5 0.67438
   21. 479 0.5 911.5 0.67438; 480 0.75 911.5 0.67438; 483 -0.7 911.5 0.87538
   22. 484 -0.5 911.5 0.87538; 485 -0.25 911.5 0.87538; 486 0 911.5 0.87538
   23. 487 0.25 911.5 0.87538; 488 0.5 911.5 0.87538; 489 0.75 911.5 0.87538
   24. 498 0 911.5 -0.62362; 499 0 911.5 -0.87362; 500 0.25 911.5 -0.62362
   25. 501 0.25 911.5 -0.87362; 502 0.5 911.5 -0.62362; 503 0.5 911.5 -0.87362
   26. 504 0.75 911.5 -0.62362; 505 0.75 911.5 -0.87362; 512 -0.25 911.5 -0.37362
   27. 513 0 911.5 -0.37362; 514 0.25 911.5 -0.37362; 515 0.5 911.5 -0.37362
   28. 516 0.75 911.5 -0.37362; 521 -0.25 911.5 -0.12462; 522 0 911.5 -0.12462
   29. 523 0.25 911.5 -0.12462; 524 0.5 911.5 -0.12462; 525 0.75 911.5 -0.12462
   30. 528 0 913 -0.125; 803 0 911.5 -1.12462; 805 0.25 911.5 -1.12462
   31. 807 0.5 911.5 -1.12462; 809 0.75 911.5 -1.12462; 837 1 911.5 0.42438
   32. 838 1 911.5 0.17538; 839 1.25 911.5 0.42438; 840 1.25 911.5 0.17538
   33. 841 1.5 911.5 0.42438; 842 1.5 911.5 0.17538; 843 1 911.5 0.67438
   34. 844 1.25 911.5 0.67438; 845 1.5 911.5 0.67438; 846 1 911.5 0.87538
   35. 847 1.25 911.5 0.87538; 848 1.5 911.5 0.87538; 849 1 911.5 -0.62362
   36. 850 1 911.5 -0.87362; 851 1.25 911.5 -0.62362; 852 1.25 911.5 -0.87362
   37. 853 1.5 911.5 -0.62362; 854 1.5 911.5 -0.87362; 855 1 911.5 -0.37362
   38. 856 1.25 911.5 -0.37362; 857 1.5 911.5 -0.37362; 858 1 911.5 -0.12462
```



```
   39. 859 1.25 911.5 -0.12462; 860 1.5 911.5 -0.12462; 861 1 911.5 -1.12462
   40. 862 1.25 911.5 -1.12462; 863 1.5 911.5 -1.12462; 864 1.8 911.5 0.42438
   41. 865 1.8 911.5 0.17538; 866 1.8 911.5 0.67438; 867 1.8 911.5 0.87538
   42. 868 1.8 911.5 -0.62362; 869 1.8 911.5 -0.87362; 870 1.8 911.5 -0.37362
   43. 871 1.8 911.5 -0.12462; 872 1.8 911.5 -1.12462; 873 -0.075 911.5 -1.12462
   44. 874 -0.139786 911.5 -0.917303; 875 -0.210961 911.5 -0.689545
   45. 876 -0.304377 911.5 -0.390613; 877 -0.156584 911.5 -0.672552
   46. 878 -0.375267 911.5 -0.163766; 879 -0.460676 911.5 0.109544
   47. 880 -0.553808 911.5 0.407565; 881 -0.406868 911.5 0.126359
   48. 882 -0.624982 911.5 0.635323
   49. MEMBER INCIDENCES
   50. 529 522 528
   51. ELEMENT INCIDENCES SHELL
   52. 467 460 462 463 461; 468 462 464 465 463; 469 464 466 467 465
   53. 470 466 468 469 467; 474 475 476 460 457; 475 476 477 462 460
   54. 476 477 478 464 462; 477 478 479 466 464; 478 479 480 468 466
   55. 482 484 485 476 475; 483 485 486 477 476; 484 486 487 478 477
   56. 485 487 488 479 478; 486 488 489 480 479; 500 498 500 501 499
   57. 501 500 502 503 501; 502 502 504 505 503; 508 513 514 500 498
   58. 509 514 515 502 500; 510 515 516 504 502; 515 521 522 513 512
   59. 516 522 523 514 513; 517 523 524 515 514; 518 524 525 516 515
   60. 523 461 463 522 521; 524 463 465 523 522; 525 465 467 524 523
   61. 526 467 469 525 524; 788 499 501 805 803; 789 501 503 807 805
   62. 790 503 505 809 807; 815 468 837 838 469; 816 837 839 840 838
   63. 817 839 841 842 840; 818 480 843 837 468; 819 843 844 839 837
   64. 820 844 845 841 839; 821 489 846 843 480; 822 846 847 844 843
   65. 823 847 848 845 844; 824 504 849 850 505; 825 849 851 852 850
   66. 826 851 853 854 852; 827 516 855 849 504; 828 855 856 851 849
   67. 829 856 857 853 851; 830 525 858 855 516; 831 858 859 856 855
   68. 832 859 860 857 856; 833 469 838 858 525; 834 838 840 859 858
   69. 835 840 842 860 859; 836 505 850 861 809; 837 850 852 862 861
   70. 838 852 854 863 862; 839 841 864 865 842; 840 845 866 864 841
   71. 841 848 867 866 845; 842 853 868 869 854; 843 857 870 868 853
   72. 844 860 871 870 857; 845 842 865 871 860; 846 854 869 872 863
   73. 867 483 484 475 882; 868 882 475 457 880; 869 457 460 461 881
   74. 870 880 457 881 879; 871 881 461 521; 872 879 881 521 878; 873 878 521 512 876
   75. 874 876 512 877 875; 875 512 513 498 877; 876 877 498 499; 877 875 877 499 874
   76. 878 874 499 803 873
   77. START GROUP DEFINITION
   78. ELEMENT
   79. _RAFT 467 TO 470 474 TO 478 482 TO 486 500 TO 502 508 TO 510 515 TO 518 523 -
   80. 524 TO 526 788 TO 790 815 TO 846
   81. _RAFT_MOM 467 TO 470 474 TO 478 482 TO 486 500 TO 502 508 TO 510 517 518 525 -
   82. 526 788 TO 790 815 TO 829 831 832 834 TO 846 867 TO 878
   83. _RAFT_SHEAR 470 474 TO 478 482 TO 486 500 TO 502 510 518 526 788 TO 790 815 -
   84. 816 TO 846 867 868 876 TO 878
   85. MEMBER
   86. END GROUP DEFINITION
   87. ELEMENT PROPERTY
   88. 467 TO 470 474 TO 478 482 TO 486 500 TO 502 508 TO 510 515 TO 518 523 TO 526 -
   89. 788 TO 790 815 TO 846 867 TO 878 THICKNESS 0.5
   90. DEFINE MATERIAL START
   91. ISOTROPIC CONCRETE
   92. E 2.17185E+007
   93. POISSON 0.17
   94. DENSITY 25
```

```
 95. ALPHA 1E-005
 96. DAMP 0.05
 97. TYPE CONCRETE
 98. STRENGTH FCU 27579
 99. END DEFINE MATERIAL
100. MEMBER PROPERTY AMERICAN
101. 529 PRIS YD 0.6 ZD 0.6
102. CONSTANTS
103. MATERIAL CONCRETE ALL
104. SUPPORTS
105. 467 TO 470 474 TO 478 482 TO 486 500 TO 502 508 TO 510 515 TO 518 523 TO 526 -
106. 788 TO 790 815 TO 846 867 TO 878 PLATE MAT DIRECT Y SUBGRADE 57600 PRINT
```

JOINT	INFLUENCE AREA
457	0.040912
460	0.061692
461	0.054136
462	0.062375
463	0.068625
464	0.062375
465	0.068625
466	0.062375
467	0.068625
468	0.062375
469	0.068625
475	0.042598
476	0.056375
477	0.056375
478	0.056375
479	0.056375
480	0.056375
483	0.010572
484	0.022158
485	0.025125
486	0.025125
487	0.025125
488	0.025125
489	0.025125
498	0.049785
499	0.052789
500	0.062500
501	0.062625
502	0.062500
503	0.062625
504	0.062500
505	0.062625
512	0.040970
513	0.061680
514	0.062375
515	0.062375
516	0.062375
521	0.056906
522	0.068625
523	0.068625

524	0.068625
525	0.068625
803	0.021205
805	0.031375
807	0.031375
809	0.031375
837	0.062375
838	0.068625
839	0.062375
840	0.068625
841	0.068612
842	0.075487
843	0.056375
844	0.056375
845	0.062012
846	0.025125
847	0.025125
848	0.027637
849	0.062500
850	0.062625
851	0.062500
852	0.062625
853	0.068750
854	0.068887
855	0.062375
856	0.062375
857	0.068612
858	0.068625
859	0.068625
860	0.075487
861	0.031375
862	0.031375
863	0.034512
864	0.037425
865	0.041175
866	0.033825
867	0.015075
868	0.037500
869	0.037575
870	0.037425
871	0.041175
872	0.018825
873	0.005108
874	0.014543
875	0.009194
876	0.008949
877	0.029258
878	0.014750
879	0.009776
880	0.008876
881	0.031132
882	0.015387
TOTAL	4.375000

```
107. ********************************************************************************
108. LOAD 1 LOADTYPE NONE  TITLE DEAD LOAD ( DL )
109. SELFWEIGHT Y -1
```

```
110. ELEMENT LOAD
111. * LOAD DUE TO BACKFILL
112. 467 TO 470 474 TO 478 482 TO 486 500 TO 502 508 TO 510 515 TO 518 523 TO 526 -
113. 788 TO 790 815 TO 846 867 TO 878 PR GY -15
114. * LOAD DUE TO SURCHARGE
115. 467 TO 470 474 TO 478 482 TO 486 500 TO 502 508 TO 510 515 TO 518 523 TO 526 -
116. 788 TO 790 815 TO 846 867 TO 878 PR GY -20
117. JOINT LOAD
118. 528 FX -0.034 FY -27.7 FZ 5
119. ********************************************************************************
120. LOAD 2 LOADTYPE NONE  TITLE LIVE LOAD ( LL )
121. JOINT LOAD
122. 528 FX 0.13 FY -133.12 FZ 25
123. ********************************************************************************
124. LOAD 3 LOADTYPE NONE  TITLE WIND LOAD +X  ( WLX1 )
125. JOINT LOAD
126. 528 FX 8.7 FY -15.2 FZ 0.02
127. ********************************************************************************
128. LOAD 4 LOADTYPE NONE  TITLE WIND LOAD -X  ( WLX2 )
129. JOINT LOAD
130. 528 FX -8.7 FY 15.2 FZ -0.02
131. ********************************************************************************
132. LOAD 5 LOADTYPE NONE  TITLE WIND LOAD +Z  ( WLZ1 )
133. JOINT LOAD
134. 528 FX -0.44 FY 6 FZ 5.8
135. ********************************************************************************
136. LOAD 6 LOADTYPE NONE  TITLE WIND LOAD -Z  ( WLZ2 )
137. JOINT LOAD
138. 528 FX 0.44 FY -6 FZ -5.8
139. ********************************************************************************
140. LOAD 7 LOADTYPE NONE  TITLE TEMPERATURE LOAD +25  ( TL1 )
141. JOINT LOAD
142. 528 FX -1.1 FY 2 FZ -67.4
143. ********************************************************************************
144. LOAD 8 LOADTYPE NONE  TITLE TEMPERATURE LOAD -25  ( TL2 )
145. JOINT LOAD
146. 528 FX 1.1 FY -2 FZ 67.4
147. ********************************************************************************
148. LOAD 9 LOADTYPE NONE  TITLE SEISMIC LOAD  ( ELX )
149. JOINT LOAD
150. 528 FX 1 FY -3.35 FZ 0.008
151. ********************************************************************************
152. LOAD 10 LOADTYPE NONE  TITLE SEISMIC LOAD  ( ELZ )
153. JOINT LOAD
154. 528 FX -0.127 FY 1.428 FZ 0.75
155. ********************************************************************************
156. PERFORM ANALYSIS
```


P R O B L E M S T A T I S T I C S

NUMBER OF JOINTS 93 NUMBER OF MEMBERS 1
NUMBER OF PLATES 75 NUMBER OF SOLIDS 0
NUMBER OF SURFACES 0 NUMBER OF SUPPORTS 92

SOLVER USED IS THE OUT-OF-CORE BASIC SOLVER

ORIGINAL/FINAL BAND-WIDTH= 92/ 15/ 48 DOF
TOTAL PRIMARY LOAD CASES = 10, TOTAL DEGREES OF FREEDOM = 282
TOTAL LOAD COMBINATION CASES = 0 SO FAR.
SIZE OF STIFFNESS MATRIX = 14 DOUBLE KILO-WORDS
REQRD/AVAIL. DISK SPACE = 12.6/ 142645.2 MB

```
157. CHANGE
158. SUPPORTS
159. 467 TO 470 474 TO 478 482 TO 486 500 TO 502 508 TO 510 515 TO 518 523 TO 526 -
160. 788 TO 790 815 TO 846 867 TO 877 -
161. 878 PLATE MAT DIRECT Y SUBGRADE 57600 COMPRESSION
162. ********************************************************************************
163. ***FACTORED LOAD COMBINATIONS FOR STRENGTH DESIGN
164. LOAD 101 1.4DL
165. REPEAT LOAD
166. 1 1.4
167. PERFORM ANALYSIS

**NOTE-Tension/Compression converged after  1 iterations, Case=  101

168. CHANGE
169. LOAD 102 1.4DL + 1.7LL
170. REPEAT LOAD
171. 1 1.4 2 1.7
172. PERFORM ANALYSIS
**START ITERATION NO.           2

**NOTE-Tension/Compression converged after  2 iterations, Case=  102

173. CHANGE
174. LOAD 103 1.2DL + 1.6WLX1 + 1.0 LL
```

```
 175. REPEAT LOAD
 176. 1 1.2 3 1.6 2 1.0
 177. PERFORM ANALYSIS

**NOTE-Tension/Compression converged after  1 iterations, Case=  103



 178. CHANGE
 179. LOAD 104 1.2DL + 1.6WLX2 + 1.0 LL
 180. REPEAT LOAD
 181. 1 1.2 4 1.6 2 1.0
 182. PERFORM ANALYSIS

**NOTE-Tension/Compression converged after  1 iterations, Case=  104



 183. CHANGE
 184. LOAD 105 1.2DL + 1.6WLZ1 + 1.0 LL
 185. REPEAT LOAD
 186. 1 1.2 5 1.6 2 1.0
 187. PERFORM ANALYSIS

**NOTE-Tension/Compression converged after  1 iterations, Case=  105



 188. CHANGE
 189. LOAD 106 1.2DL + 1.6WLZ2 + 1.0 LL
 190. REPEAT LOAD
 191. 1 1.2 6 1.6 2 1.0
 192. PERFORM ANALYSIS

**NOTE-Tension/Compression converged after  1 iterations, Case=  106



 193. CHANGE
 194. LOAD 107 0.9DL + 1.6WLX1
 195. REPEAT LOAD
 196. 1 0.9 3 1.6
 197. PERFORM ANALYSIS
```



```
**NOTE-Tension/Compression converged after  1 iterations, Case=  107



 198. CHANGE
 199. LOAD 108 0.9DL + 1.6WLX2
 200. REPEAT LOAD
 201. 1 0.9 4 1.6
 202. PERFORM ANALYSIS

**NOTE-Tension/Compression converged after  1 iterations, Case=  108



 203. CHANGE
 204. LOAD 109 0.9DL + 1.6WLZ1
 205. REPEAT LOAD
 206. 1 0.9 5 1.6
 207. PERFORM ANALYSIS

**NOTE-Tension/Compression converged after  1 iterations, Case=  109



 208. CHANGE
 209. LOAD 110 0.9DL + 1.6WLZ2
 210. REPEAT LOAD
 211. 1 0.9 6 1.6
 212. PERFORM ANALYSIS

**NOTE-Tension/Compression converged after  1 iterations, Case=  110



 213. CHANGE
 214. LOAD 111 1.2DL + 1.0ELX + 1.0 LL
 215. REPEAT LOAD
 216. 1 1.2 2 1.0 9 1.0
 217. PERFORM ANALYSIS
```

**NOTE-Tension/Compression converged after 1 iterations, Case= 111

218. CHANGE
219. LOAD 112 1.2DL + 1.0ELZ + 1.0 LL
220. REPEAT LOAD
221. 1 1.2 2 1.0 10 1.0
222. PERFORM ANALYSIS

**NOTE-Tension/Compression converged after 1 iterations, Case= 112

223. CHANGE
224. LOAD 113 0.9DL + 1.0ELX
225. REPEAT LOAD
226. 1 0.9 9 1.0
227. PERFORM ANALYSIS

**NOTE-Tension/Compression converged after 1 iterations, Case= 113

228. CHANGE
229. LOAD 114 0.9DL + 1.0ELZ
230. REPEAT LOAD
231. 1 0.9 10 1.0
232. PERFORM ANALYSIS

**NOTE-Tension/Compression converged after 1 iterations, Case= 114

233. CHANGE
234. LOAD 115 1.4DL + 1.7LL + 1.0 TL1
235. REPEAT LOAD
236. 1 1.4 2 1.7 7 1.0
237. PERFORM ANALYSIS
**START ITERATION NO. 2
**START ITERATION NO. 3

**NOTE-Tension/Compression converged after 3 iterations, Case= 115

238. CHANGE
239. LOAD 116 1.4DL + 1.7LL + 1.0 TL2
240. REPEAT LOAD
241. 1 1.4 2 1.7 8 1.0
242. PERFORM ANALYSIS
**START ITERATION NO. 2
**START ITERATION NO. 3

**NOTE-Tension/Compression converged after 3 iterations, Case= 116

243. CHANGE
244. **
245. * UNFACTORED LOAD COMBINATIONS FOR STABILITY CHECK
246. LOAD 201 1.0DL
247. REPEAT LOAD
248. 1 1.0
249. PERFORM ANALYSIS

**NOTE-Tension/Compression converged after 1 iterations, Case= 201

250. CHANGE
251. LOAD 202 1.0DL + 1.0LL
252. REPEAT LOAD
253. 1 1.0 2 1.0
254. PERFORM ANALYSIS

**NOTE-Tension/Compression converged after 1 iterations, Case= 202

255. CHANGE
256. LOAD 203 1.0DL + 1.0WLX1
257. REPEAT LOAD
258. 1 1.0 3 1.0
259. PERFORM ANALYSIS

**NOTE-Tension/Compression converged after 1 iterations, Case= 203

260. CHANGE
261. LOAD 204 1.0DL + 1.0WLX2
262. REPEAT LOAD
263. 1 1.0 4 1.0
264. PERFORM ANALYSIS

**NOTE-Tension/Compression converged after 1 iterations, Case= 204

265. CHANGE
266. LOAD 205 1.0DL + 1.0WLZ1
267. REPEAT LOAD
268. 1 1.0 5 1.0
269. PERFORM ANALYSIS

**NOTE-Tension/Compression converged after 1 iterations, Case= 205

270. CHANGE
271. LOAD 206 1.0DL + 1.0WLZ2
272. REPEAT LOAD
273. 1 1.0 6 1.0
274. PERFORM ANALYSIS

**NOTE-Tension/Compression converged after 1 iterations, Case= 206

275. CHANGE
276. LOAD 207 1.0DL + 0.75WLX1 + 0.75LL
277. REPEAT LOAD
278. 1 1.0 3 0.75 2 0.75
279. PERFORM ANALYSIS

**NOTE-Tension/Compression converged after 1 iterations, Case= 207

280. CHANGE
281. LOAD 208 1.0DL + 0.75WLX2 + 0.75LL
282. REPEAT LOAD
283. 1 1.0 4 0.75 2 0.75
284. PERFORM ANALYSIS

**NOTE-Tension/Compression converged after 1 iterations, Case= 208

285. CHANGE
286. LOAD 209 1.0DL + 0.75WLZ1 + 0.75LL
287. REPEAT LOAD
288. 1 1.0 5 0.75 2 0.75
289. PERFORM ANALYSIS

**NOTE-Tension/Compression converged after 1 iterations, Case= 209

290. CHANGE
291. LOAD 210 1.0DL + 0.75WLZ2 + 0.75LL
292. REPEAT LOAD
293. 1 1.0 6 0.75 2 0.75
294. PERFORM ANALYSIS

**NOTE-Tension/Compression converged after 1 iterations, Case= 210

295. CHANGE
296. LOAD 211 0.6DL + 1.0WLX1
297. REPEAT LOAD
298. 1 0.6 3 1.0
299. PERFORM ANALYSIS

**NOTE-Tension/Compression converged after 1 iterations, Case= 211

300. CHANGE
301. LOAD 212 0.6DL + 1.0WLX2
302. REPEAT LOAD
303. 1 0.6 4 1.0

```
 304. PERFORM ANALYSIS

**NOTE-Tension/Compression converged after  1 iterations, Case=  212



 305. CHANGE
 306. LOAD 213 0.6DL + 1.0WLZ1
 307. REPEAT LOAD
 308. 1 0.6 5 1.0
 309. PERFORM ANALYSIS

**NOTE-Tension/Compression converged after  1 iterations, Case=  213



 310. CHANGE
 311. LOAD 214 0.6DL + 1.0WLZ2
 312. REPEAT LOAD
 313. 1 0.6 6 1.0
 314. PERFORM ANALYSIS

**NOTE-Tension/Compression converged after  1 iterations, Case=  214



 315. CHANGE
 316. LOAD 215 1.0DL + 1.0LL + 1.0TL1
 317. REPEAT LOAD
 318. 1 1.0 2 1.0 7 1.0
 319. PERFORM ANALYSIS
**START ITERATION NO.           2
**START ITERATION NO.           3

**NOTE-Tension/Compression converged after  3 iterations, Case=  215



 320. CHANGE
 321. LOAD 216 1.0DL + 1.0LL + 1.0TL2
 322. REPEAT LOAD
 323. 1 1.0 2 1.0 8 1.0
 324. PERFORM ANALYSIS
```



```
**START ITERATION NO.           2
**START ITERATION NO.           3

**NOTE-Tension/Compression converged after  3 iterations, Case=  216



 325. CHANGE
 326. LOAD 217 0.6DL + 0.7ELX
 327. REPEAT LOAD
 328. 1 0.6 9 0.7
 329. PERFORM ANALYSIS

**NOTE-Tension/Compression converged after  1 iterations, Case=  217



 330. CHANGE
 331. LOAD 218 0.6DL + 0.7ELZ
 332. REPEAT LOAD
 333. 1 0.6 10 0.7
 334. PERFORM ANALYSIS

**NOTE-Tension/Compression converged after  1 iterations, Case=  218



 335. CHANGE
 336. ******************************************************************************
 337. PERFORM ANALYSIS

  ** ALL CASES BEING MADE ACTIVE BEFORE RE-ANALYSIS. **

**NOTE-Tension/Compression converged after  1 iterations, Case=    1



**START ITERATION NO.           2
**START ITERATION NO.           3
**START ITERATION NO.           4
**START ITERATION NO.           5

**NOTE-Tension/Compression converged after  5 iterations, Case=    2
```

**NOTE-Tension/Compression converged after 1 iterations, Case= 3

**START ITERATION NO. 2

***WARNING - INSTABILITY AT JOINT 872 DIRECTION = FY
PROBABLE CAUSE SINGULAR-ADDING WEAK SPRING
K-MATRIX DIAG= 1.6453036E+04 L-MATRIX DIAG= -2.0736479E-10 EQN NO 280
***NOTE - VERY WEAK SPRING ADDED FOR STABILITY

NOTE STAAD DETECTS INSTABILITIES AS EXCESSIVE LOSS OF SIGNIFICANT DIGITS
DURING DECOMPOSITION. WHEN A DECOMPOSED DIAGONAL IS LESS THAN THE
BUILT-IN REDUCTION FACTOR TIMES THE ORIGINAL STIFFNESS MATRIX DIAGONAL,
STAAD PRINTS A SINGULARITY NOTICE. THE BUILT-IN REDUCTION FACTOR
IS 1.000E-09

THE ABOVE CONDITIONS COULD ALSO BE CAUSED BY VERY STIFF OR VERY WEAK
ELEMENTS AS WELL AS TRUE SINGULARITIES.

***WARNING - INSTABILITY AT JOINT 872 DIRECTION = MX
PROBABLE CAUSE SINGULAR-ADDING WEAK SPRING
K-MATRIX DIAG= 1.2849652E+06 L-MATRIX DIAG= -6.5588392E-07 EQN NO 281
***NOTE - VERY WEAK SPRING ADDED FOR STABILITY

***WARNING - INSTABILITY AT JOINT 872 DIRECTION = MZ
PROBABLE CAUSE SINGULAR-ADDING WEAK SPRING
K-MATRIX DIAG= 1.1596877E+06 L-MATRIX DIAG= -3.7089922E-07 EQN NO 282
***NOTE - VERY WEAK SPRING ADDED FOR STABILITY

**NOTE-Tension/Compression converged after 2 iterations, Case= 4

**START ITERATION NO. 2
**START ITERATION NO. 3
**START ITERATION NO. 4
**START ITERATION NO. 5

***WARNING - INSTABILITY AT JOINT 872 DIRECTION = MX
PROBABLE CAUSE SINGULAR-ADDING WEAK SPRING
K-MATRIX DIAG= 1.2849652E+06 L-MATRIX DIAG= -2.7916394E-07 EQN NO 281
***NOTE - VERY WEAK SPRING ADDED FOR STABILITY

NOTE STAAD DETECTS INSTABILITIES AS EXCESSIVE LOSS OF SIGNIFICANT DIGITS
DURING DECOMPOSITION. WHEN A DECOMPOSED DIAGONAL IS LESS THAN THE
BUILT-IN REDUCTION FACTOR TIMES THE ORIGINAL STIFFNESS MATRIX DIAGONAL,
STAAD PRINTS A SINGULARITY NOTICE. THE BUILT-IN REDUCTION FACTOR
IS 1.000E-09

THE ABOVE CONDITIONS COULD ALSO BE CAUSED BY VERY STIFF OR VERY WEAK
ELEMENTS AS WELL AS TRUE SINGULARITIES.

**START ITERATION NO. 6

***WARNING - INSTABILITY AT JOINT 872 DIRECTION = MX
PROBABLE CAUSE SINGULAR-ADDING WEAK SPRING
K-MATRIX DIAG= 1.2849652E+06 L-MATRIX DIAG= -2.0582229E-07 EQN NO 281
***NOTE - VERY WEAK SPRING ADDED FOR STABILITY

NOTE STAAD DETECTS INSTABILITIES AS EXCESSIVE LOSS OF SIGNIFICANT DIGITS
DURING DECOMPOSITION. WHEN A DECOMPOSED DIAGONAL IS LESS THAN THE
BUILT-IN REDUCTION FACTOR TIMES THE ORIGINAL STIFFNESS MATRIX DIAGONAL,
STAAD PRINTS A SINGULARITY NOTICE. THE BUILT-IN REDUCTION FACTOR
IS 1.000E-09

THE ABOVE CONDITIONS COULD ALSO BE CAUSED BY VERY STIFF OR VERY WEAK
ELEMENTS AS WELL AS TRUE SINGULARITIES.

**START ITERATION NO. 7

***WARNING - INSTABILITY AT JOINT 872 DIRECTION = MX
PROBABLE CAUSE SINGULAR-ADDING WEAK SPRING
K-MATRIX DIAG= 1.2849652E+06 L-MATRIX DIAG= -2.1513551E-07 EQN NO 281
***NOTE - VERY WEAK SPRING ADDED FOR STABILITY

NOTE STAAD DETECTS INSTABILITIES AS EXCESSIVE LOSS OF SIGNIFICANT DIGITS
DURING DECOMPOSITION. WHEN A DECOMPOSED DIAGONAL IS LESS THAN THE
BUILT-IN REDUCTION FACTOR TIMES THE ORIGINAL STIFFNESS MATRIX DIAGONAL,
STAAD PRINTS A SINGULARITY NOTICE. THE BUILT-IN REDUCTION FACTOR
IS 1.000E-09

THE ABOVE CONDITIONS COULD ALSO BE CAUSED BY VERY STIFF OR VERY WEAK
ELEMENTS AS WELL AS TRUE SINGULARITIES.

***WARNING - INSTABILITY AT JOINT 872 DIRECTION = MZ
PROBABLE CAUSE SINGULAR-ADDING WEAK SPRING
K-MATRIX DIAG= 1.1596877E+06 L-MATRIX DIAG= -3.7183054E-07 EQN NO 282
***NOTE - VERY WEAK SPRING ADDED FOR STABILITY
**START ITERATION NO. 8

***WARNING - INSTABILITY AT JOINT 872 DIRECTION = FY
PROBABLE CAUSE SINGULAR-ADDING WEAK SPRING
K-MATRIX DIAG= 1.6453036E+04 L-MATRIX DIAG= -2.0736479E-10 EQN NO 280
***NOTE - VERY WEAK SPRING ADDED FOR STABILITY

NOTE STAAD DETECTS INSTABILITIES AS EXCESSIVE LOSS OF SIGNIFICANT DIGITS
DURING DECOMPOSITION. WHEN A DECOMPOSED DIAGONAL IS LESS THAN THE
BUILT-IN REDUCTION FACTOR TIMES THE ORIGINAL STIFFNESS MATRIX DIAGONAL,
STAAD PRINTS A SINGULARITY NOTICE. THE BUILT-IN REDUCTION FACTOR
IS 1.000E-09

THE ABOVE CONDITIONS COULD ALSO BE CAUSED BY VERY STIFF OR VERY WEAK
ELEMENTS AS WELL AS TRUE SINGULARITIES.

***WARNING - INSTABILITY AT JOINT 872 DIRECTION = MX
PROBABLE CAUSE SINGULAR-ADDING WEAK SPRING
K-MATRIX DIAG= 1.2849652E+06 L-MATRIX DIAG= -6.5588392E-07 EQN NO 281

***NOTE - VERY WEAK SPRING ADDED FOR STABILITY

***WARNING - INSTABILITY AT JOINT 872 DIRECTION = MZ
PROBABLE CAUSE SINGULAR-ADDING WEAK SPRING
K-MATRIX DIAG= 1.1596877E+06 L-MATRIX DIAG= -3.7089922E-07 EQN NO 282
***NOTE - VERY WEAK SPRING ADDED FOR STABILITY
**START ITERATION NO. 9
**START ITERATION NO. 10

***WARNING - INSTABILITY AT JOINT 872 DIRECTION = MZ
PROBABLE CAUSE SINGULAR-ADDING WEAK SPRING
K-MATRIX DIAG= 1.1596877E+06 L-MATRIX DIAG= -3.7183054E-07 EQN NO 282
***NOTE - VERY WEAK SPRING ADDED FOR STABILITY

NOTE STAAD DETECTS INSTABILITIES AS EXCESSIVE LOSS OF SIGNIFICANT DIGITS
DURING DECOMPOSITION. WHEN A DECOMPOSED DIAGONAL IS LESS THAN THE
BUILT-IN REDUCTION FACTOR TIMES THE ORIGINAL STIFFNESS MATRIX DIAGONAL,
STAAD PRINTS A SINGULARITY NOTICE. THE BUILT-IN REDUCTION FACTOR
IS 1.000E-09

THE ABOVE CONDITIONS COULD ALSO BE CAUSED BY VERY STIFF OR VERY WEAK
ELEMENTS AS WELL AS TRUE SINGULARITIES.

**START ITERATION NO. 11

***WARNING - INSTABILITY AT JOINT 872 DIRECTION = MZ
PROBABLE CAUSE SINGULAR-ADDING WEAK SPRING
K-MATRIX DIAG= 1.1596877E+06 L-MATRIX DIAG= -3.7159771E-07 EQN NO 282
***NOTE - VERY WEAK SPRING ADDED FOR STABILITY

NOTE STAAD DETECTS INSTABILITIES AS EXCESSIVE LOSS OF SIGNIFICANT DIGITS
DURING DECOMPOSITION. WHEN A DECOMPOSED DIAGONAL IS LESS THAN THE
BUILT-IN REDUCTION FACTOR TIMES THE ORIGINAL STIFFNESS MATRIX DIAGONAL,
STAAD PRINTS A SINGULARITY NOTICE. THE BUILT-IN REDUCTION FACTOR
IS 1.000E-09

THE ABOVE CONDITIONS COULD ALSO BE CAUSED BY VERY STIFF OR VERY WEAK
ELEMENTS AS WELL AS TRUE SINGULARITIES.

**START ITERATION NO. 12

***WARNING - INSTABILITY AT JOINT 872 DIRECTION = MZ
PROBABLE CAUSE SINGULAR-ADDING WEAK SPRING
K-MATRIX DIAG= 1.1596877E+06 L-MATRIX DIAG= -3.7229620E-07 EQN NO 282
***NOTE - VERY WEAK SPRING ADDED FOR STABILITY

NOTE STAAD DETECTS INSTABILITIES AS EXCESSIVE LOSS OF SIGNIFICANT DIGITS
DURING DECOMPOSITION. WHEN A DECOMPOSED DIAGONAL IS LESS THAN THE
BUILT-IN REDUCTION FACTOR TIMES THE ORIGINAL STIFFNESS MATRIX DIAGONAL,
STAAD PRINTS A SINGULARITY NOTICE. THE BUILT-IN REDUCTION FACTOR
IS 1.000E-09

THE ABOVE CONDITIONS COULD ALSO BE CAUSED BY VERY STIFF OR VERY WEAK
ELEMENTS AS WELL AS TRUE SINGULARITIES.

**START ITERATION NO. 13

***WARNING - INSTABILITY AT JOINT 872 DIRECTION = MX
PROBABLE CAUSE SINGULAR-ADDING WEAK SPRING
K-MATRIX DIAG= 1.2849652E+06 L-MATRIX DIAG= -2.1513551E-07 EQN NO 281
***NOTE - VERY WEAK SPRING ADDED FOR STABILITY

NOTE STAAD DETECTS INSTABILITIES AS EXCESSIVE LOSS OF SIGNIFICANT DIGITS
DURING DECOMPOSITION. WHEN A DECOMPOSED DIAGONAL IS LESS THAN THE
BUILT-IN REDUCTION FACTOR TIMES THE ORIGINAL STIFFNESS MATRIX DIAGONAL,
STAAD PRINTS A SINGULARITY NOTICE. THE BUILT-IN REDUCTION FACTOR
IS 1.000E-09

THE ABOVE CONDITIONS COULD ALSO BE CAUSED BY VERY STIFF OR VERY WEAK
ELEMENTS AS WELL AS TRUE SINGULARITIES.

***WARNING - INSTABILITY AT JOINT 872 DIRECTION = MZ
PROBABLE CAUSE SINGULAR-ADDING WEAK SPRING
K-MATRIX DIAG= 1.1596877E+06 L-MATRIX DIAG= -3.7183054E-07 EQN NO 282
***NOTE - VERY WEAK SPRING ADDED FOR STABILITY
**START ITERATION NO. 14

***WARNING - INSTABILITY AT JOINT 872 DIRECTION = FY
PROBABLE CAUSE SINGULAR-ADDING WEAK SPRING
K-MATRIX DIAG= 1.6453036E+04 L-MATRIX DIAG= -2.0736479E-10 EQN NO 280
***NOTE - VERY WEAK SPRING ADDED FOR STABILITY

NOTE STAAD DETECTS INSTABILITIES AS EXCESSIVE LOSS OF SIGNIFICANT DIGITS
DURING DECOMPOSITION. WHEN A DECOMPOSED DIAGONAL IS LESS THAN THE
BUILT-IN REDUCTION FACTOR TIMES THE ORIGINAL STIFFNESS MATRIX DIAGONAL,
STAAD PRINTS A SINGULARITY NOTICE. THE BUILT-IN REDUCTION FACTOR
IS 1.000E-09

THE ABOVE CONDITIONS COULD ALSO BE CAUSED BY VERY STIFF OR VERY WEAK
ELEMENTS AS WELL AS TRUE SINGULARITIES.

***WARNING - INSTABILITY AT JOINT 872 DIRECTION = MX
PROBABLE CAUSE SINGULAR-ADDING WEAK SPRING
K-MATRIX DIAG= 1.2849652E+06 L-MATRIX DIAG= -6.5588392E-07 EQN NO 281
***NOTE - VERY WEAK SPRING ADDED FOR STABILITY

***WARNING - INSTABILITY AT JOINT 872 DIRECTION = MZ
PROBABLE CAUSE SINGULAR-ADDING WEAK SPRING
K-MATRIX DIAG= 1.1596877E+06 L-MATRIX DIAG= -3.7089922E-07 EQN NO 282
***NOTE - VERY WEAK SPRING ADDED FOR STABILITY
**START ITERATION NO. 15

**WARNING-Tension/Compression did not converge after 15 iterations, Case= 5
** If results are not satisfactory then enter a
** SET ITERLIM i command (i>10) before loads to get more iterations.

**START ITERATION NO. 2
**START ITERATION NO. 3
**START ITERATION NO. 4

**START ITERATION NO. 5
**START ITERATION NO. 6
**START ITERATION NO. 7

***WARNING - INSTABILITY AT JOINT 872 DIRECTION = MX
PROBABLE CAUSE SINGULAR-ADDING WEAK SPRING
K-MATRIX DIAG= 1.2849652E+06 L-MATRIX DIAG= -5.0803646E-07 EQN NO 281
***NOTE - VERY WEAK SPRING ADDED FOR STABILITY

NOTE STAAD DETECTS INSTABILITIES AS EXCESSIVE LOSS OF SIGNIFICANT DIGITS
DURING DECOMPOSITION. WHEN A DECOMPOSED DIAGONAL IS LESS THAN THE
BUILT-IN REDUCTION FACTOR TIMES THE ORIGINAL STIFFNESS MATRIX DIAGONAL,
STAAD PRINTS A SINGULARITY NOTICE. THE BUILT-IN REDUCTION FACTOR
IS 1.000E-09

THE ABOVE CONDITIONS COULD ALSO BE CAUSED BY VERY STIFF OR VERY WEAK
ELEMENTS AS WELL AS TRUE SINGULARITIES.

**START ITERATION NO. 8

***WARNING - INSTABILITY AT JOINT 872 DIRECTION = MX
PROBABLE CAUSE SINGULAR-ADDING WEAK SPRING
K-MATRIX DIAG= 1.2849652E+06 L-MATRIX DIAG= -6.7707151E-07 EQN NO 281
***NOTE - VERY WEAK SPRING ADDED FOR STABILITY

NOTE STAAD DETECTS INSTABILITIES AS EXCESSIVE LOSS OF SIGNIFICANT DIGITS
DURING DECOMPOSITION. WHEN A DECOMPOSED DIAGONAL IS LESS THAN THE
BUILT-IN REDUCTION FACTOR TIMES THE ORIGINAL STIFFNESS MATRIX DIAGONAL,
STAAD PRINTS A SINGULARITY NOTICE. THE BUILT-IN REDUCTION FACTOR
IS 1.000E-09

THE ABOVE CONDITIONS COULD ALSO BE CAUSED BY VERY STIFF OR VERY WEAK
ELEMENTS AS WELL AS TRUE SINGULARITIES.

**NOTE-Tension/Compression converged after 8 iterations, Case= 6

**START ITERATION NO. 2
**START ITERATION NO. 3
**START ITERATION NO. 4
**START ITERATION NO. 5
**START ITERATION NO. 6

***WARNING - INSTABILITY AT JOINT 872 DIRECTION = MX
PROBABLE CAUSE SINGULAR-ADDING WEAK SPRING
K-MATRIX DIAG= 1.2849652E+06 L-MATRIX DIAG= -5.6368299E-07 EQN NO 281
***NOTE - VERY WEAK SPRING ADDED FOR STABILITY

NOTE STAAD DETECTS INSTABILITIES AS EXCESSIVE LOSS OF SIGNIFICANT DIGITS
DURING DECOMPOSITION. WHEN A DECOMPOSED DIAGONAL IS LESS THAN THE
BUILT-IN REDUCTION FACTOR TIMES THE ORIGINAL STIFFNESS MATRIX DIAGONAL,
STAAD PRINTS A SINGULARITY NOTICE. THE BUILT-IN REDUCTION FACTOR
IS 1.000E-09

THE ABOVE CONDITIONS COULD ALSO BE CAUSED BY VERY STIFF OR VERY WEAK
ELEMENTS AS WELL AS TRUE SINGULARITIES.

**START ITERATION NO. 7

***WARNING - INSTABILITY AT JOINT 872 DIRECTION = MX
PROBABLE CAUSE SINGULAR-ADDING WEAK SPRING
K-MATRIX DIAG= 1.2849652E+06 L-MATRIX DIAG= -5.3807162E-07 EQN NO 281
***NOTE - VERY WEAK SPRING ADDED FOR STABILITY

NOTE STAAD DETECTS INSTABILITIES AS EXCESSIVE LOSS OF SIGNIFICANT DIGITS
DURING DECOMPOSITION. WHEN A DECOMPOSED DIAGONAL IS LESS THAN THE
BUILT-IN REDUCTION FACTOR TIMES THE ORIGINAL STIFFNESS MATRIX DIAGONAL,
STAAD PRINTS A SINGULARITY NOTICE. THE BUILT-IN REDUCTION FACTOR
IS 1.000E-09

THE ABOVE CONDITIONS COULD ALSO BE CAUSED BY VERY STIFF OR VERY WEAK
ELEMENTS AS WELL AS TRUE SINGULARITIES.

**START ITERATION NO. 8

***WARNING - INSTABILITY AT JOINT 872 DIRECTION = MX
PROBABLE CAUSE SINGULAR-ADDING WEAK SPRING
K-MATRIX DIAG= 1.2849652E+06 L-MATRIX DIAG= -7.0780516E-07 EQN NO 281
***NOTE - VERY WEAK SPRING ADDED FOR STABILITY

NOTE STAAD DETECTS INSTABILITIES AS EXCESSIVE LOSS OF SIGNIFICANT DIGITS
DURING DECOMPOSITION. WHEN A DECOMPOSED DIAGONAL IS LESS THAN THE
BUILT-IN REDUCTION FACTOR TIMES THE ORIGINAL STIFFNESS MATRIX DIAGONAL,
STAAD PRINTS A SINGULARITY NOTICE. THE BUILT-IN REDUCTION FACTOR
IS 1.000E-09

THE ABOVE CONDITIONS COULD ALSO BE CAUSED BY VERY STIFF OR VERY WEAK
ELEMENTS AS WELL AS TRUE SINGULARITIES.

***WARNING - INSTABILITY AT JOINT 872 DIRECTION = MZ
PROBABLE CAUSE SINGULAR-ADDING WEAK SPRING
K-MATRIX DIAG= 1.1596877E+06 L-MATRIX DIAG= -4.1350722E-07 EQN NO 282
***NOTE - VERY WEAK SPRING ADDED FOR STABILITY
**START ITERATION NO. 9

***WARNING - INSTABILITY AT JOINT 872 DIRECTION = FY
PROBABLE CAUSE SINGULAR-ADDING WEAK SPRING
K-MATRIX DIAG= 1.6453036E+04 L-MATRIX DIAG= -2.0736479E-10 EQN NO 280
***NOTE - VERY WEAK SPRING ADDED FOR STABILITY

NOTE STAAD DETECTS INSTABILITIES AS EXCESSIVE LOSS OF SIGNIFICANT DIGITS
DURING DECOMPOSITION. WHEN A DECOMPOSED DIAGONAL IS LESS THAN THE
BUILT-IN REDUCTION FACTOR TIMES THE ORIGINAL STIFFNESS MATRIX DIAGONAL,
STAAD PRINTS A SINGULARITY NOTICE. THE BUILT-IN REDUCTION FACTOR
IS 1.000E-09

THE ABOVE CONDITIONS COULD ALSO BE CAUSED BY VERY STIFF OR VERY WEAK
ELEMENTS AS WELL AS TRUE SINGULARITIES.

***WARNING - INSTABILITY AT JOINT 872 DIRECTION = MX
PROBABLE CAUSE SINGULAR-ADDING WEAK SPRING
K-MATRIX DIAG= 1.2849652E+06 L-MATRIX DIAG= -6.5588392E-07 EQN NO 281
***NOTE - VERY WEAK SPRING ADDED FOR STABILITY

***WARNING - INSTABILITY AT JOINT 872 DIRECTION = MZ
PROBABLE CAUSE SINGULAR-ADDING WEAK SPRING
K-MATRIX DIAG= 1.1596877E+06 L-MATRIX DIAG= -3.7089922E-07 EQN NO 282
***NOTE - VERY WEAK SPRING ADDED FOR STABILITY

**NOTE-Tension/Compression converged after 9 iterations, Case= 7

**START ITERATION NO. 2
**START ITERATION NO. 3
**START ITERATION NO. 4
**START ITERATION NO. 5

***WARNING - INSTABILITY AT JOINT 872 DIRECTION = MX
PROBABLE CAUSE SINGULAR-ADDING WEAK SPRING
K-MATRIX DIAG= 1.2849652E+06 L-MATRIX DIAG= -5.0128438E-07 EQN NO 281
***NOTE - VERY WEAK SPRING ADDED FOR STABILITY

NOTE STAAD DETECTS INSTABILITIES AS EXCESSIVE LOSS OF SIGNIFICANT DIGITS
DURING DECOMPOSITION. WHEN A DECOMPOSED DIAGONAL IS LESS THAN THE
BUILT-IN REDUCTION FACTOR TIMES THE ORIGINAL STIFFNESS MATRIX DIAGONAL,
STAAD PRINTS A SINGULARITY NOTICE. THE BUILT-IN REDUCTION FACTOR
IS 1.000E-09

THE ABOVE CONDITIONS COULD ALSO BE CAUSED BY VERY STIFF OR VERY WEAK
ELEMENTS AS WELL AS TRUE SINGULARITIES.

**NOTE-Tension/Compression converged after 5 iterations, Case= 8

**NOTE-Tension/Compression converged after 1 iterations, Case= 9

**START ITERATION NO. 2
**START ITERATION NO. 3
**START ITERATION NO. 4
**START ITERATION NO. 5

***WARNING - INSTABILITY AT JOINT 872 DIRECTION = MX
PROBABLE CAUSE SINGULAR-ADDING WEAK SPRING
K-MATRIX DIAG= 1.2849652E+06 L-MATRIX DIAG= -1.2107193E-07 EQN NO 281
***NOTE - VERY WEAK SPRING ADDED FOR STABILITY

NOTE STAAD DETECTS INSTABILITIES AS EXCESSIVE LOSS OF SIGNIFICANT DIGITS
DURING DECOMPOSITION. WHEN A DECOMPOSED DIAGONAL IS LESS THAN THE
BUILT-IN REDUCTION FACTOR TIMES THE ORIGINAL STIFFNESS MATRIX DIAGONAL,
STAAD PRINTS A SINGULARITY NOTICE. THE BUILT-IN REDUCTION FACTOR
IS 1.000E-09

THE ABOVE CONDITIONS COULD ALSO BE CAUSED BY VERY STIFF OR VERY WEAK
ELEMENTS AS WELL AS TRUE SINGULARITIES.

**START ITERATION NO. 6

***WARNING - INSTABILITY AT JOINT 872 DIRECTION = MX
PROBABLE CAUSE SINGULAR-ADDING WEAK SPRING
K-MATRIX DIAG= 1.2849652E+06 L-MATRIX DIAG= -2.0582229E-07 EQN NO 281
***NOTE - VERY WEAK SPRING ADDED FOR STABILITY

NOTE STAAD DETECTS INSTABILITIES AS EXCESSIVE LOSS OF SIGNIFICANT DIGITS
DURING DECOMPOSITION. WHEN A DECOMPOSED DIAGONAL IS LESS THAN THE
BUILT-IN REDUCTION FACTOR TIMES THE ORIGINAL STIFFNESS MATRIX DIAGONAL,
STAAD PRINTS A SINGULARITY NOTICE. THE BUILT-IN REDUCTION FACTOR
IS 1.000E-09

THE ABOVE CONDITIONS COULD ALSO BE CAUSED BY VERY STIFF OR VERY WEAK
ELEMENTS AS WELL AS TRUE SINGULARITIES.

**START ITERATION NO. 7

***WARNING - INSTABILITY AT JOINT 872 DIRECTION = MX
PROBABLE CAUSE SINGULAR-ADDING WEAK SPRING
K-MATRIX DIAG= 1.2849652E+06 L-MATRIX DIAG= -2.1513551E-07 EQN NO 281
***NOTE - VERY WEAK SPRING ADDED FOR STABILITY

NOTE STAAD DETECTS INSTABILITIES AS EXCESSIVE LOSS OF SIGNIFICANT DIGITS
DURING DECOMPOSITION. WHEN A DECOMPOSED DIAGONAL IS LESS THAN THE
BUILT-IN REDUCTION FACTOR TIMES THE ORIGINAL STIFFNESS MATRIX DIAGONAL,
STAAD PRINTS A SINGULARITY NOTICE. THE BUILT-IN REDUCTION FACTOR
IS 1.000E-09

THE ABOVE CONDITIONS COULD ALSO BE CAUSED BY VERY STIFF OR VERY WEAK
ELEMENTS AS WELL AS TRUE SINGULARITIES.

***WARNING - INSTABILITY AT JOINT 872 DIRECTION = MZ
PROBABLE CAUSE SINGULAR-ADDING WEAK SPRING
K-MATRIX DIAG= 1.1596877E+06 L-MATRIX DIAG= -3.7183054E-07 EQN NO 282
***NOTE - VERY WEAK SPRING ADDED FOR STABILITY
**START ITERATION NO. 8

***WARNING - INSTABILITY AT JOINT 872 DIRECTION = FY
PROBABLE CAUSE SINGULAR-ADDING WEAK SPRING
K-MATRIX DIAG= 1.6453036E+04 L-MATRIX DIAG= -2.0736479E-10 EQN NO 280
***NOTE - VERY WEAK SPRING ADDED FOR STABILITY

NOTE STAAD DETECTS INSTABILITIES AS EXCESSIVE LOSS OF SIGNIFICANT DIGITS
DURING DECOMPOSITION. WHEN A DECOMPOSED DIAGONAL IS LESS THAN THE
BUILT-IN REDUCTION FACTOR TIMES THE ORIGINAL STIFFNESS MATRIX DIAGONAL,
STAAD PRINTS A SINGULARITY NOTICE. THE BUILT-IN REDUCTION FACTOR
IS 1.000E-09

THE ABOVE CONDITIONS COULD ALSO BE CAUSED BY VERY STIFF OR VERY WEAK
ELEMENTS AS WELL AS TRUE SINGULARITIES.

***WARNING - INSTABILITY AT JOINT 872 DIRECTION = MX
PROBABLE CAUSE SINGULAR-ADDING WEAK SPRING
K-MATRIX DIAG= 1.2849652E+06 L-MATRIX DIAG= -6.5588392E-07 EQN NO 281
***NOTE - VERY WEAK SPRING ADDED FOR STABILITY

***WARNING - INSTABILITY AT JOINT 872 DIRECTION = MZ
PROBABLE CAUSE SINGULAR-ADDING WEAK SPRING
K-MATRIX DIAG= 1.1596877E+06 L-MATRIX DIAG= -3.7089922E-07 EQN NO 282
***NOTE - VERY WEAK SPRING ADDED FOR STABILITY

**NOTE-Tension/Compression converged after 8 iterations, Case= 10

**NOTE-Tension/Compression converged after 1 iterations, Case= 101

**START ITERATION NO. 2

**NOTE-Tension/Compression converged after 2 iterations, Case= 102

**NOTE-Tension/Compression converged after 1 iterations, Case= 103

**NOTE-Tension/Compression converged after 1 iterations, Case= 104

**NOTE-Tension/Compression converged after 1 iterations, Case= 105

**NOTE-Tension/Compression converged after 1 iterations, Case= 106

**NOTE-Tension/Compression converged after 1 iterations, Case= 107

**NOTE-Tension/Compression converged after 1 iterations, Case= 108

**NOTE-Tension/Compression converged after 1 iterations, Case= 109

**NOTE-Tension/Compression converged after 1 iterations, Case= 110

**NOTE-Tension/Compression converged after 1 iterations, Case= 111

**NOTE-Tension/Compression converged after 1 iterations, Case= 112

**NOTE-Tension/Compression converged after 1 iterations, Case= 113

**NOTE-Tension/Compression converged after 1 iterations, Case= 114

**START ITERATION NO. 2
**START ITERATION NO. 3

**NOTE-Tension/Compression converged after 3 iterations, Case= 115

**START ITERATION NO. 2
**START ITERATION NO. 3

**NOTE-Tension/Compression converged after 3 iterations, Case= 116

**NOTE-Tension/Compression converged after 1 iterations, Case= 201

**NOTE-Tension/Compression converged after 1 iterations, Case= 202

**NOTE-Tension/Compression converged after 1 iterations, Case= 203

**NOTE-Tension/Compression converged after 1 iterations, Case= 204

**NOTE-Tension/Compression converged after 1 iterations, Case= 205

**NOTE-Tension/Compression converged after 1 iterations, Case= 206

**NOTE-Tension/Compression converged after 1 iterations, Case= 207

**NOTE-Tension/Compression converged after 1 iterations, Case= 208

**NOTE-Tension/Compression converged after 1 iterations, Case= 209

**NOTE-Tension/Compression converged after 1 iterations, Case= 210

**NOTE-Tension/Compression converged after 1 iterations, Case= 211

**NOTE-Tension/Compression converged after 1 iterations, Case= 212

**NOTE-Tension/Compression converged after 1 iterations, Case= 213

**NOTE-Tension/Compression converged after 1 iterations, Case= 214

**START ITERATION NO. 2
**START ITERATION NO. 3

**NOTE-Tension/Compression converged after 3 iterations, Case= 215

**START ITERATION NO. 2
**START ITERATION NO. 3

**NOTE-Tension/Compression converged after 3 iterations, Case= 216

**NOTE-Tension/Compression converged after 1 iterations, Case= 217

**NOTE-Tension/Compression converged after 1 iterations, Case= 218

338. LOAD LIST 101 TO 116
339. START CONCRETE DESIGN

```
340. CODE ACI
341. TRACK 2 ALL
342. FC 35000 MEMB 529
343. CLB 0.05 MEMB 529
344. CLS 0.05 ALL
345. MAXMAIN 25 ALL
346. MINMAIN 25 ALL
347. MINSEC 10 ALL
348. DESIGN COLUMN 529
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================================================================

     COLUMN  NO.   529  DESIGN PER ACI 318-08 - AXIAL + BENDING


FY - 413.7  FC -  35.0 MPA,  SQRE SIZE - 600.0 X 600.0 MMS, TIED
        ONLY MINIMUM STEEL IS REQUIRED.
        AREA OF STEEL REQUIRED =  3600.0  SQ. MM

BAR CONFIGURATION       REINF PCT.   LOAD   LOCATION   PHI
-------------------------------------------------------------

  8 - 25 MM               1.091      101      END     0.650
(PROVIDE EQUAL NUMBER OF BARS ON EACH FACE)
TIE BAR NUMBER   10 SPACING 400.00 MM

    COLUMN INTERACTION: MOMENT ABOUT Z -AXIS (KN-MET)

 ------------------------------------------------------------
   P0        Pn max    P-bal.    M-bal.     e-bal. (MM)
12217.70   9774.16   4481.49   1062.73     237.1
   M0        P-tens.   Des.Pn    Des.Mn     e/h
  420.14  -1624.53     59.66      0.00    0.00000
 ------------------------------------------------------------

    COLUMN INTERACTION: MOMENT ABOUT Y -AXIS (KN-MET)

 ------------------------------------------------------------
   P0        Pn max    P-bal.    M-bal.     e-bal. (MM)
12217.70   9774.16   4481.49   1062.73     237.1
   M0        P-tens.   Des.Pn    Des.Mn     e/h
  420.14  -1624.53     59.66      0.00    0.00000
 ------------------------------------------------------------
                          Pn       Mn       Pn       Mn    (@ Z )
          |              9022.30   709.00  4511.15  1061.99
       P0 |*             8270.44   812.98  3759.29  1019.91
          | *            7518.58   897.94  3007.43   956.53
   Pn,max|__*           6766.73   962.48  2255.58   875.37
          |   *          6014.87  1009.20  1503.72   752.23
   Pn     |    *         5263.01  1041.19   751.86   595.66
  NOMINAL|     *          Pn       Mn       Pn       Mn    (@ Y )
    AXIAL|      *       9022.30   709.00  4511.15  1061.99
COMPRESSION|       *     8270.44   812.98  3759.29  1019.91
        Pb|-------*Mb    7518.58   897.94  3007.43   956.53
          |      *       6766.73   962.48  2255.58   875.37
__________|____*_______  6014.87  1009.20  1503.72   752.23
          |  * M0   Mn,  5263.01  1041.19   751.86   595.66
          | *   BENDING
   P-tens |*     MOMENT
          |


********************END OF COLUMN DESIGN RESULTS********************
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